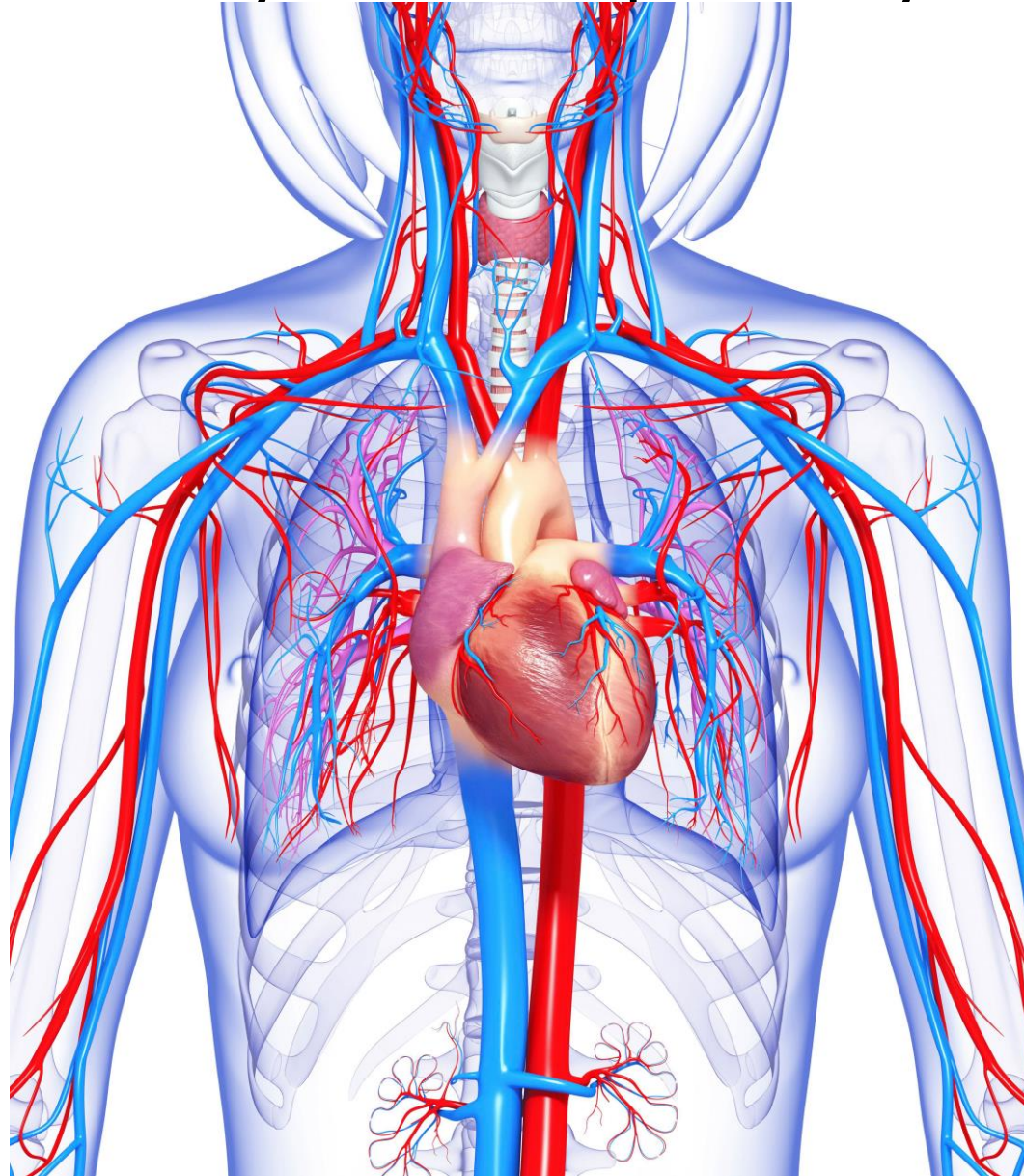
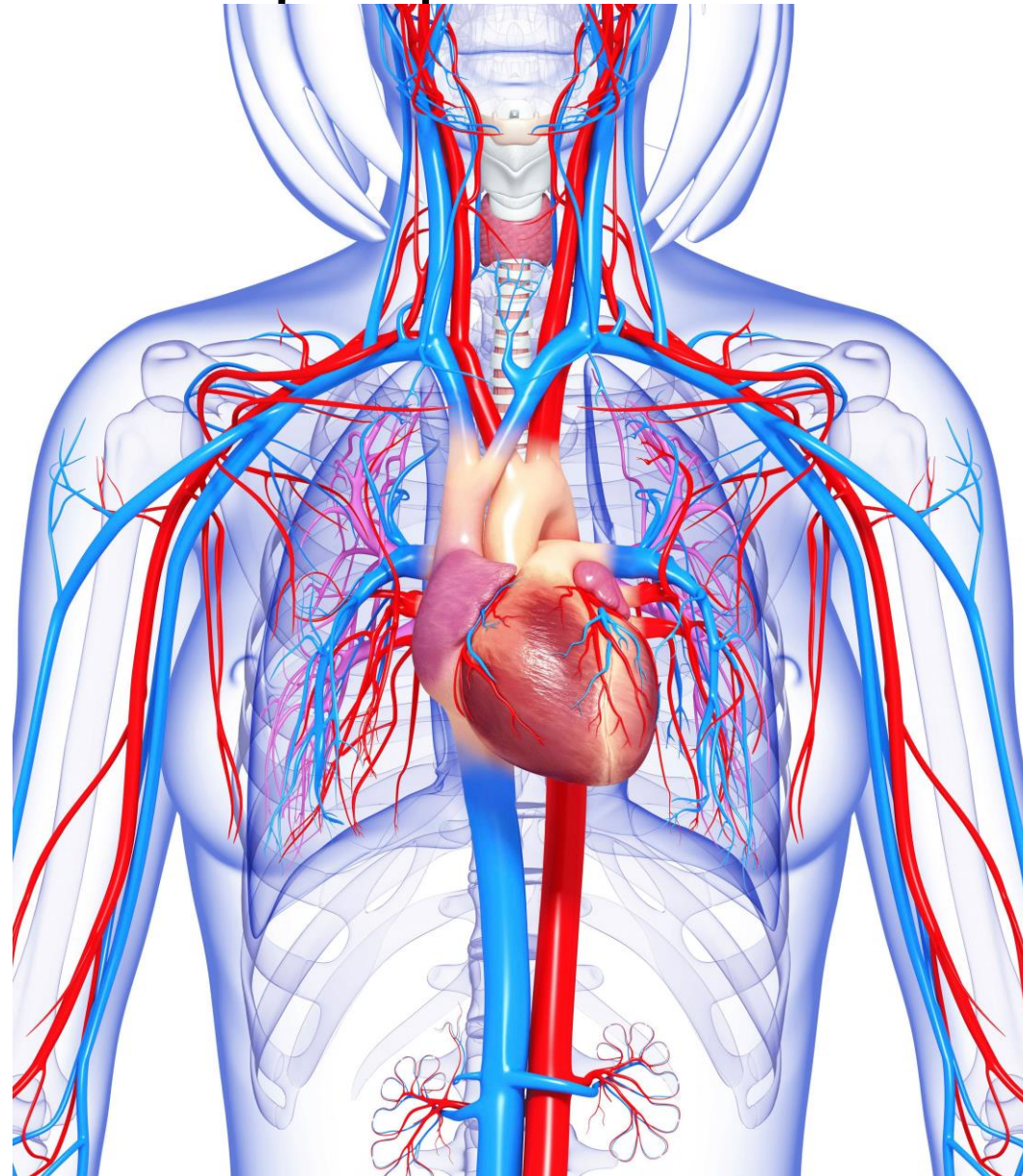


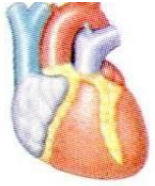
THE CARDIOVASCULAR SYSTEM

The Body's Transport System



The Body's Transport

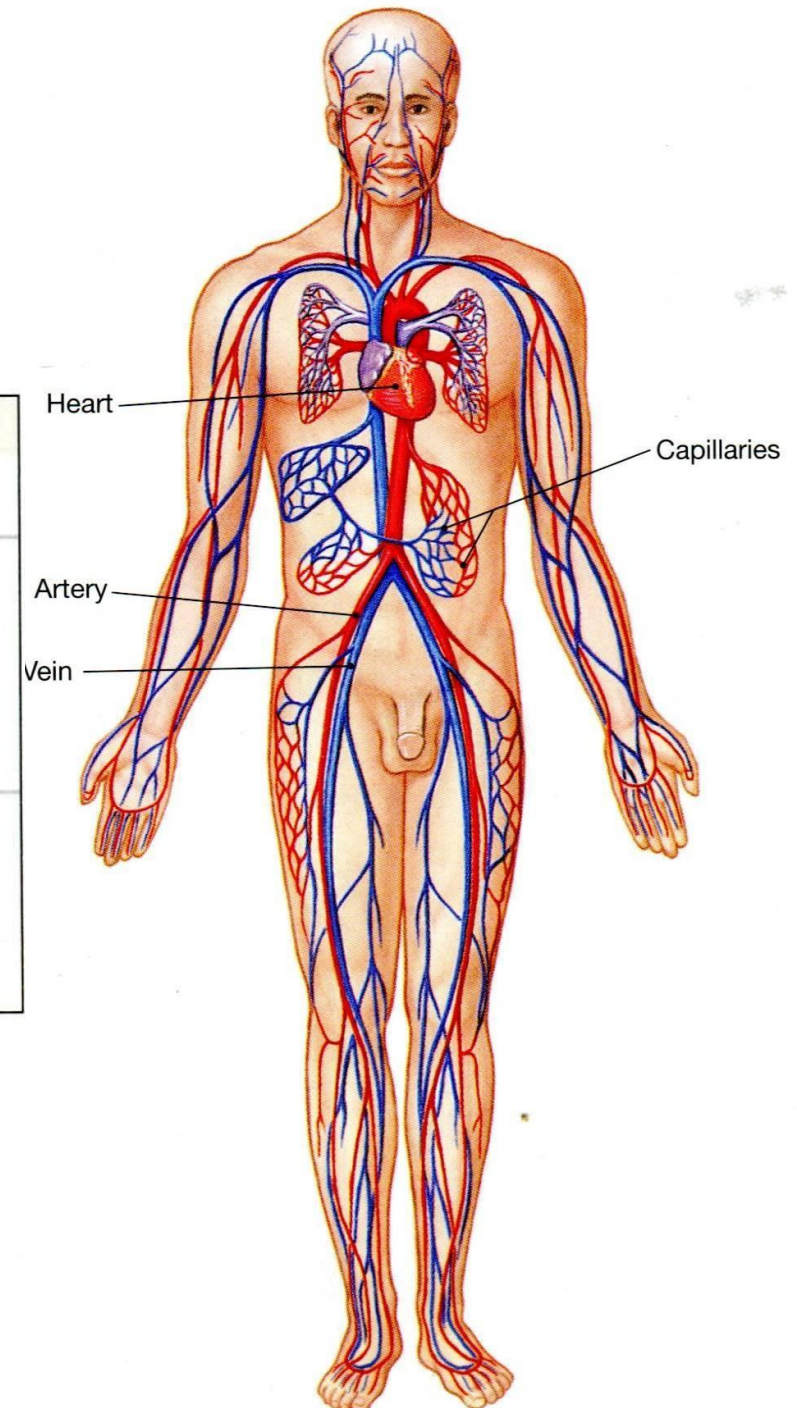




(f) THE CARDIOVASCULAR SYSTEM

Transports cells and dissolved materials, including nutrients, wastes, and gases

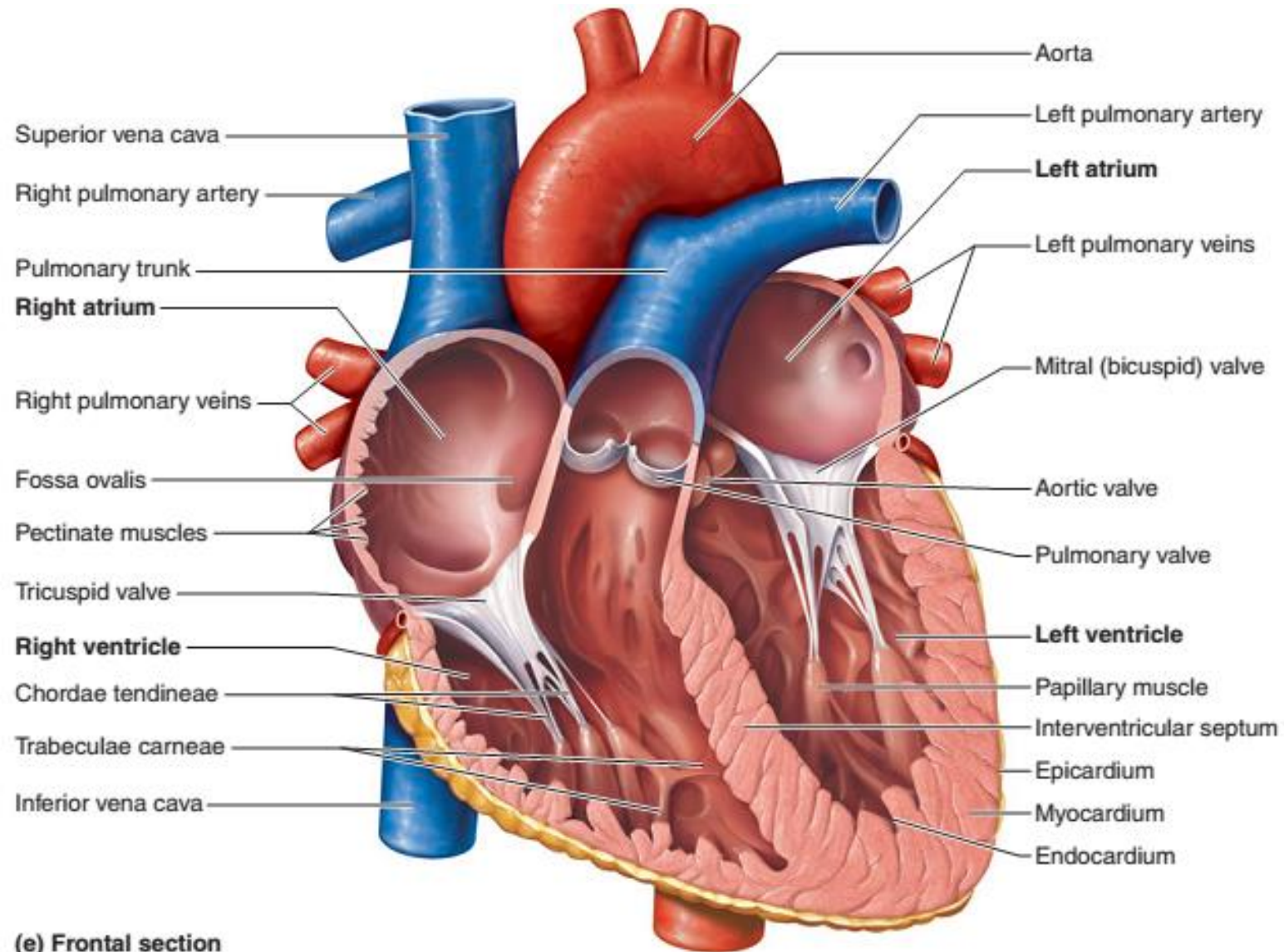
Organ/Component	Primary Functions
Heart	Propels blood; maintains blood pressure
Blood Vessels	Distribute blood around the body
Arteries	Carry blood from heart to capillaries
Capillaries	Permit diffusion between blood and interstitial fluids
Veins	Return blood from capillaries to the heart
Blood	Transports oxygen, carbon dioxide, and blood cells; delivers nutrients and hormones; removes waste products; assists in temperature regulation and defense against disease



The circulatory system

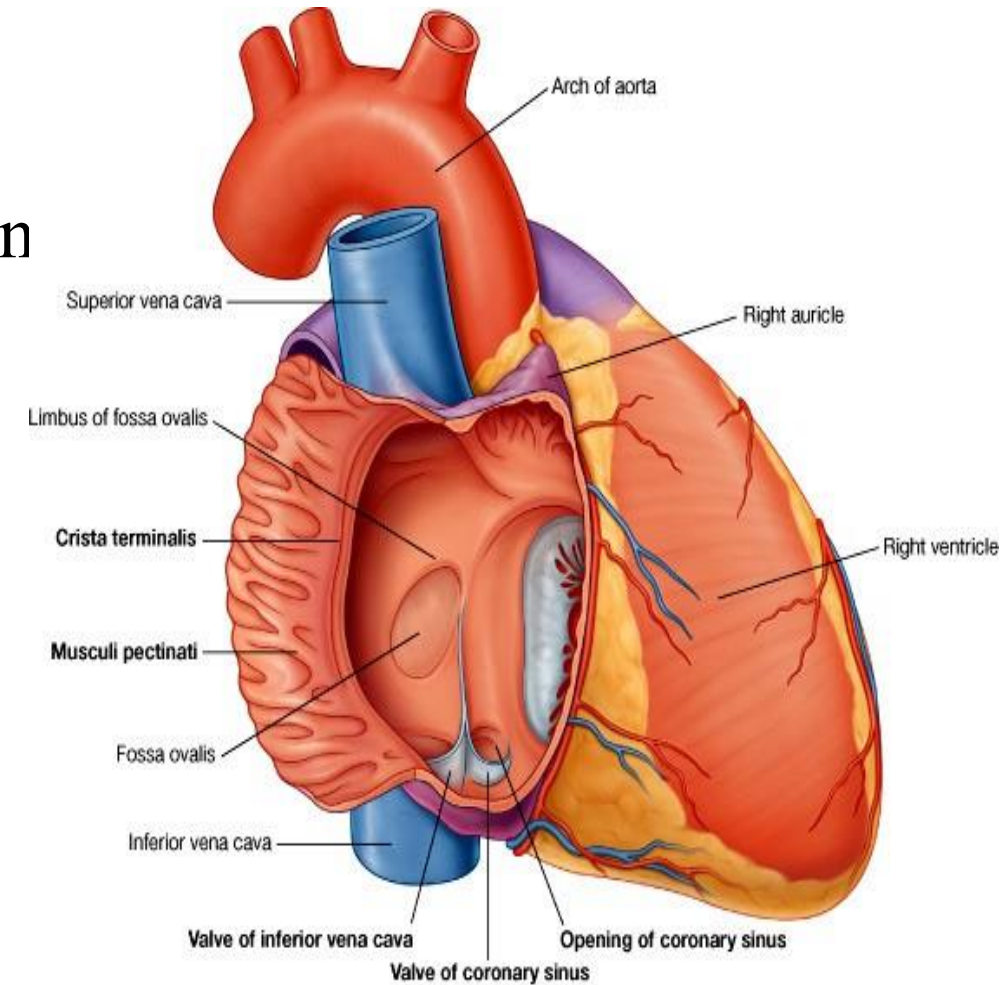
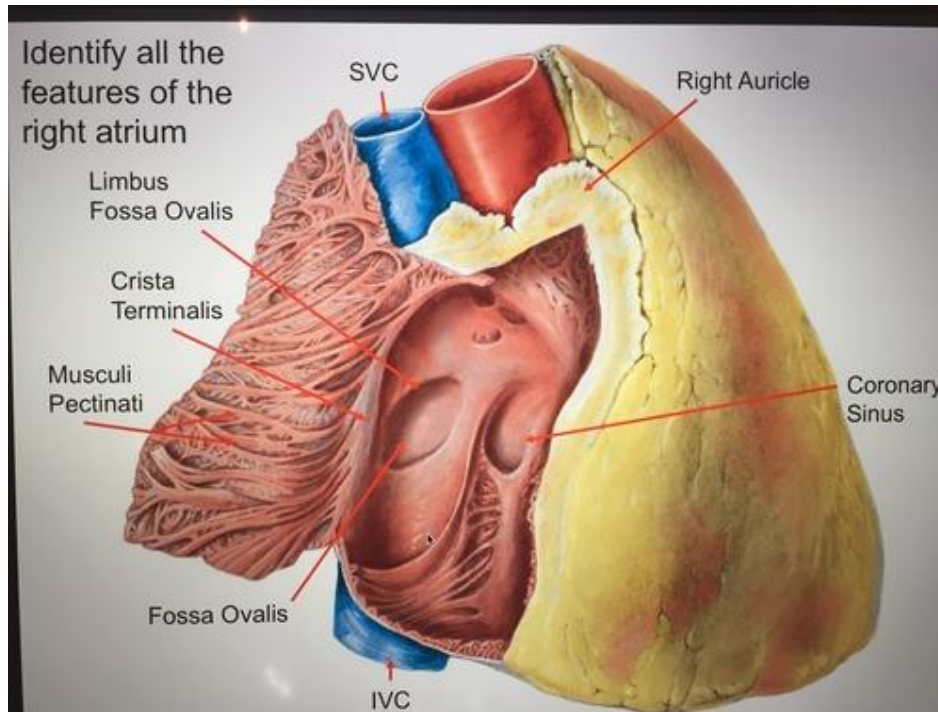
- ✓ Sometimes called the cardiovascular system
- ✓ Consists of:
 - i. Heart
 - ii. blood vessels
 - iii. blood.
- ✓ It transports oxygen, hormones and nutrients to all the cells in the body
- ✓ It picks up waste products generated by metabolic processes and delivers them to other organs for disposal.
- ✓ The heart provides the "muscle" needed to pump blood throughout the body.
- ✓ The system circulates blood in two circuits:
 - i. Pulmonary circuit
 - ii. Systemic circuit.

THE HEART



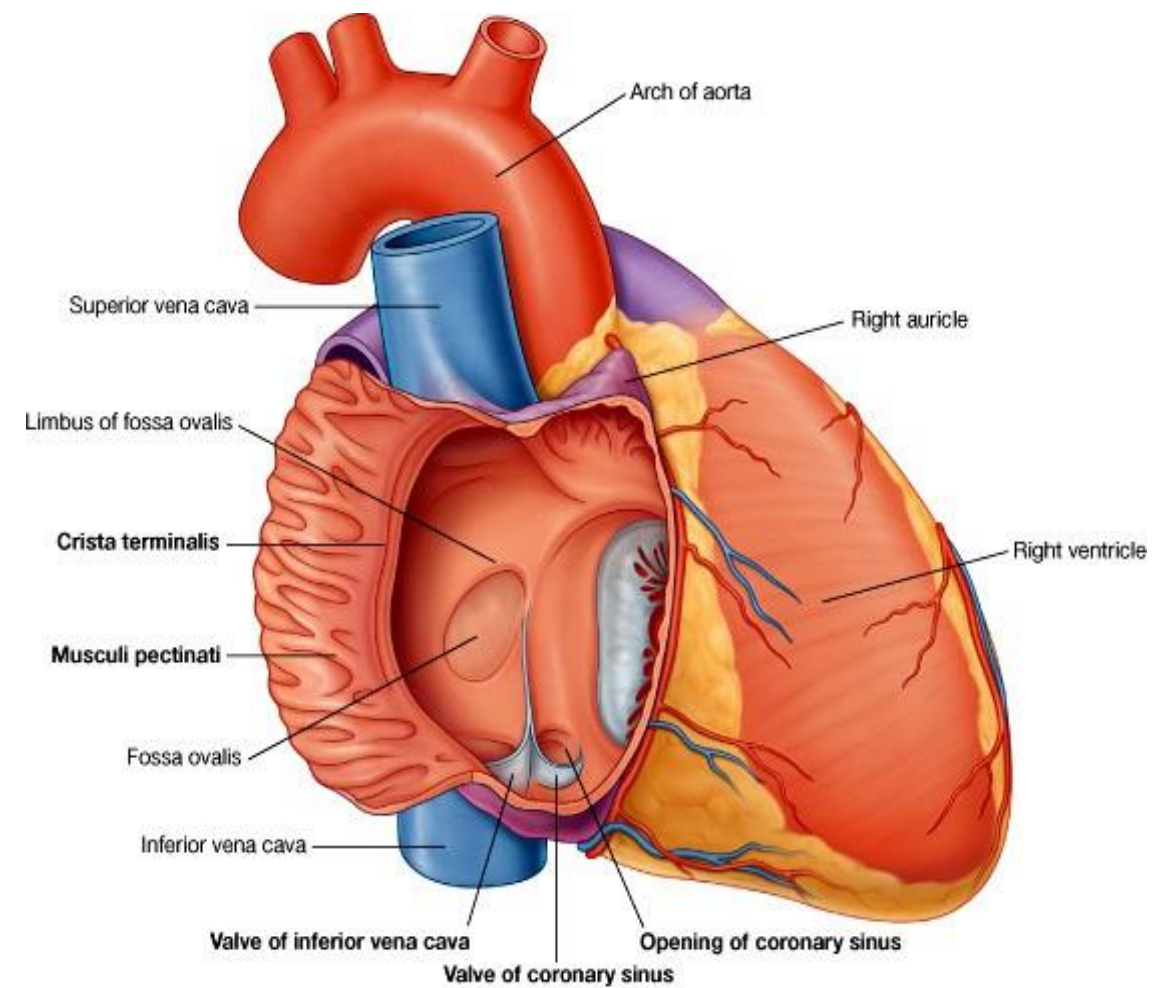
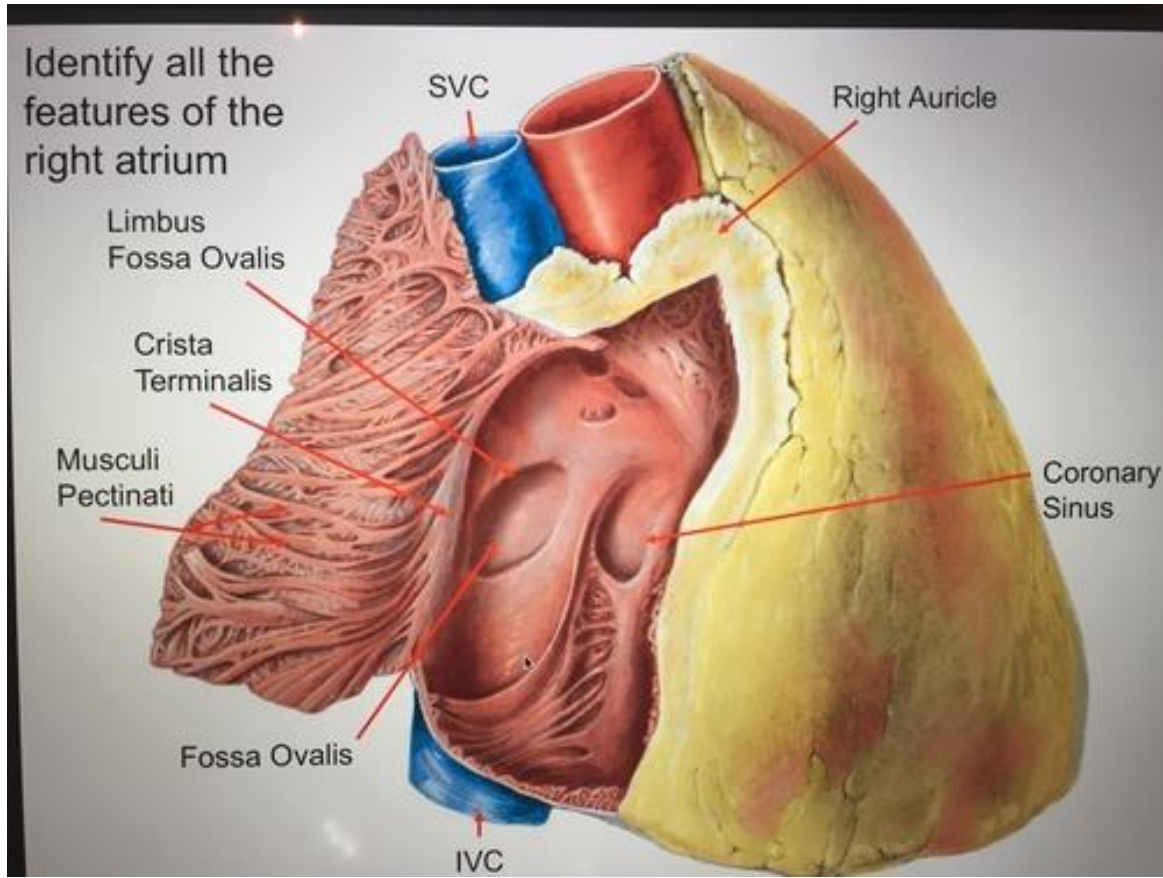
Right Atrium

- ✓ Consists of a main cavity & auricle (a small outpouching)
- ✓ On the outside of the heart at the junction between the right atrium & the right auricle is a vertical groove, the **sulcus terminalis**
- ✓ Which on the inside forms a ridge, the **crista terminalis**.



Right Atrium

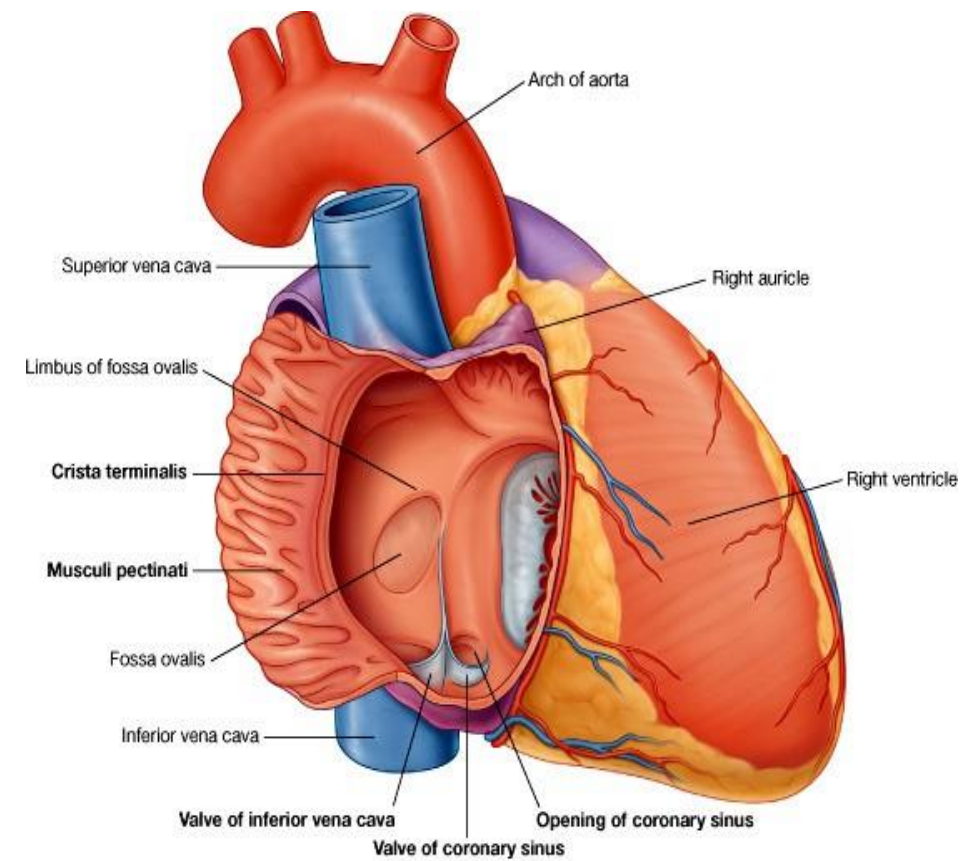
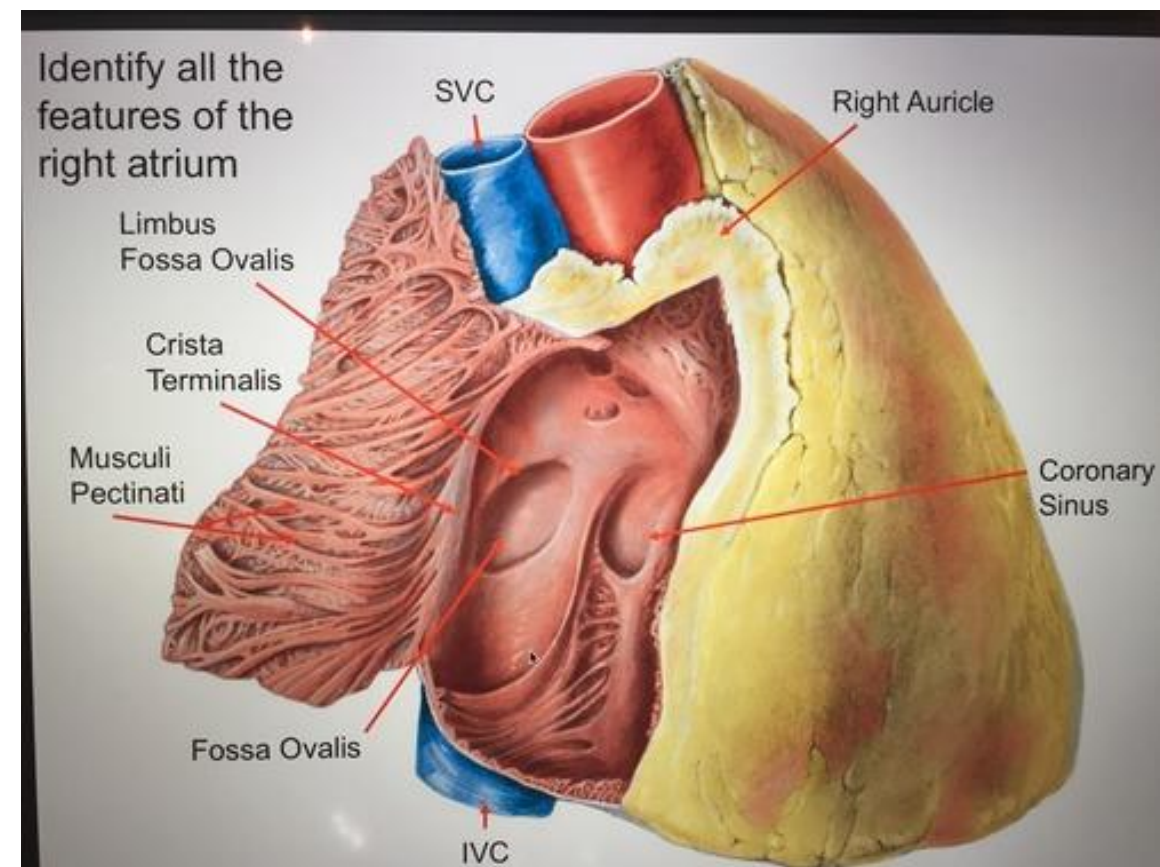
- ✓ Main part of the atrium
- ✓ Lies posterior to the ridge
- ✓ Is smooth walled



- ✓ Is derived embryologically from the **sinus venosus**.

Right Atrium

- ✓ The part of the atrium in front of the ridge is roughened
- ✓ Or trabeculated by bundles of muscle fibers, the **musculi pectinati**

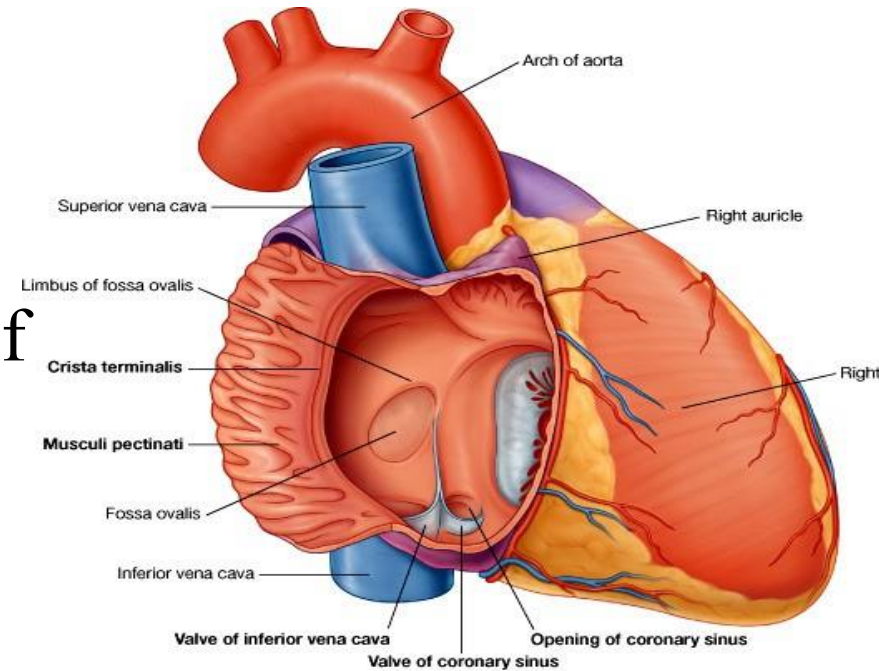


- ✓ Musculi pectinati run from the crista terminalis to the auricle.
- ✓ This anterior part is derived embryologically from the **primitive atrium**

Openings into the Right Atrium

Superior vena cava

- ✓ Opens into the upper part of the right atrium
- ✓ It has no valve.
- ✓ It returns the blood to the heart from the upper half of the body.



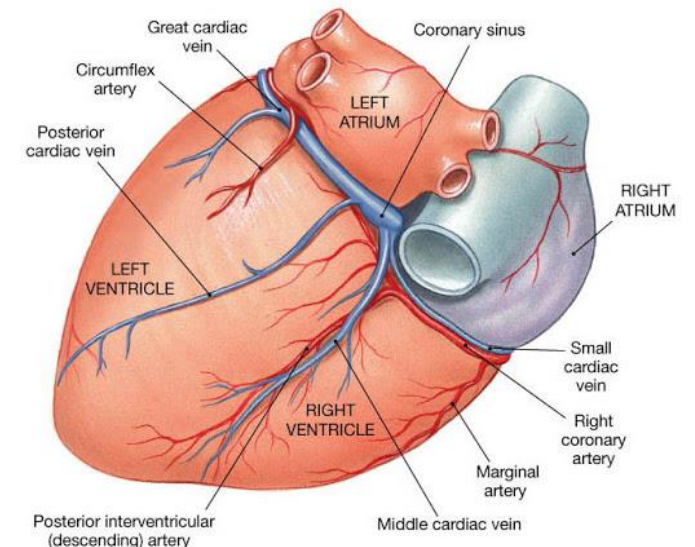
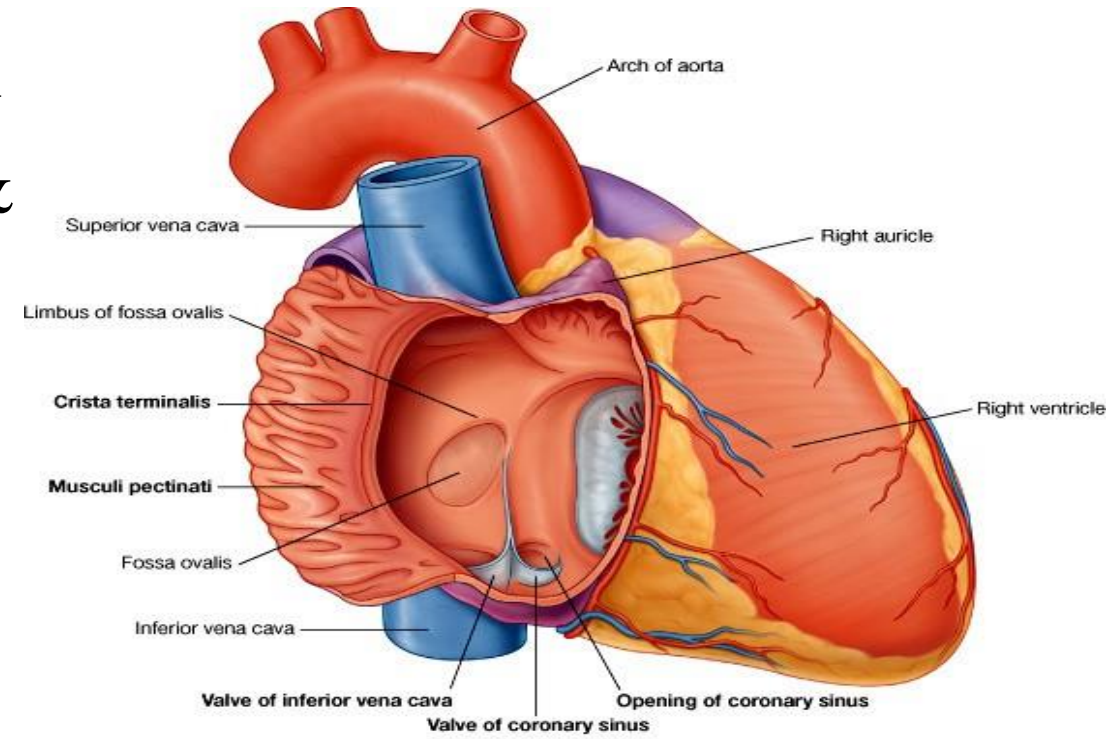
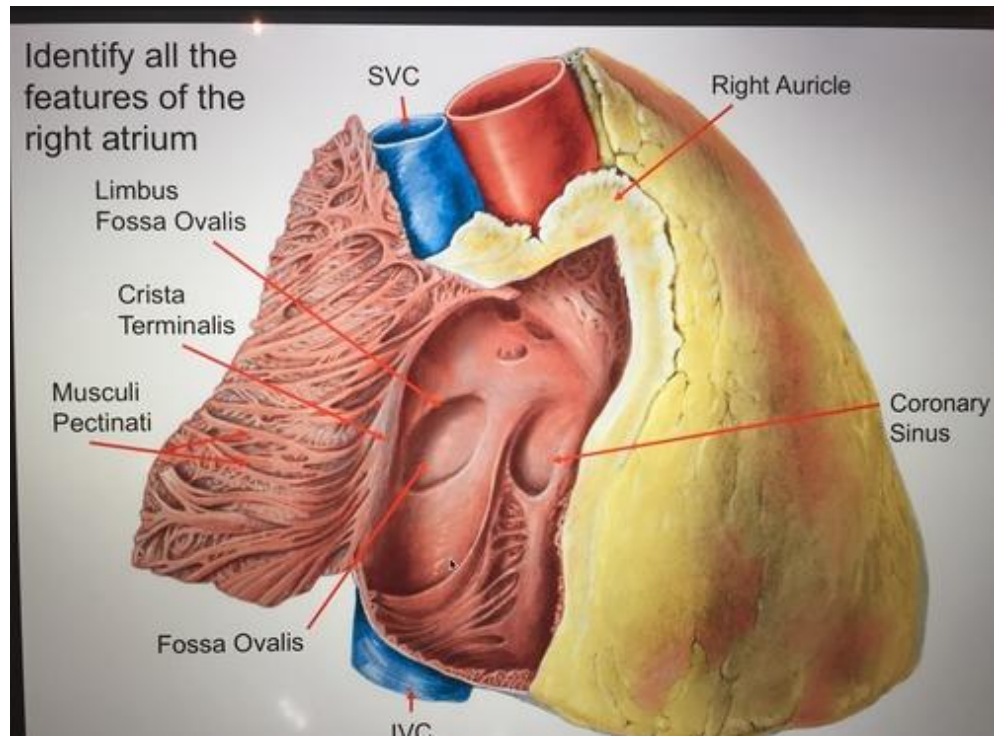
Inferior vena cava

- ✓ larger than the superior vena cava
- ✓ opens into the lower part of the right atrium
- ✓ It is guarded by a rudimentary, non-functioning valve (Eustachian valve).
- ✓ It returns the blood to the heart from the lower half of the body.

Openings into the Right Atrium

Coronary sinus

- ✓ Drains most of the blood from the heart wall
- ✓ Opens into the right atrium between the IVC & the atrioventricular orifice.
- ✓ It is guarded by a rudimentary non-functioning valve (**Thebesian valve**).



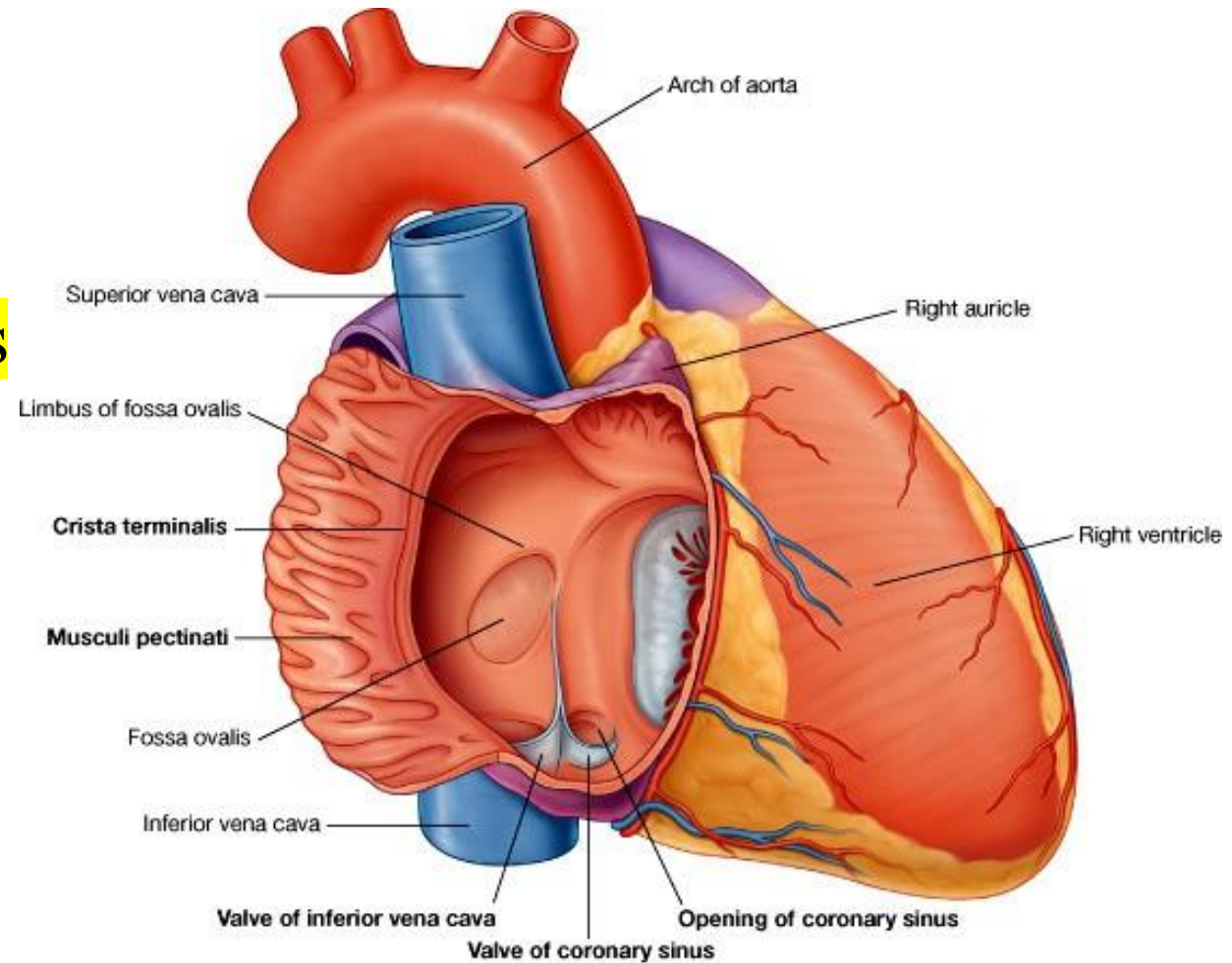
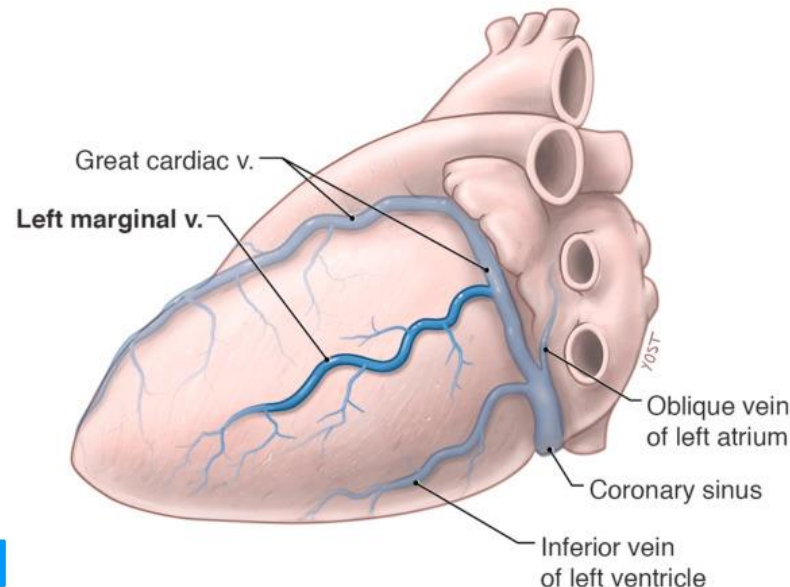
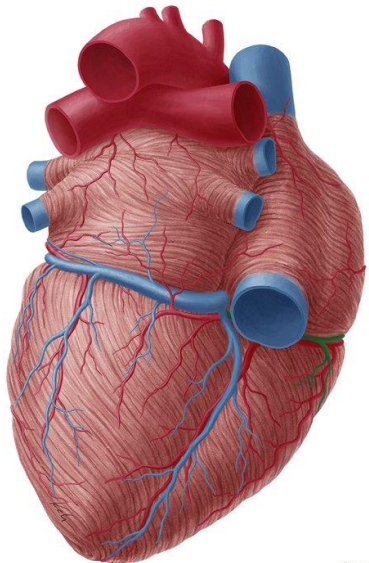
Openings into the Right Atrium

Right atrioventricular orifice

- ✓ Lies anterior to the IVC opening & is guarded by the tricuspid valve.

Many small orifices of small cardiac veins

- ✓ Drain the wall of the heart and open directly into the right atrium.

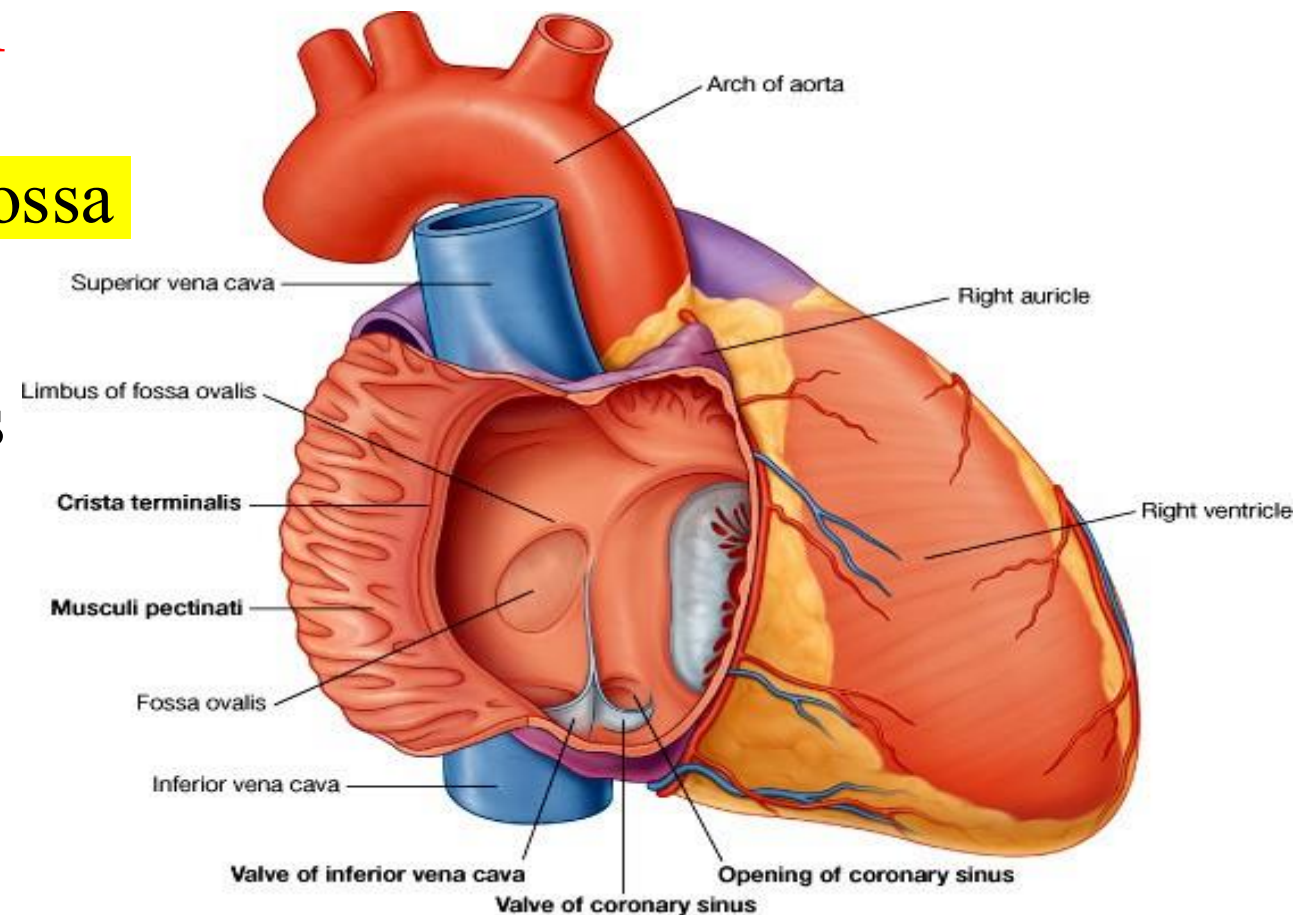
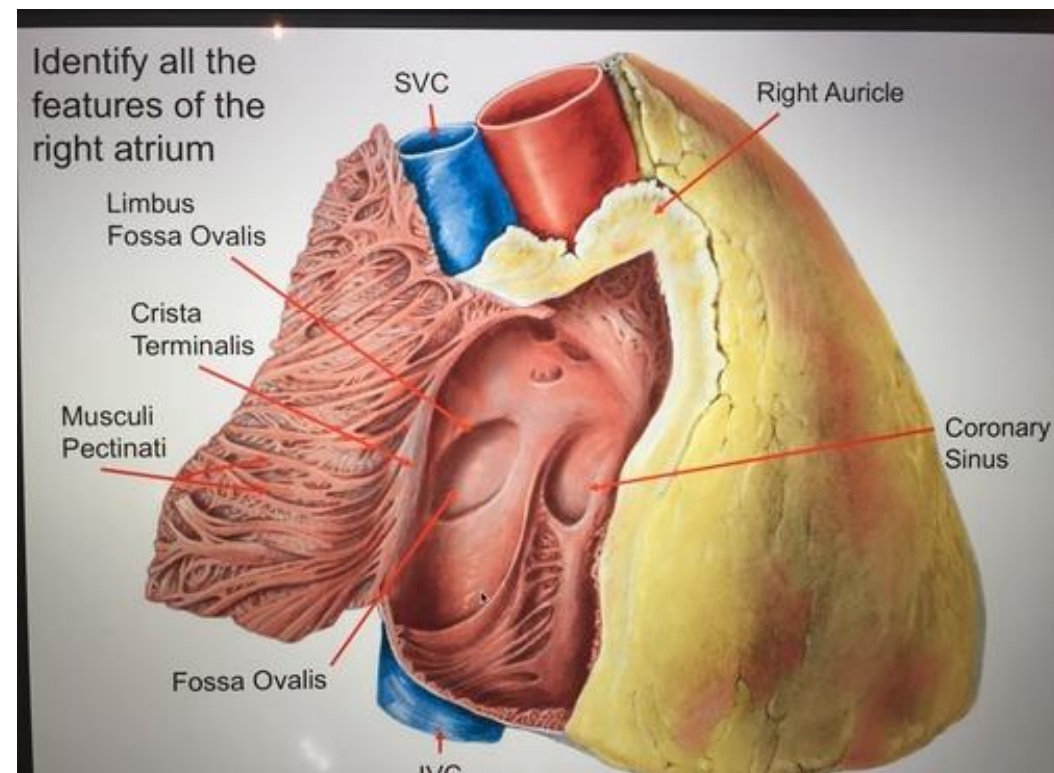


Right Atrium

Fetal Remnants

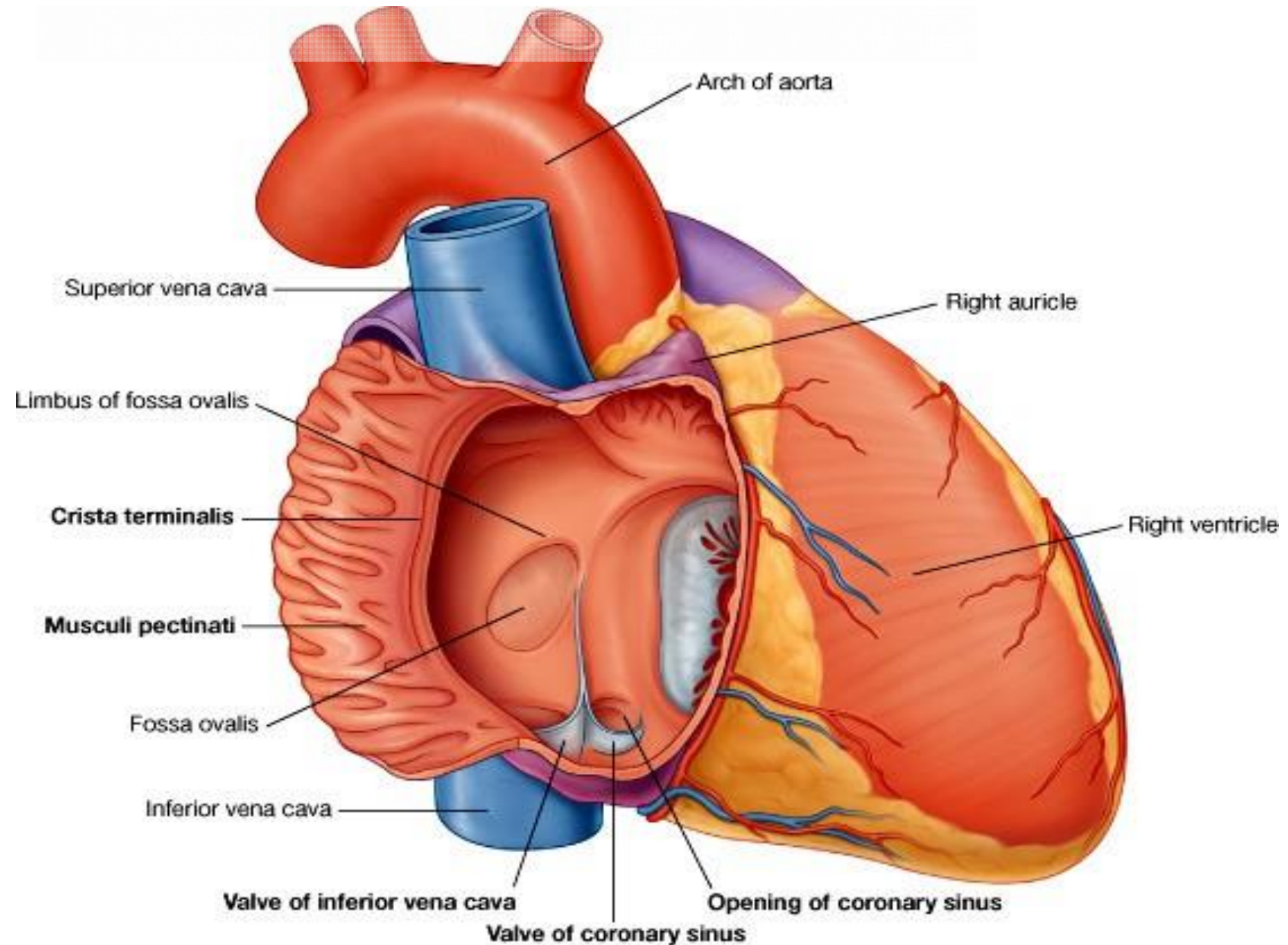
Fossa ovalis & anulus ovalis (limbus of fossa ovalis).

✓ lie on the atrial septum which separates the right atrium from the left atrium .



✓ The fossa ovalis is a shallow depression which is the site of the **foramen ovale** in the fetus .

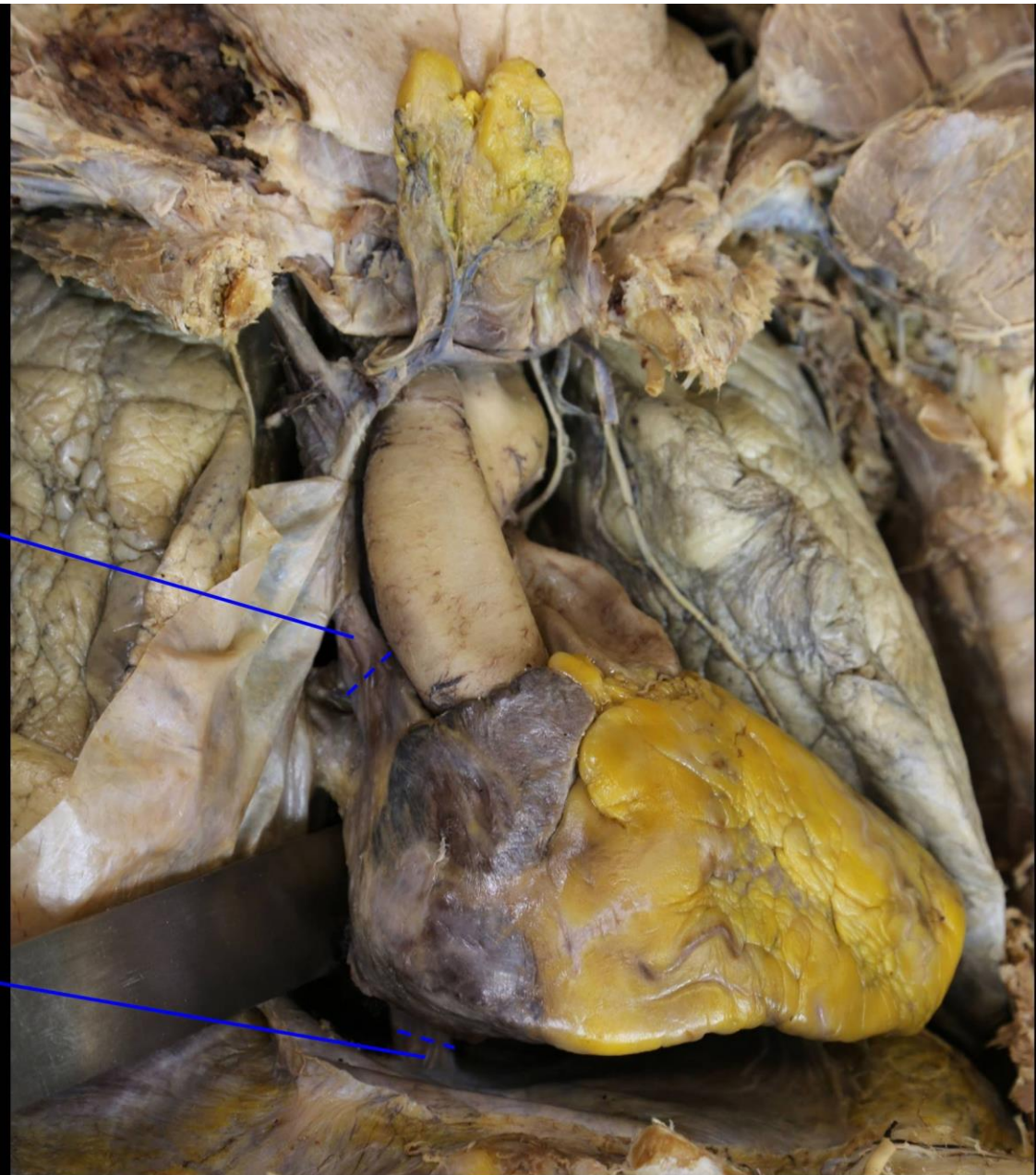
Internal view of right atrium.



Right Atrium

Superior vena
cava

Inferior vena
cava

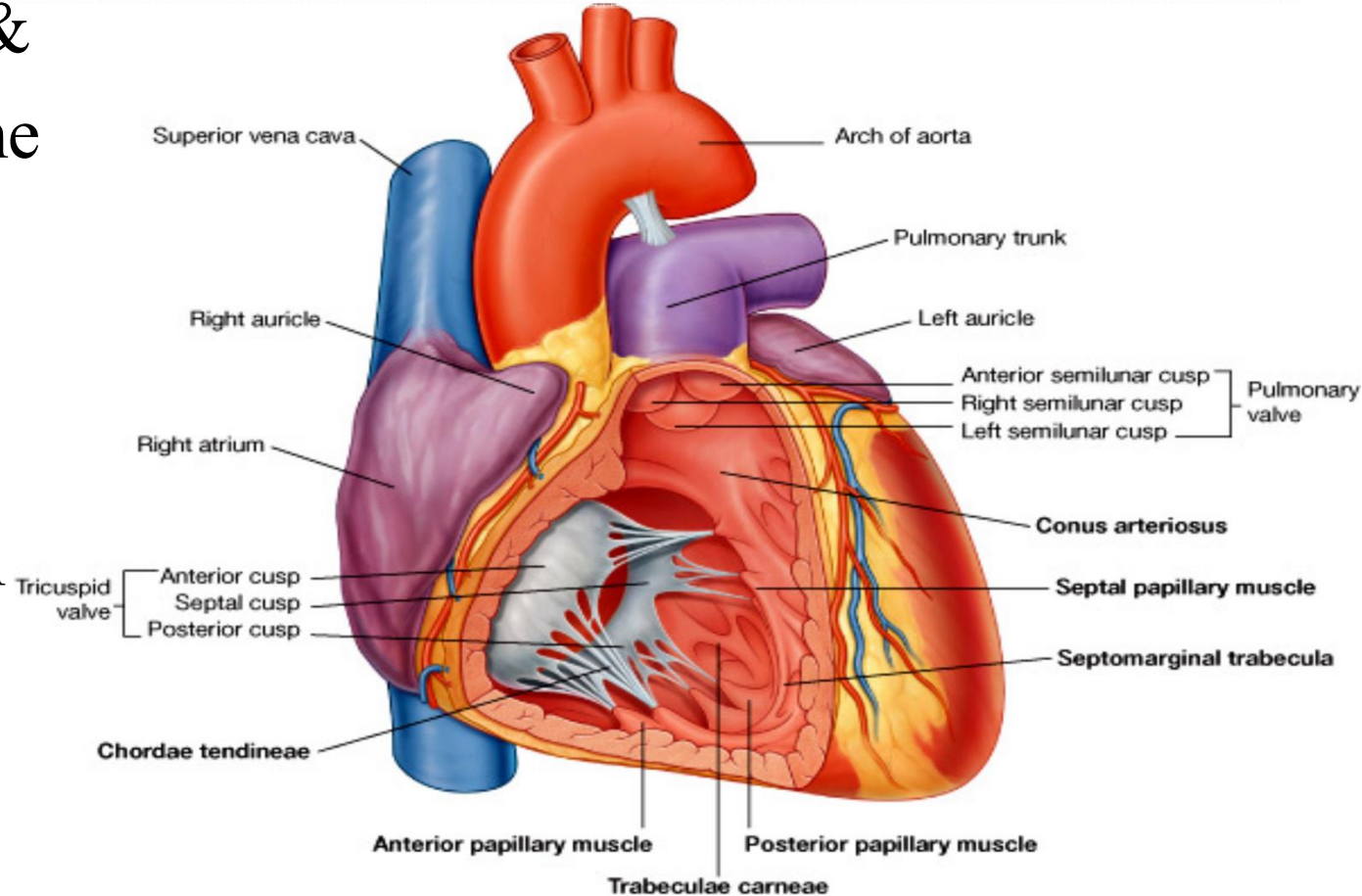


Heart, anterolateral

Right Ventricle

✓ Communicates with the right atrium through **the atrioventricular orifice** & with the pulmonary trunk through the **pulmonary orifice**.

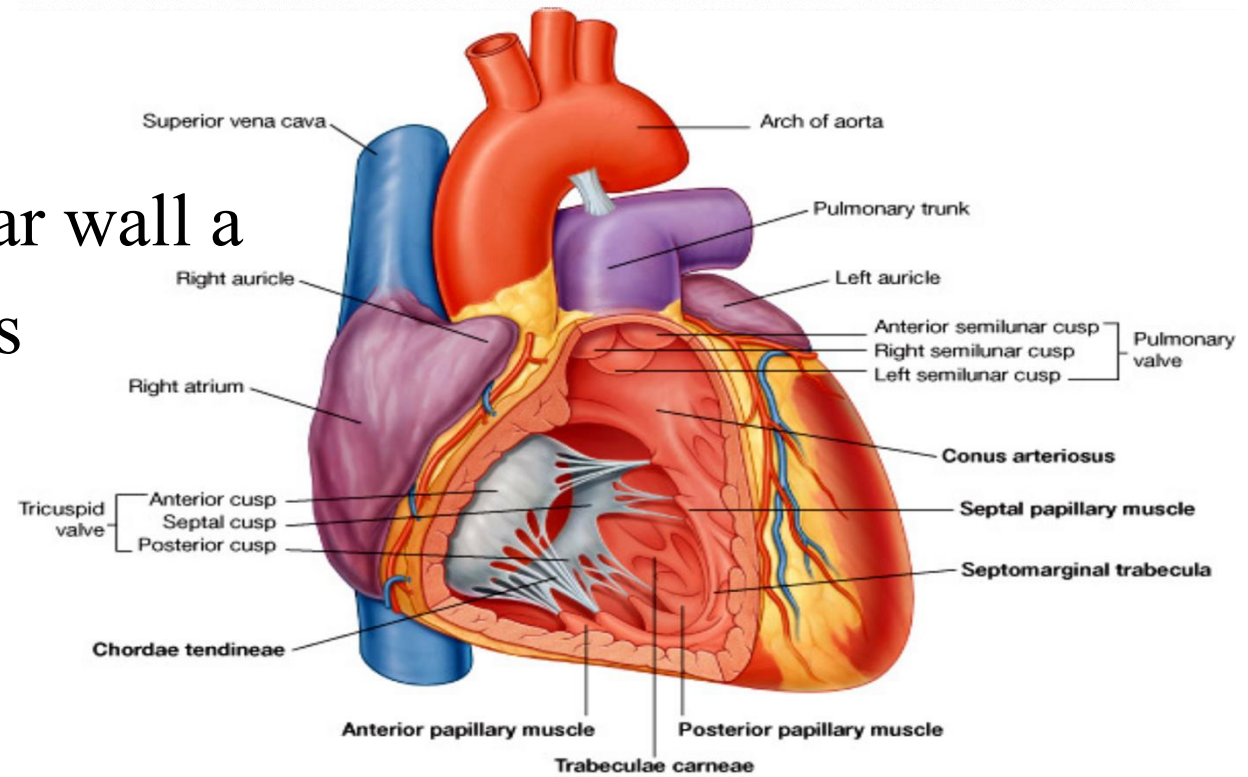
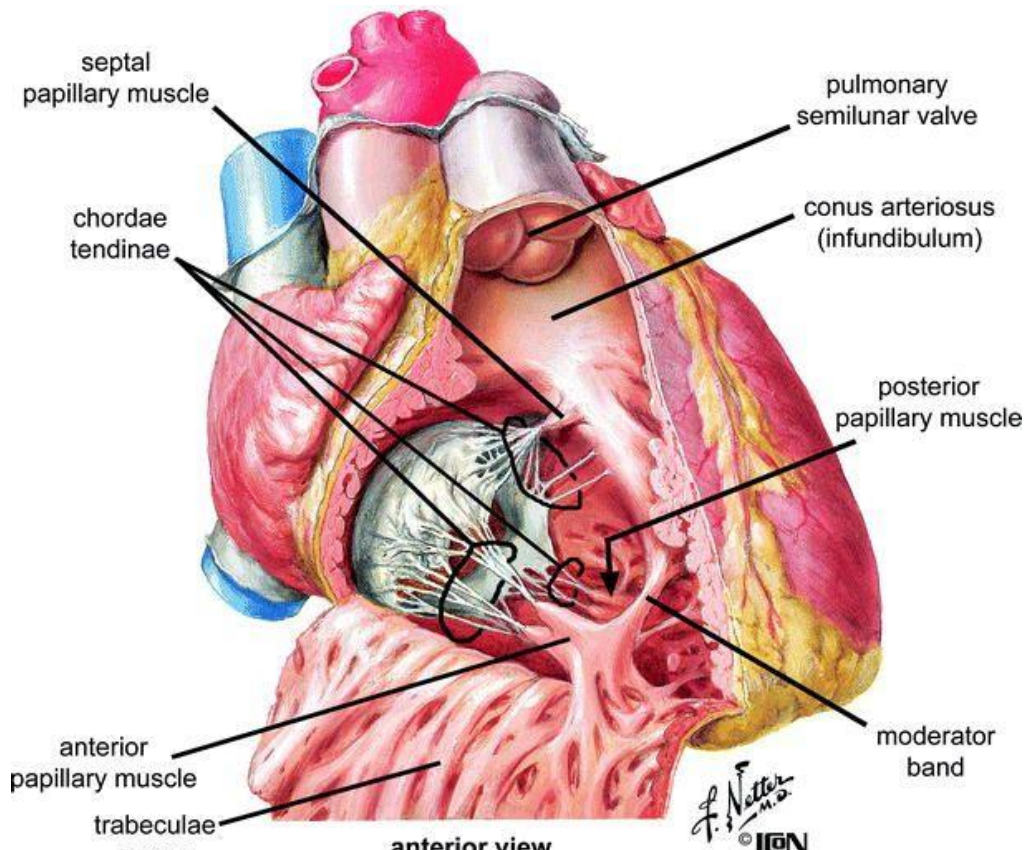
✓ As the cavity approaches the pulmonary orifice it becomes funnel shaped (**infundibulum**).



✓ The walls of the right ventricle are much thicker than those of the right atrium & show several internal projecting ridges formed of muscle bundles

Right Ventricle

- ✓ The projecting ridges give the ventricular wall a spongelike appearance and are known as **trabeculae carneae**



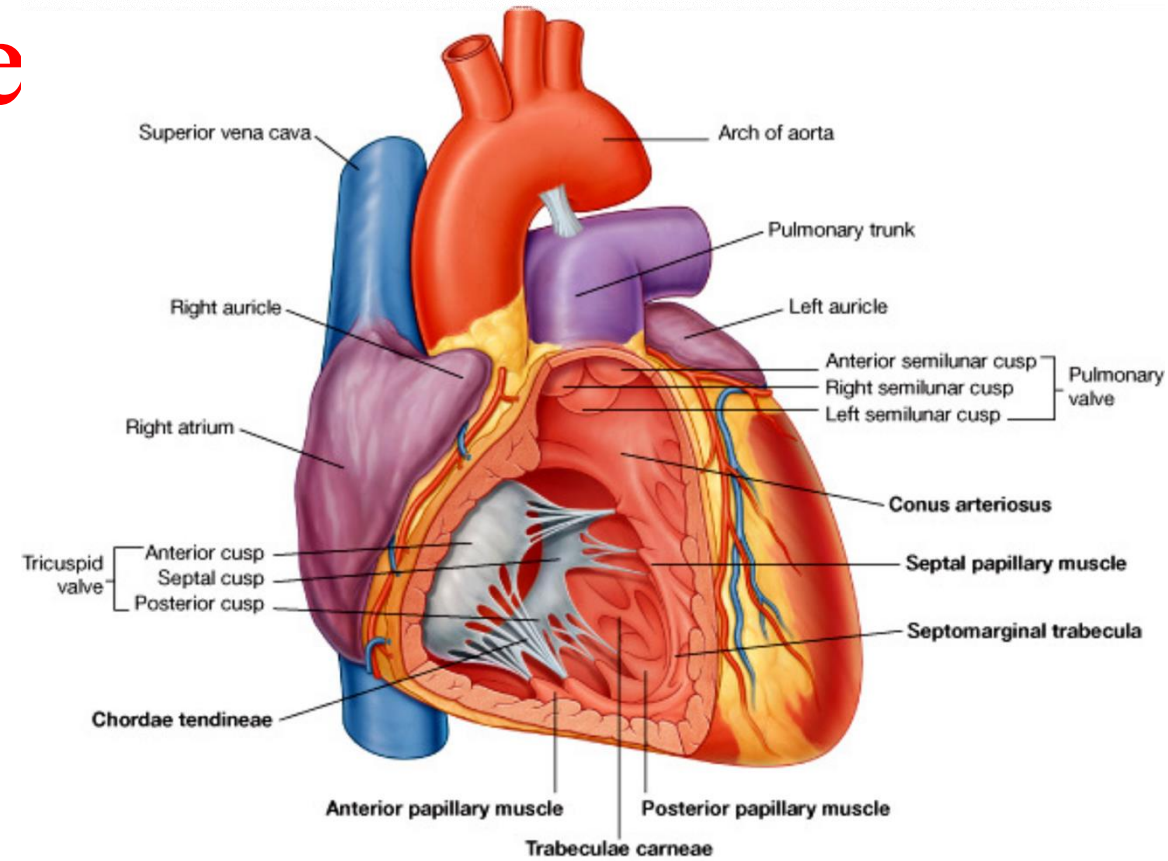
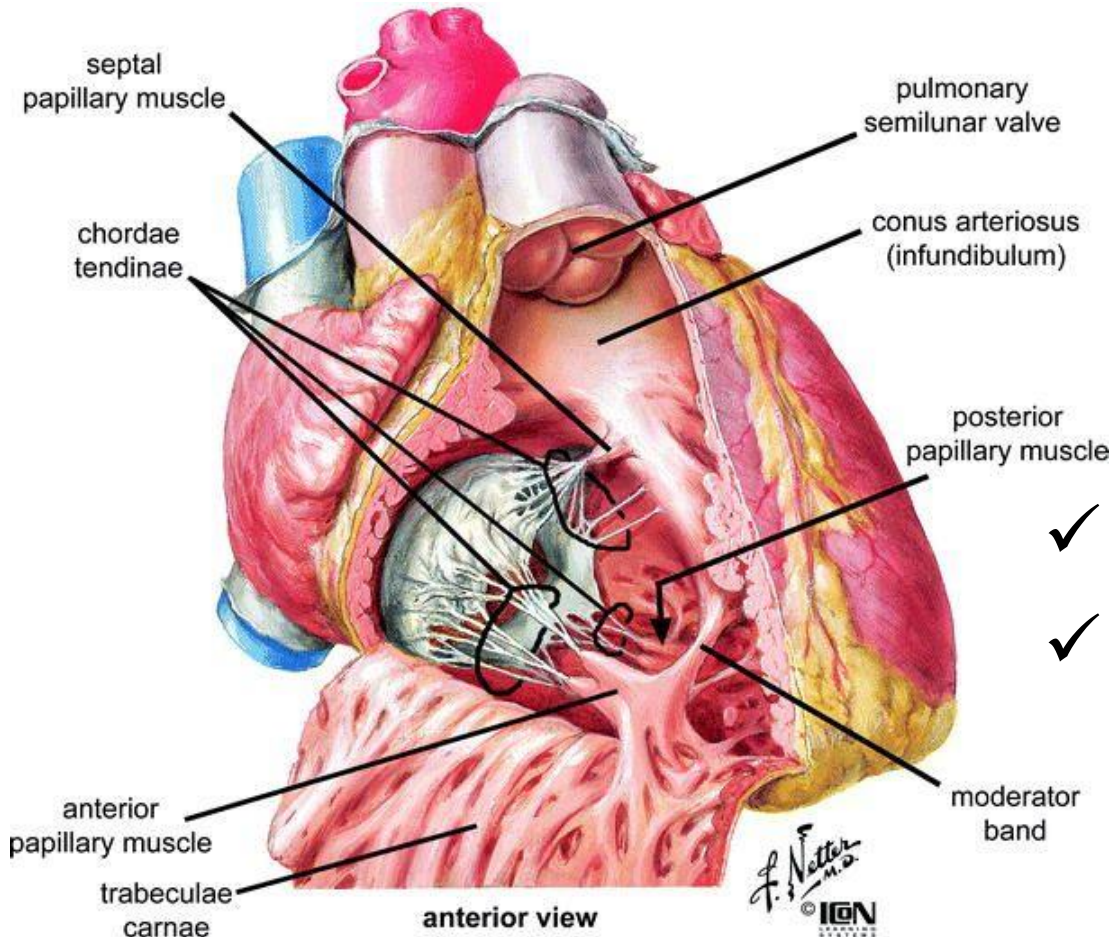
3 types of trabeculae carneae:

- papillary muscles
- Septomarginal trabecula (moderator band)
- prominent ridges

Right Ventricle

Papillary muscles

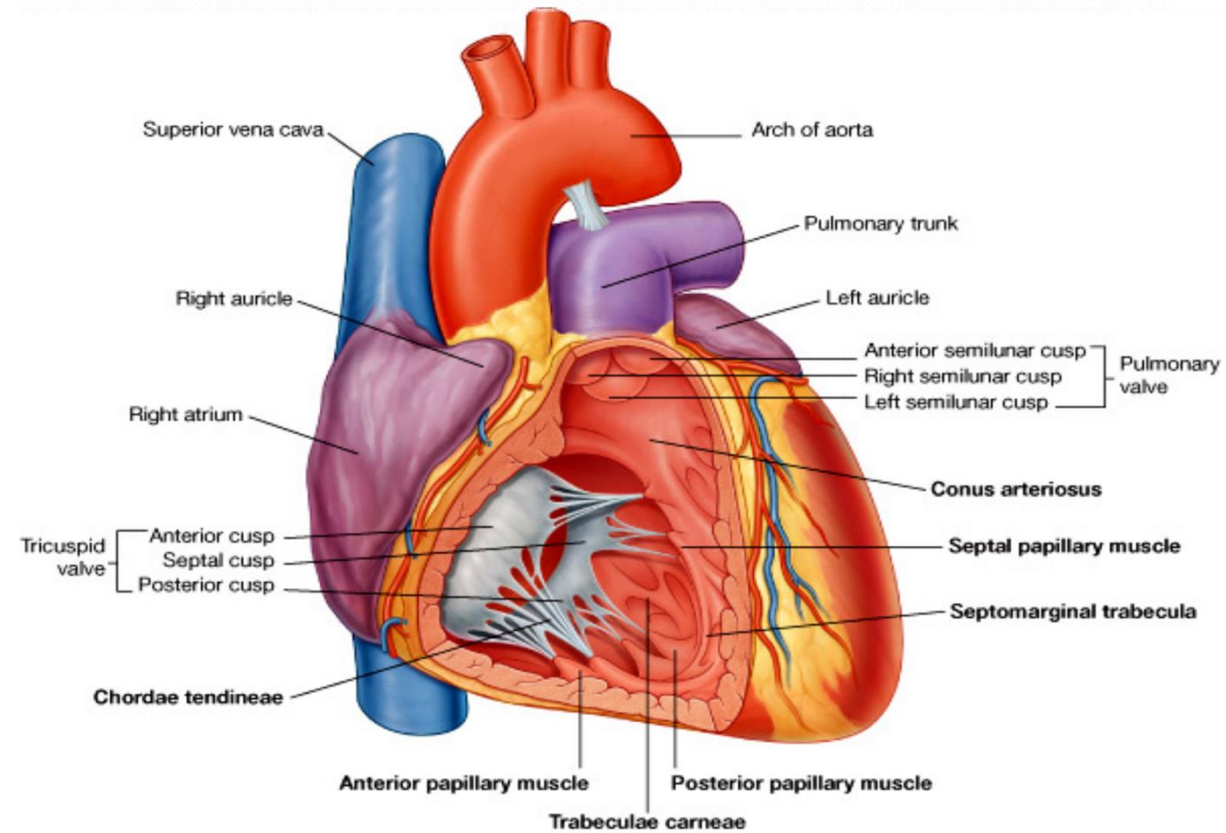
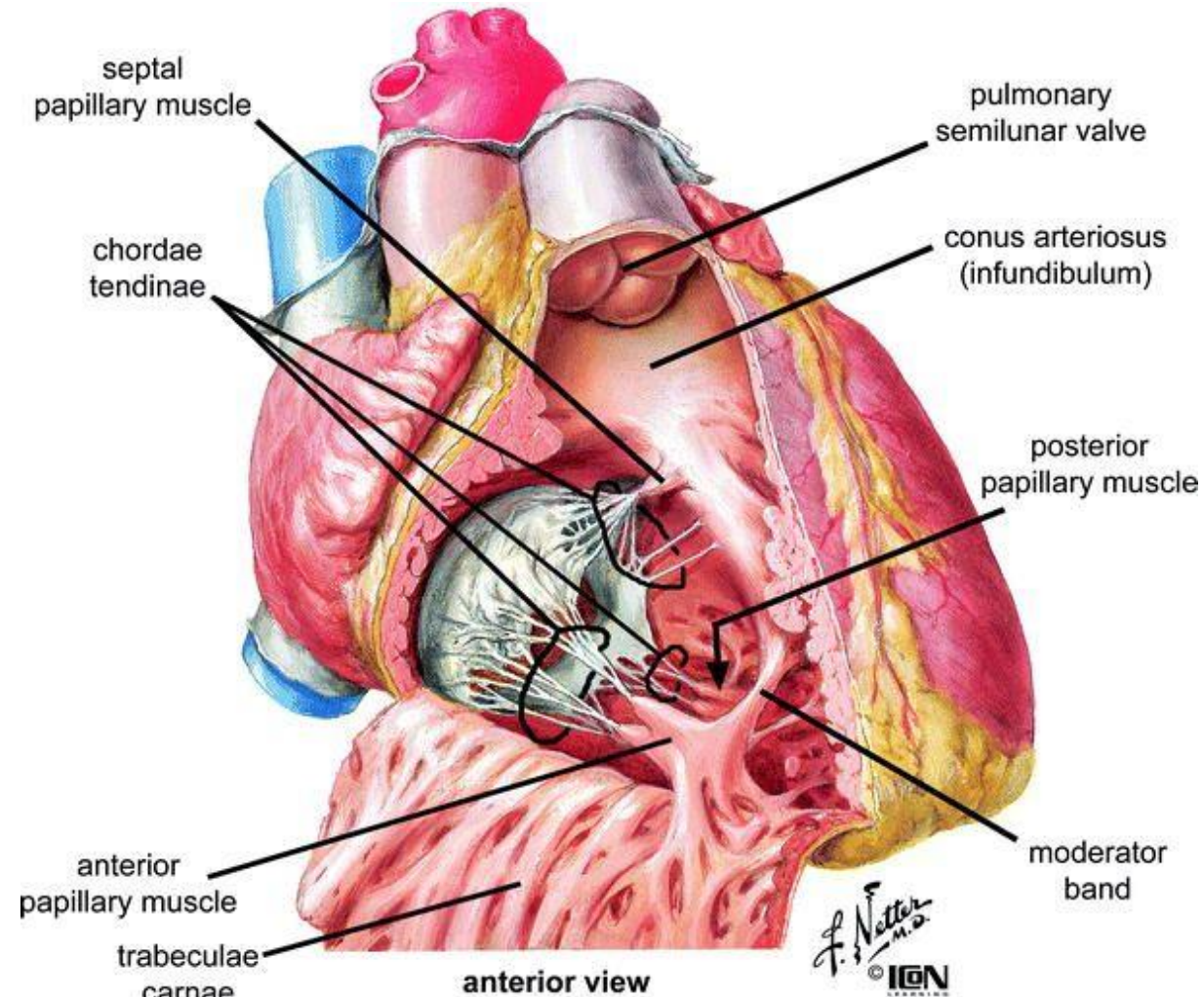
- ✓ Project inward
- ✓ Cone shaped



- ✓ Bases are attached to the wall of the ventricle
- ✓ Their apices are connected by fibrous cords (**chordae tendineae**) to the cusps of the tricuspid valve.

Right Ventricle

Papillary muscles



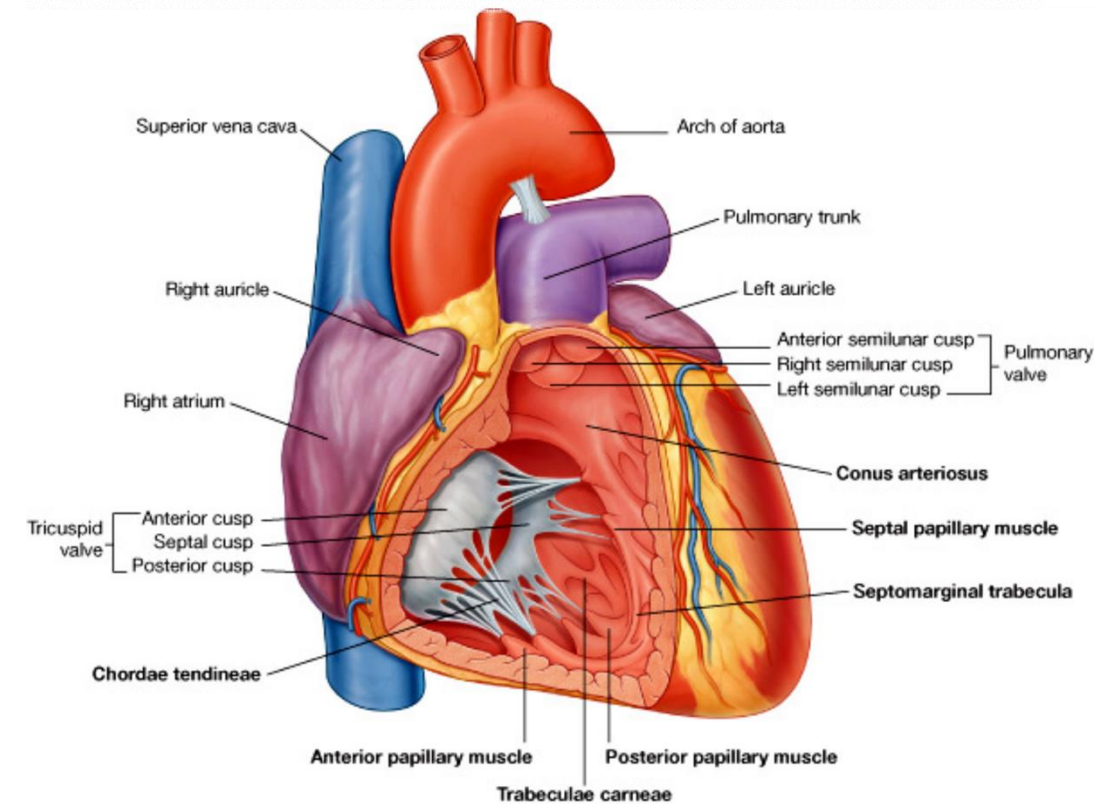
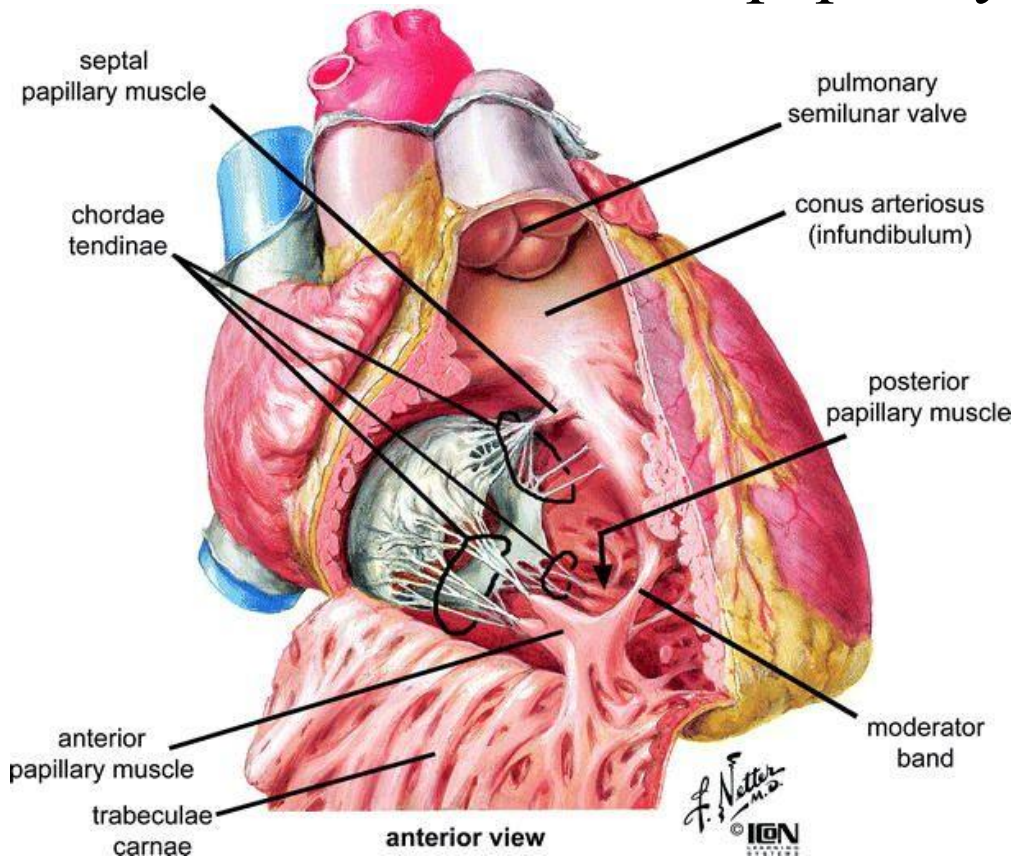
Named:

- Anterior papillary muscle
- Posterior papillary muscle
- Septal papillary muscle

Right Ventricle

Septomarginal trabecula (moderator band)

- ✓ Occurs in 60% of the population
- ✓ Connects the interventricular septum to the base of the anterior papillary muscle

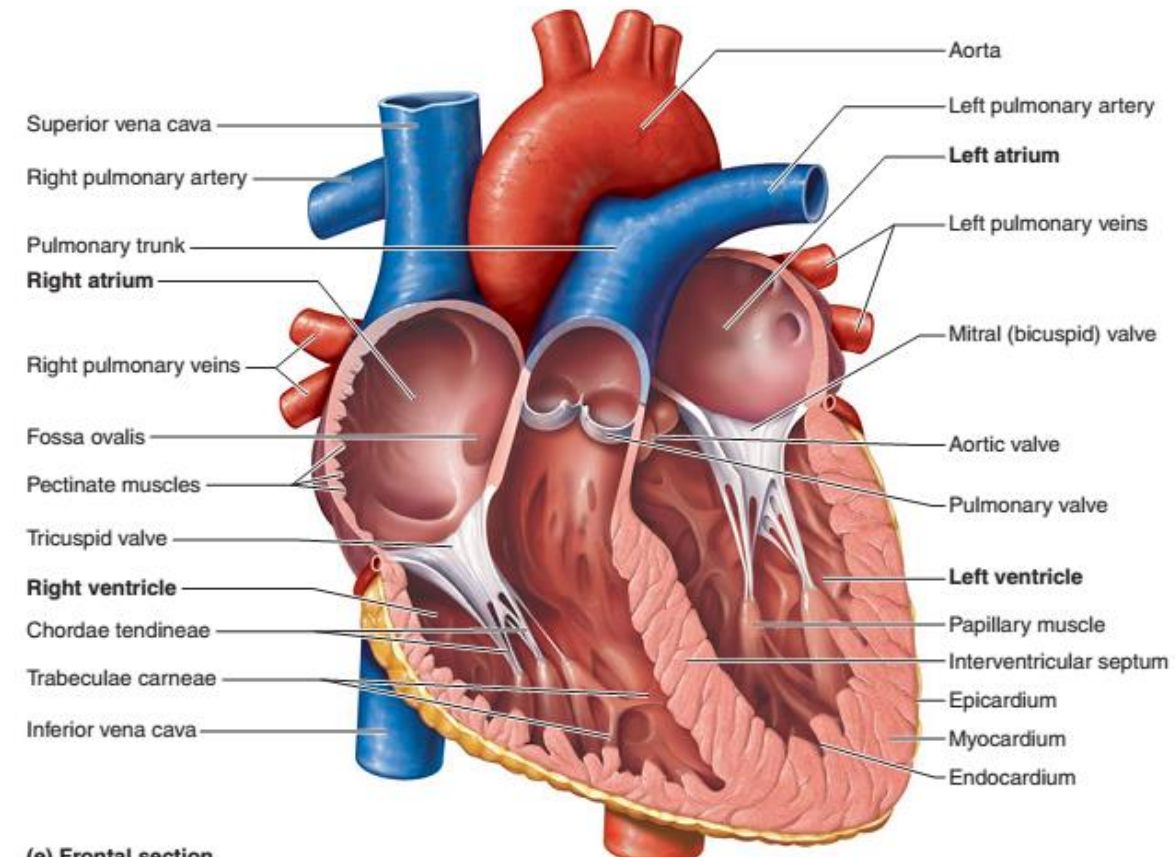
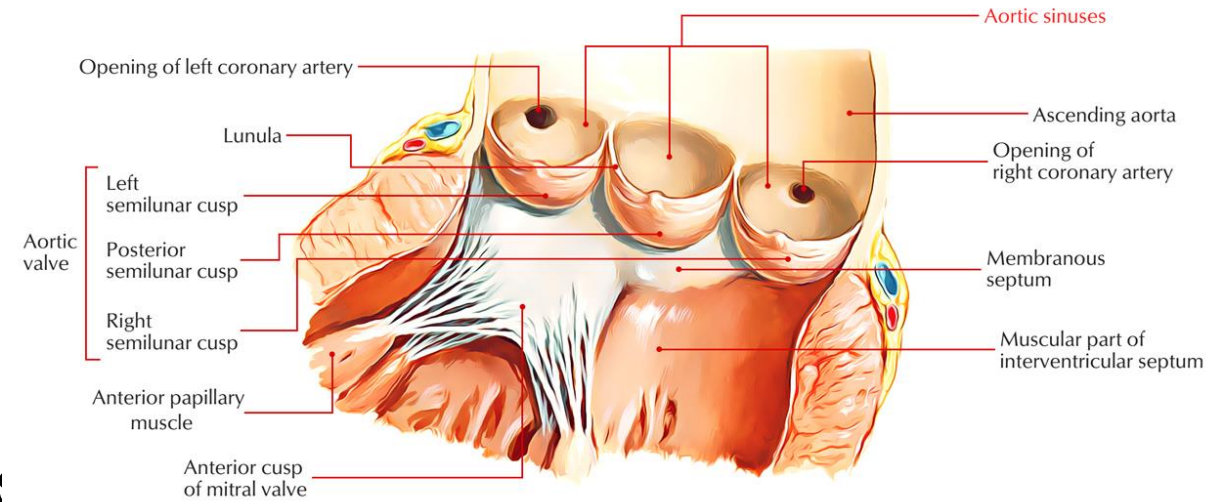


- ✓ Crosses the ventricular cavity from the septal to the anterior wall
- ✓ Contraction prevents overdistention of the ventricle.
- ✓ Also, conveys the right branch of the AV bundle (bundle of His) to the ventricular muscle

Right Ventricle

Interventricular septum

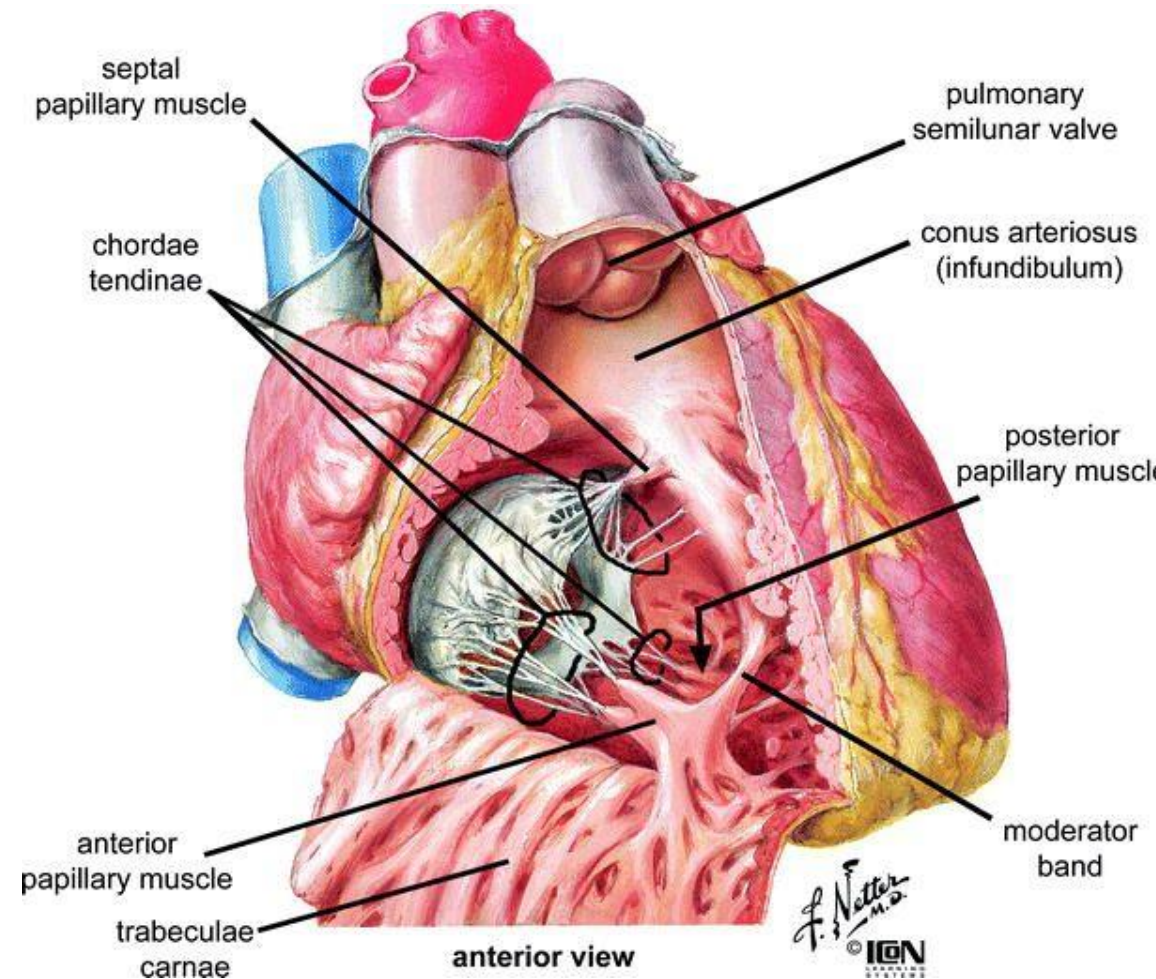
- **Membranous part**
 - ✓ Btwn the infundibulum & the vestibule
 - ✓ common site for ventricular septal defect (hole in heart)
- **Muscular part**
 - ✓ RV Pumps blood into pulmonary circuit via **pulmonary trunk**.
- **Pulmonary orifice**
 - ✓ Is guarded by the *pulmonary semilunar valves*, comprising three semilunar cusps
 - ✓ The outflow tract of the ventricle or *infundibulum* is smooth-walled



Right Ventricle

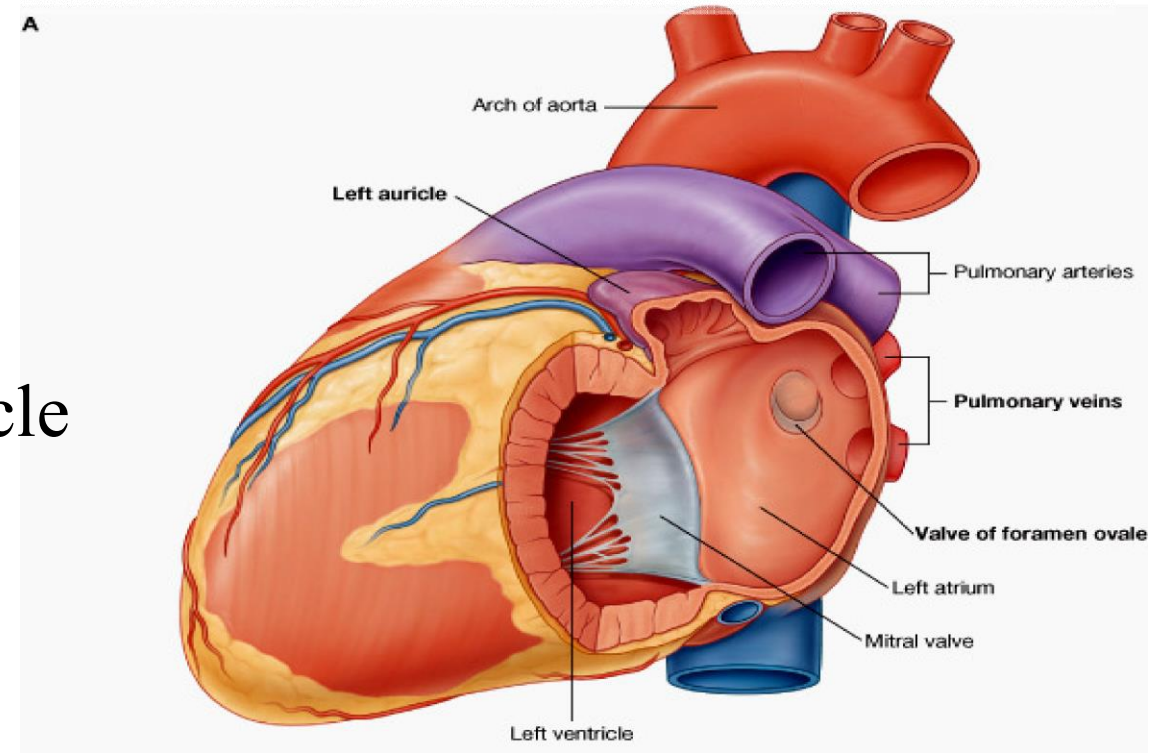
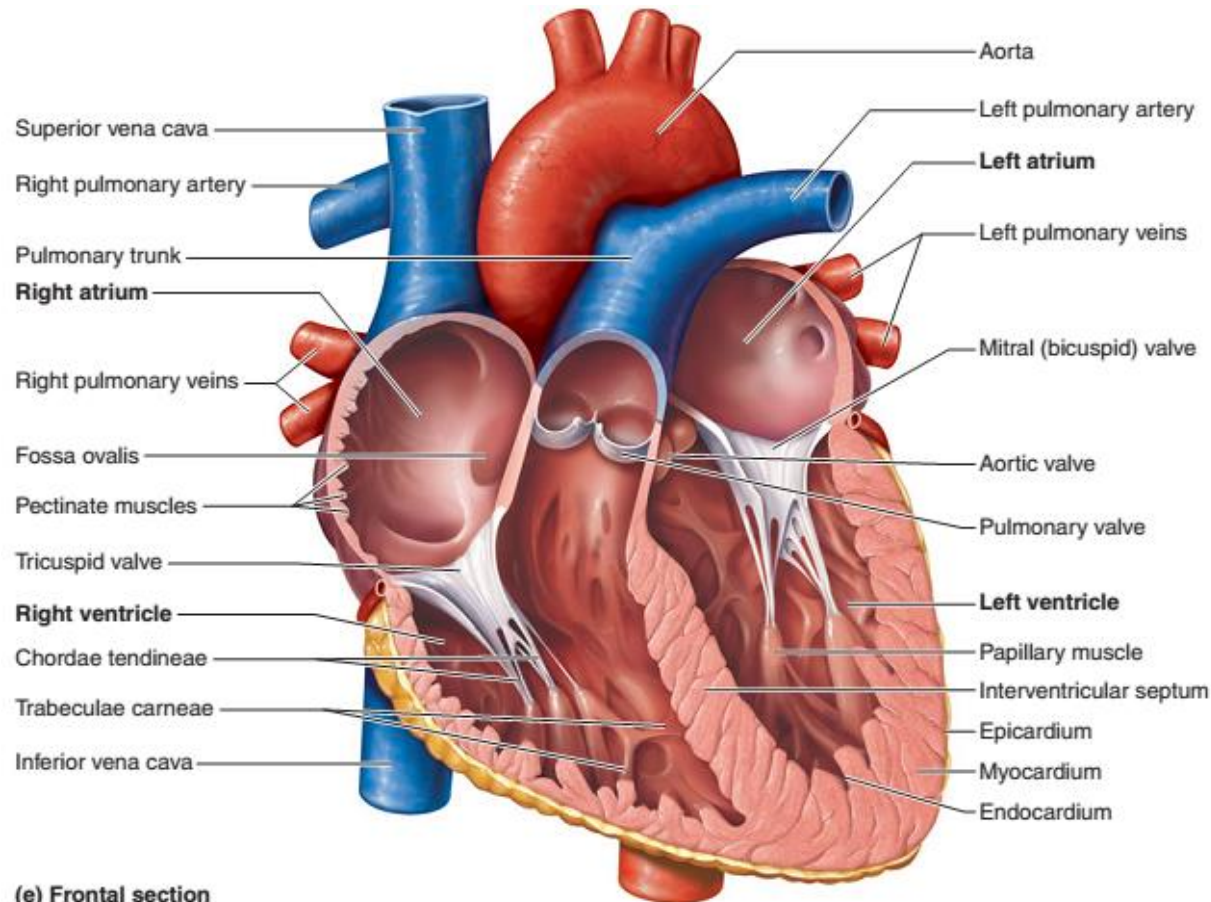
Interventricular septum

- **Membranous part**
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Left atrium

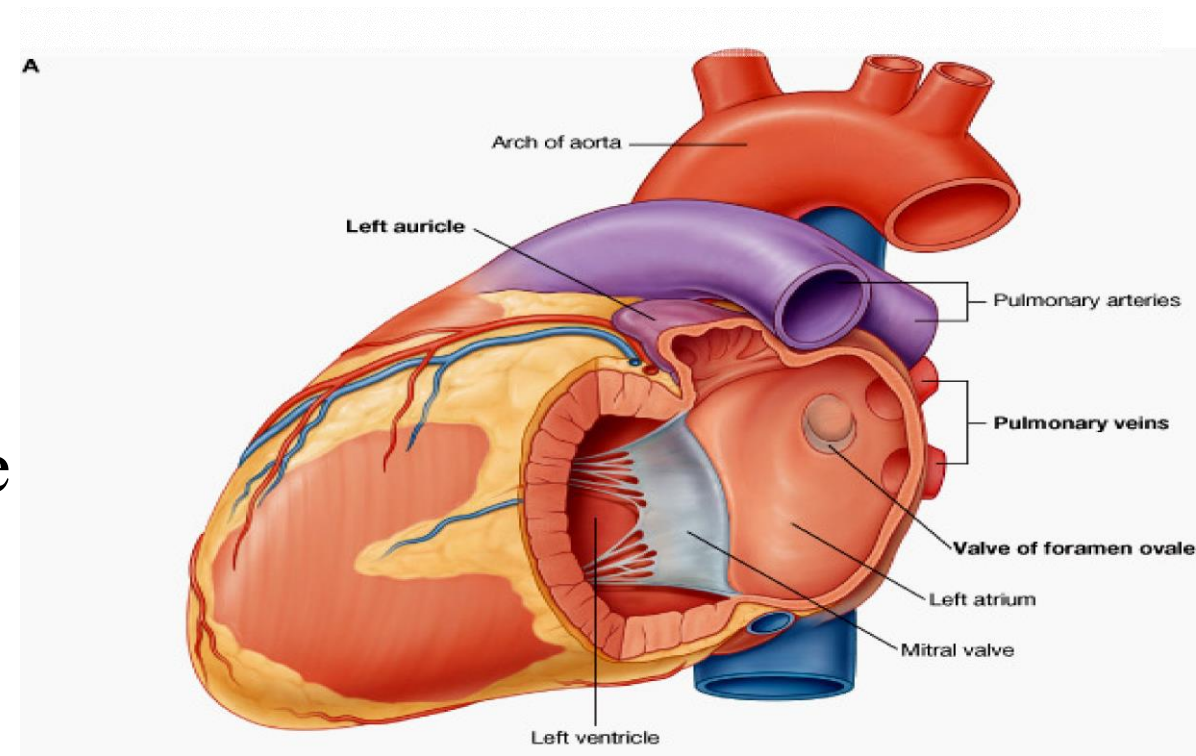
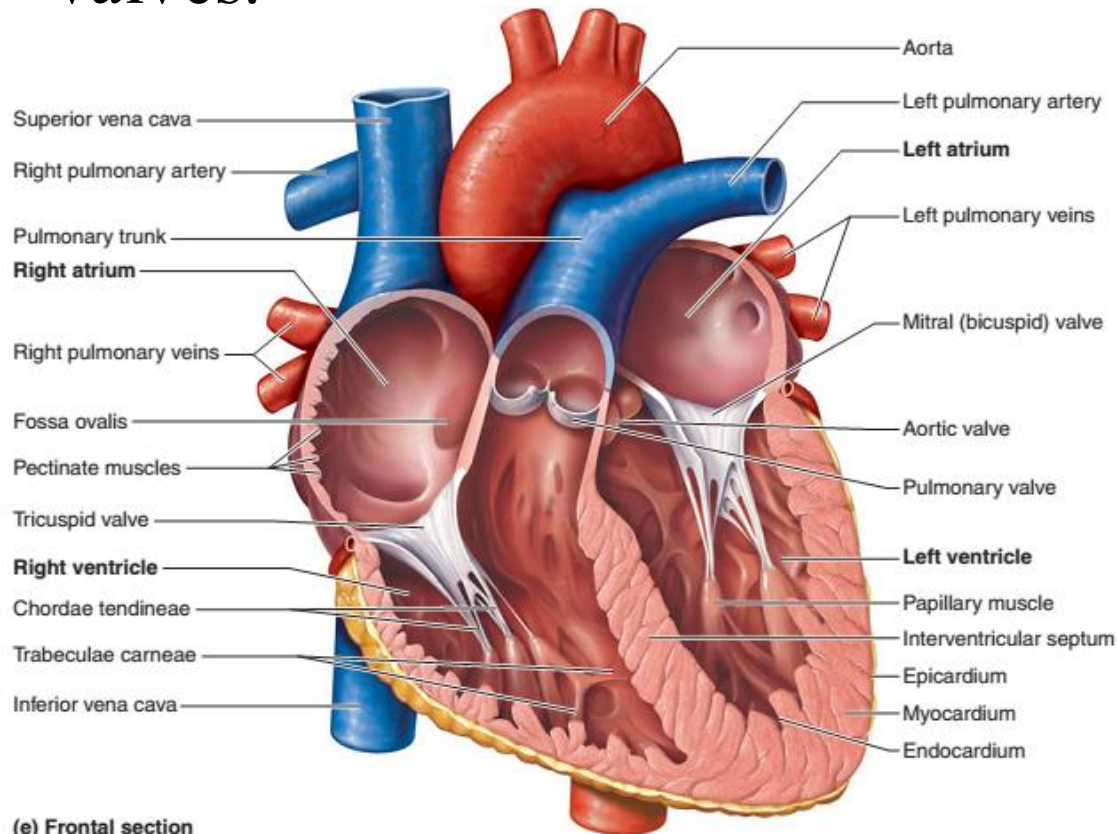
- ✓ Similar to the right atrium
- ✓ Consists of a main cavity & a left auricle.
- ✓ The wall is entirely smooth except the auricle



Left atrium

Openings into the Left Atrium

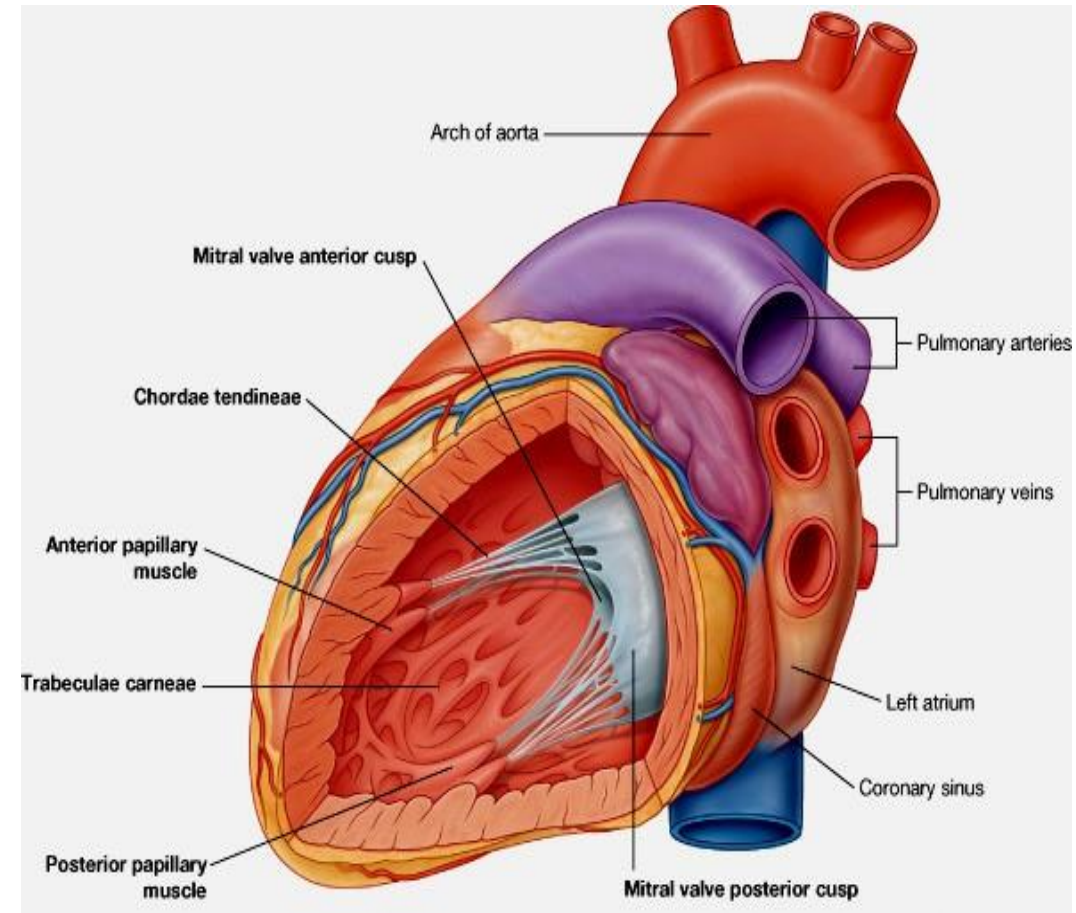
- ✓ 4 pulmonary veins, 2 from each lung,
- ✓ open through the posterior wall and have valves.



- ✓ The left atrioventricular orifice is guarded by the mitral valve.

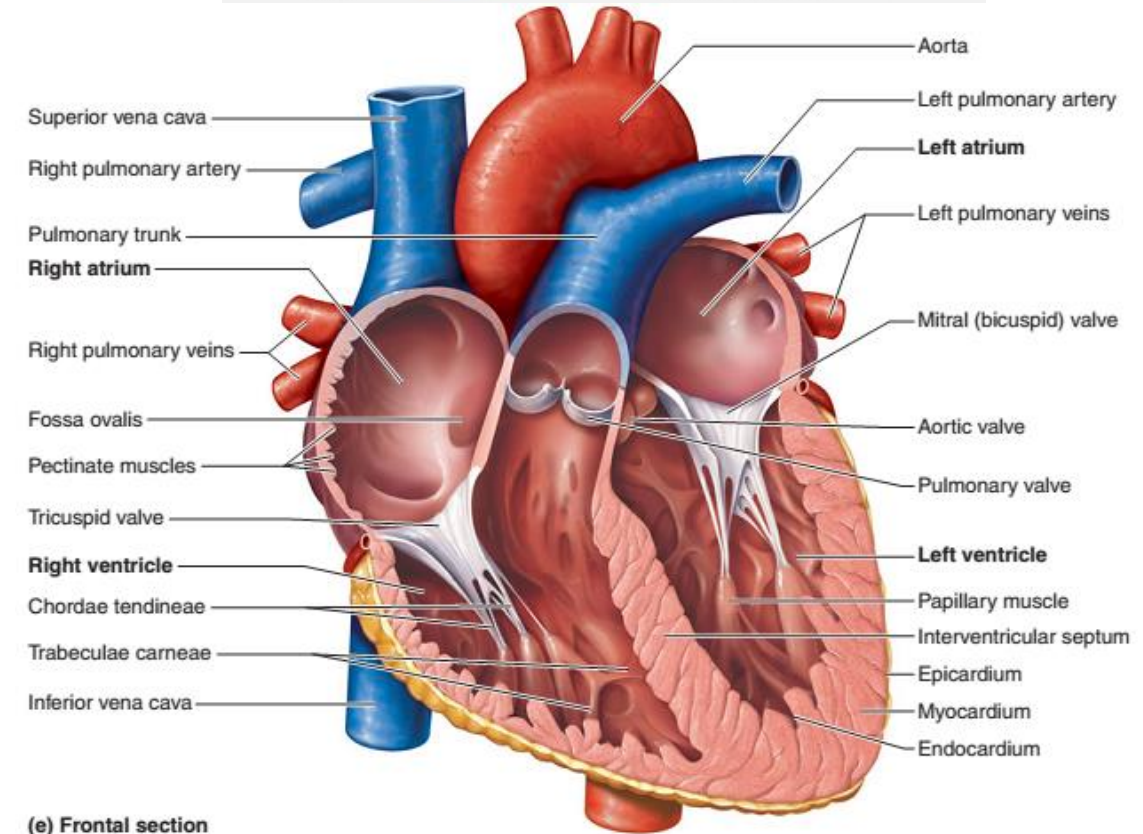
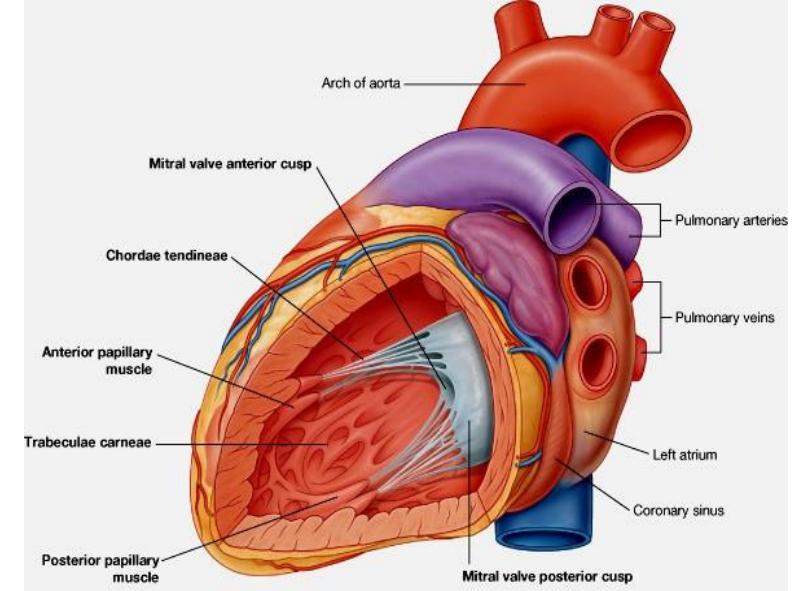
Left Ventricle

- ✓ Communicates with the:
 - i. left atrium through the atrioventricular orifice
 - ii. Aorta through the aortic orifice.
- ✓ The walls of the left ventricle are 3 times thicker than those of the right ventricle.
- ✓ The left intraventricular blood pressure is six times higher than that inside the right ventricle.



Left Ventricle

- ✓ In cross section, the left ventricle is circular; the right is crescentic because of the bulging of the ventricular septum into the cavity of the right ventricle.
- ✓ There are well-developed trabeculae carneae, two large papillary muscles but no moderator band.
- ✓ The part of the ventricle below the aortic orifice is called **aortic vestibule**



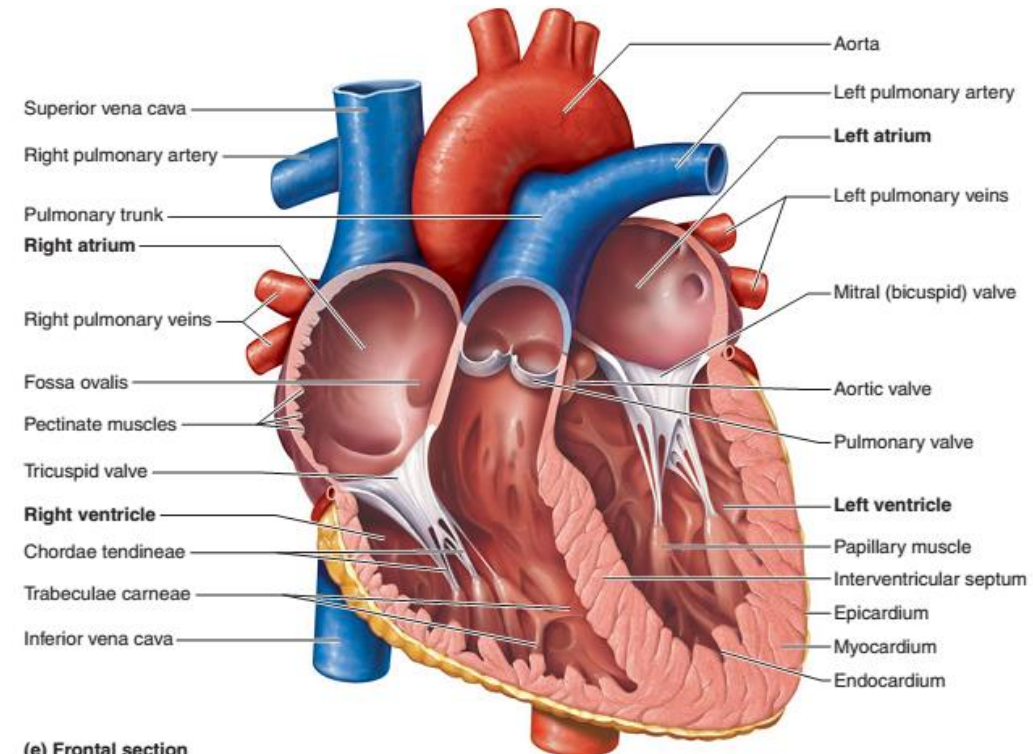
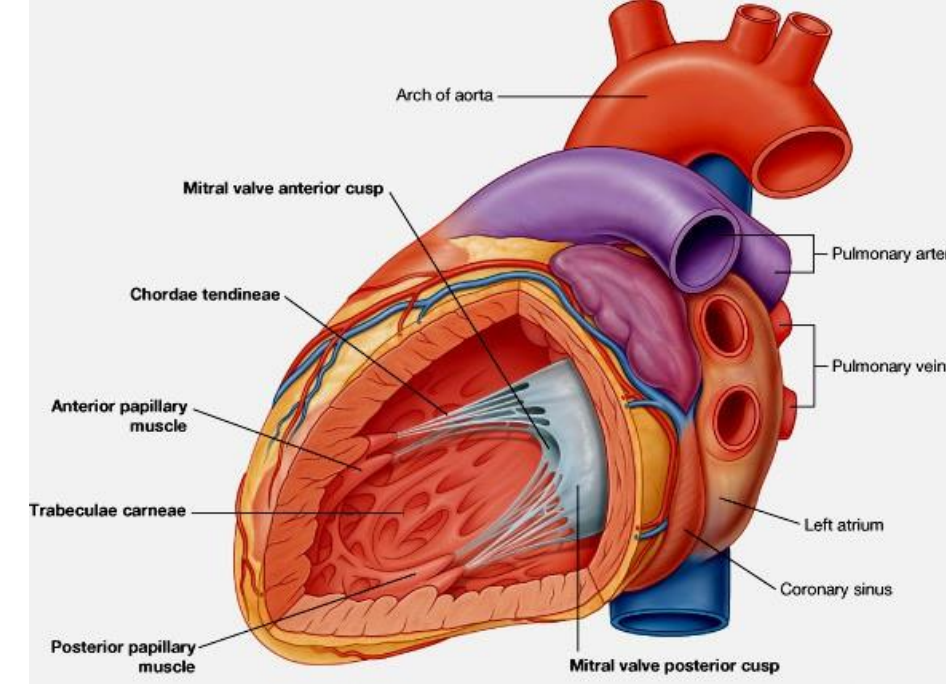
Left Ventricle

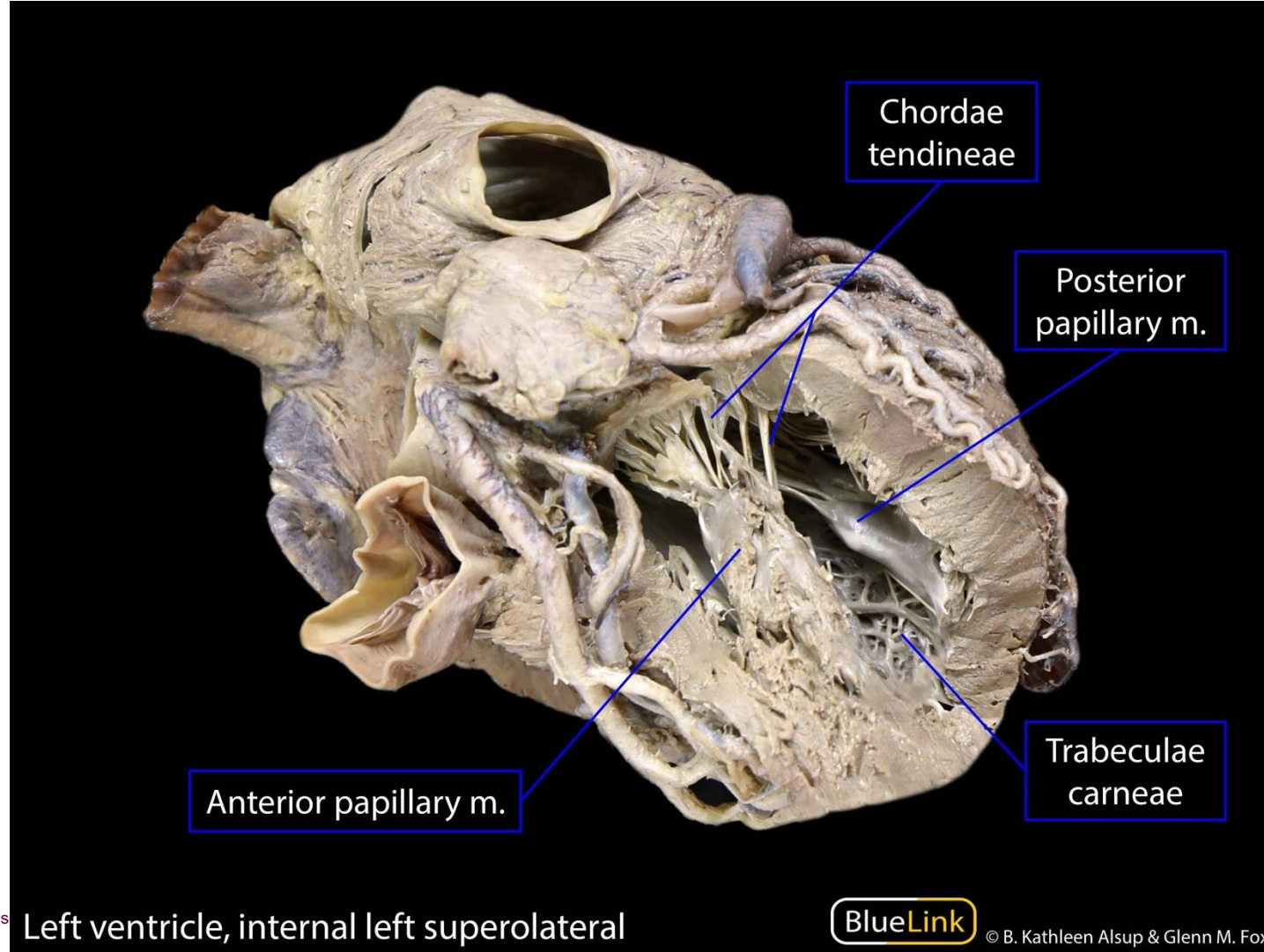
Mitral valve

- ✓ Guards the atrioventricular orifice

Aortic valve

- ✓ Guards the aortic orifice and is precisely similar in structure to the pulmonary valve .





Heart Valves – Valve Structure

Heart Valves – Valve Structure

✓ Prevent regurgitation of blood.

Consists of:

Atrioventricular (AV) valves

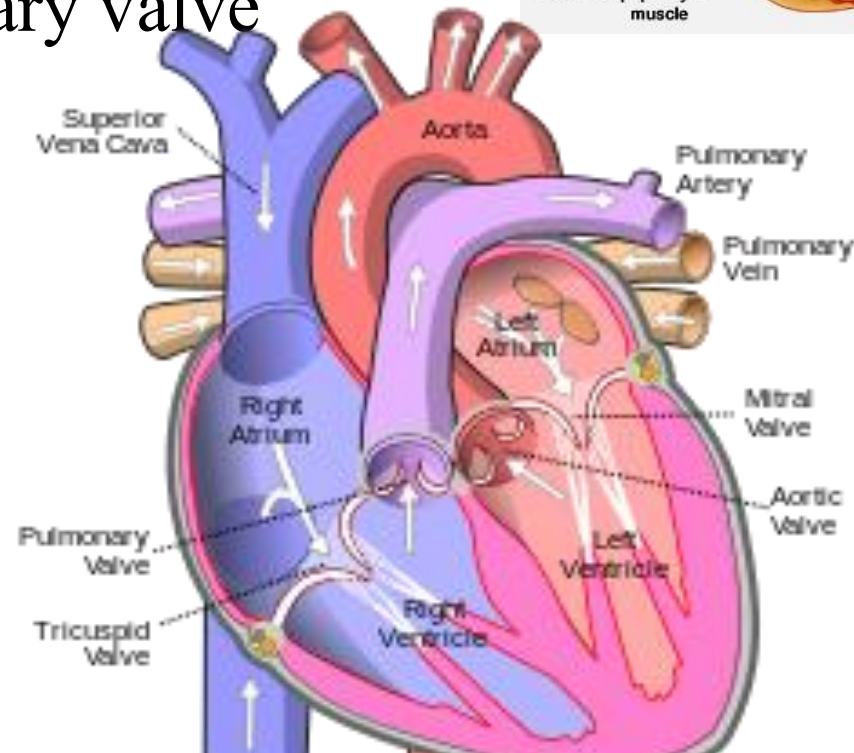
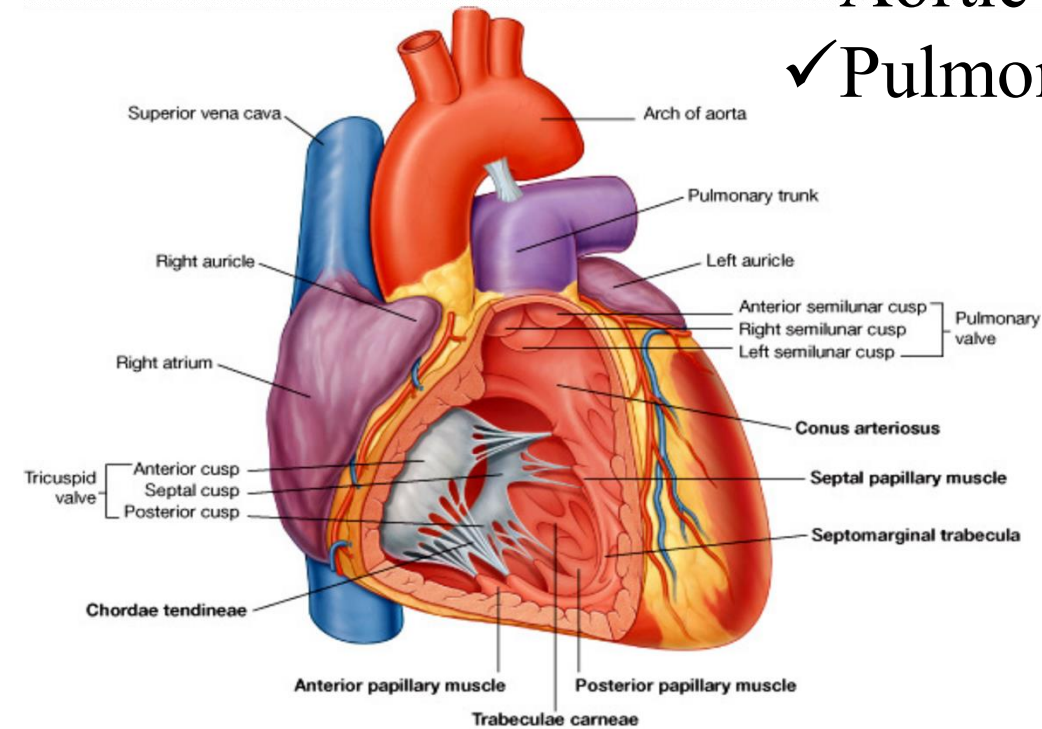
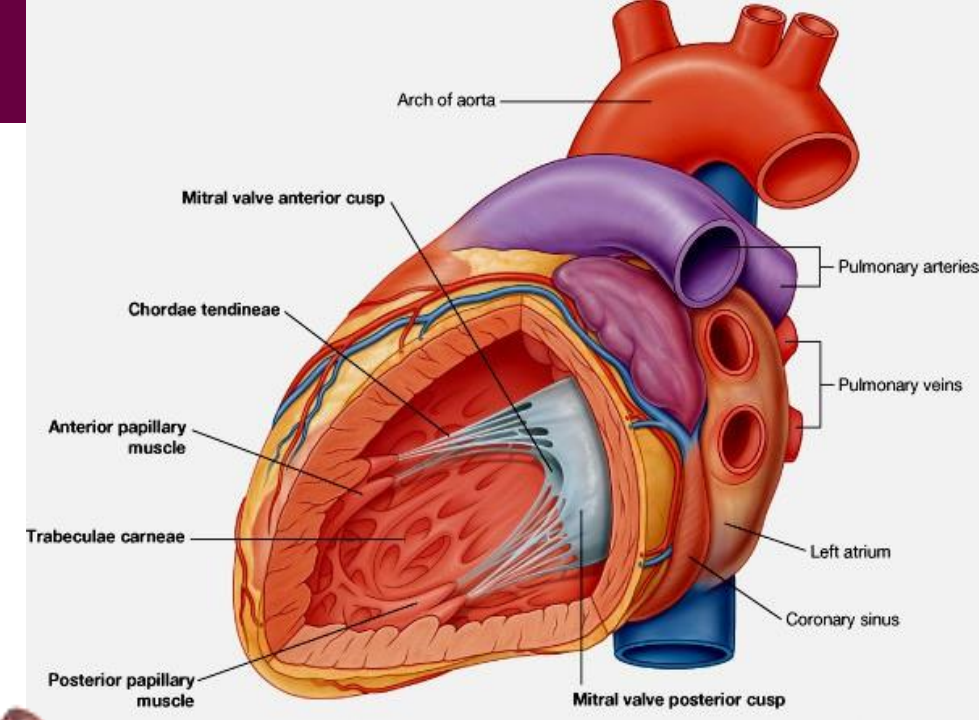
✓ Bicuspid valve

✓ Tricuspid valve

Semilunar valves

✓ Aortic valve

✓ Pulmonary valve





Heart Valves – Valve Structure

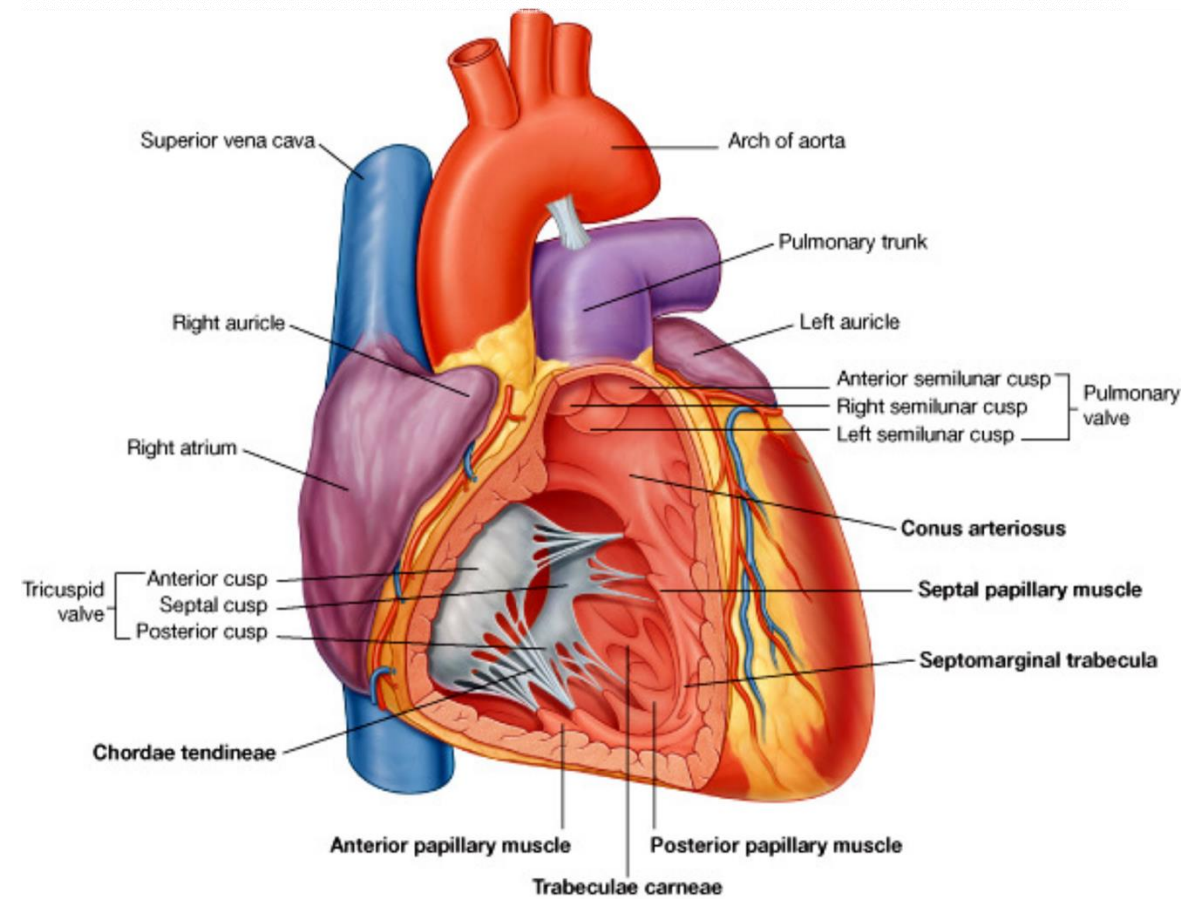
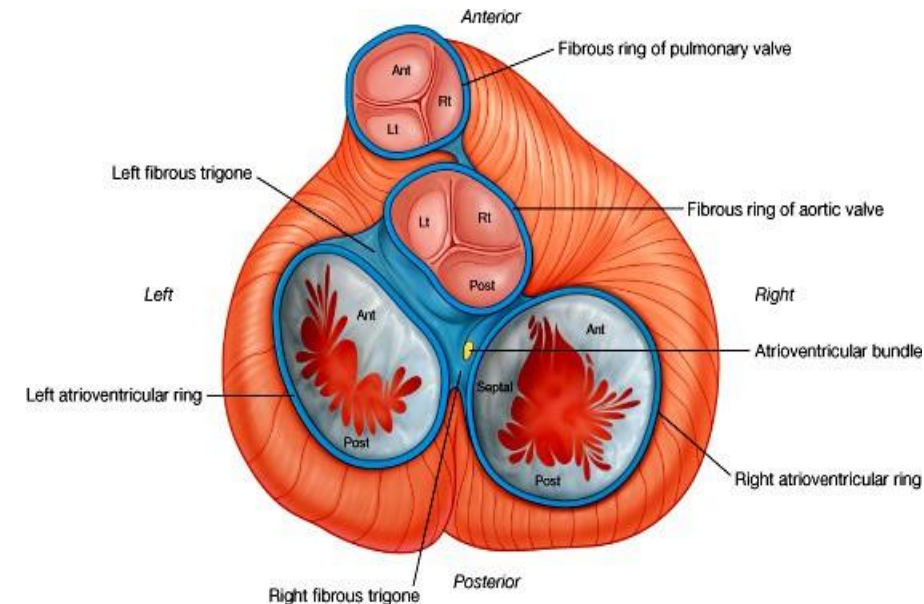
Atrioventricular (AV) valves

- ✓ Prevent backflow of blood from the ventricles to the atria when the ventricles contract (systole).
- i. Right atrioventricular valve = Tricuspid valve
- ii. Left atrioventricular valve = Bicuspid or mitral valve
- ✓ Cusps of mitral valve are smaller but thicker than those of tricuspid valve

Tricuspid valve

- ✓ Guards the right atrioventricular orifice
- ✓ Consists of 3 cusps formed by a fold of endocardium with some connective tissue enclosed:

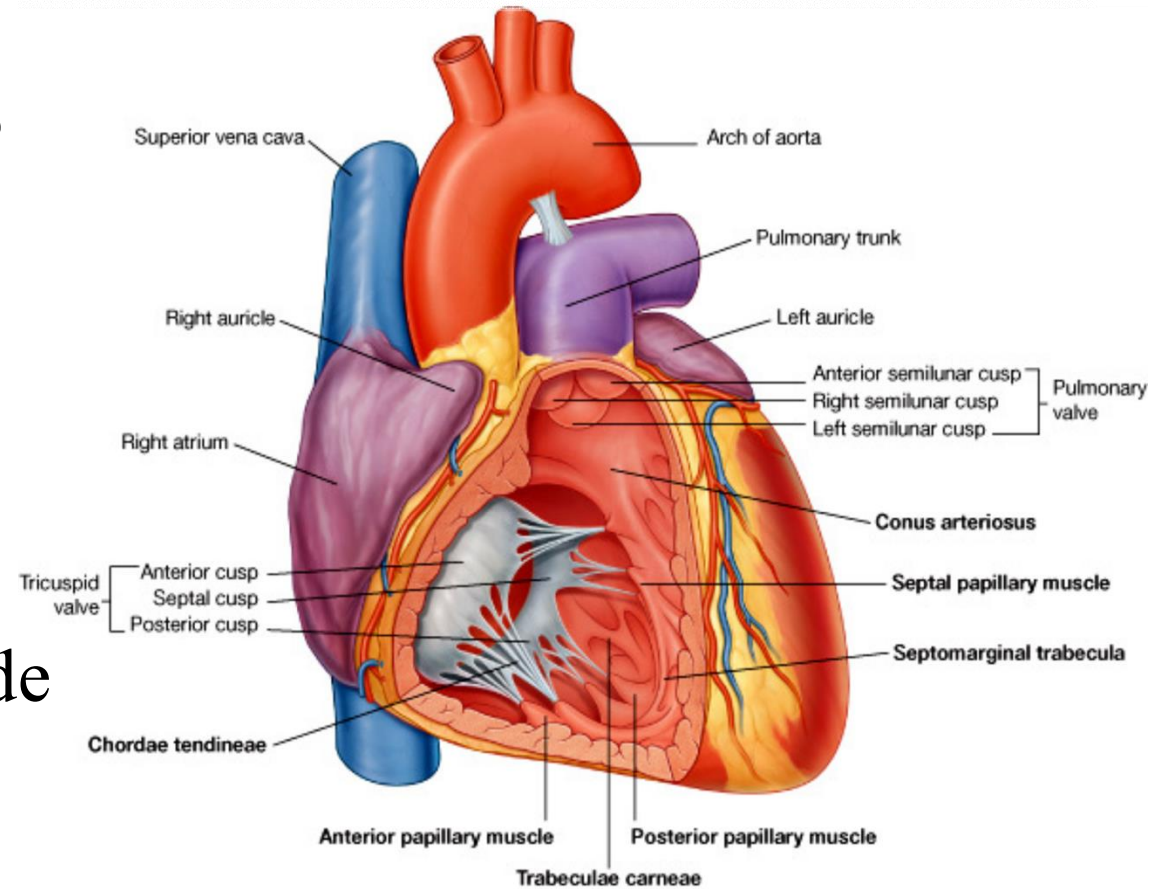
- Anterior cusp
- Septal cusp
- Posterior (inferior) cusps



- ✓ The bases of the cusps are attached to the fibrous ring of the skeleton of the heart
- ✓ whereas their free edges and ventricular surfaces are attached to the chordae tendineae

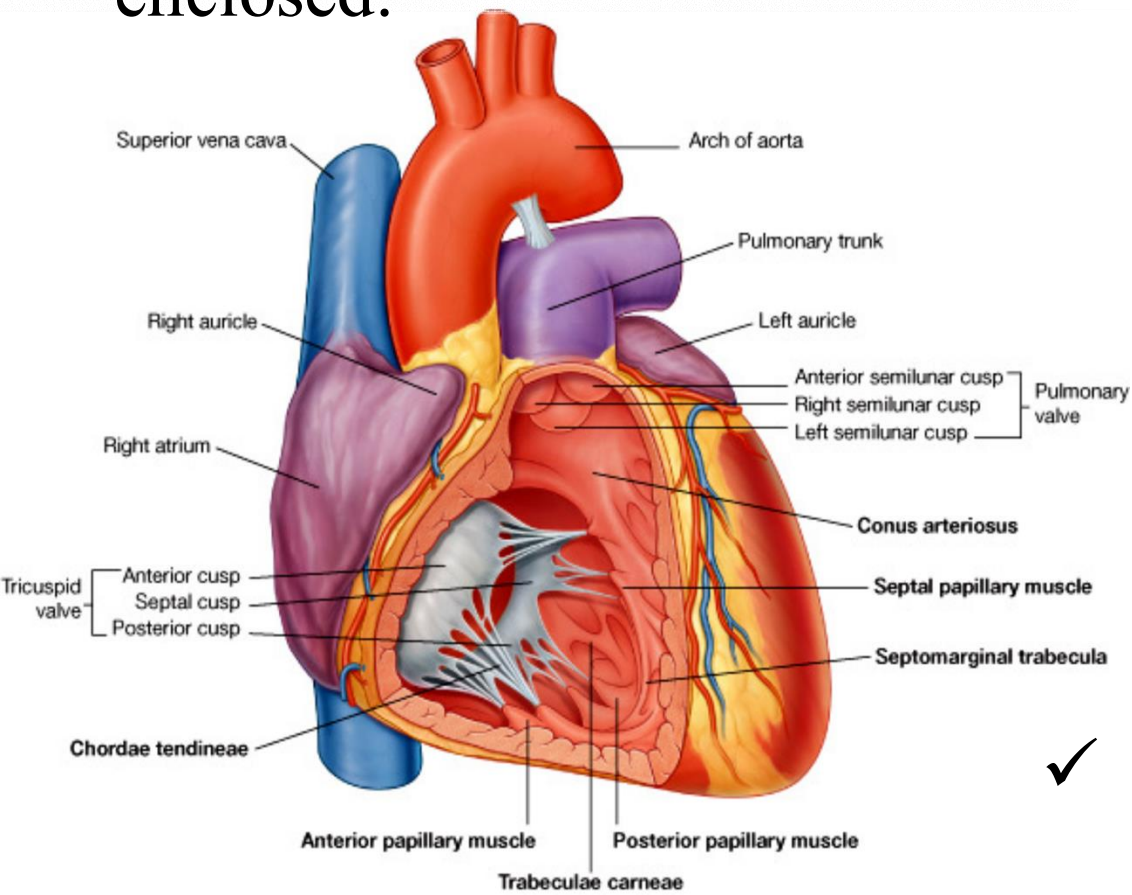
Tricuspid valve

- ✓ The chordae tendineae connect the cusps to the papillary muscles
- ✓ When the ventricle contracts, the papillary muscles contract & prevent the cusps from being forced into the atrium & turning inside out as the intraventricular pressure rises.
- ✓ To assist in this process, the chordae tendineae of one papillary muscle are connected to the adjacent parts of two cusps.

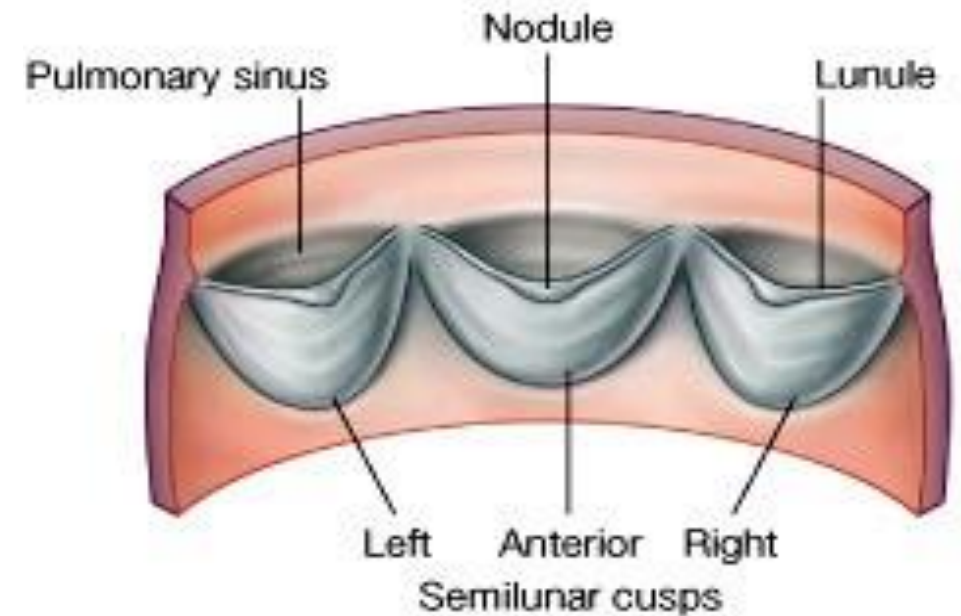
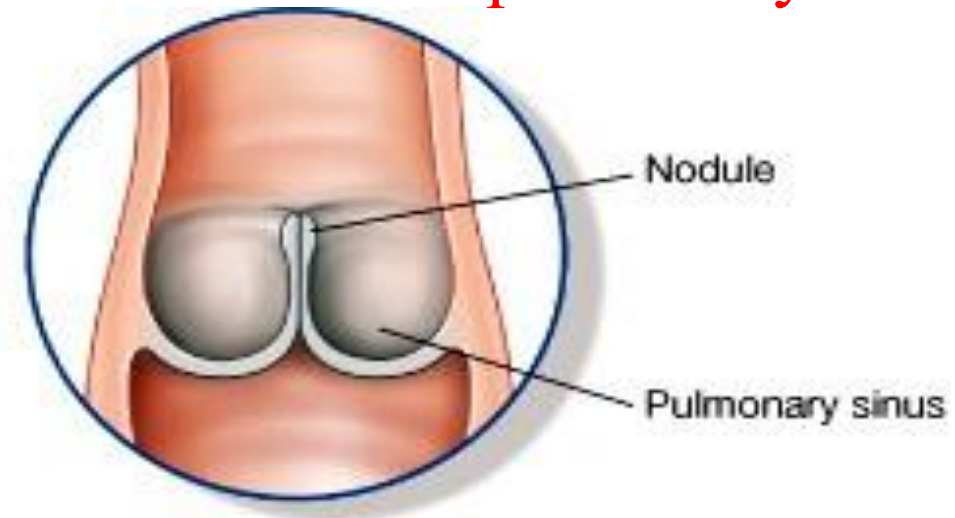


Pulmonary valve

- ✓ Guards the pulmonary orifice
- ✓ Consists of 3 semilunar cusps formed by folds of endocardium with some connective tissue enclosed.



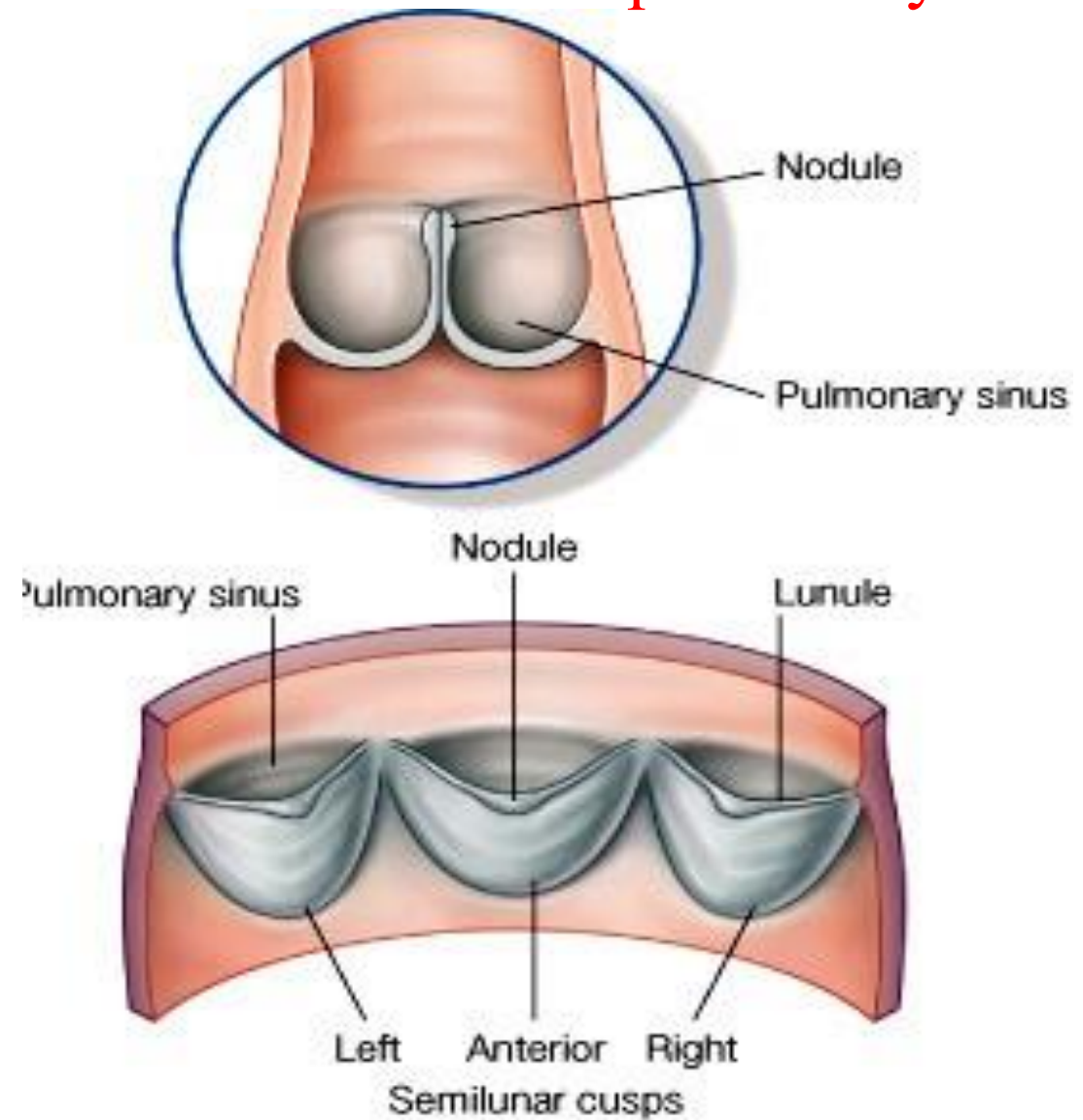
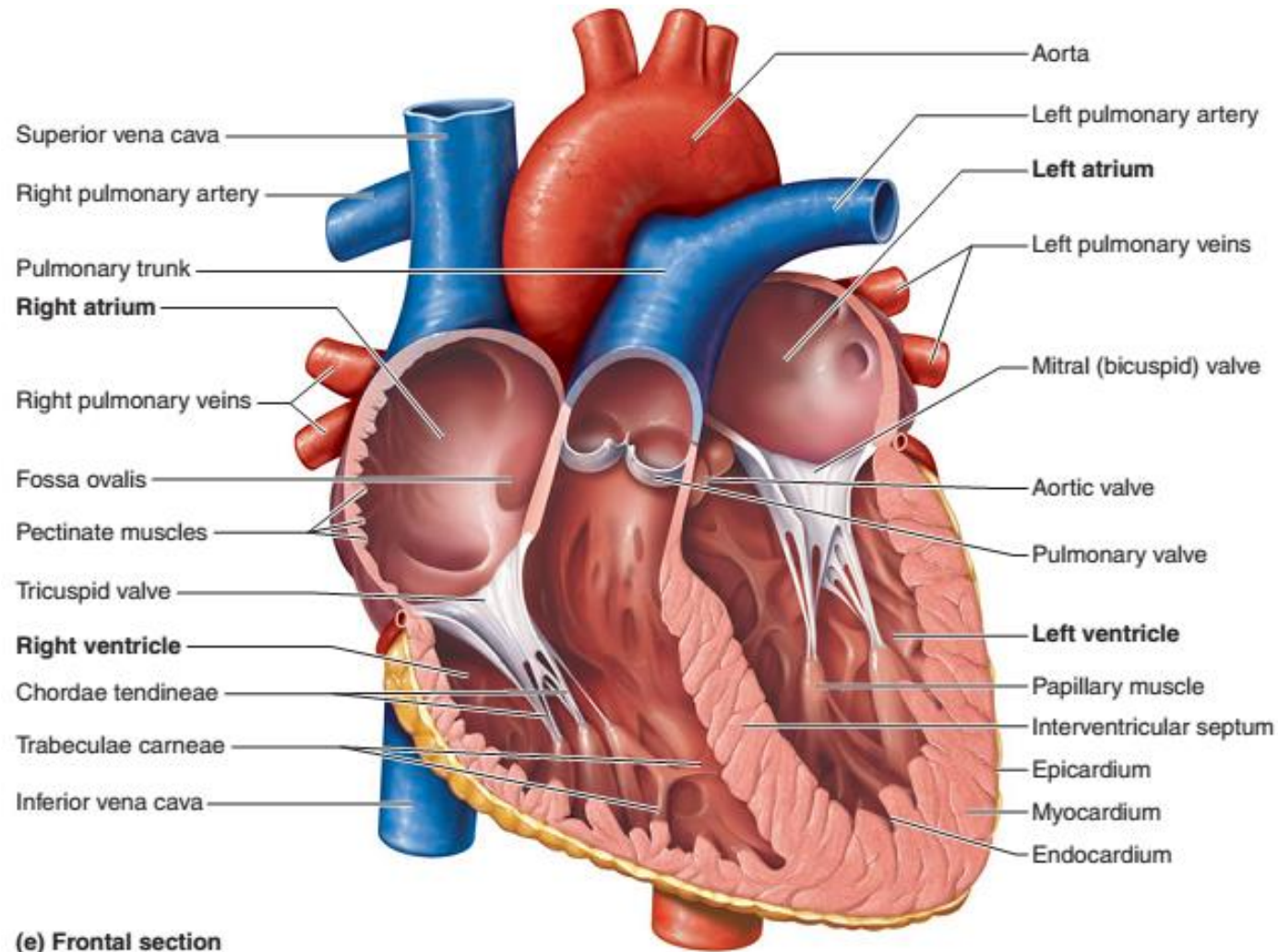
Posterior view of the pulmonary valve



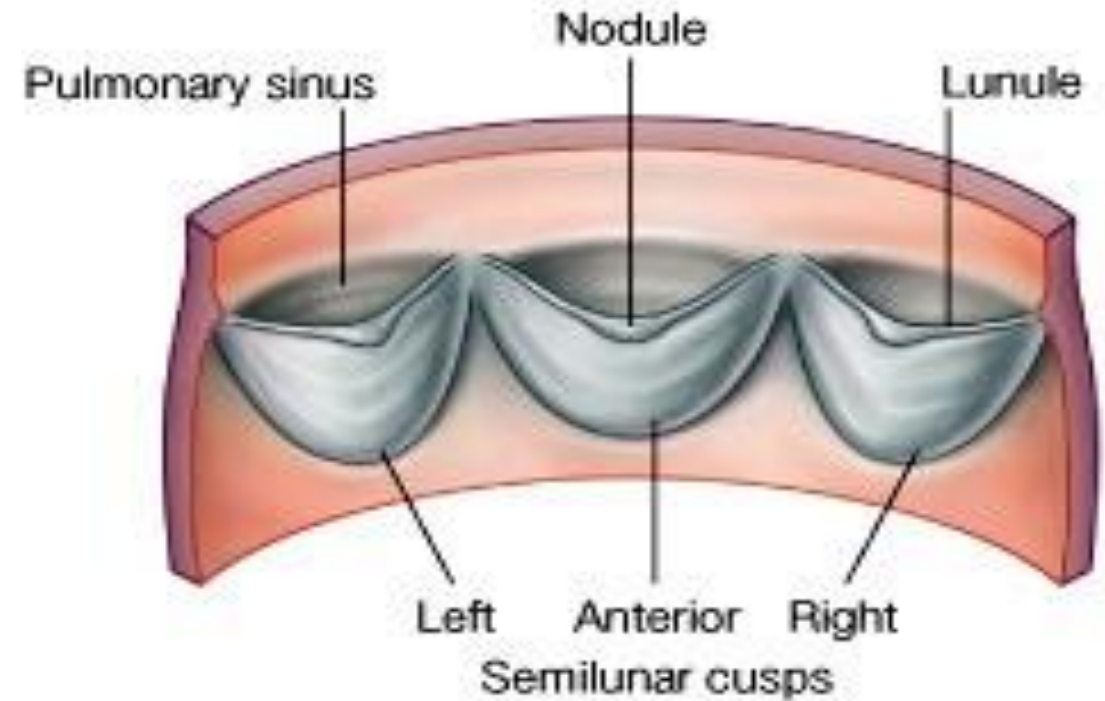
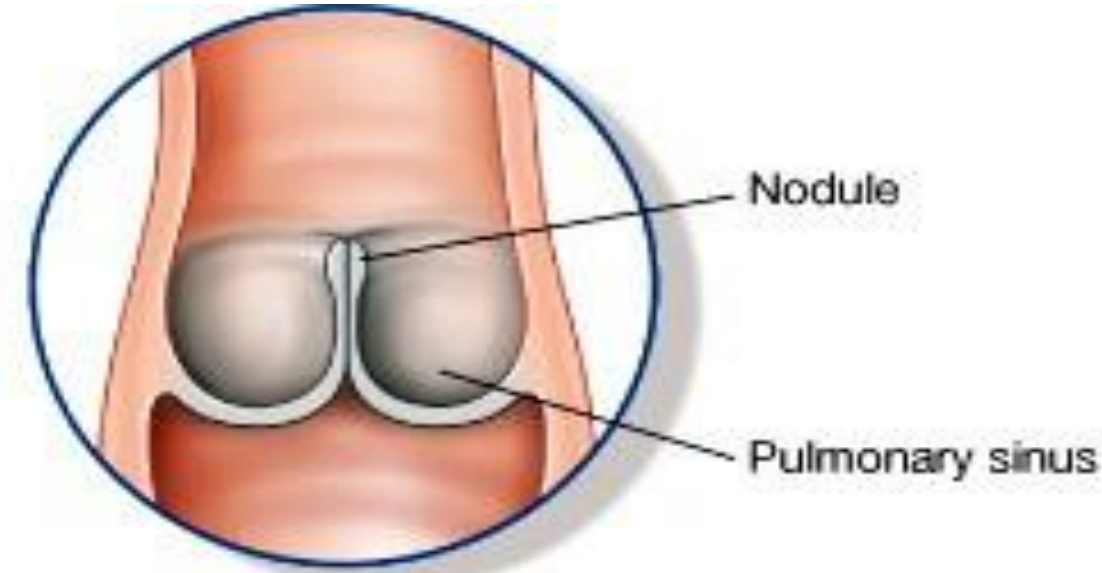
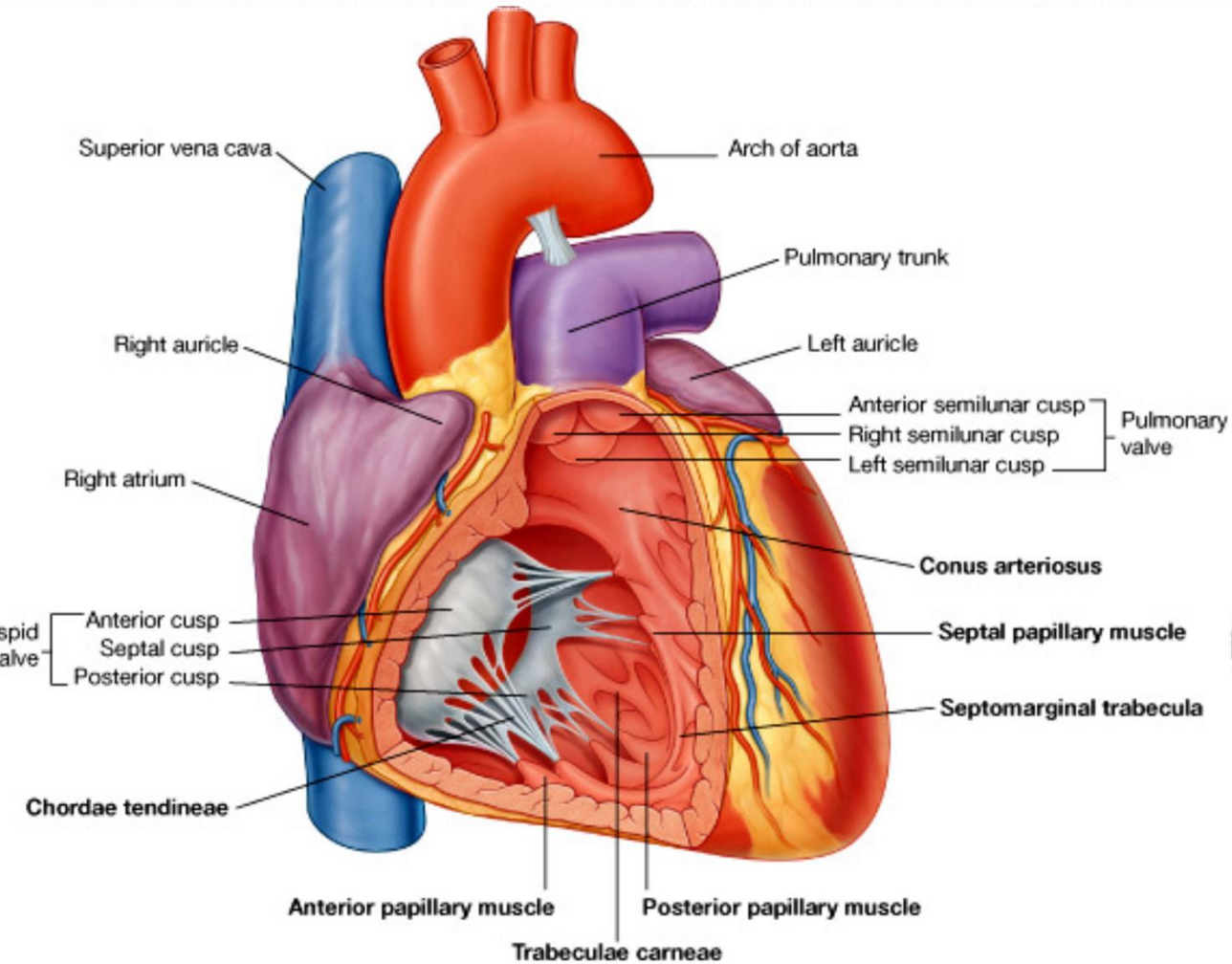
- ✓ The open mouths of the cusps are directed upward into the pulmonary trunk.

Pulmonary valve

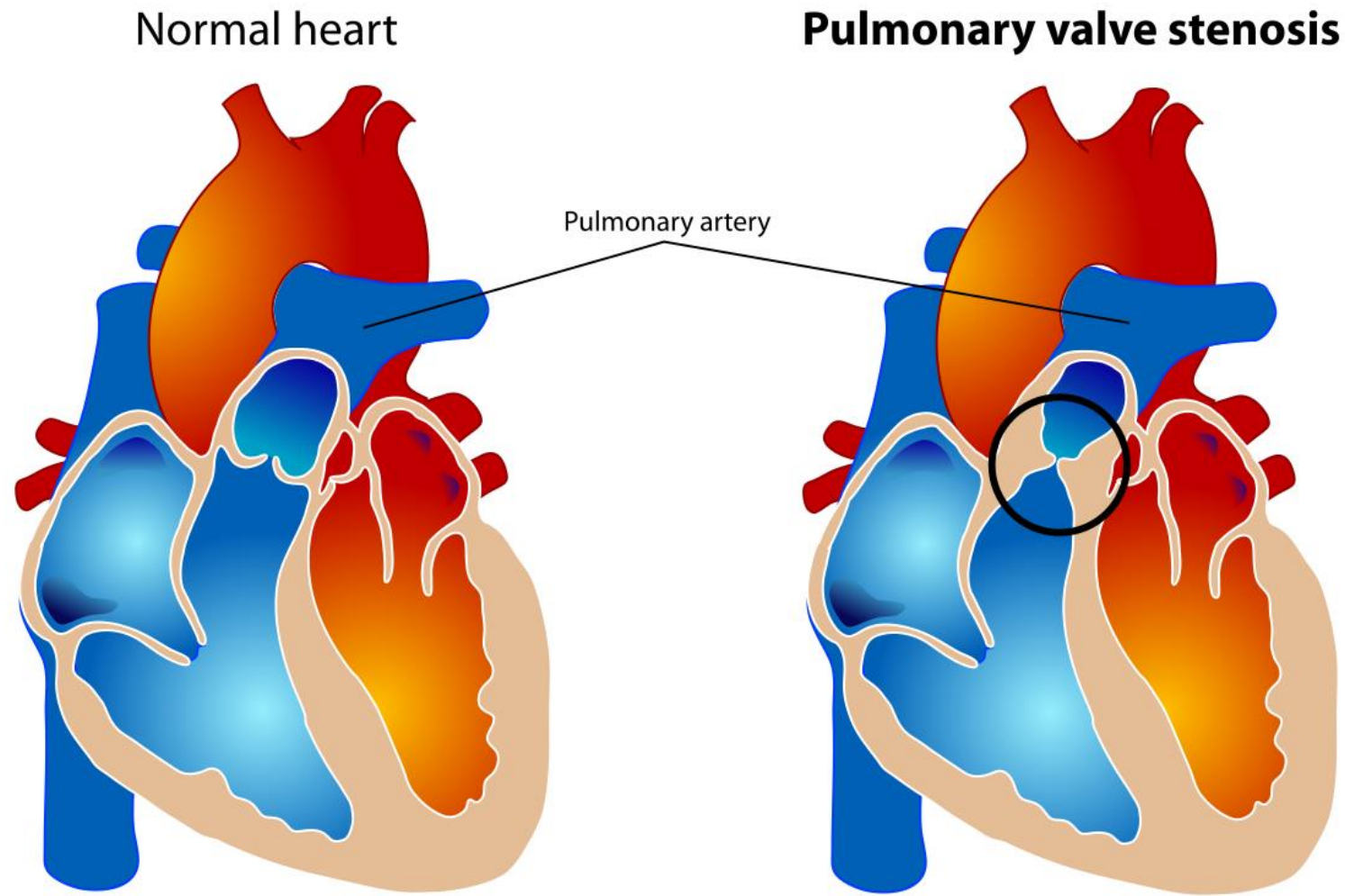
Posterior view of the pulmonary valve



Posterior view of the pulmonary valve



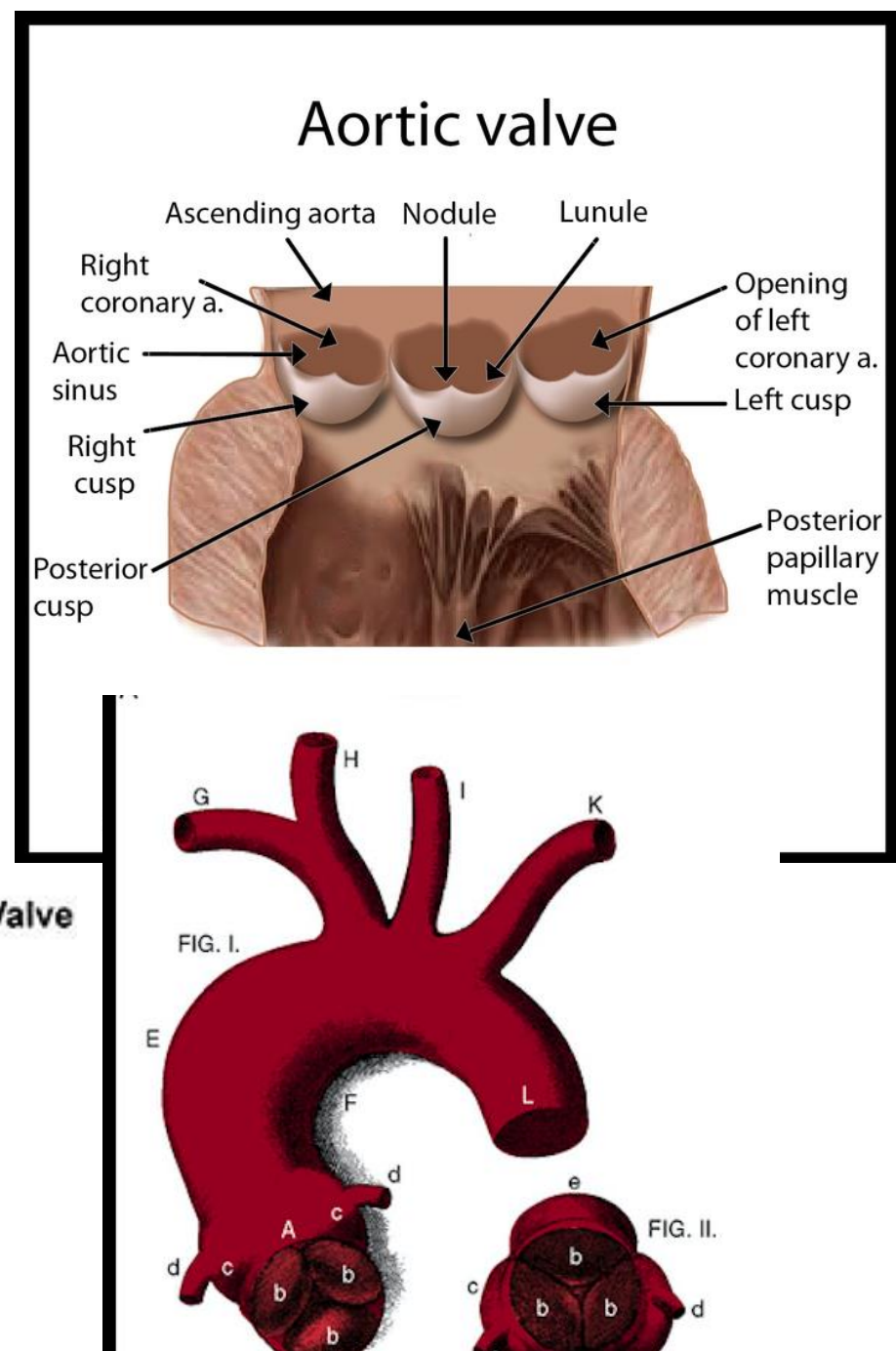
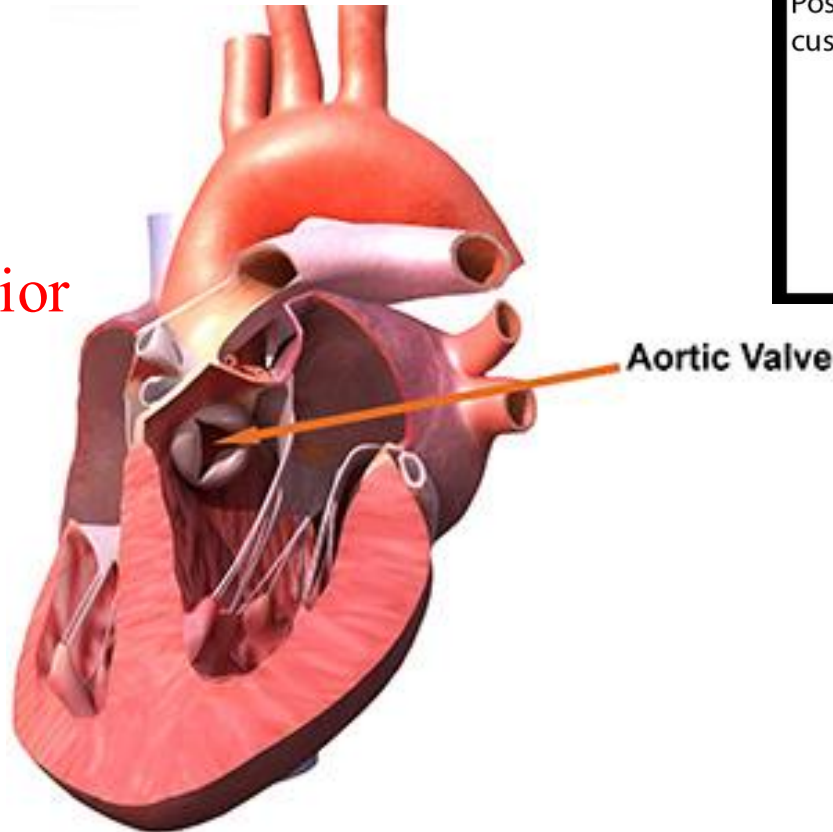
Pulmonary Valve stenosis?

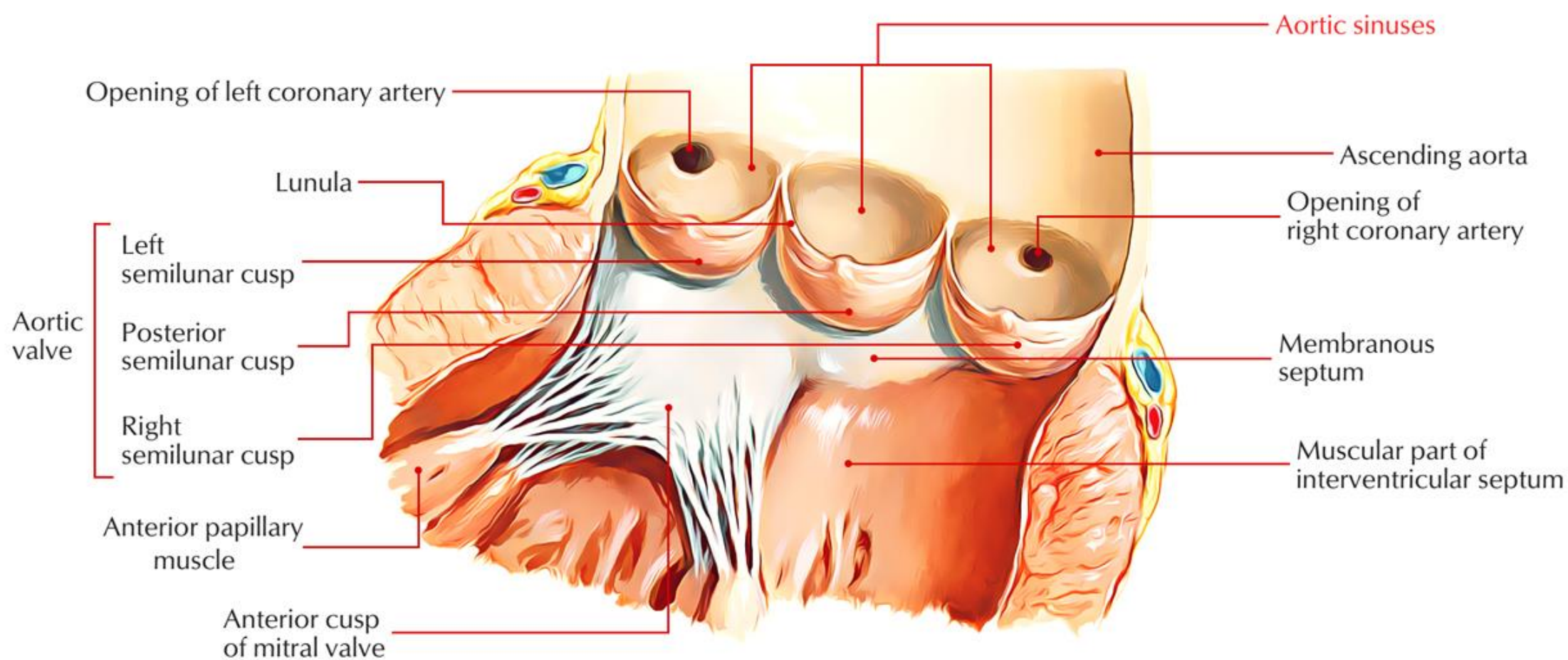


Aortic valve

- ✓ Guards the aortic orifice
- ✓ Is precisely similar in structure to the pulmonary valve.
- ✓ Consists of 3 semilunar cusps
 - i. Anterior (right) cusp
 - ii. left cusp
 - iii. Posterior cusp

} Posterior



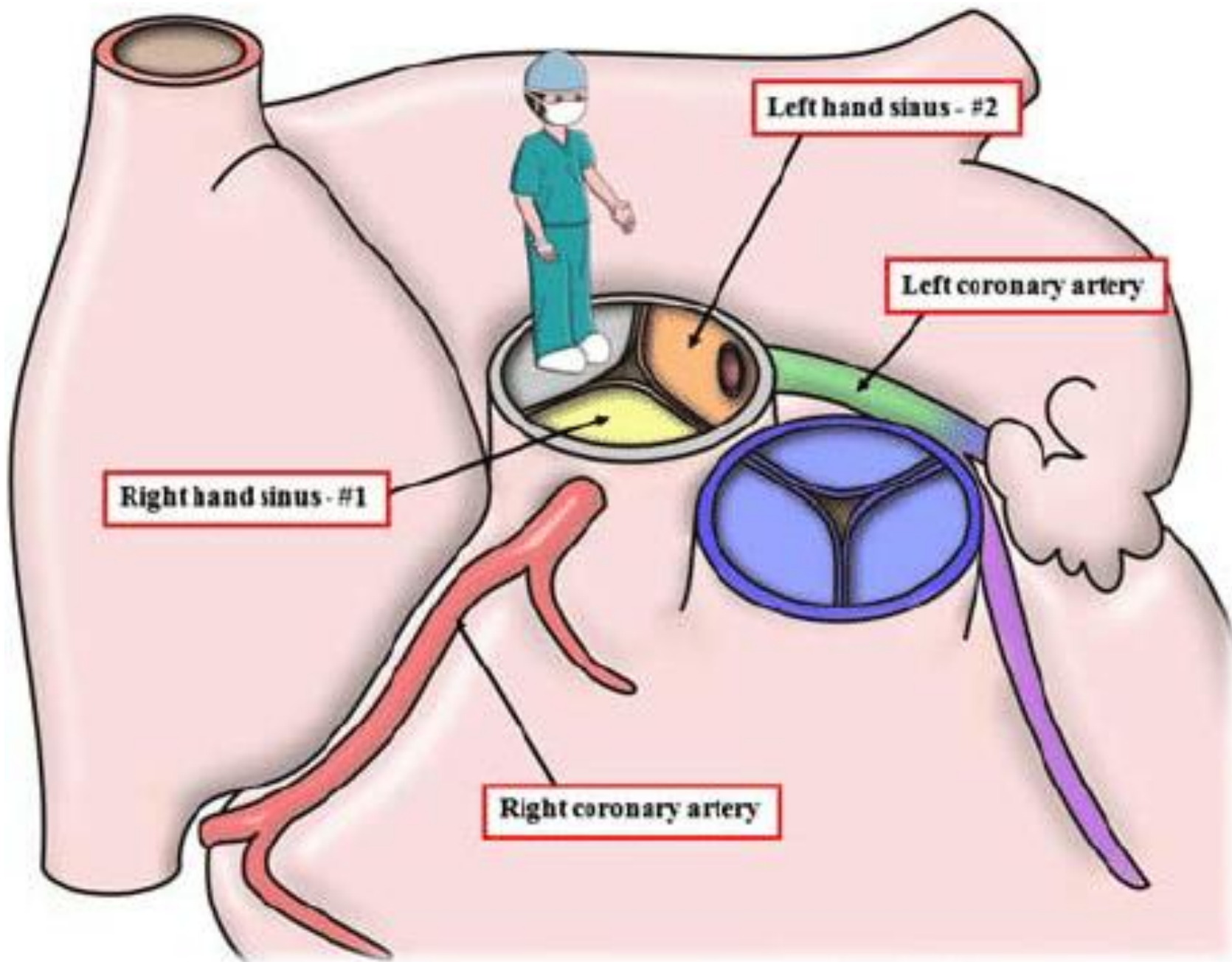
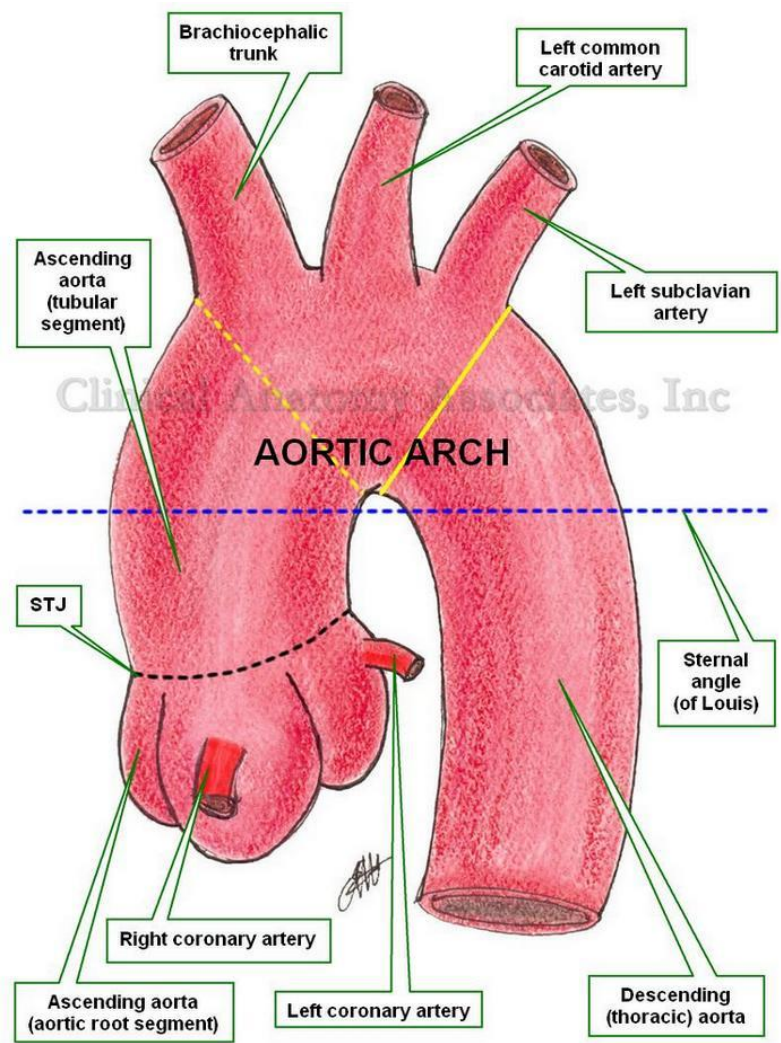


Aortic valve

- ✓ Behind each cusp the aortic wall bulges to form an **aortic sinus**

- ✓ Anterior aortic sinus gives origin to the **right coronary artery**
- ✓ left posterior sinus gives origin to the **left coronary artery**.

Aortic valve

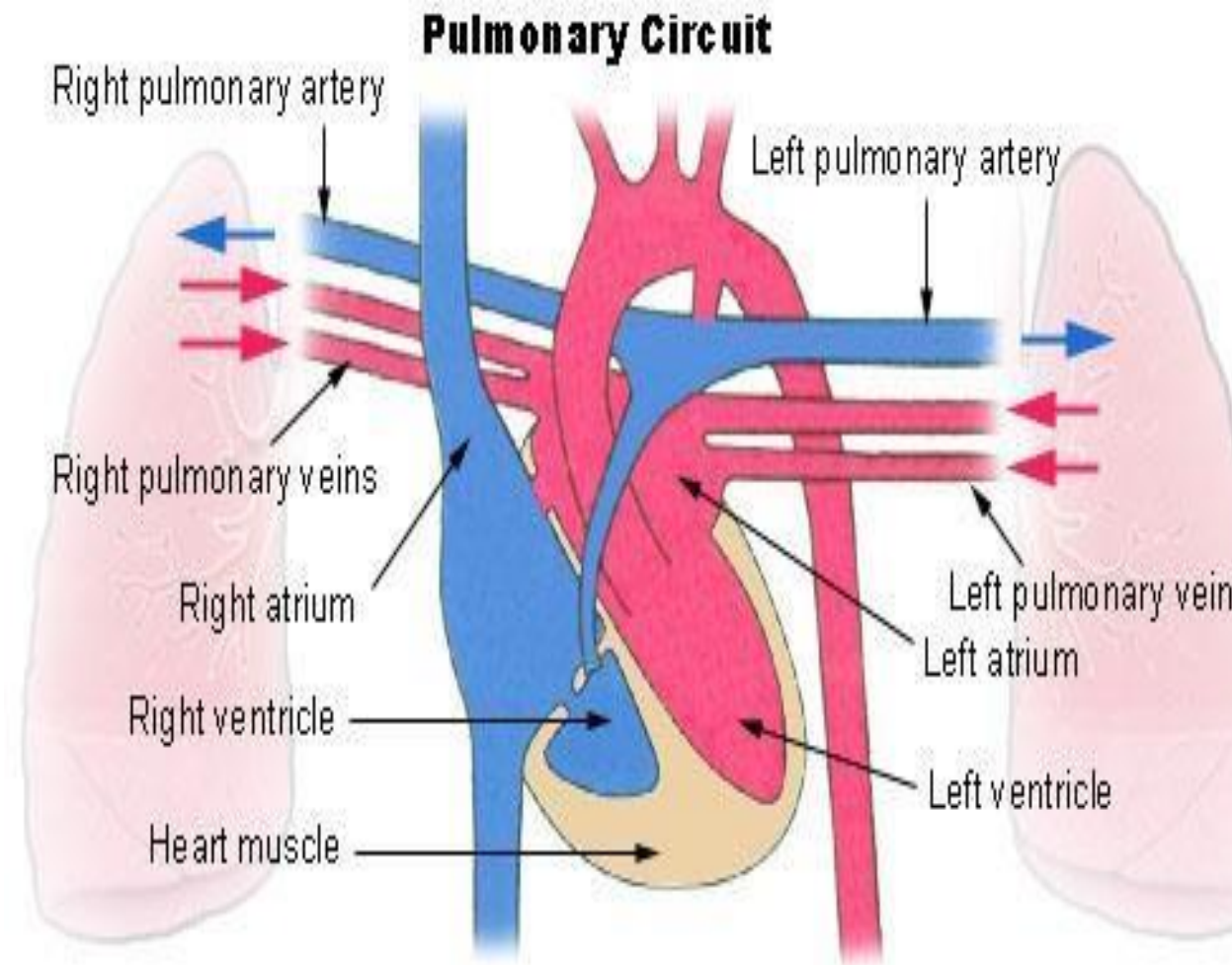


PULMONARY CIRCULATION

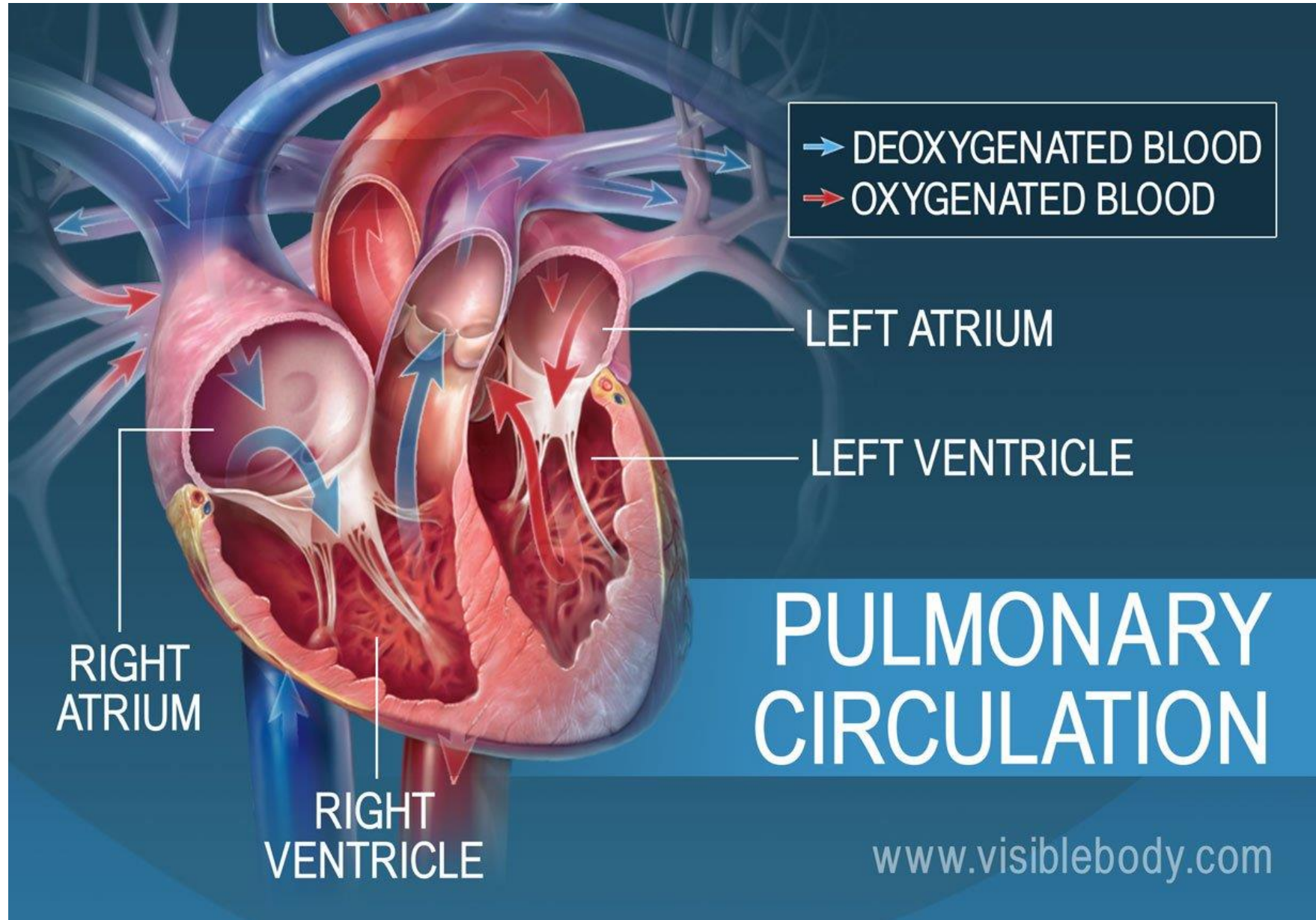
Heart: heart pumps blood through two major pathways.

Pulmonary circulation

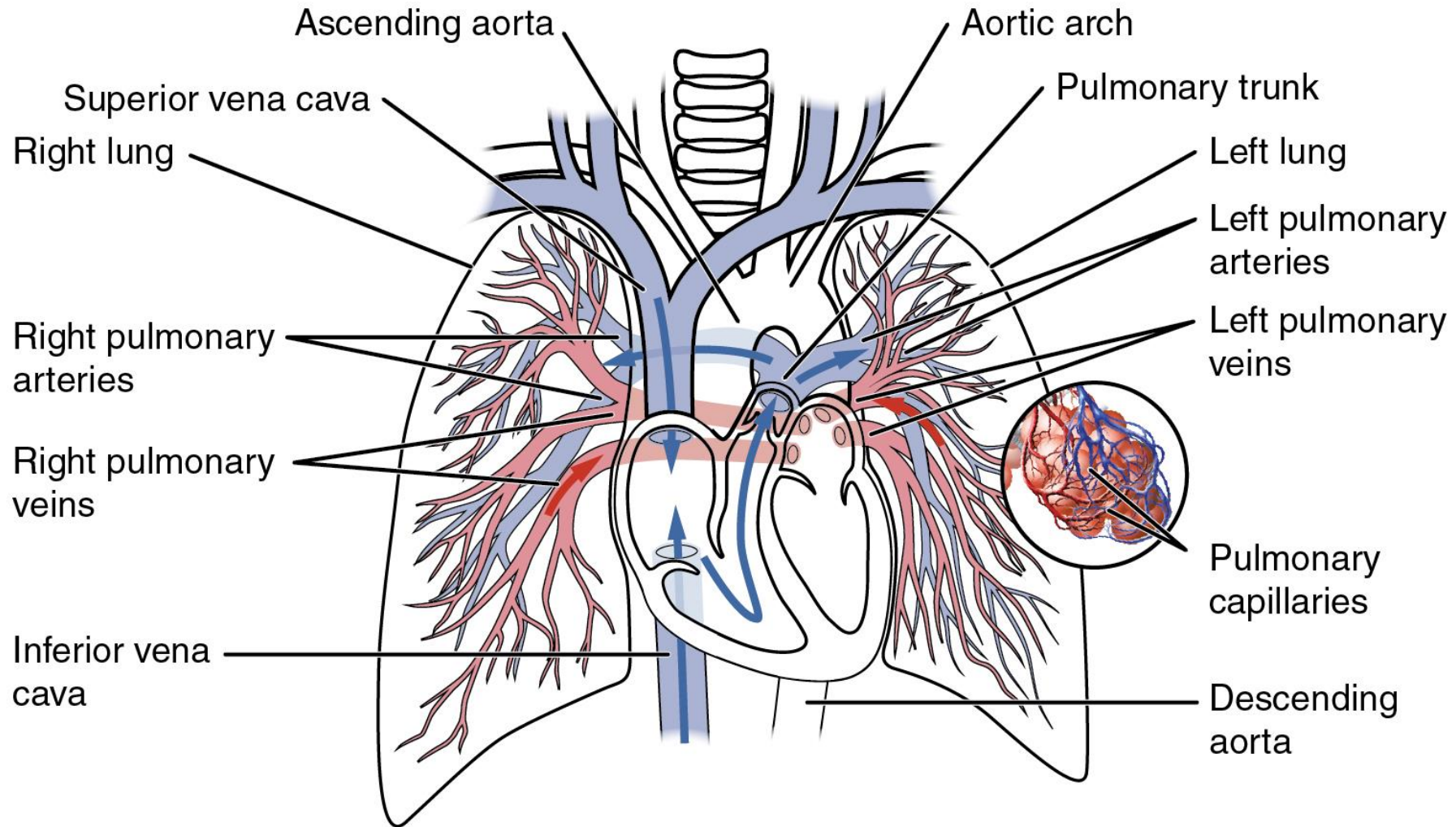
- ✓ Transports oxygen-poor blood from the right ventricle to the lungs where blood picks up a new oxygen supply
- ✓ & back to the left atrium thru the left pulmonary veins



PULMONARY CIRCULATION

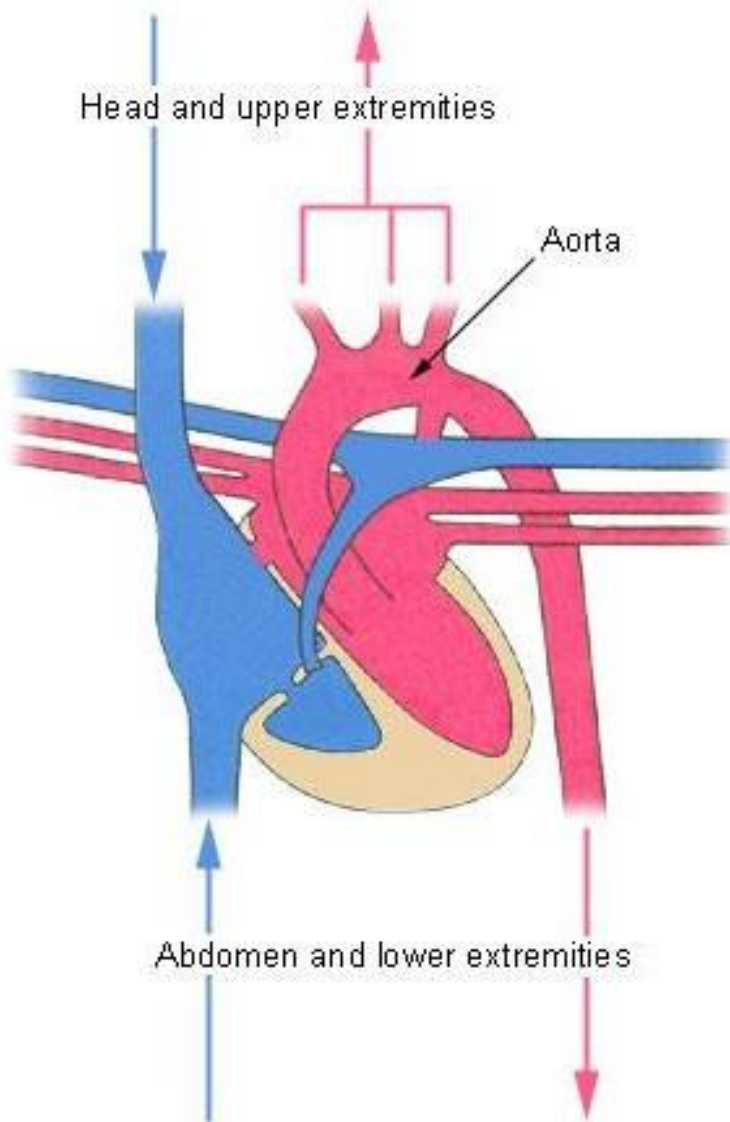


PULMONARY CIRCULATION



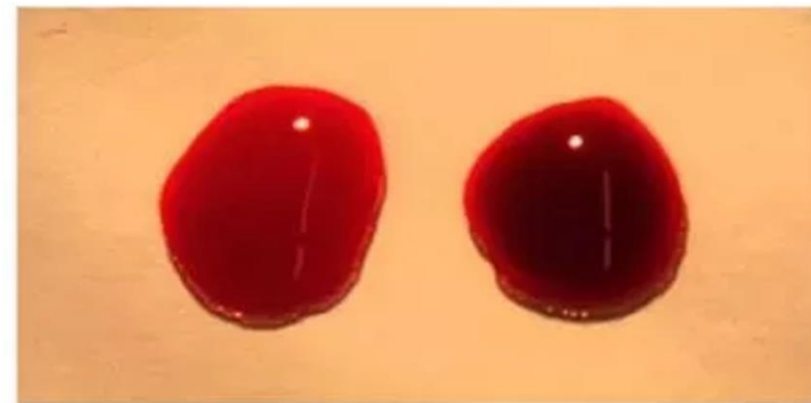
SYSTEMIC CIRCULATION

Systemic Circuit



2) Systemic circulation

- ✓ Distribute oxygen rich blood to all parts of the body
- ✓ It also picks up carbon dioxide and other waste products.



Oxygenated
blood: **Bright red**

Deoxygenated
blood:
Deep red-purple

Systemic circulation

- ✓ Carry oxygen-rich blood away from
- ✓ the left ventricle of the heart to
- ✓ the tissues of the body and back to the right atrium.

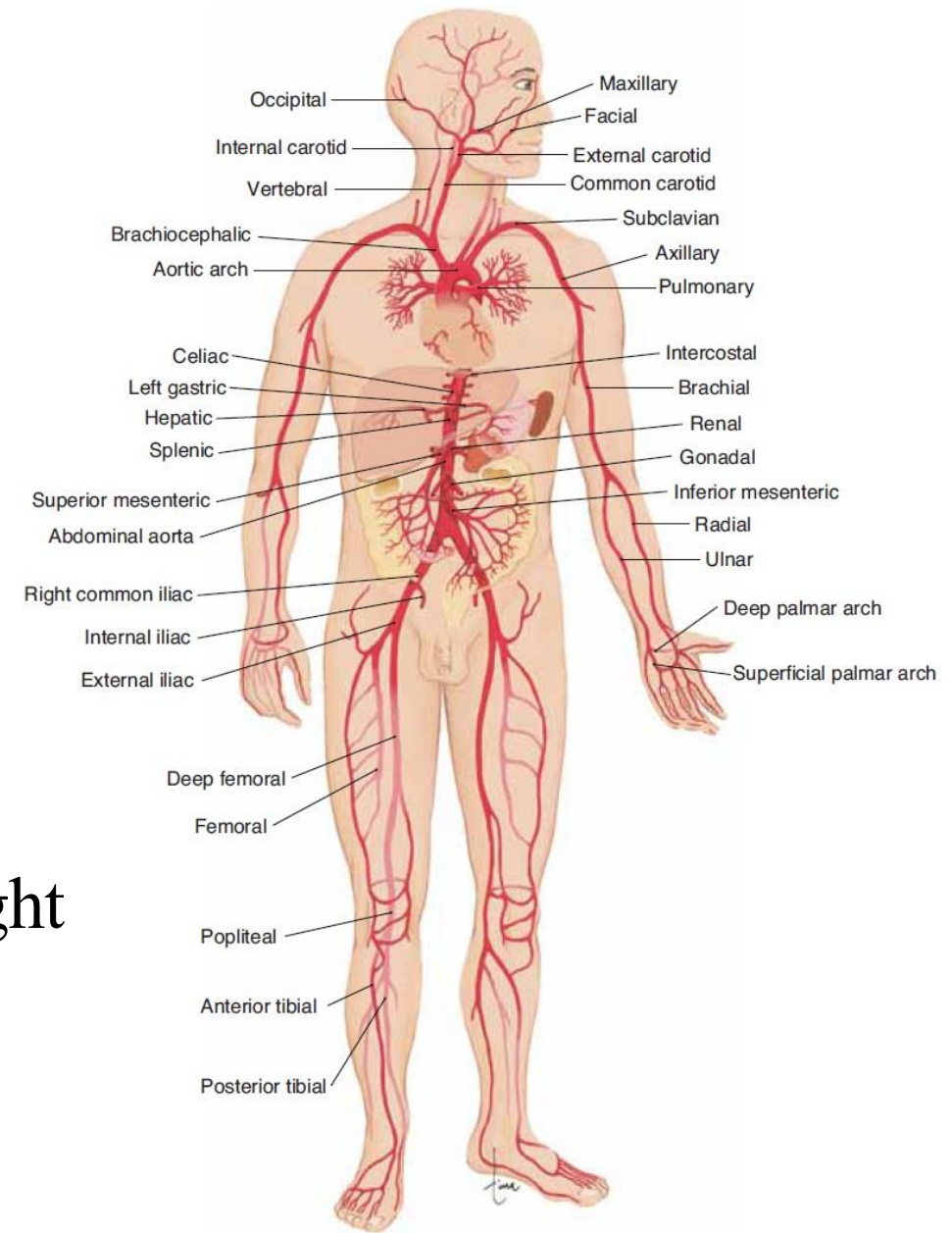
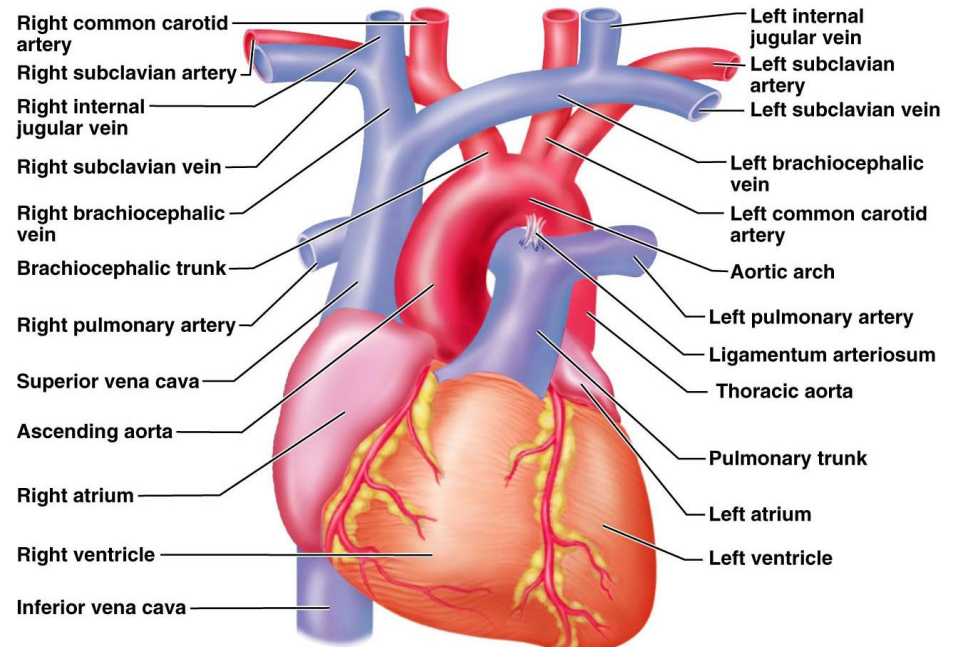
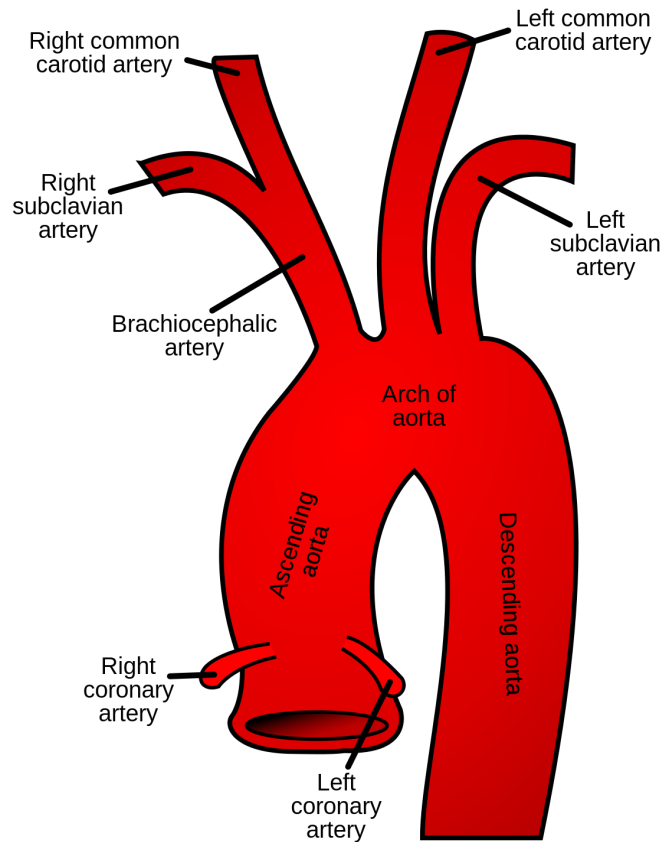


Figure 13-3. Systemic arteries. The aorta and its major branches are shown in anterior view.

QUESTION: Can you find three arteries named after bones? After organs?

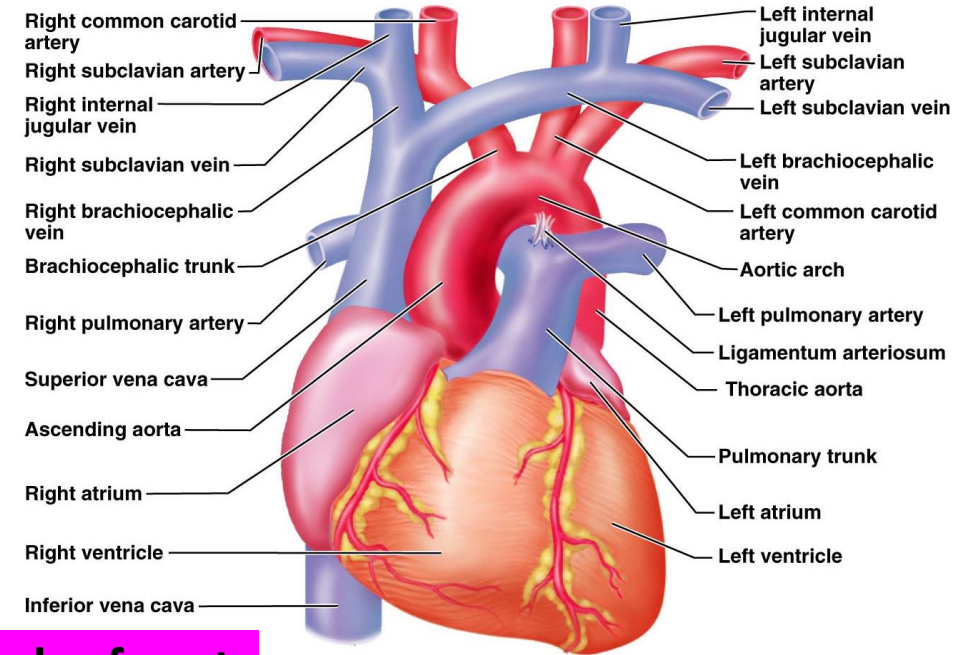
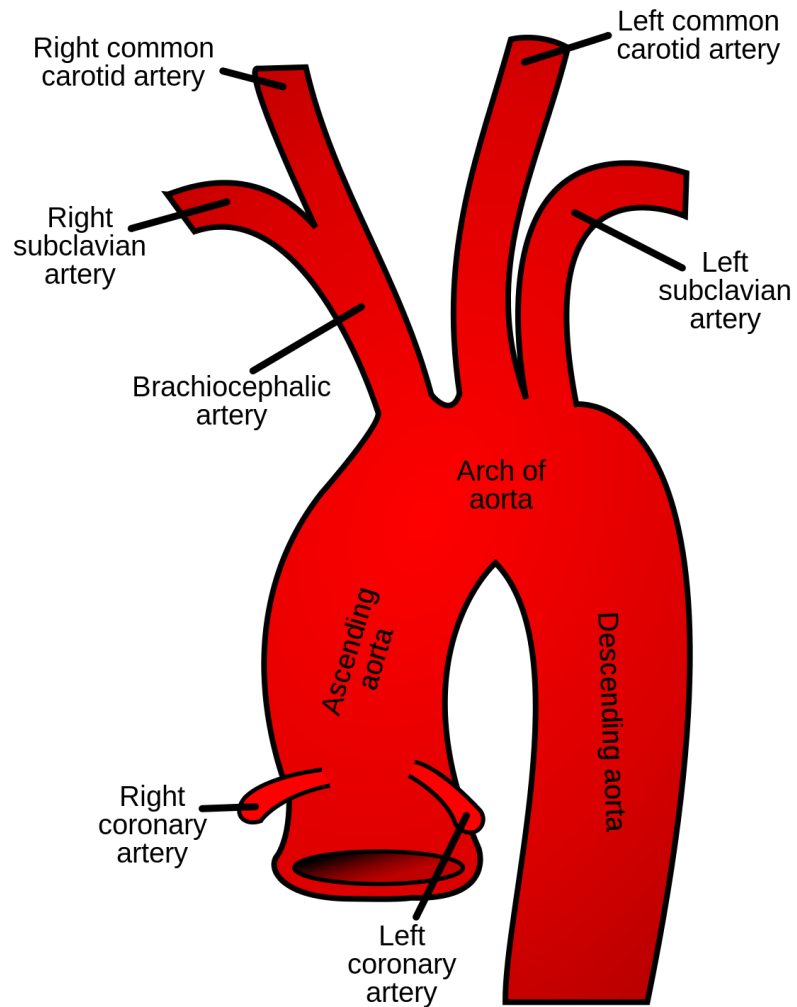
Aorta

✓ largest artery in the body



Ascending aorta

- ✓ Give rise to
 - i. right coronary artery
 - ii. left coronary artery
- ✓ supply the heart

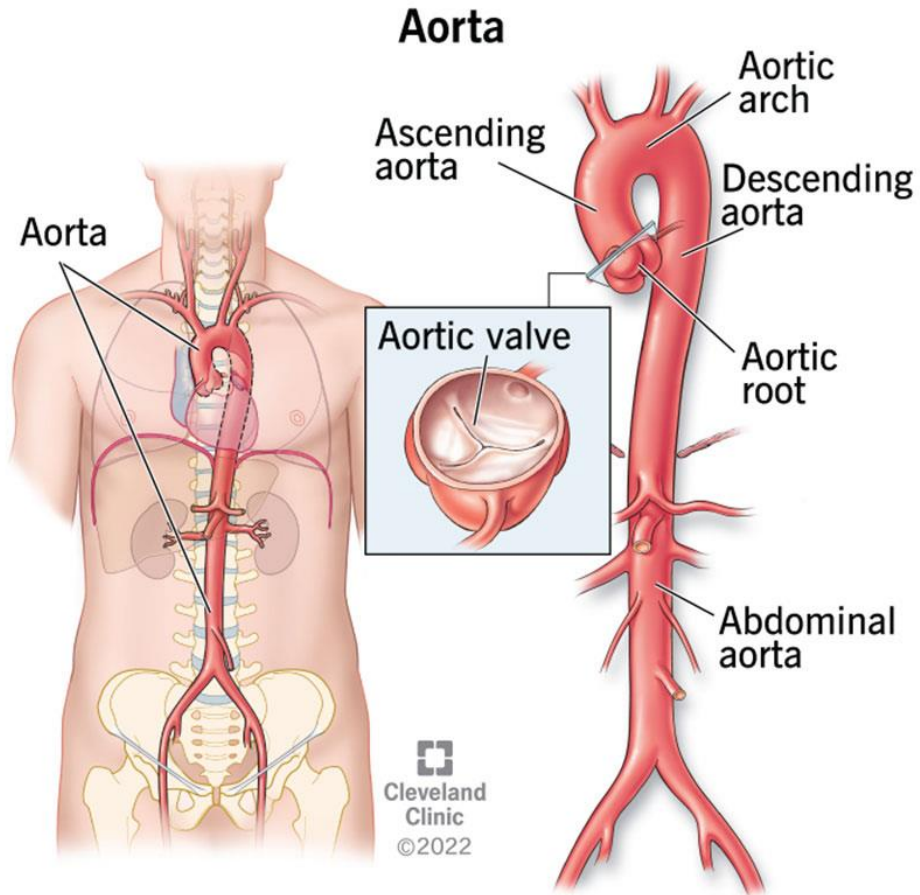


Arch of aorta

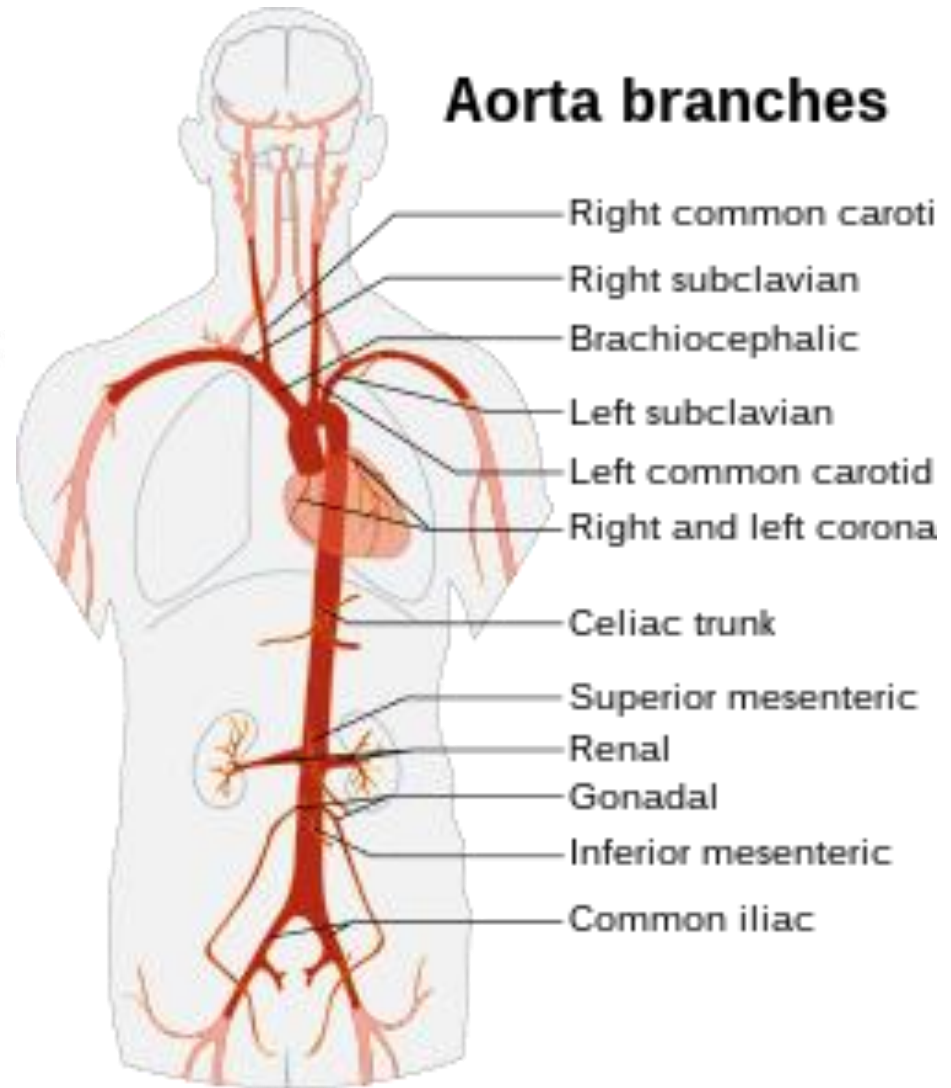
✓ 3 main branches

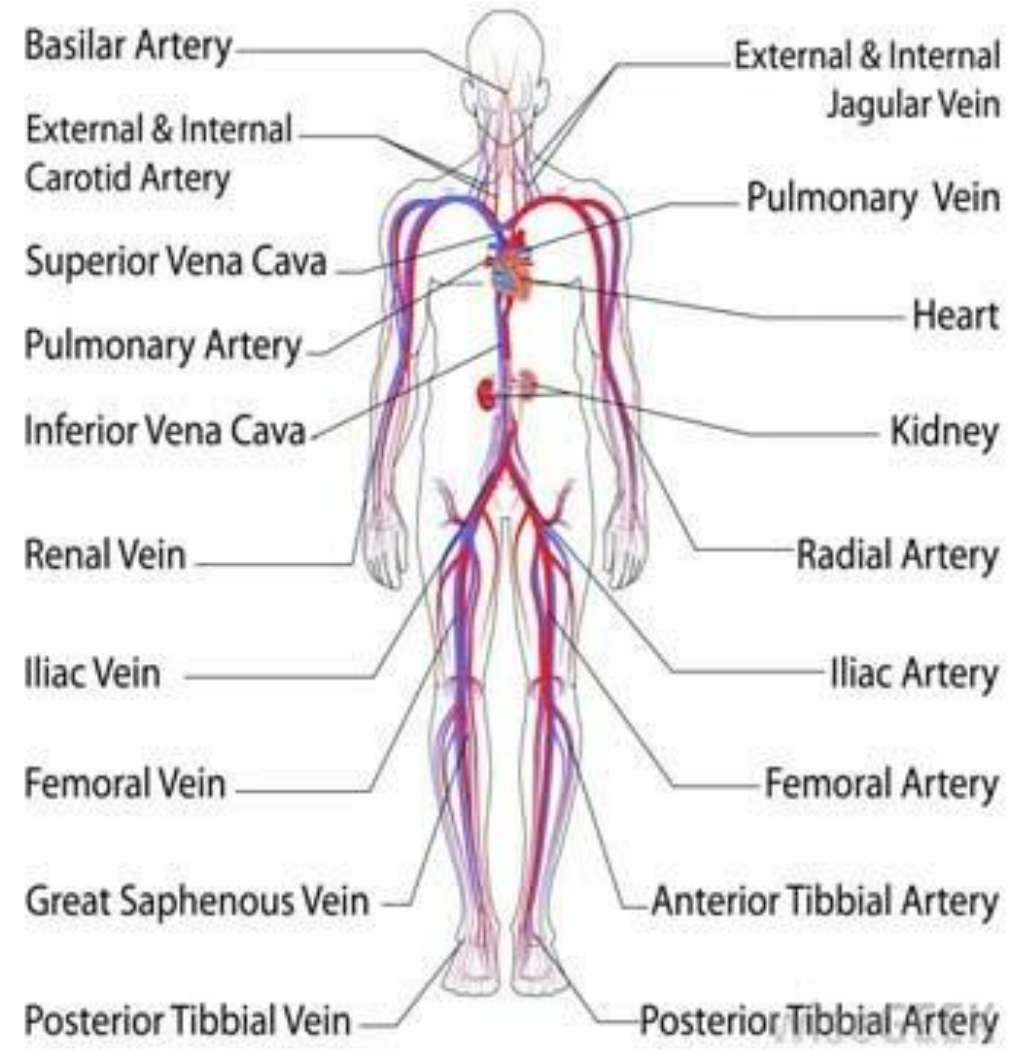
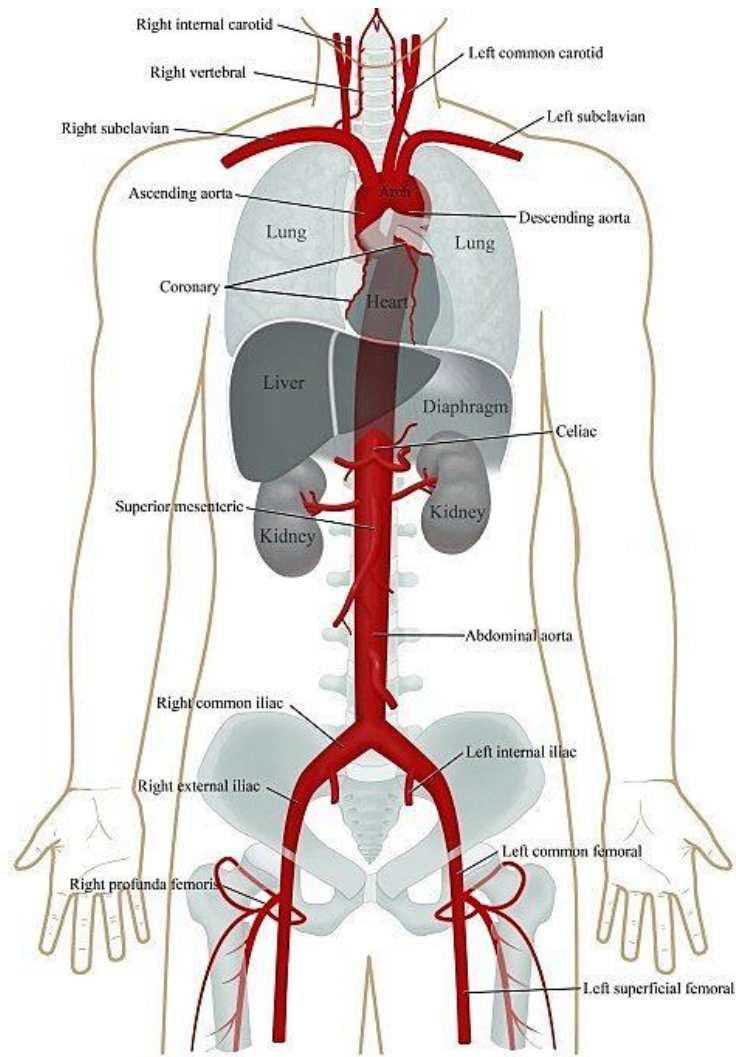
- i. **Brachiocephalic** trunk (innominate artery)
- ii. Left common carotid artery
- iii. Left subclavian artery

Descending aorta



Aorta branches





Oxygen-poor blood enters the heart from the body (blue, top left) and is pumped out to the lungs (blue, top right)



Oxygen-rich blood enters the heart from the lungs (red, right side) and is pumped out to the body (top)

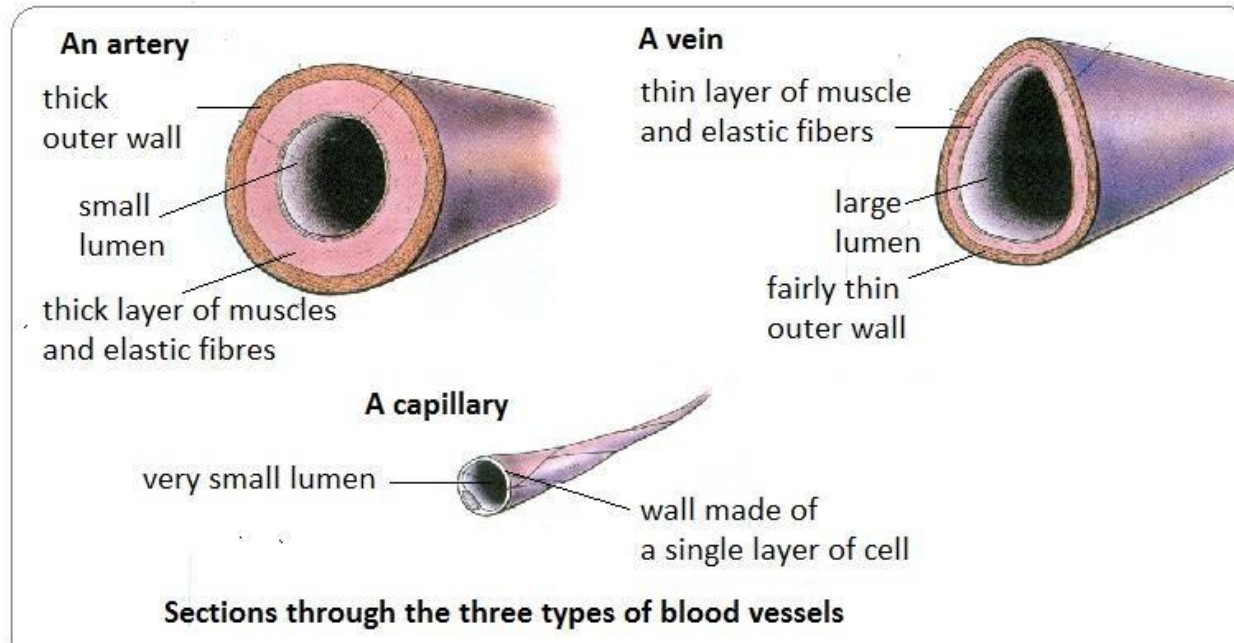
- **Hormones** from glands help regulate cell activity.
- **Oxygen** from the lungs combines with nutrients to provide energy.
- **Nutrients** from the digestive system provide food for the cells.

THE CIRCULATORY SYSTEM WORKS IN CONJUNCTION WITH OTHER BODY SYSTEMS, TO KEEP THE IT WORKING PROPERLY.

- ❖ When your blood circulates through your **digestive system**, it picks up nutrients your body absorbed from your last meal.
- ❖ **Your blood also carries oxygen inhaled by the lungs.** Your circulatory system delivers oxygen and nutrients to your heart and the other cells of your body then picks up any waste products created by these cells, including carbon dioxide, and **delivers these waste products to the kidneys and lungs for disposal.**
- ❖ The circulatory system carries **hormones from the endocrine system**, and the immune system's white blood cells that fight off infection.
- ❖ The circulatory system provides your **brain** with a constant supply of oxygen-rich blood while your brain regulates your heart rate and blood pressure.
- ❖ Your circulatory system delivers oxygen-rich blood to your **bones**. Meanwhile, your bones are busy making new blood cells.

BLOOD VESSELS - transport your blood throughout your body. There are 3 types of blood vessels.

- **Arteries**: Blood vessels that carry blood away from the heart to other parts of the body. They are much thicker than Veins because of the high pressure of blood coming from the heart.
- **Veins**: Blood vessels that carry blood from the body back to the heart.
- **Capillaries**: Tiny tubes that carry blood from the arteries to the body's cells, and then back to the veins.

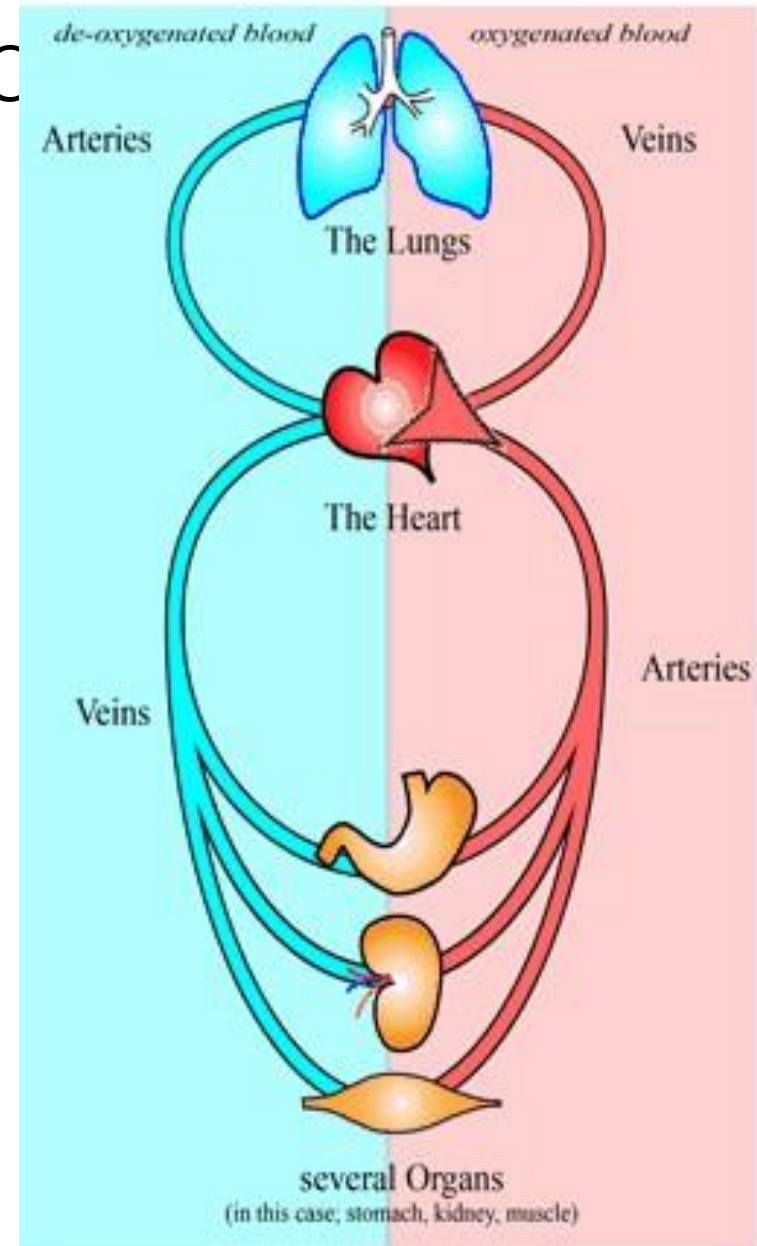
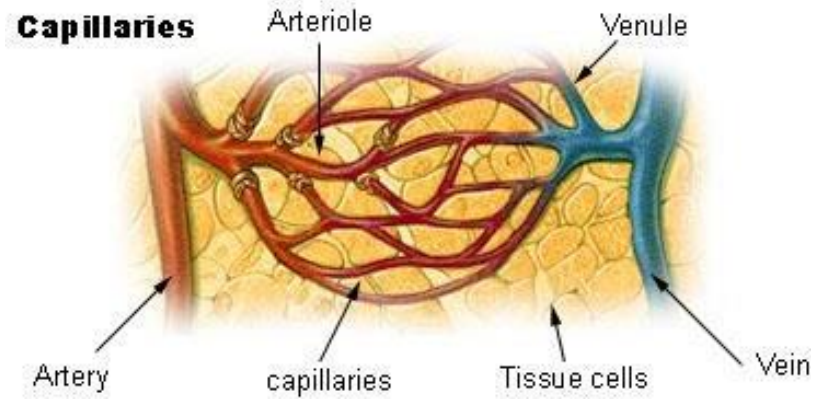
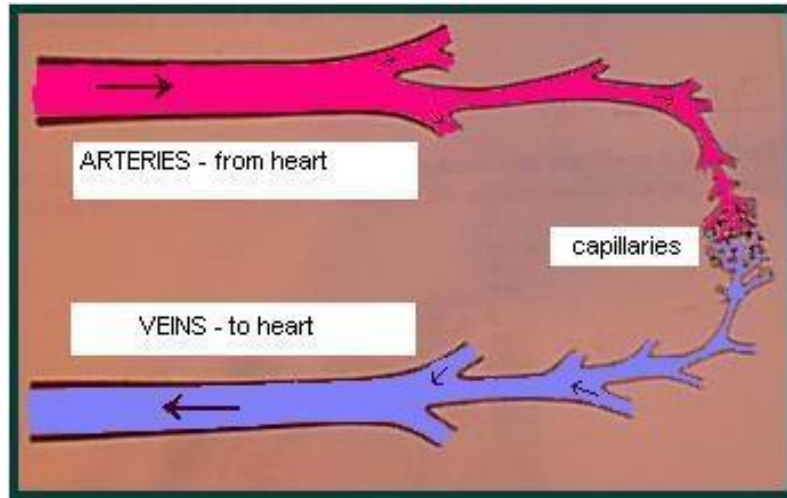


CAPILLARIES

- Body tissues contain a vast network of thin capillaries.
- Capillary walls are only one cell thick, allowing exchange of gases, nutrients, and wastes.
- Capillaries are so fine that red blood cells must line up single-file to go through them.



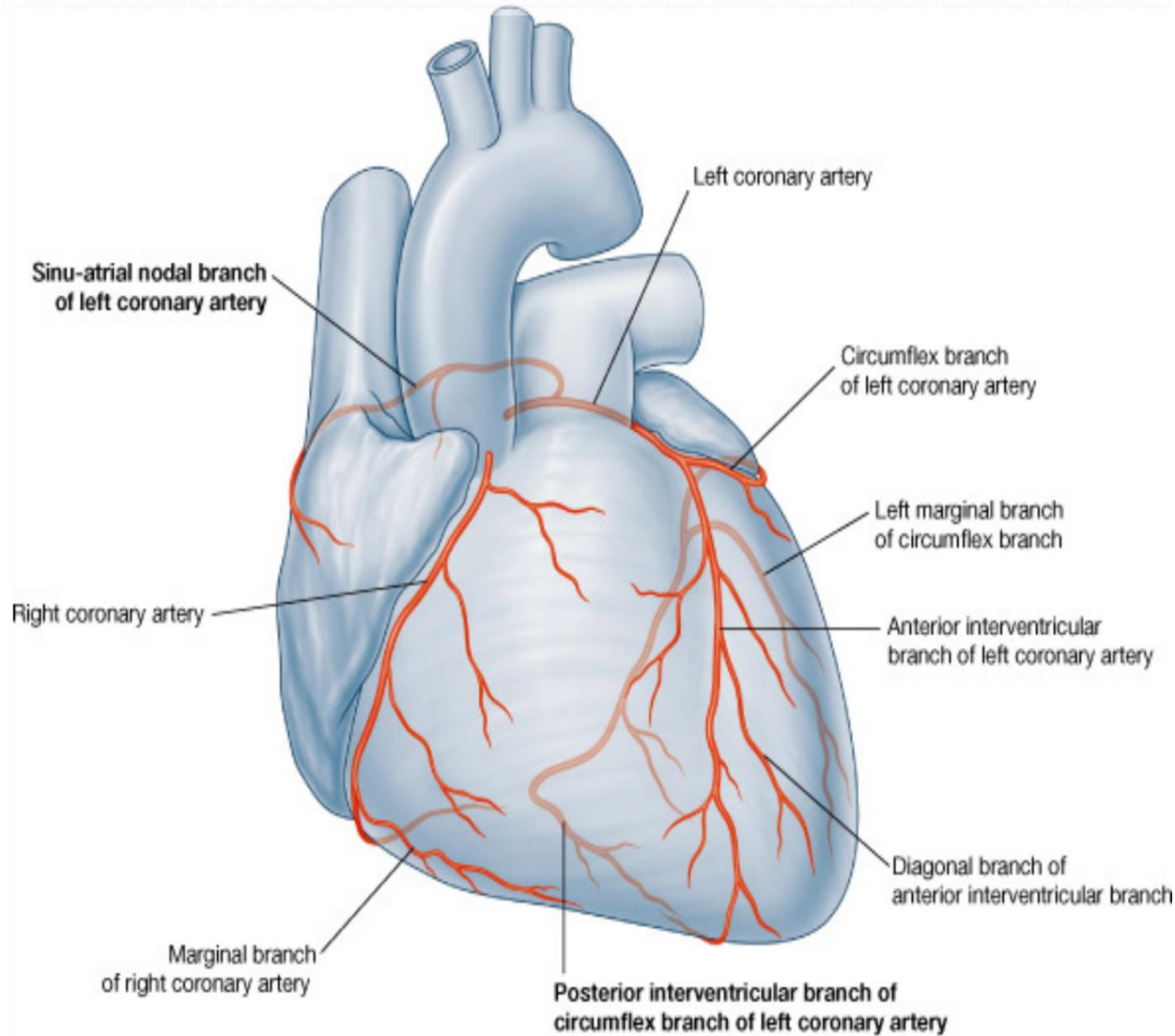
ARTERIES, VEINS, AND CAPILLARIES



Blood supply of the heart

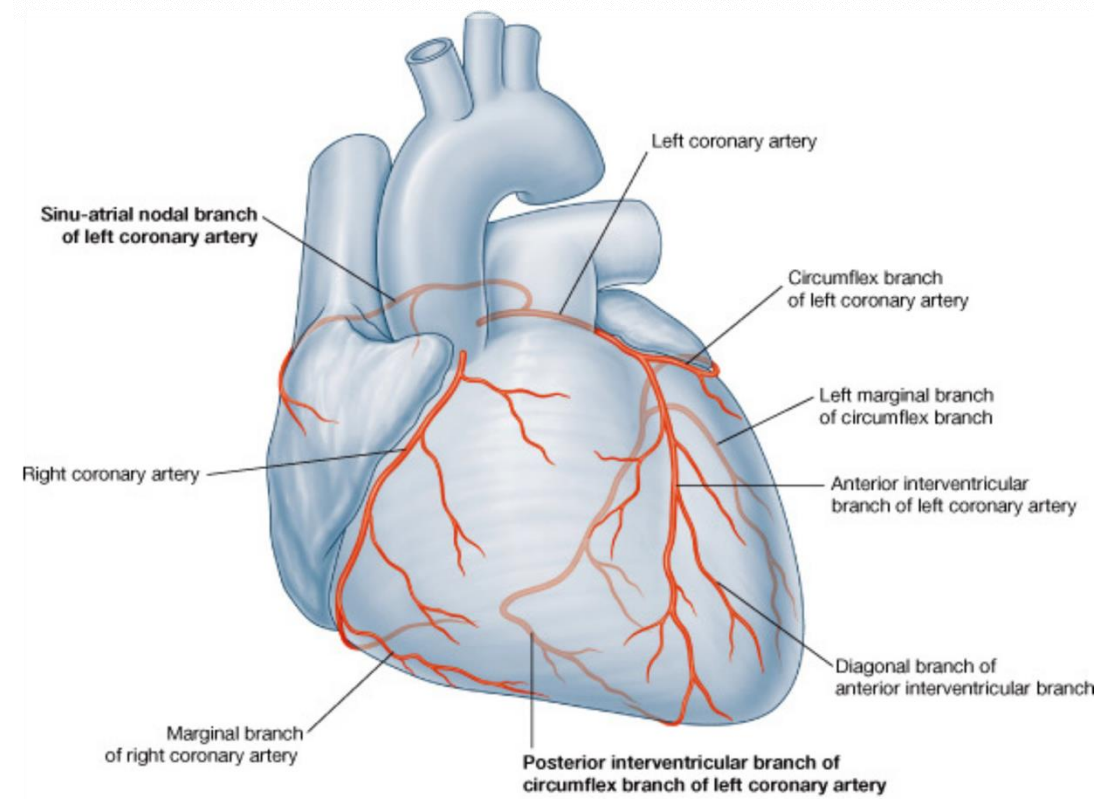
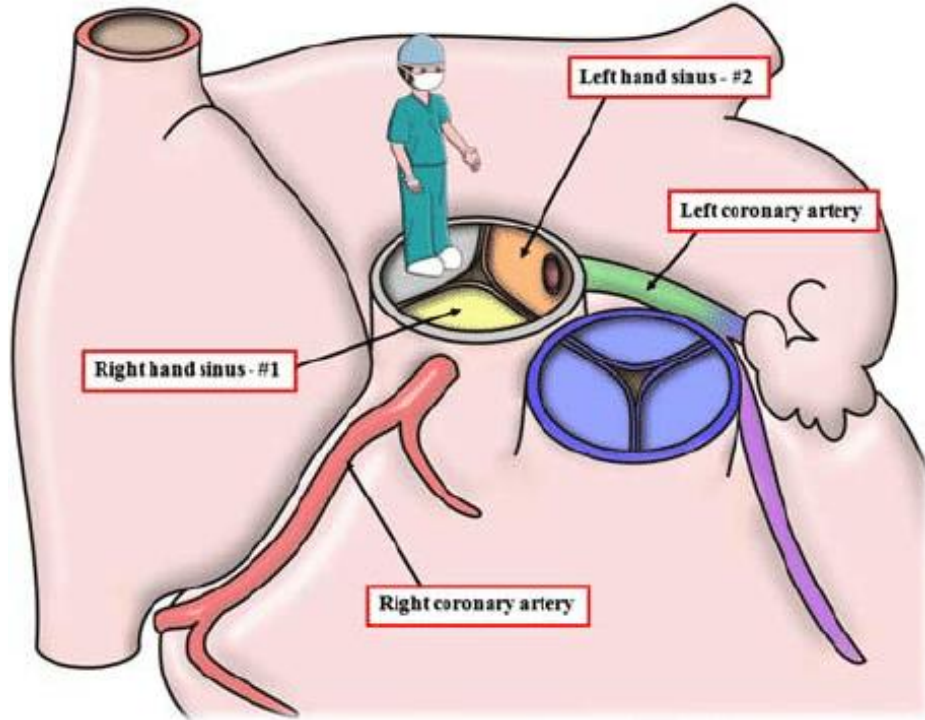
Arterial Supply of the Heart

- ✓ Provided by the:
 - i. right coronary artery
 - ii. left coronary artery
- ✓ Arise from the ascending aorta immediately above the aortic valve
- ✓ The coronary arteries & their major branches are distributed over the surface of the heart

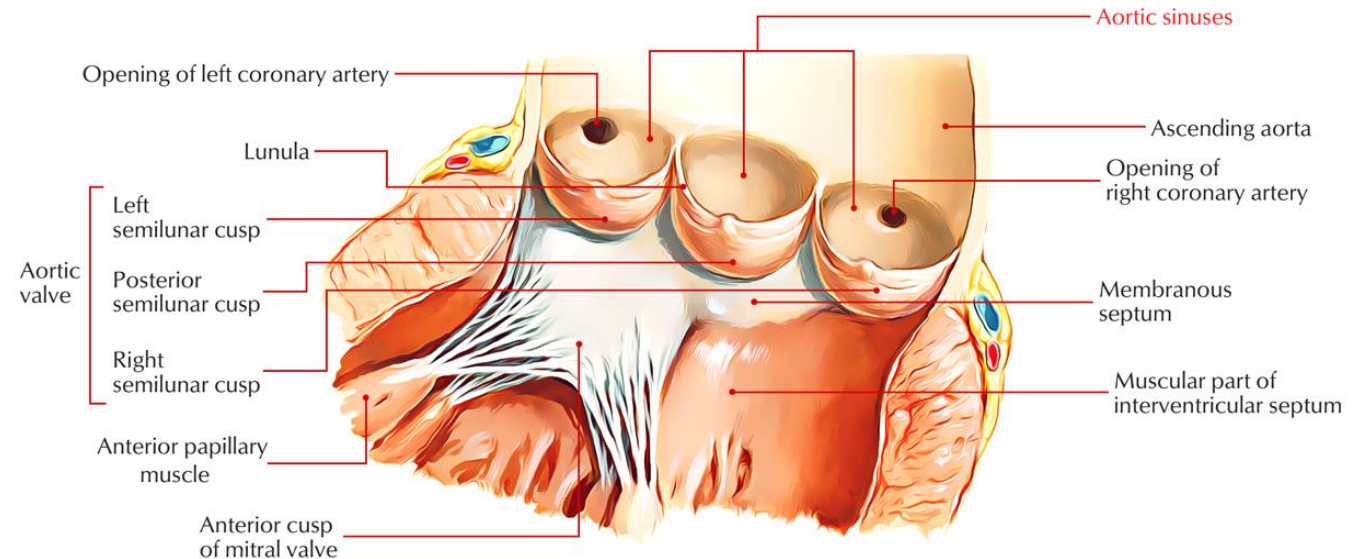


Right coronary artery

- ✓ Arises from the anterior aortic sinus of the ascending aorta
- ✓ And runs forward btwn the pulmonary trunk & the right auricle.
- ✓ It descends almost vertically in the right atrioventricular groove.

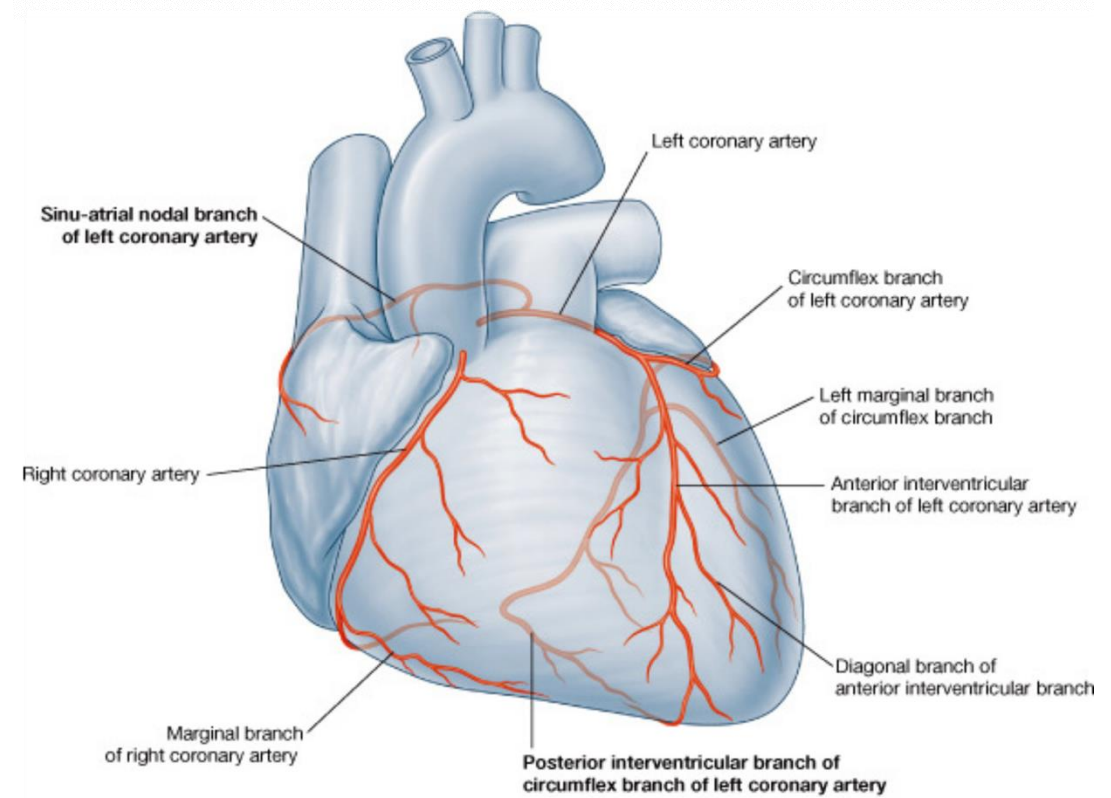
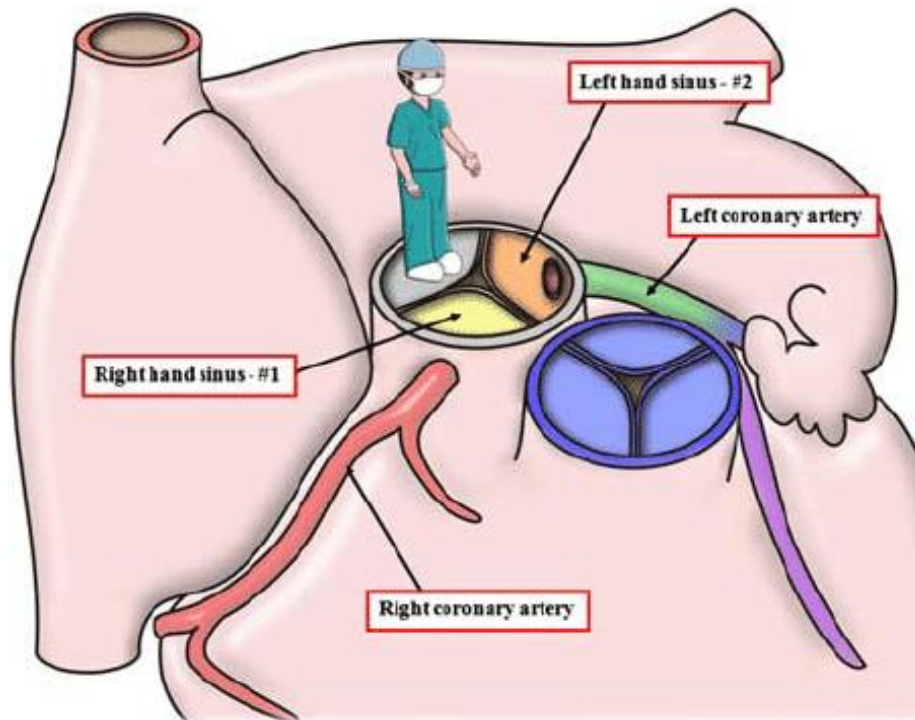


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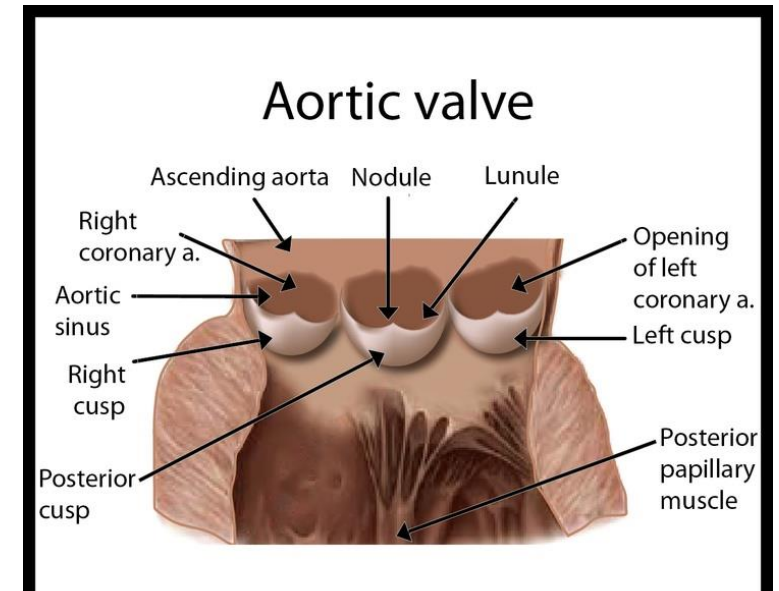


Right coronary artery

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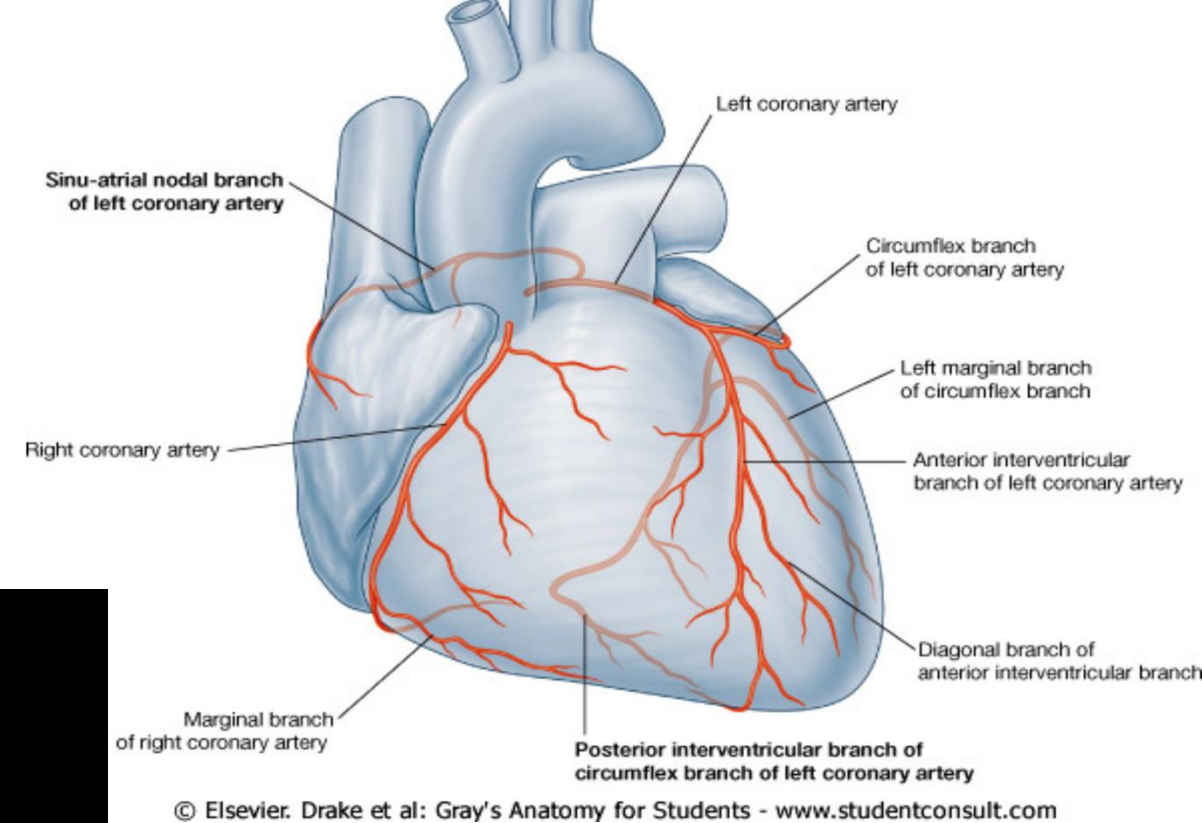
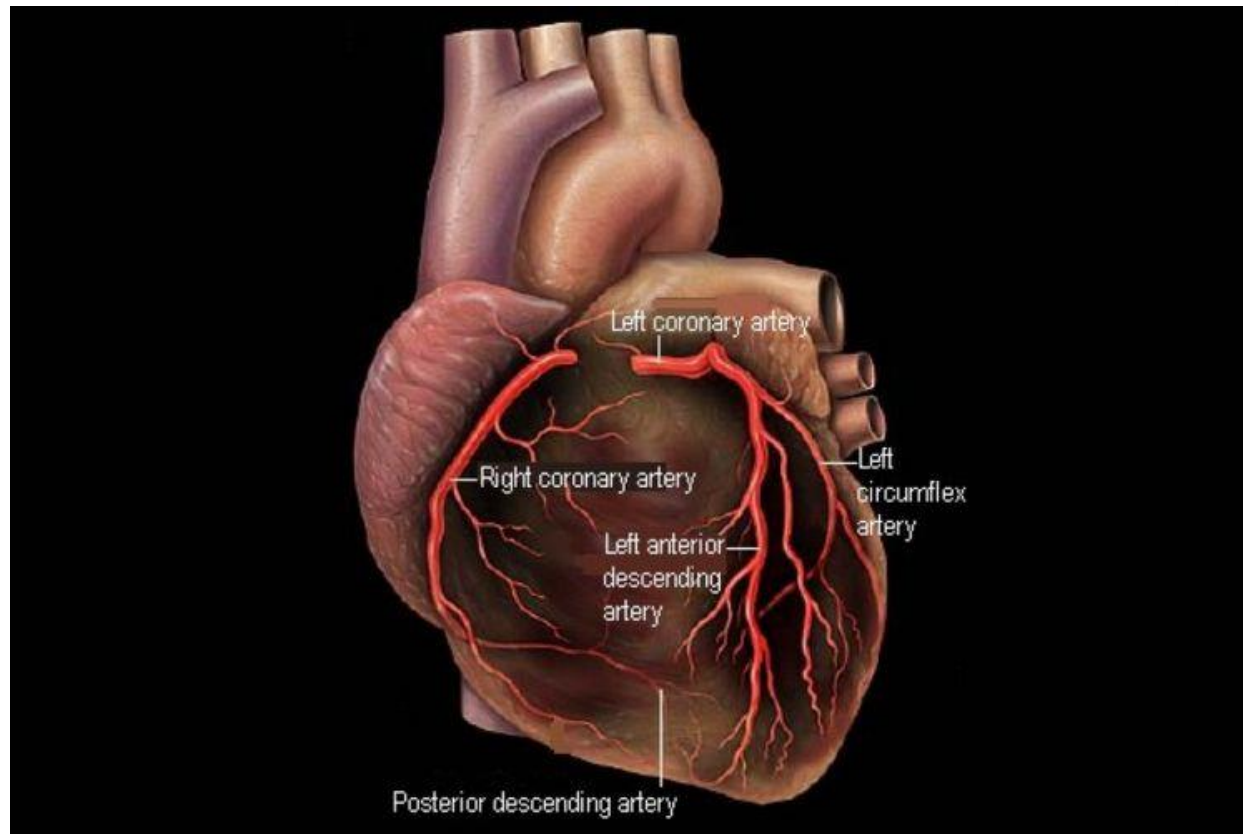


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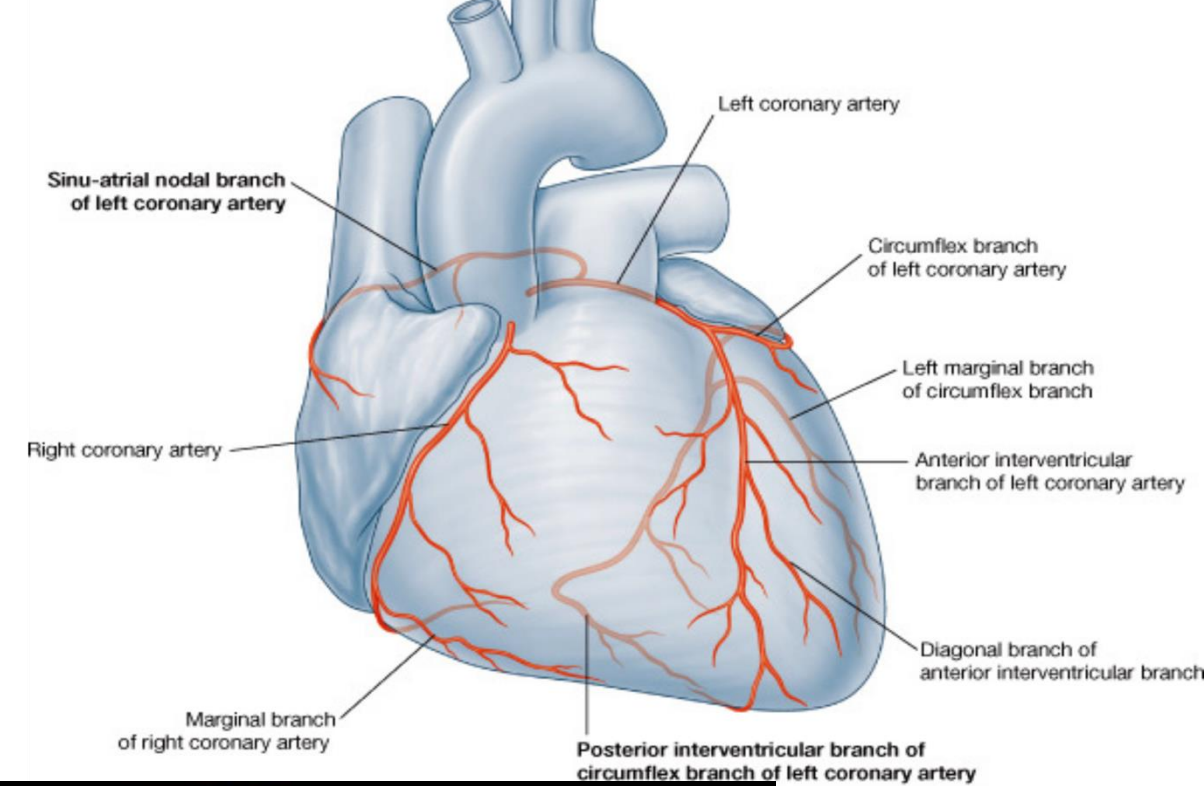
Right coronary artery

- ✓ At the inferior border of the heart it continues posteriorly along the atrioventricular groove to anastomose with the left coronary artery in the posterior interventricular groove.

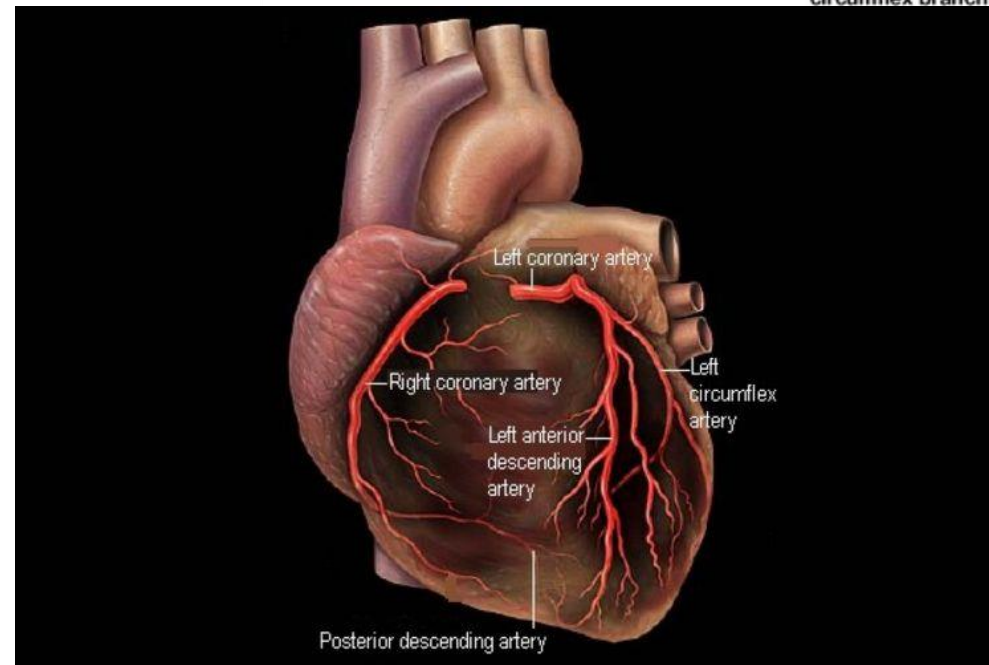
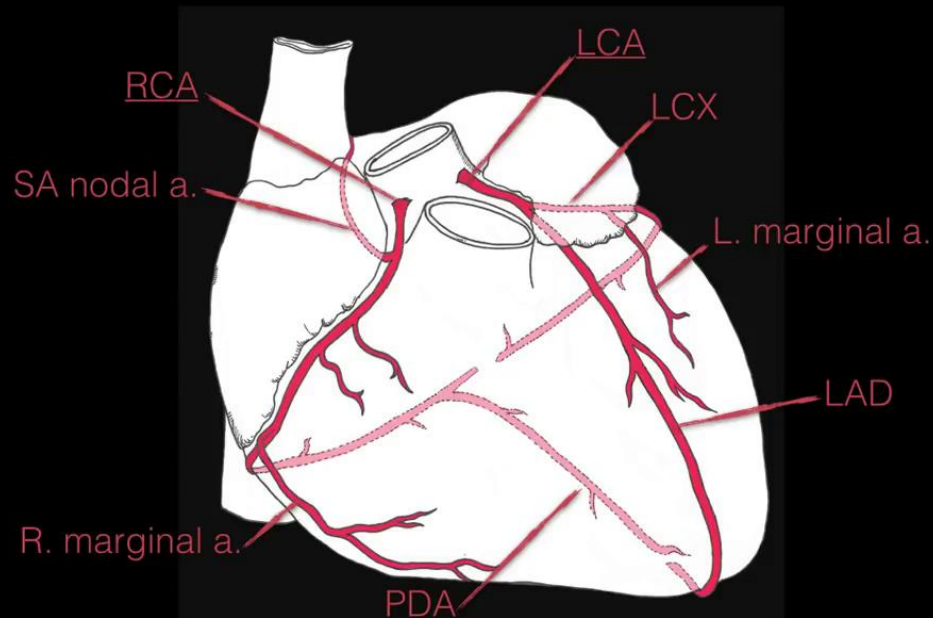


Right coronary artery

- ✓ The branches from the right coronary artery supply the:
- right atrium
 - right ventricle
 - parts of the left atrium and left ventricle
 - atrioventricular septum.

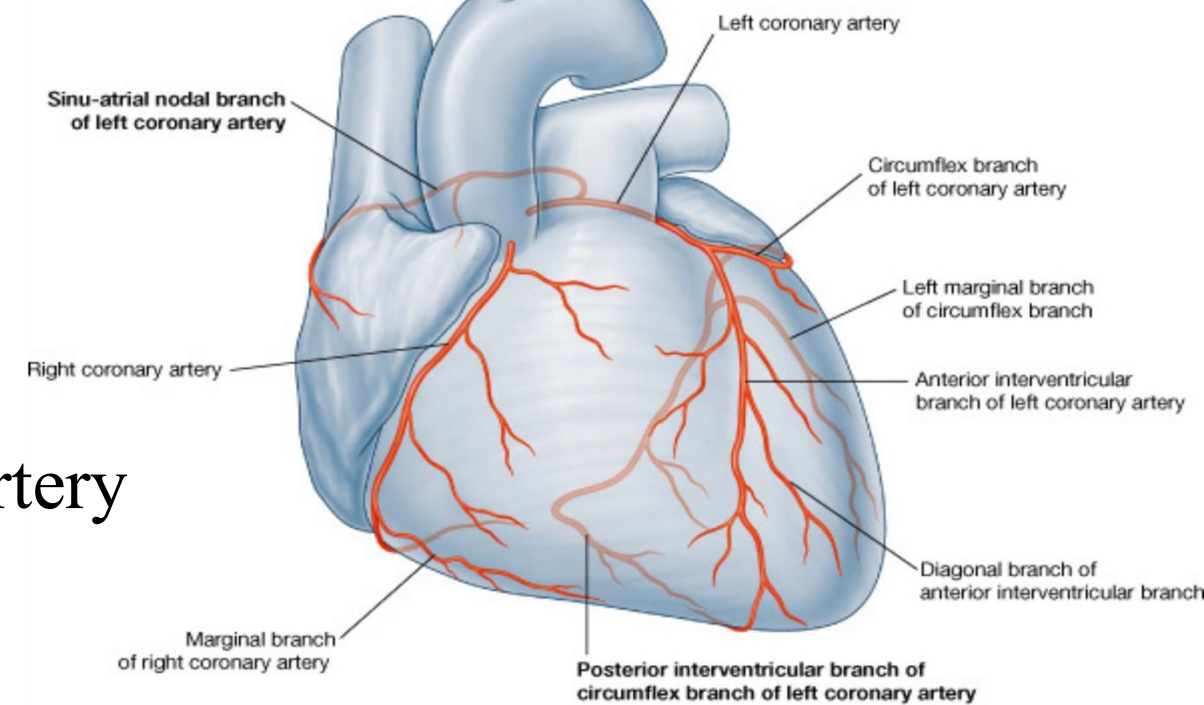
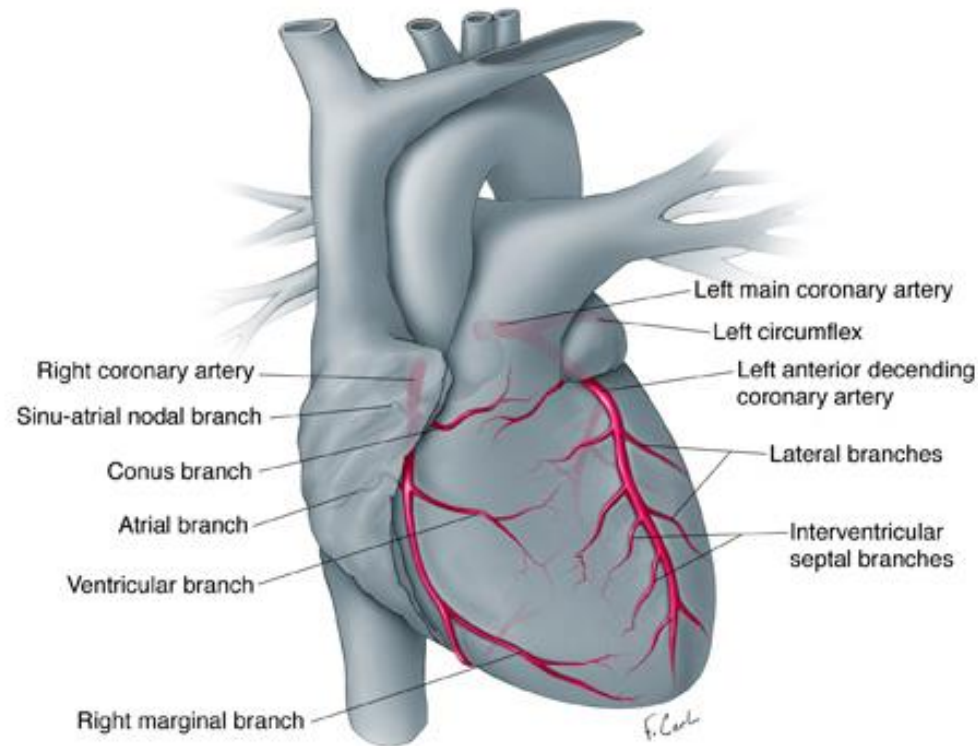


Coronary arteries

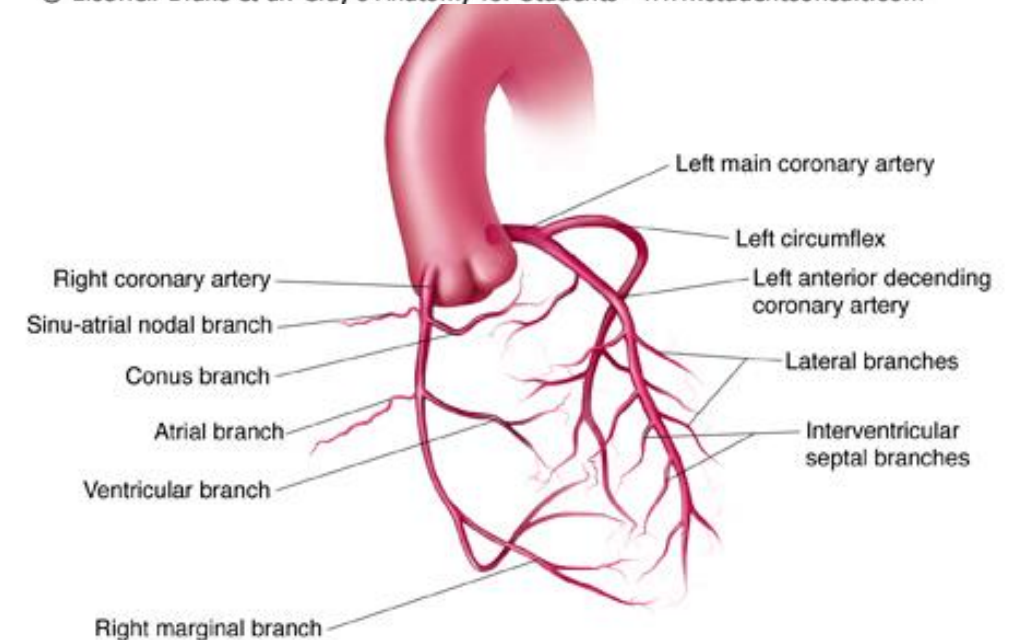


Branches of the right coronary artery

- ✓ Right conus artery
- ✓ Anterior ventricular branches
- ✓ Posterior ventricular branches
- ✓ Posterior interventricular (descending) artery
- ✓ Artery of the sinuatrial node



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Branches of the right coronary artery

Right conus artery

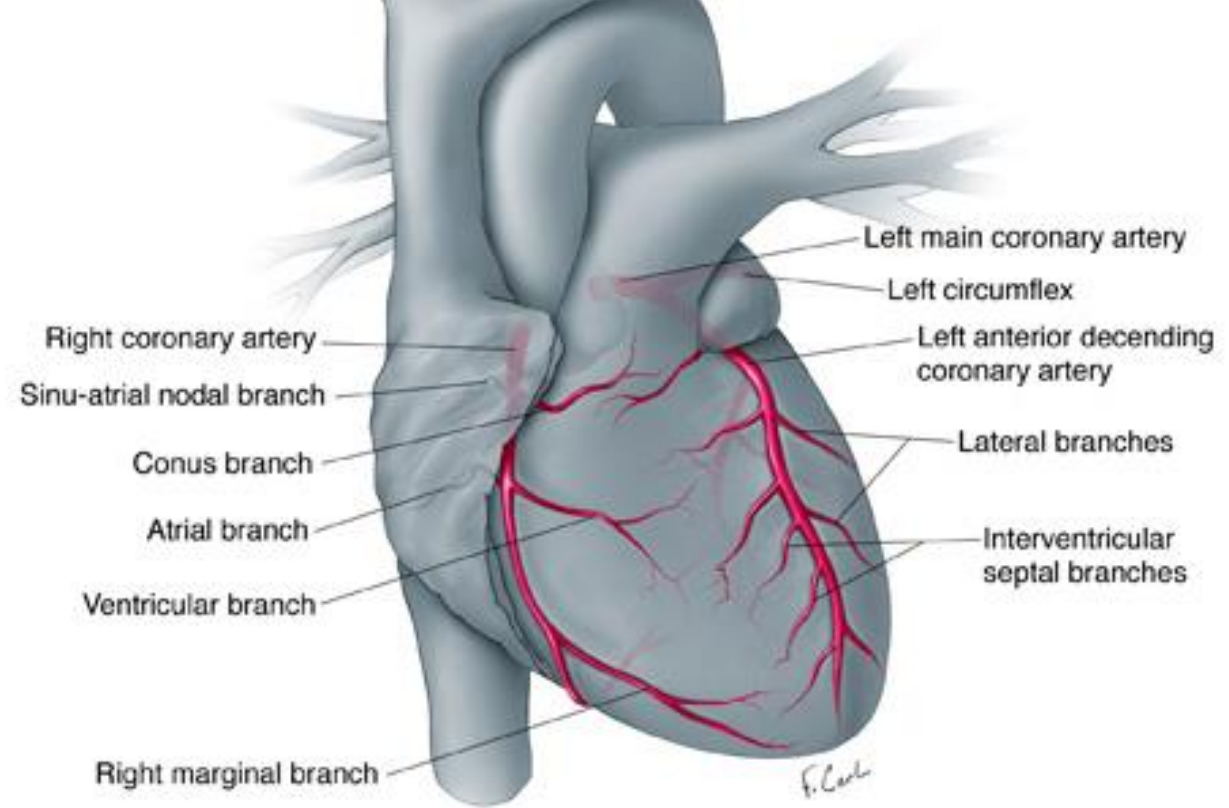
- ✓ Supplies the anterior surface of the pulmonary conus (infundibulum of the right ventricle) & the upper part of the anterior wall of the right ventricle.

Anterior ventricular branches

- ✓ Are 2 or 3 in number
- ✓ Supply the anterior surface of the right ventricle.

Marginal branch

- ✓ Is the largest & runs along the lower margin of the costal surface to reach the apex.



Posterior ventricular branches

- ✓ Are usually 2 in number
- ✓ Supply the diaphragmatic surface of the right ventricle.

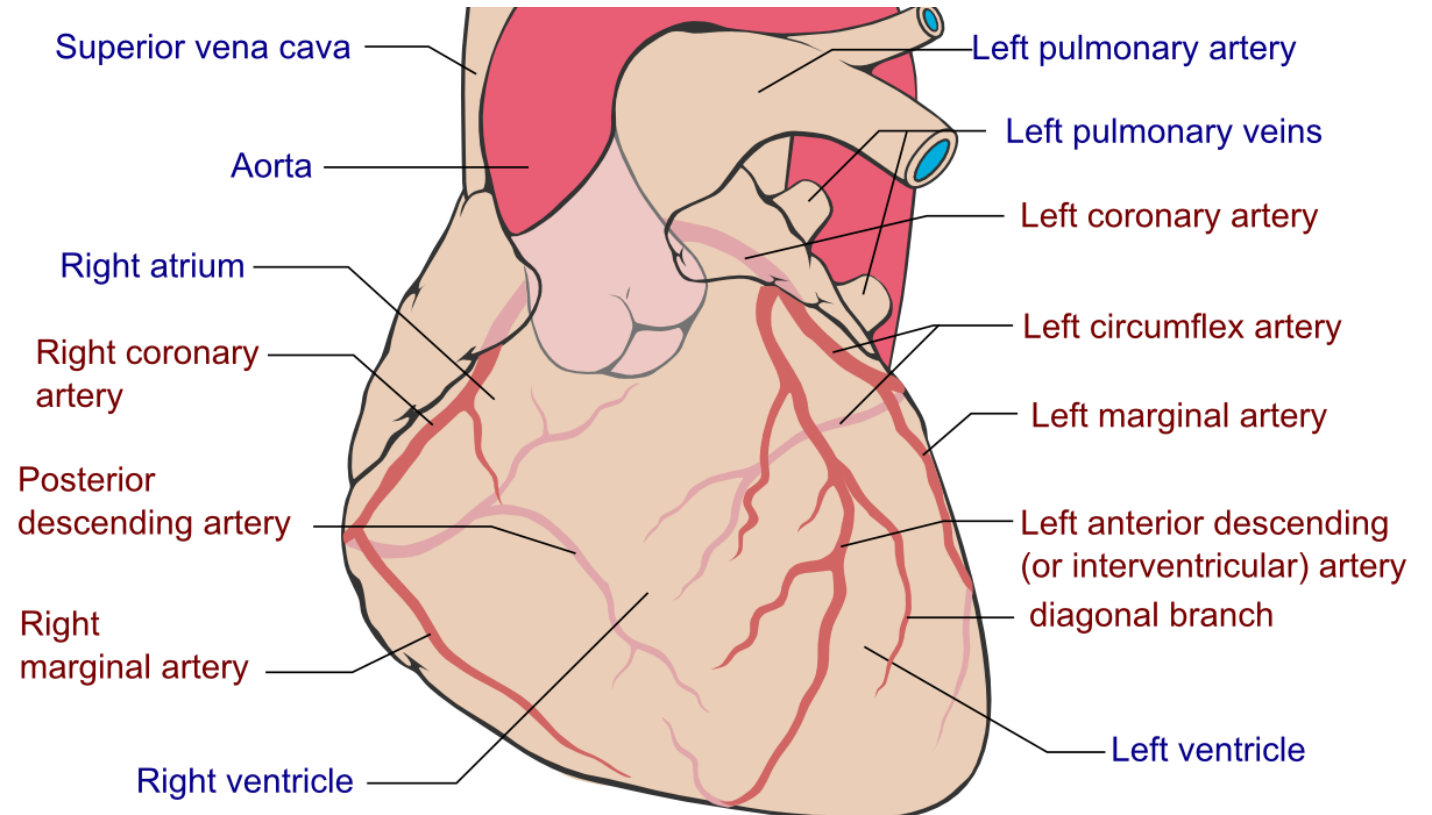
Branches of the right coronary artery

Posterior interventricular (descending) artery (PDA)

- ✓ Runs toward the apex in the posterior interventricular groove.
- ✓ It gives off branches to the right & left ventricles, including its inferior wall.
- ✓ It supplies branches to the posterior part of the ventricular septum but not to the apical part, which receives its supply from the anterior interventricular branch of the left coronary artery.

NB

- ✓ In 10% of individuals the posterior interventricular artery is replaced by a branch from the left coronary artery.



Branches of the right coronary artery

Septal branch

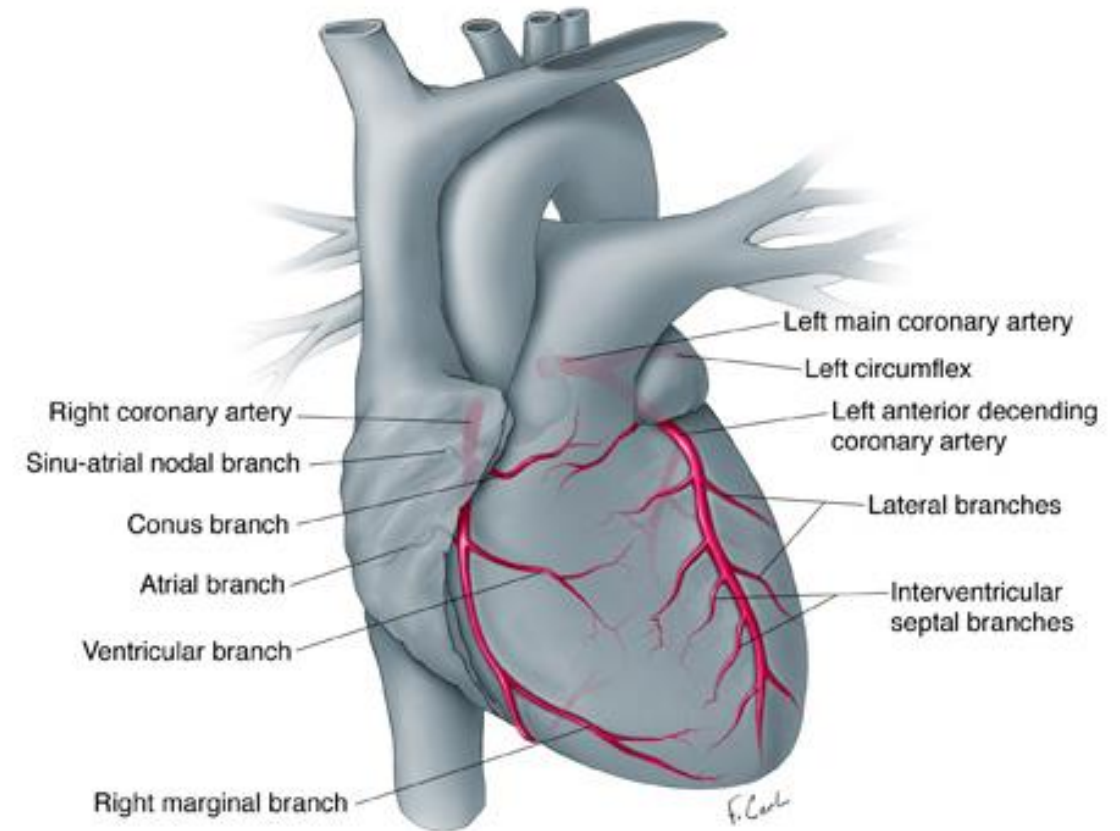
- ✓ Supplies the atrioventricular node.

Atrial branches

- ✓ Supply the anterior & lateral surfaces of the right atrium.
- ✓ One branch supplies the posterior surface of both the right & left atria.

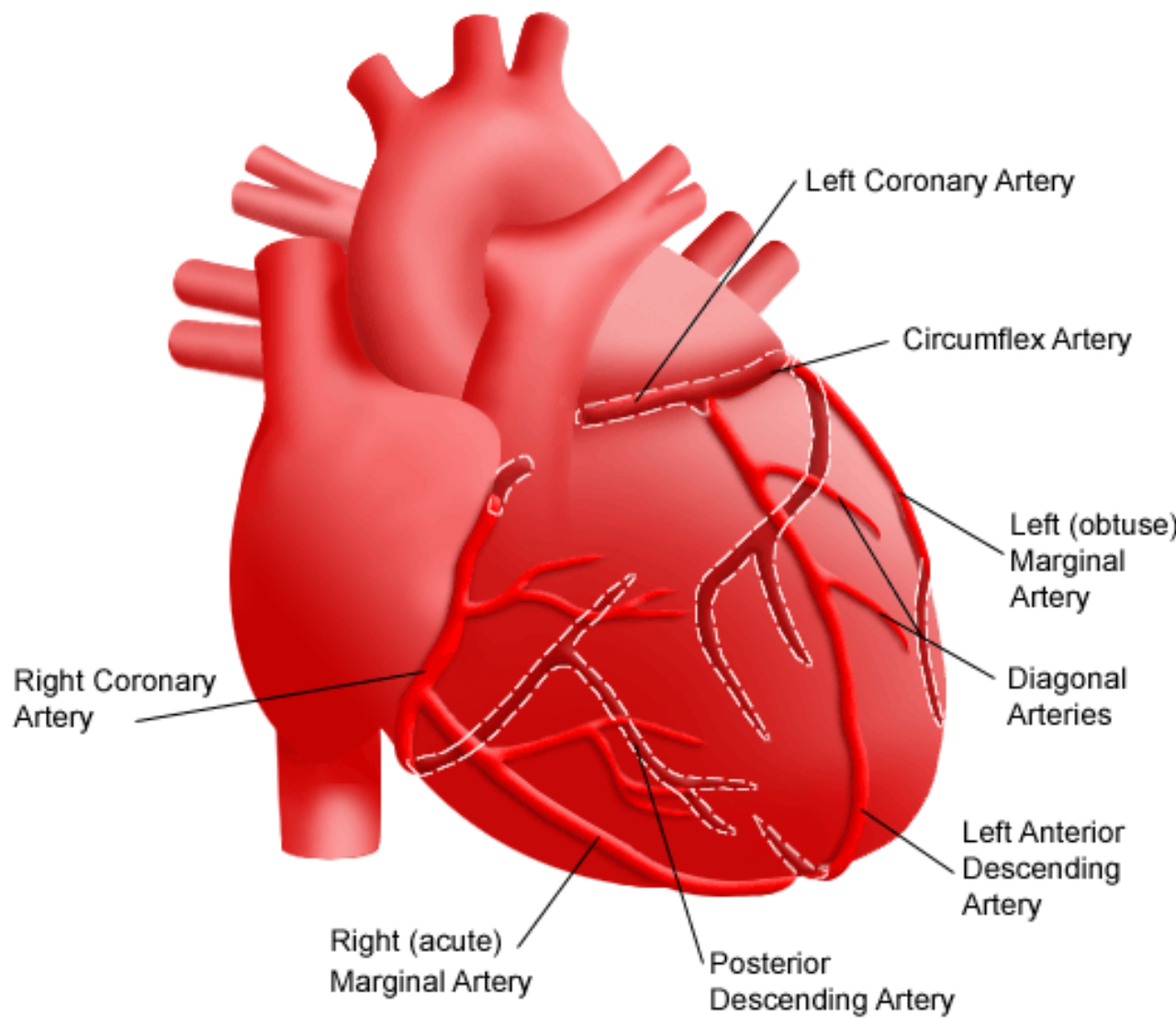
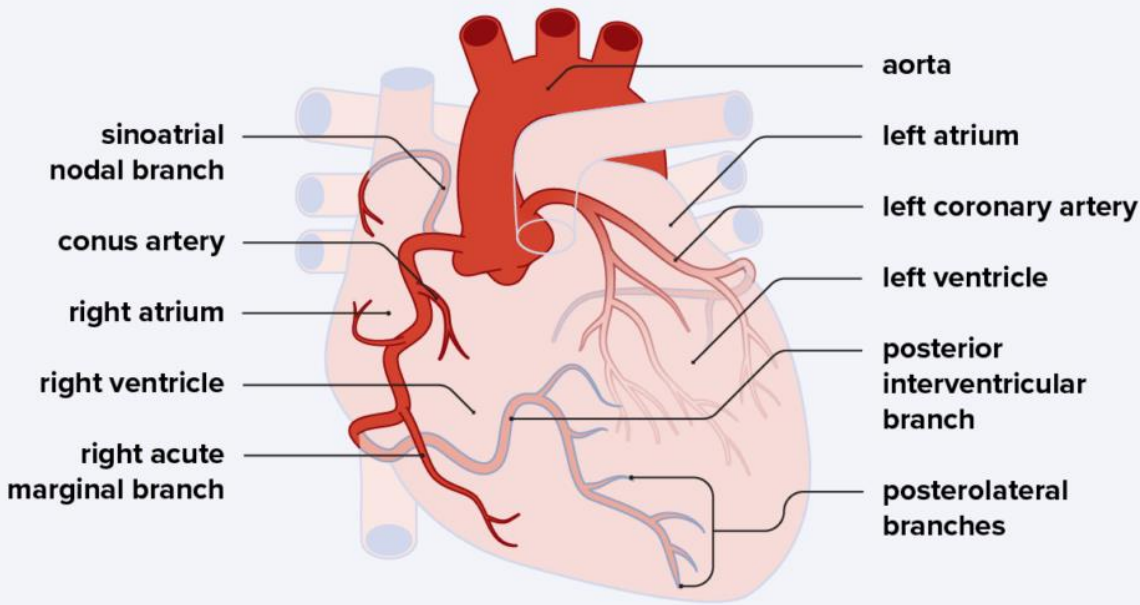
Artery of the sinuatrial node

- ✓ Supplies the node & the right & left atria
- ✓ In 35% of individuals it arises from the left coronary artery.



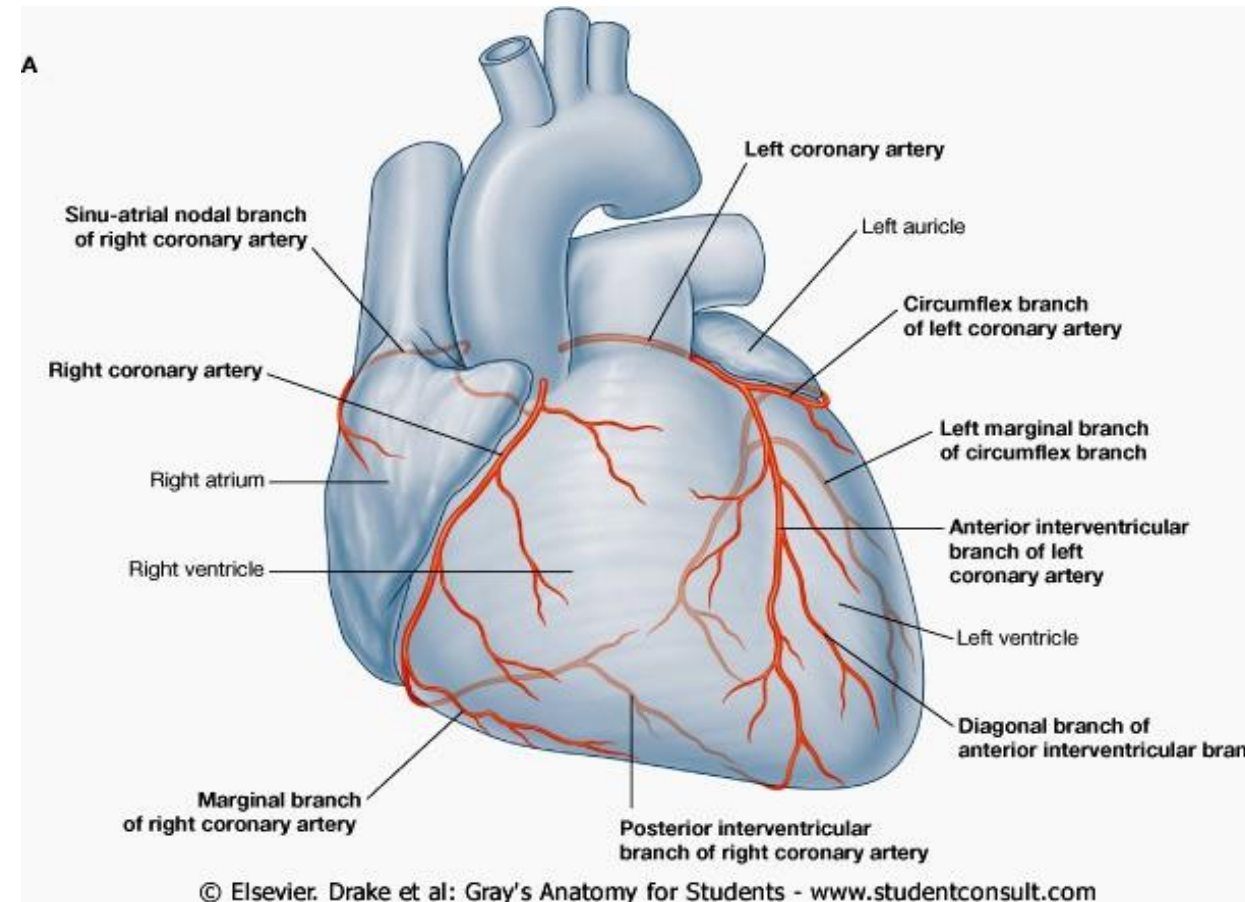
Coronary Arteries of the Heart

Branches of the right coronary artery



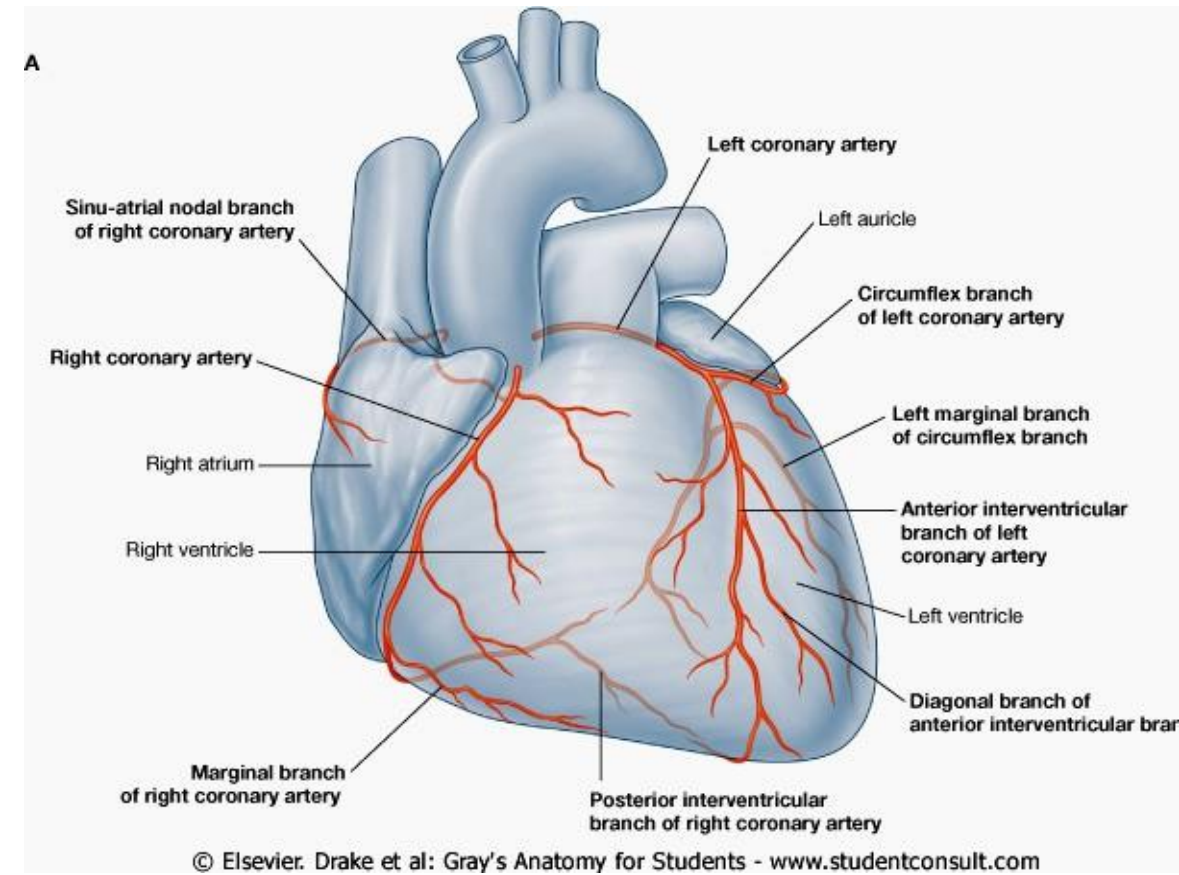
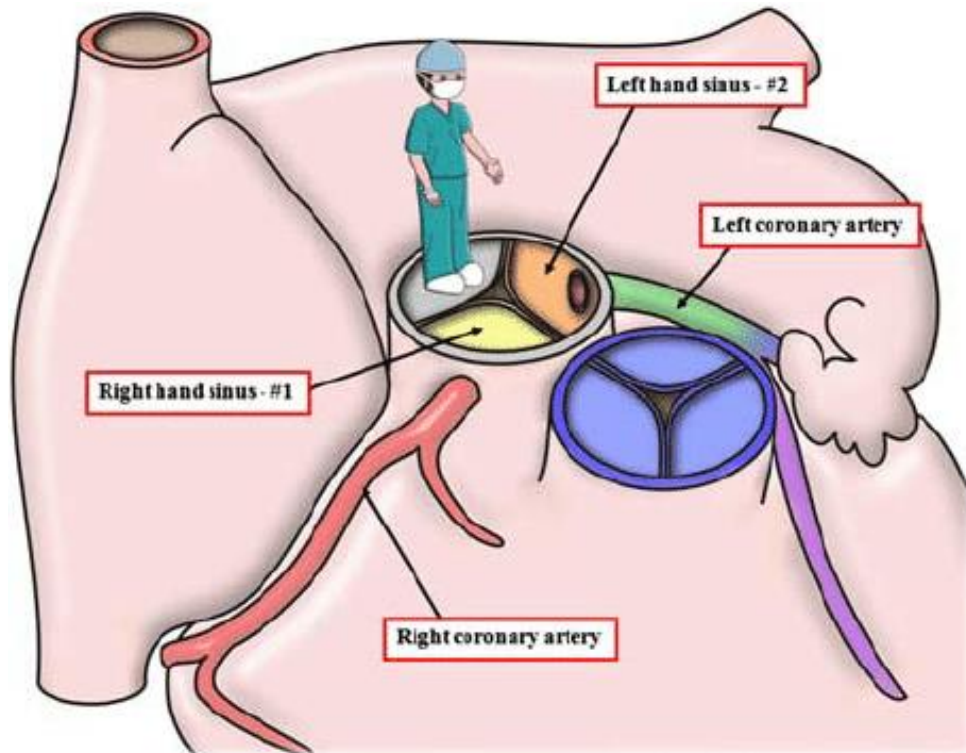
The left coronary artery

- ✓ Usually larger than the right coronary artery
- ✓ Supplies the major part of the heart including:
 - i. the greater part of the left atrium
 - ii. left ventricle
 - iii. ventricular septum.



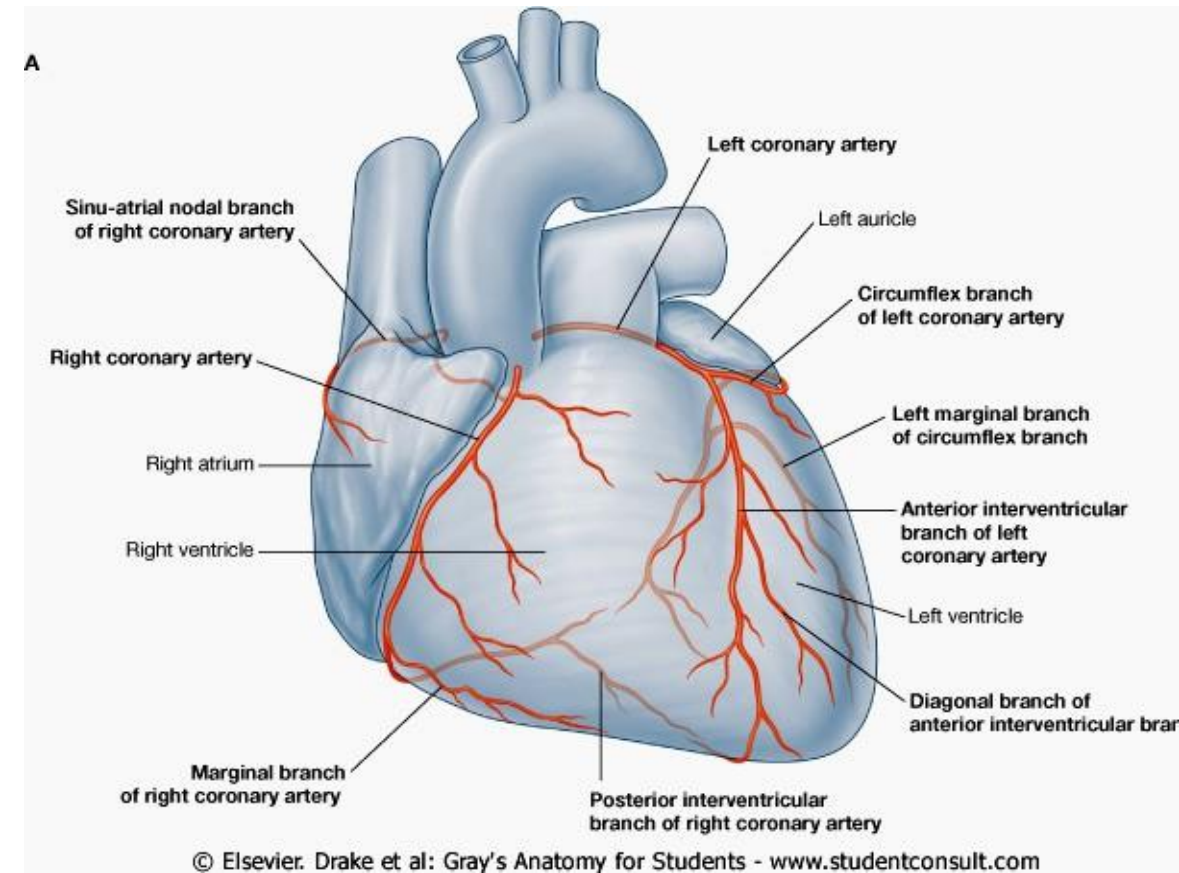
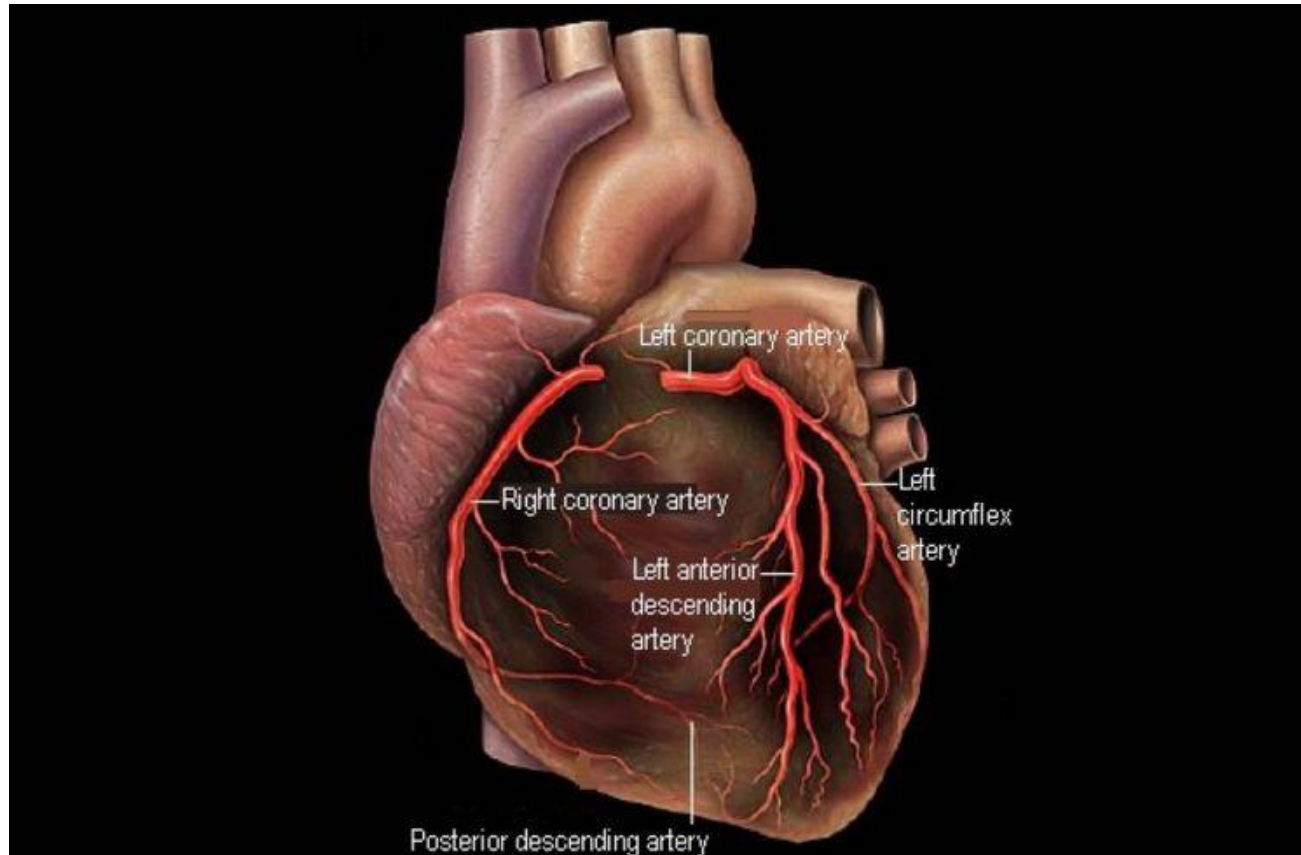
The left coronary artery

- ✓ It arises from the left posterior aortic sinus of the ascending aorta
- ✓ Passes forward btwn the pulmonary trunk & the left auricle.



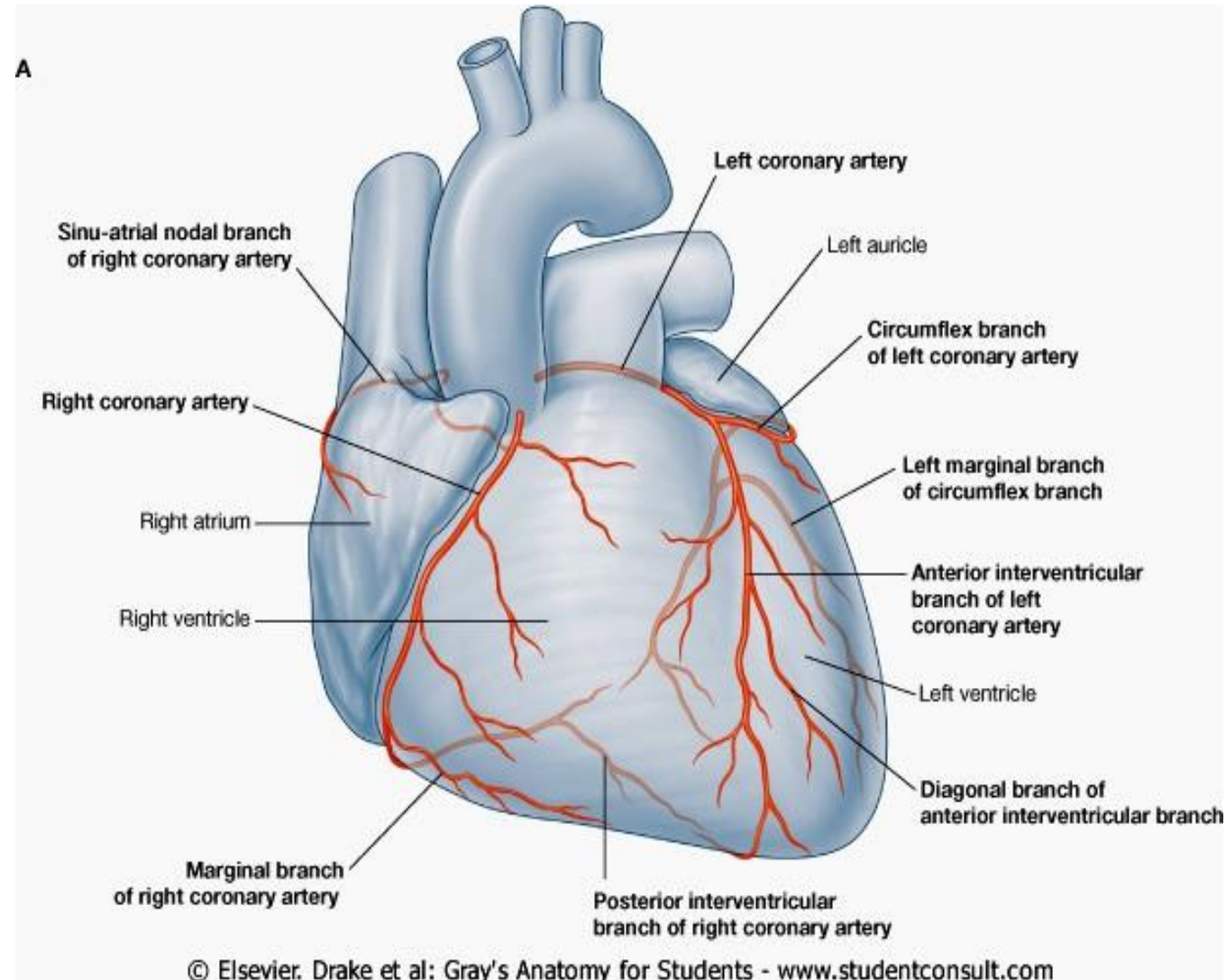
The left coronary artery

- ✓ It then enters the atrioventricular groove and divides into:
 - i. anterior interventricular branch
 - ii. circumflex branch.



Branches of the left coronary artery

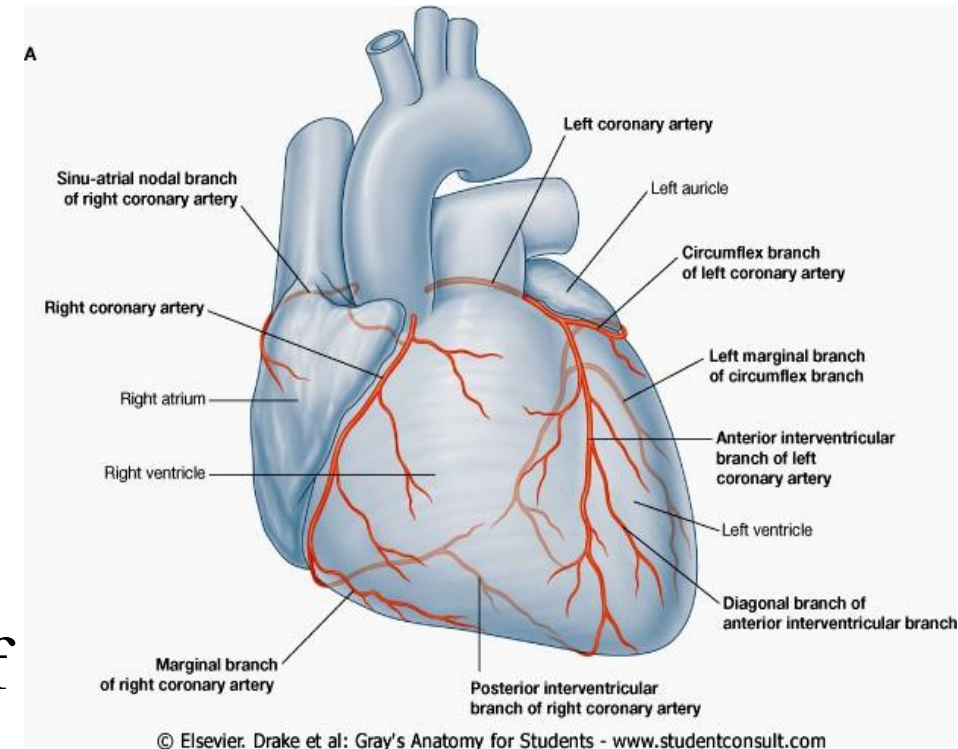
- i. Anterior interventricular (descending) branch
- ii. circumflex artery
- iii. left diagonal artery
- iv. left conus artery
- v. left marginal
- vi. Anterior ventricular
- vii. posterior ventricular branches.
- viii. Atrial branches



Branches of the left coronary artery

Anterior interventricular (descending) branch

- ✓ Runs downward in the anterior interventricular groove to the apex of the heart.
- ✓ In most individuals it then passes around the apex of the heart to enter the posterior interventricular groove & anastomoses with the terminal branches of the right coronary artery.
- ✓ In one third of individuals it ends at the apex of the heart.
- ✓ Supplies the right & left ventricles with numerous branches that also supply the anterior part of the ventricular septum.



Branches of the left coronary artery

Circumflex artery

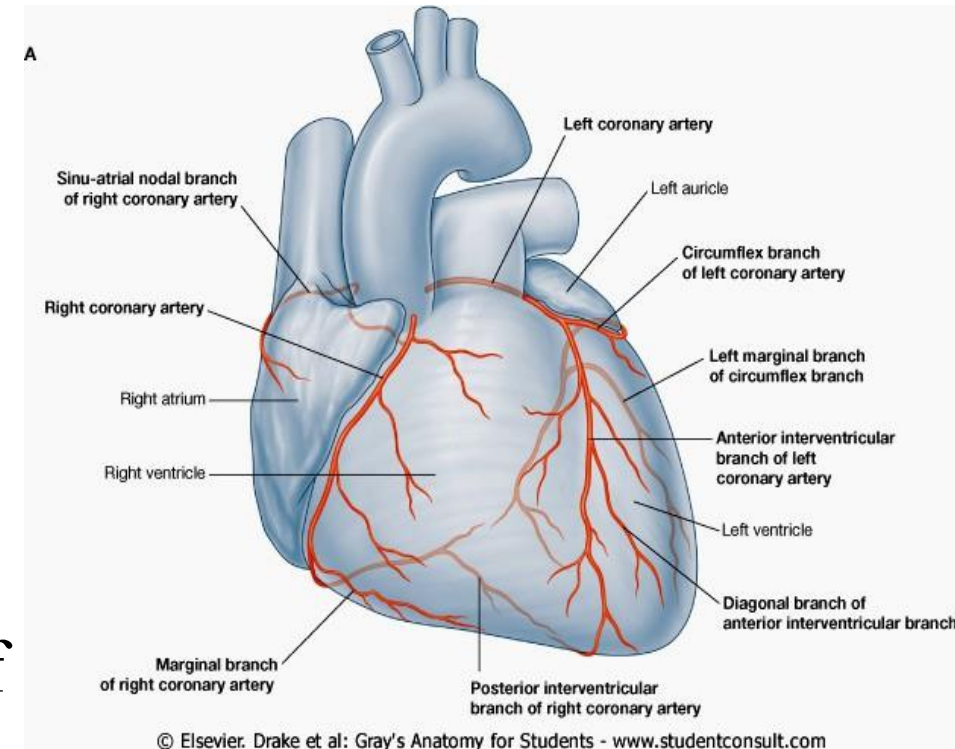
- ✓ Is the same size as the anterior interventricular artery.
- ✓ It winds around the left margin of the heart in the atrioventricular groove.

left marginal artery

- ✓ Is a large branch that supplies the left margin of the left ventricle down to the apex.

Anterior ventricular and posterior ventricular branches

- ✓ supply the left ventricle



Atrial branches

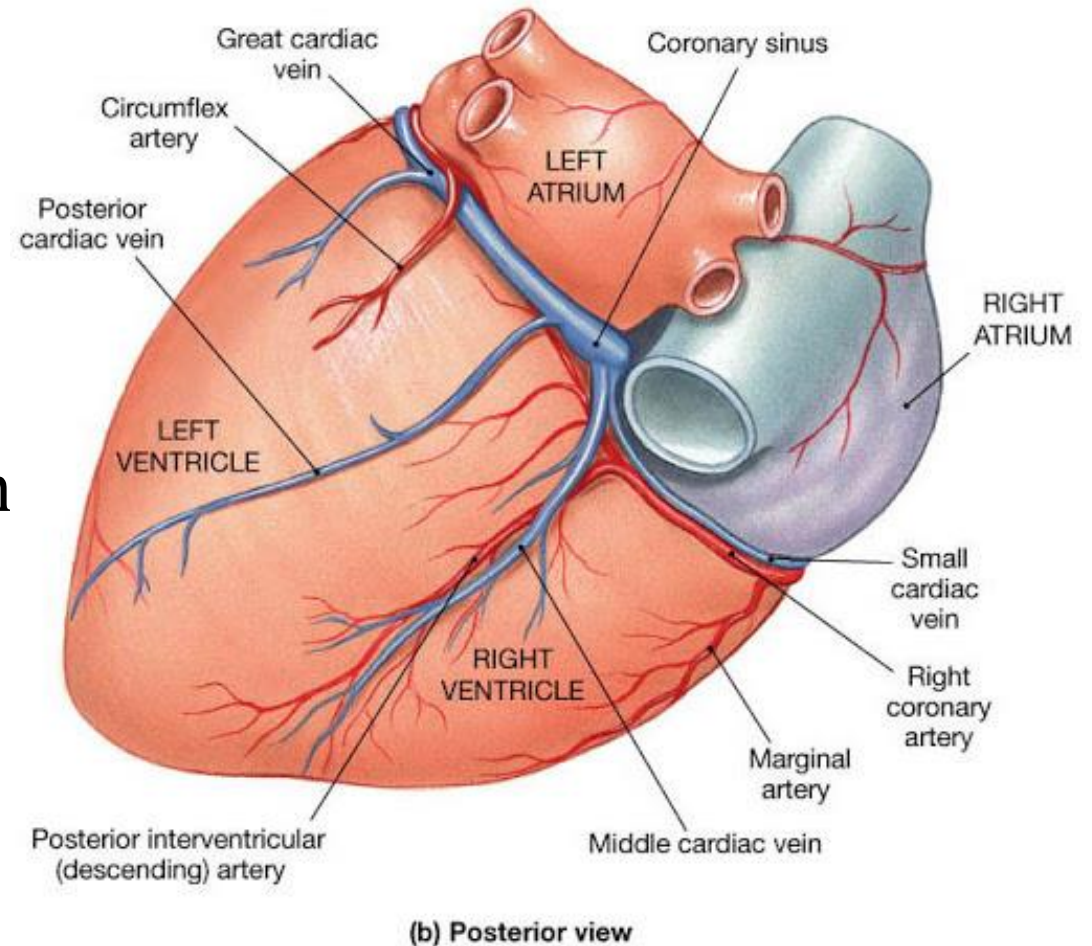
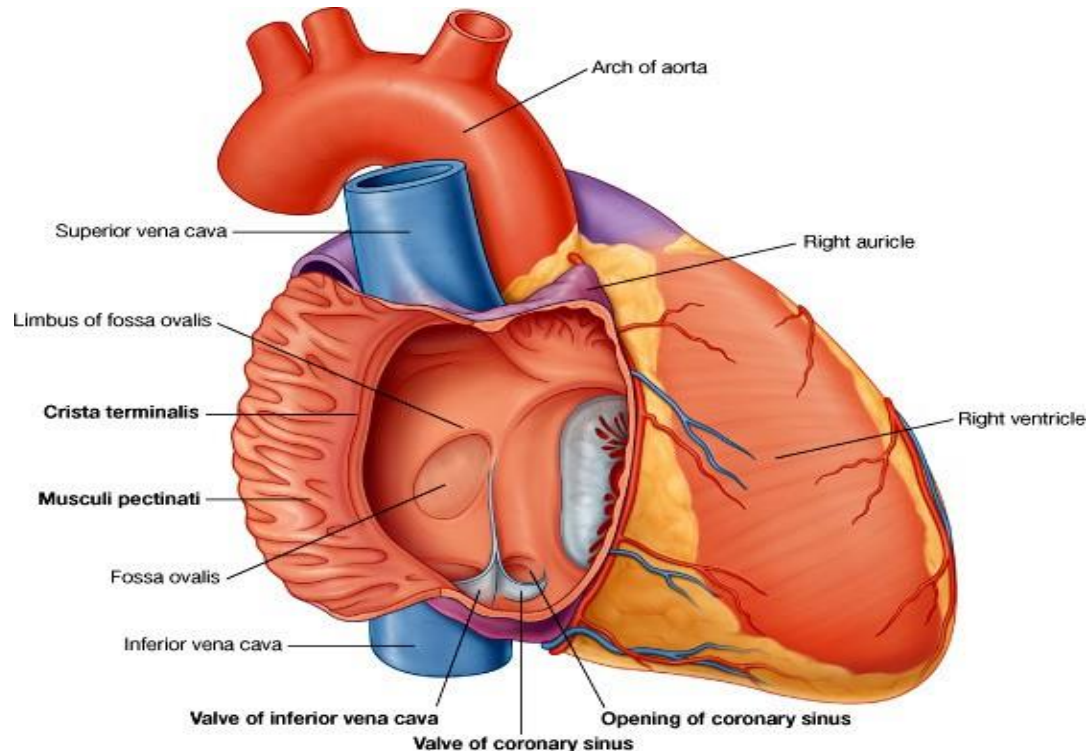
- ✓ Supply the left atrium.

Venous Drainage of the Heart

Venous Drainage of the Heart

Coronary sinus

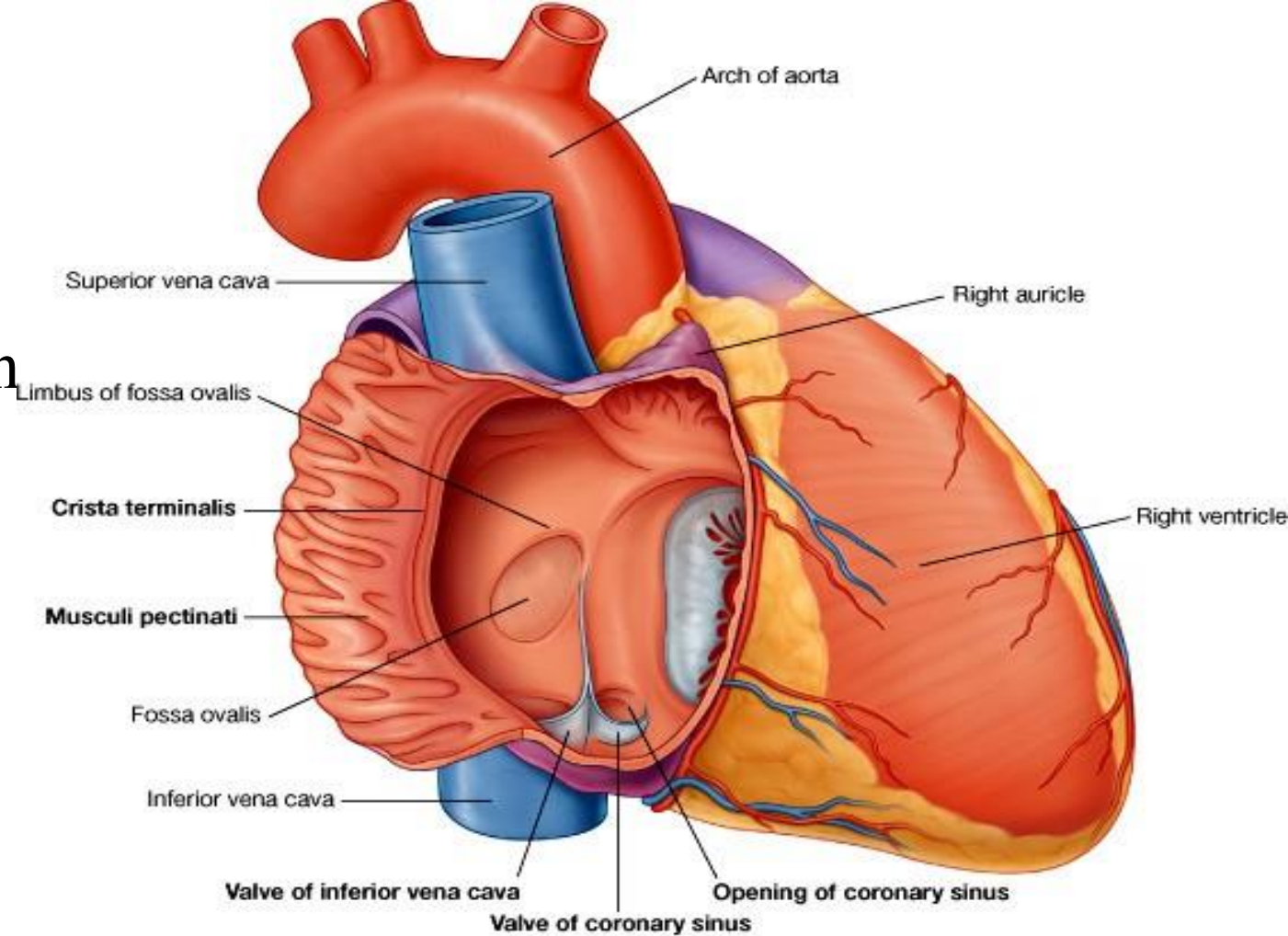
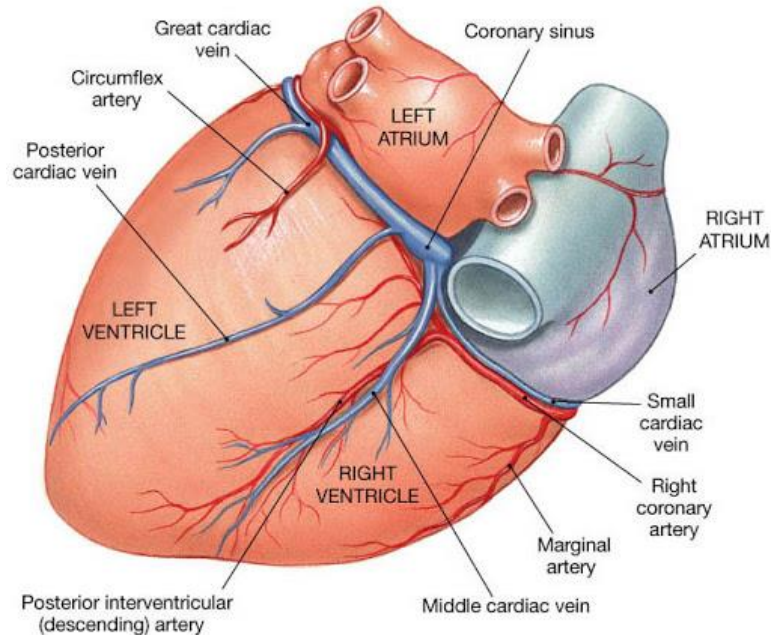
- ✓ Most blood from the heart wall drains into the right atrium thru the coronary sinus
- ✓ Lies in the posterior part of the atrioventricular groove & is a continuation of the **great cardiac vein**.



- ✓ It opens into the right atrium to the left of the IVC

Coronary sinus

- ✓ Drains most of the blood from the heart wall
- ✓ Opens into the right atrium between the IVC & the atrioventricular orifice.
- ✓ It is guarded by a rudimentary non-functioning valve.



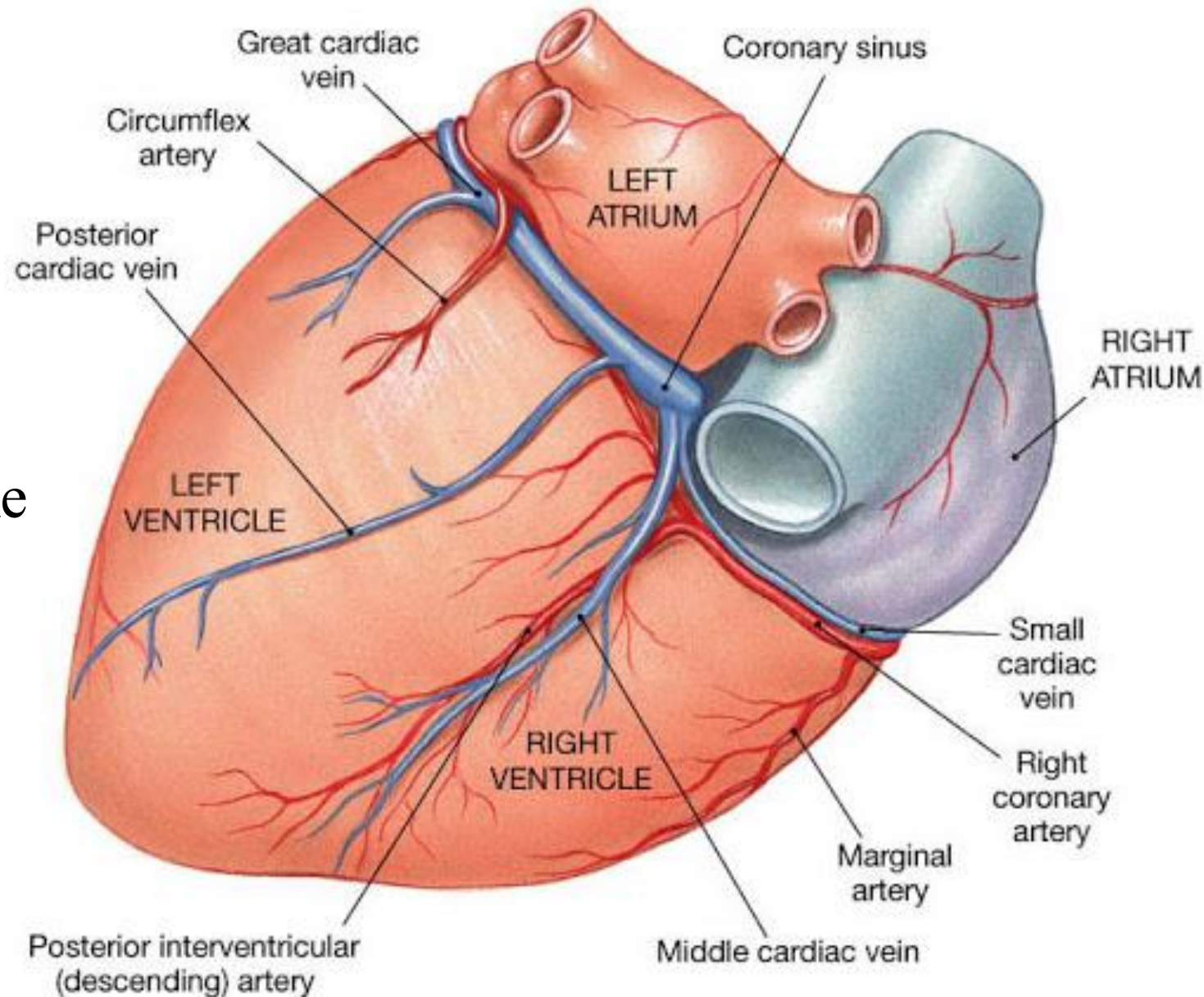
Venous Drainage of the Heart

Small & middle cardiac veins

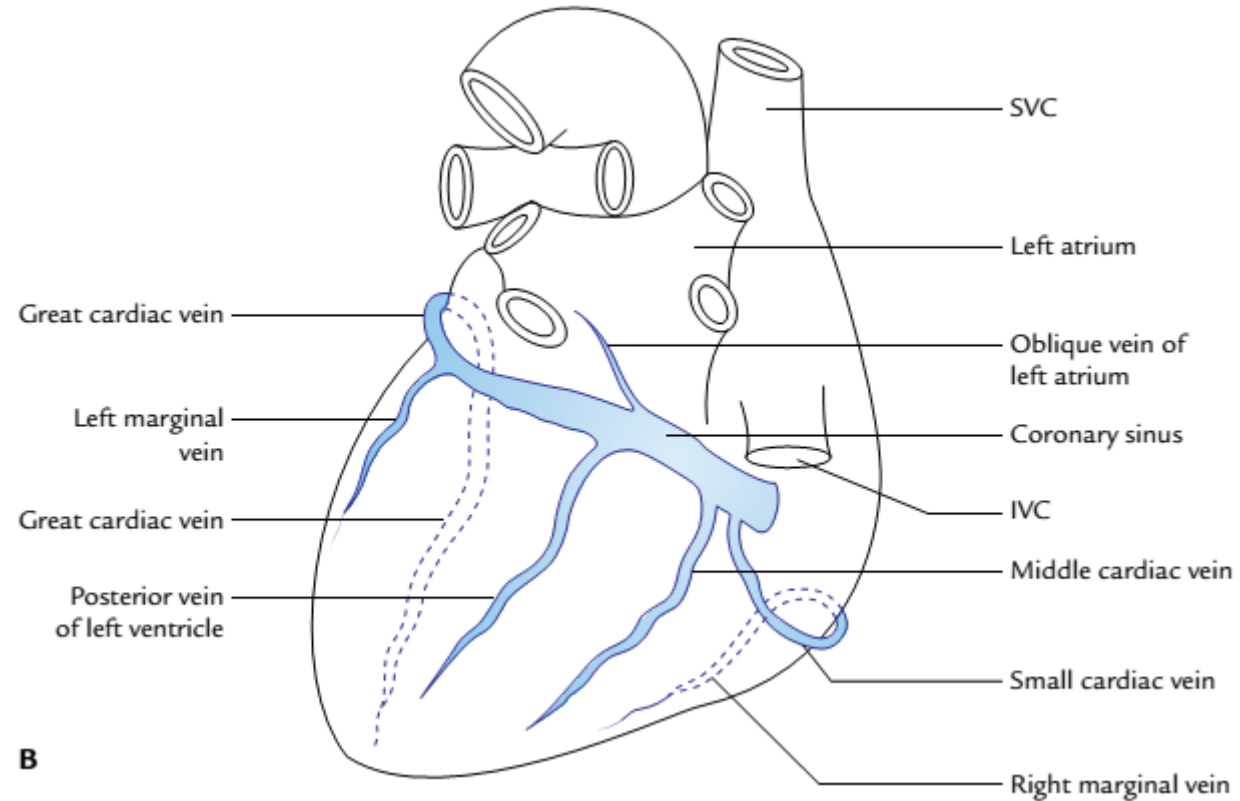
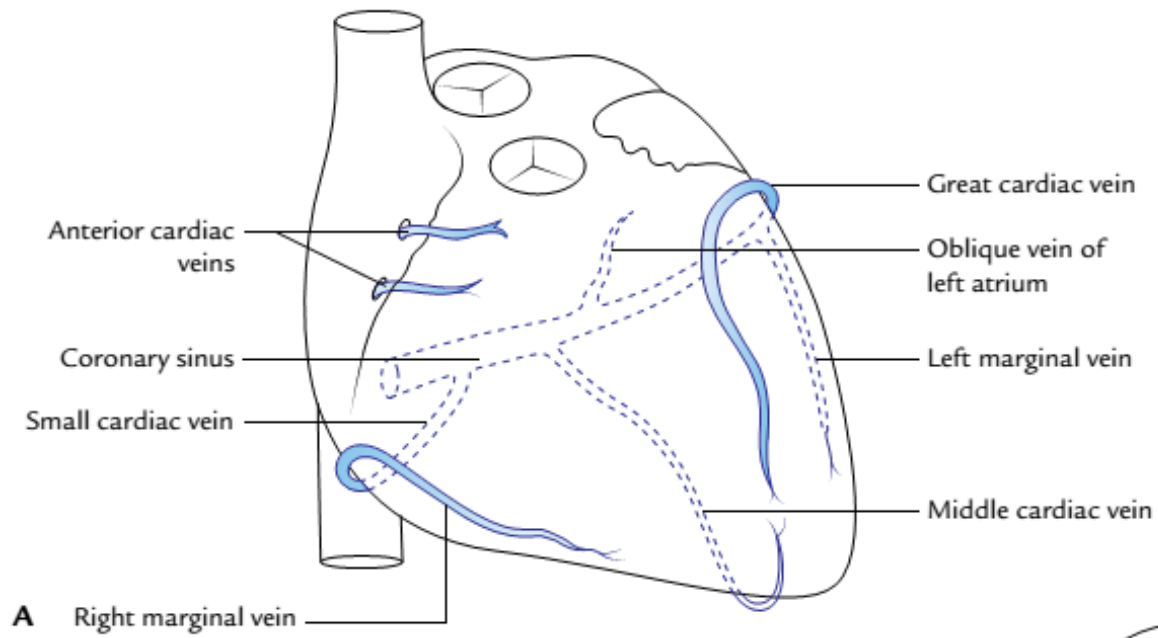
- ✓ Are tributaries of the coronary sinus.

Anterior cardiac vein

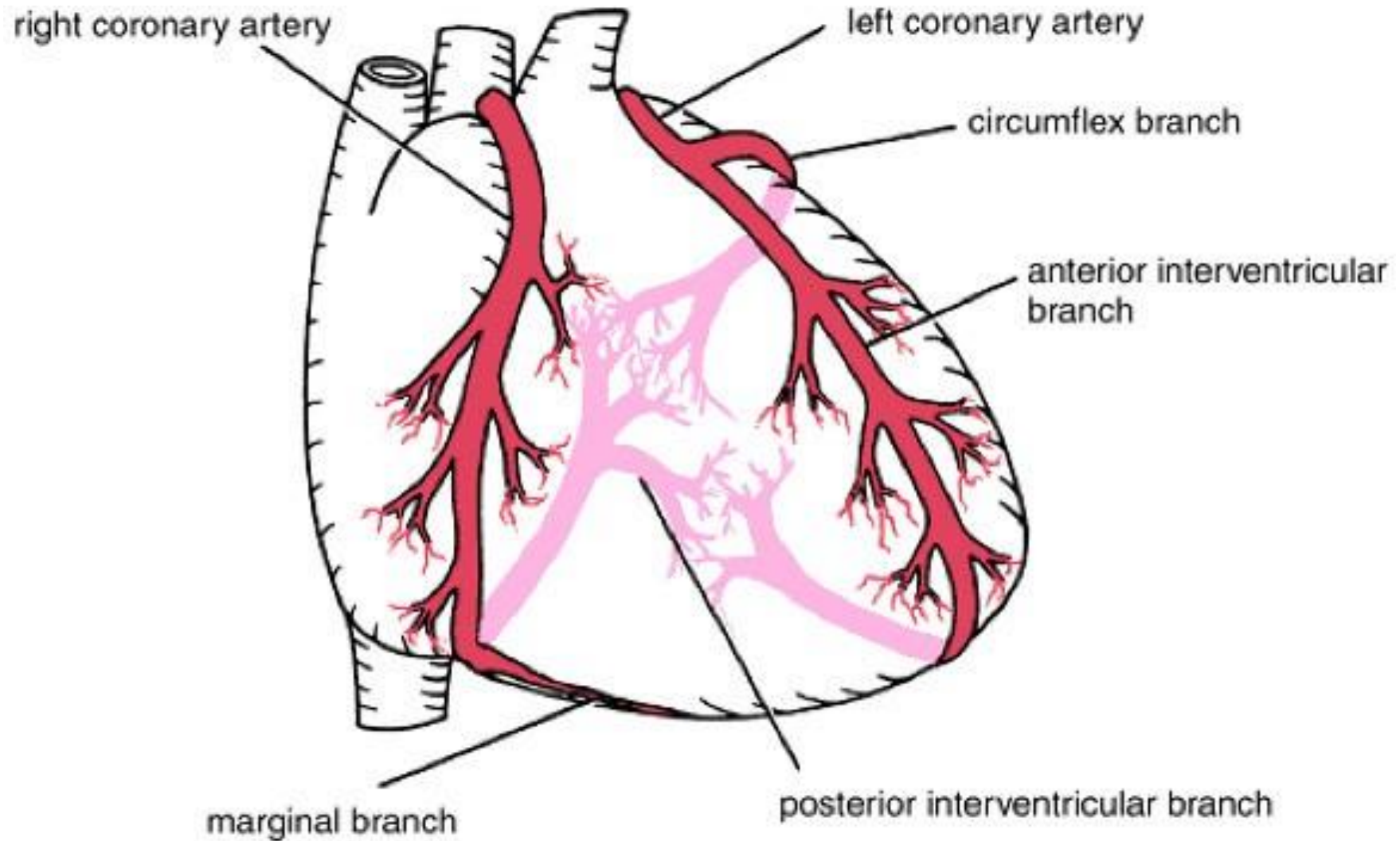
- ✓ The remainder of the blood is returned to the right atrium by the anterior cardiac vein & by small veins that open directly into the heart chambers.



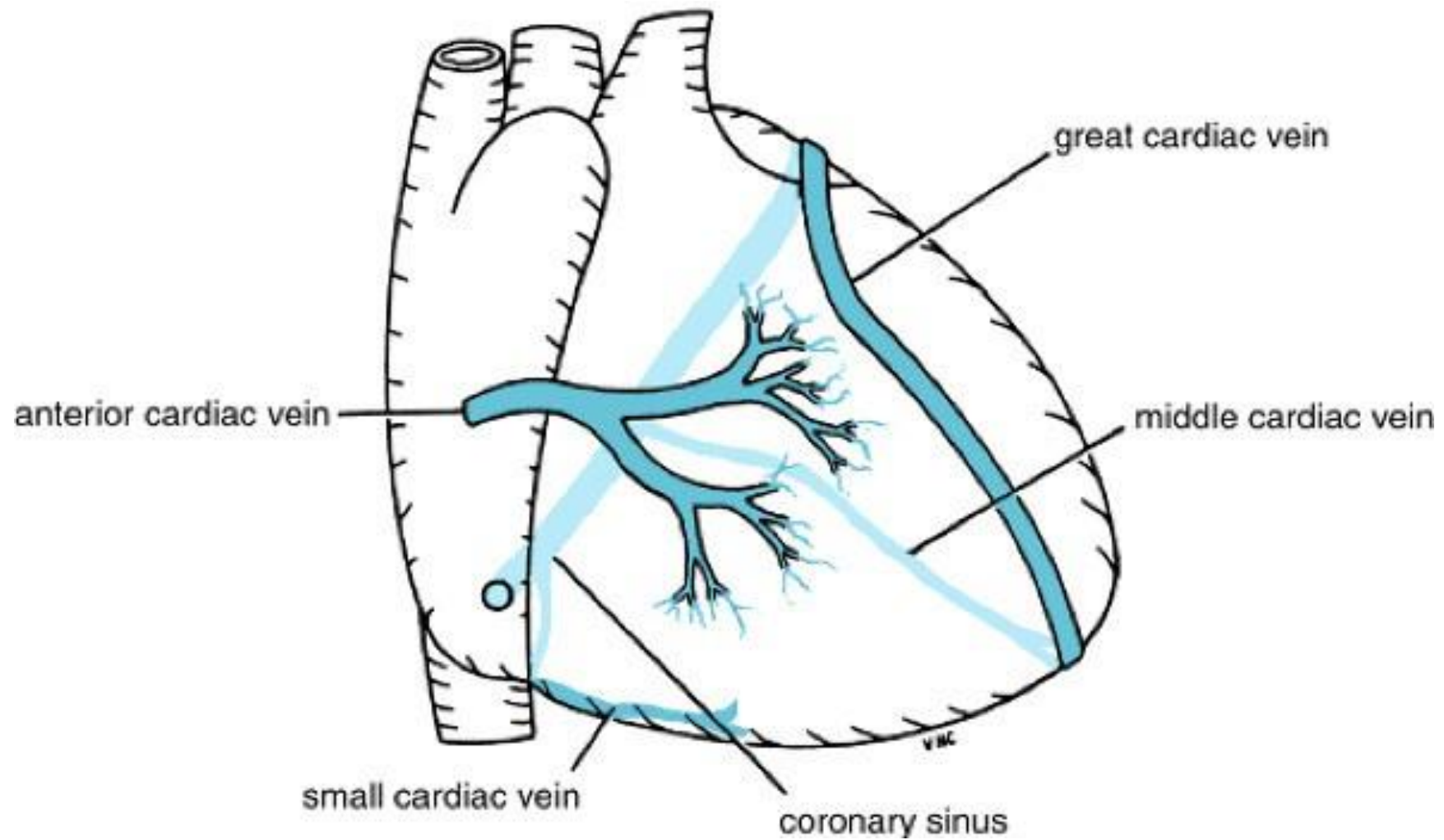
(b) Posterior view



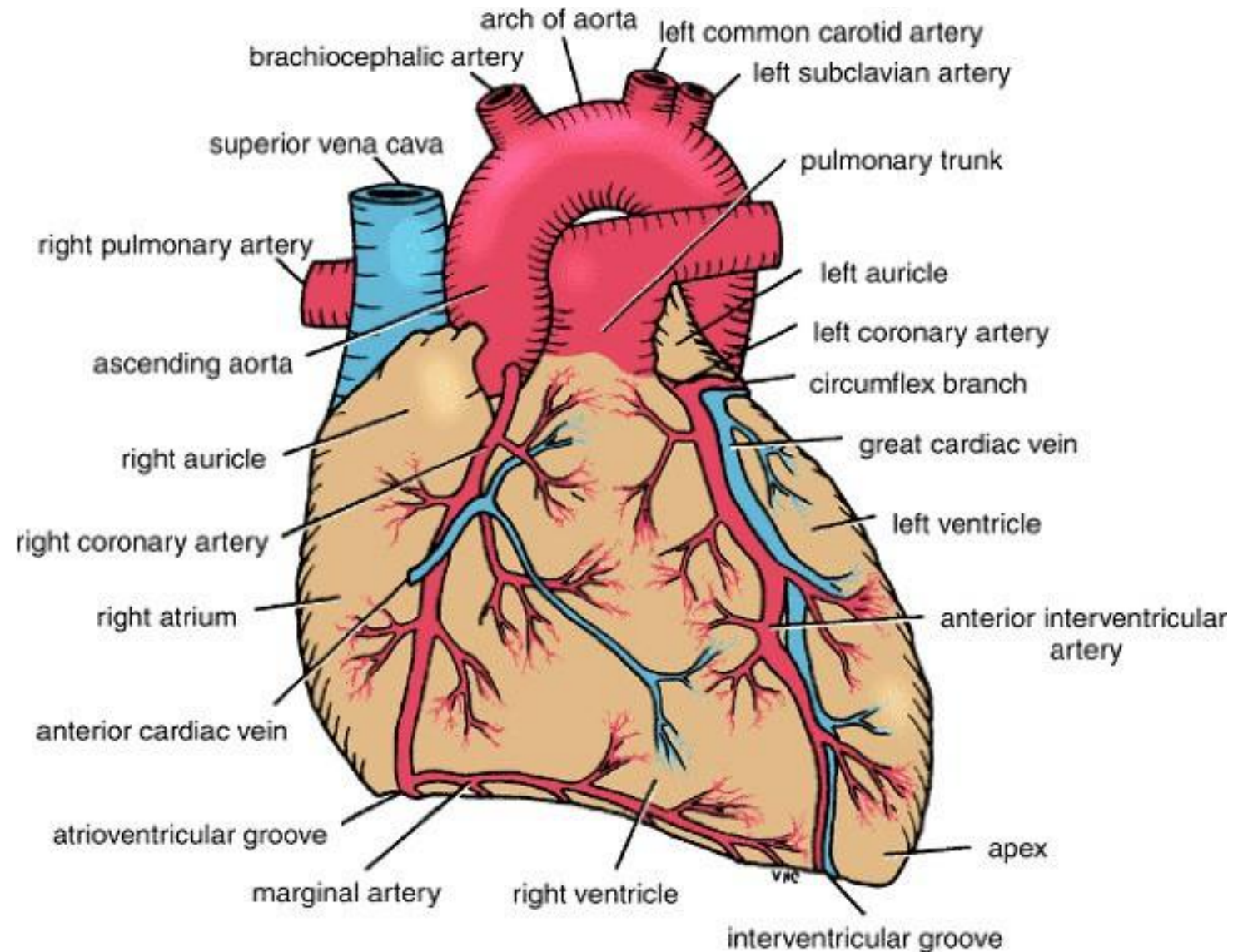
Coronary arteries



Cardiac veins .



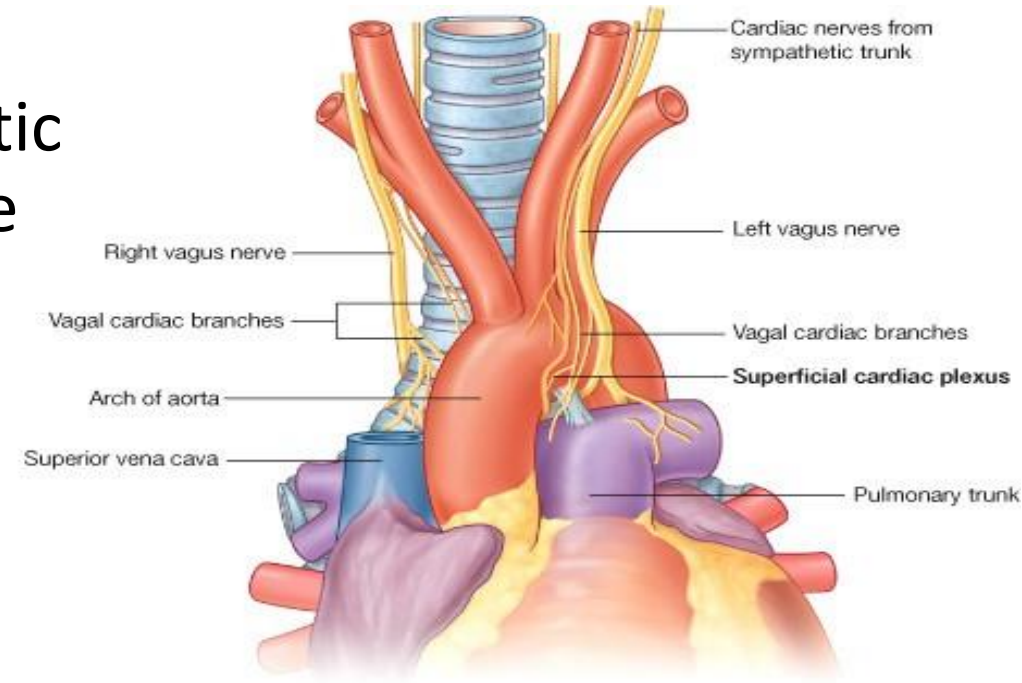
The anterior surface of the heart and the great blood vessels. Note the course of the coronary arteries and the cardiac veins .



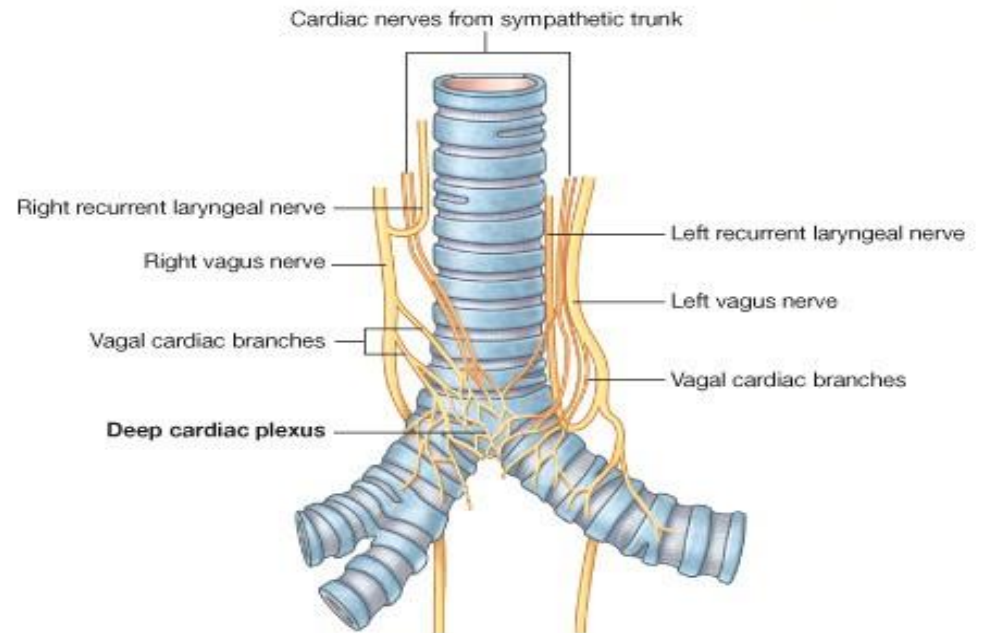
Nerve Supply of the Heart

- ✓ Is innervated by sympathetic & parasympathetic fibers of the autonomic nervous system via the **cardiac plexuses**
- ✓ Situated below the arch of the aorta.
- **Sympathetic supply**
 - ✓ Arises from the cervical & upper thoracic portions of the sympathetic trunks
- **Parasympathetic supply**
 - ✓ Comes from the vagus nerves.

A



B

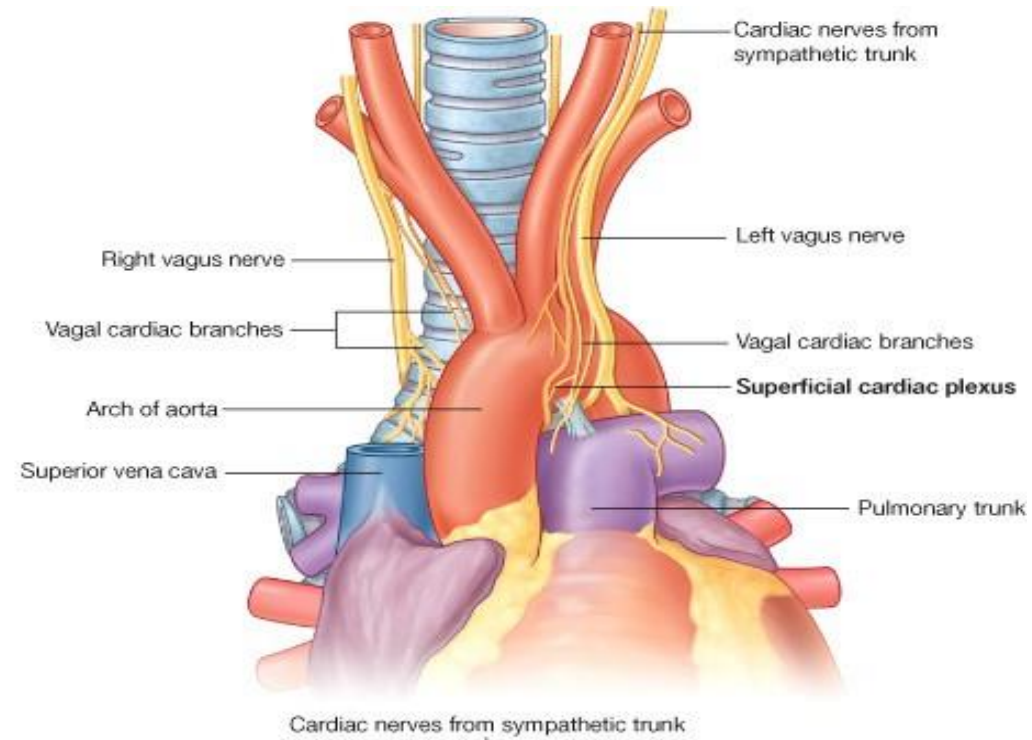


Cardiac plexus.

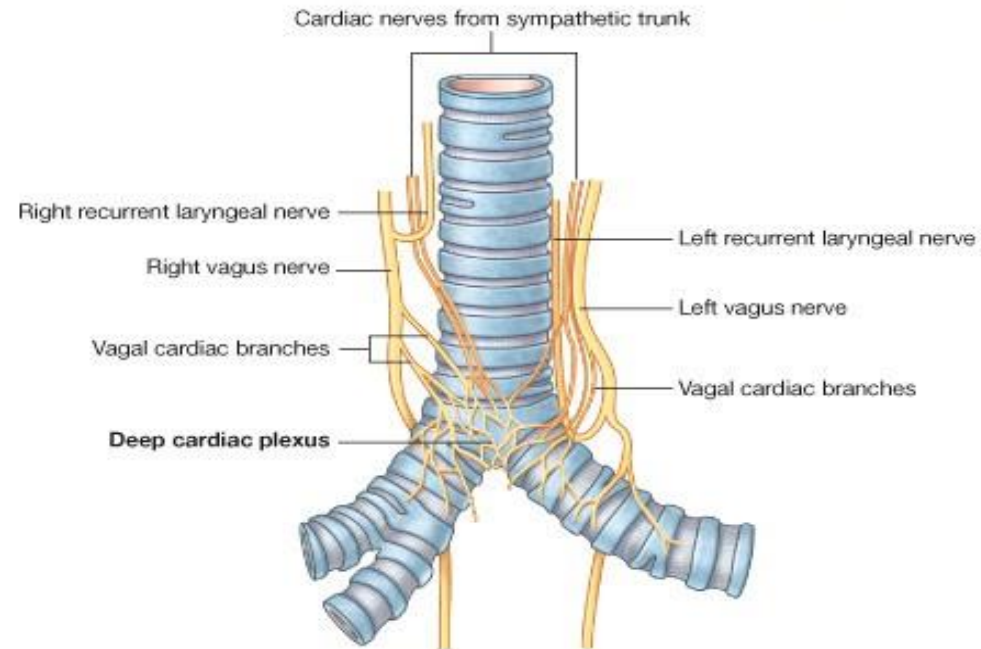
A. Superficial.

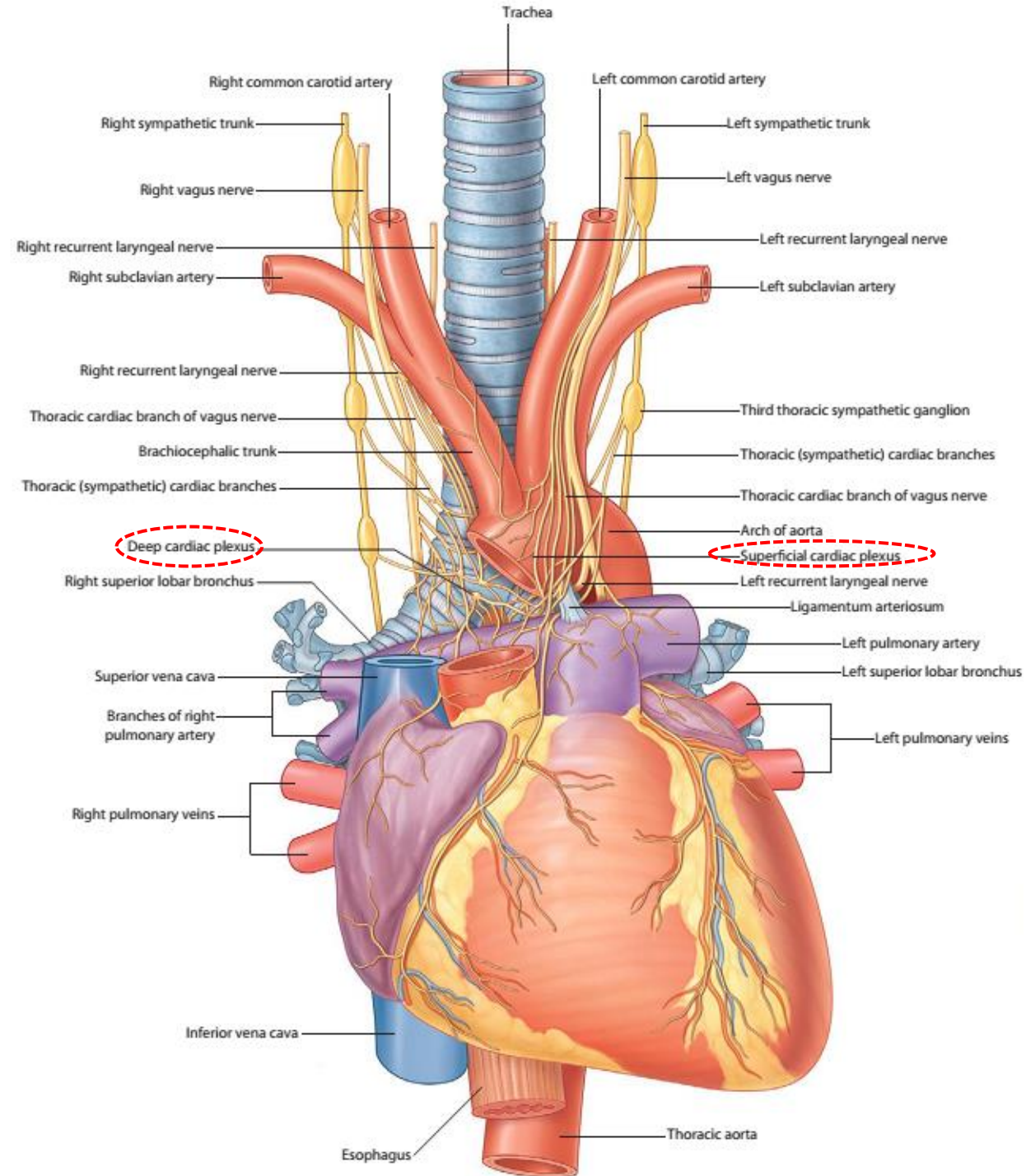
B. Deep.

A

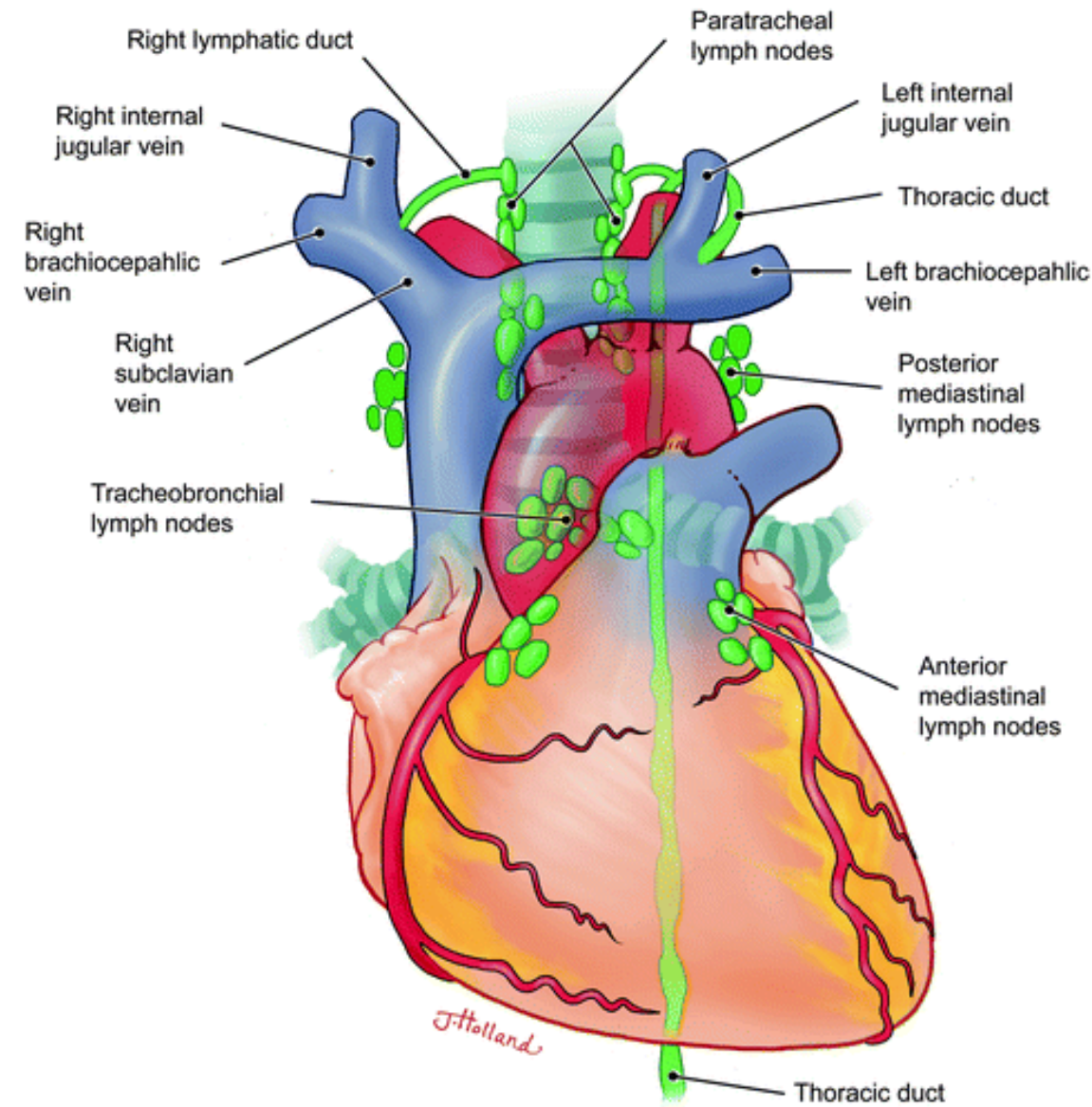


B





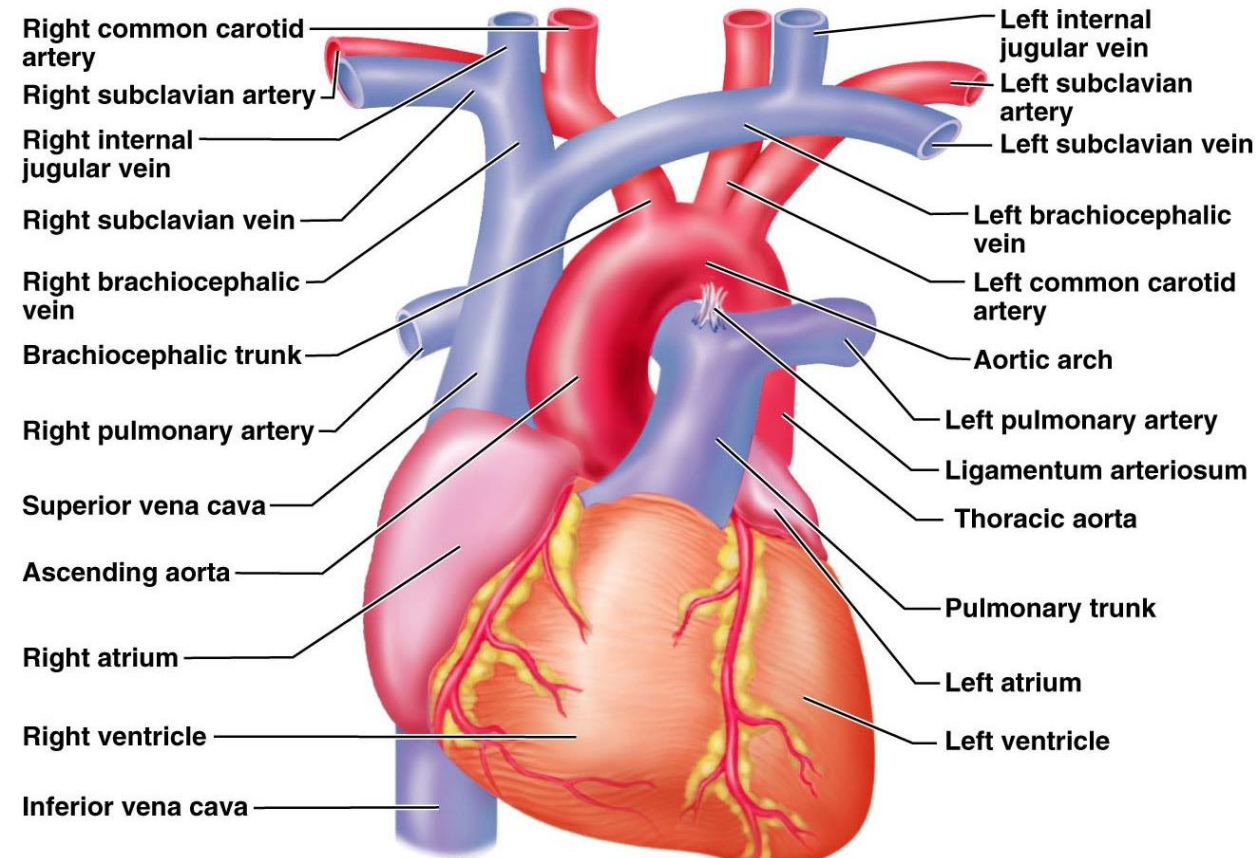
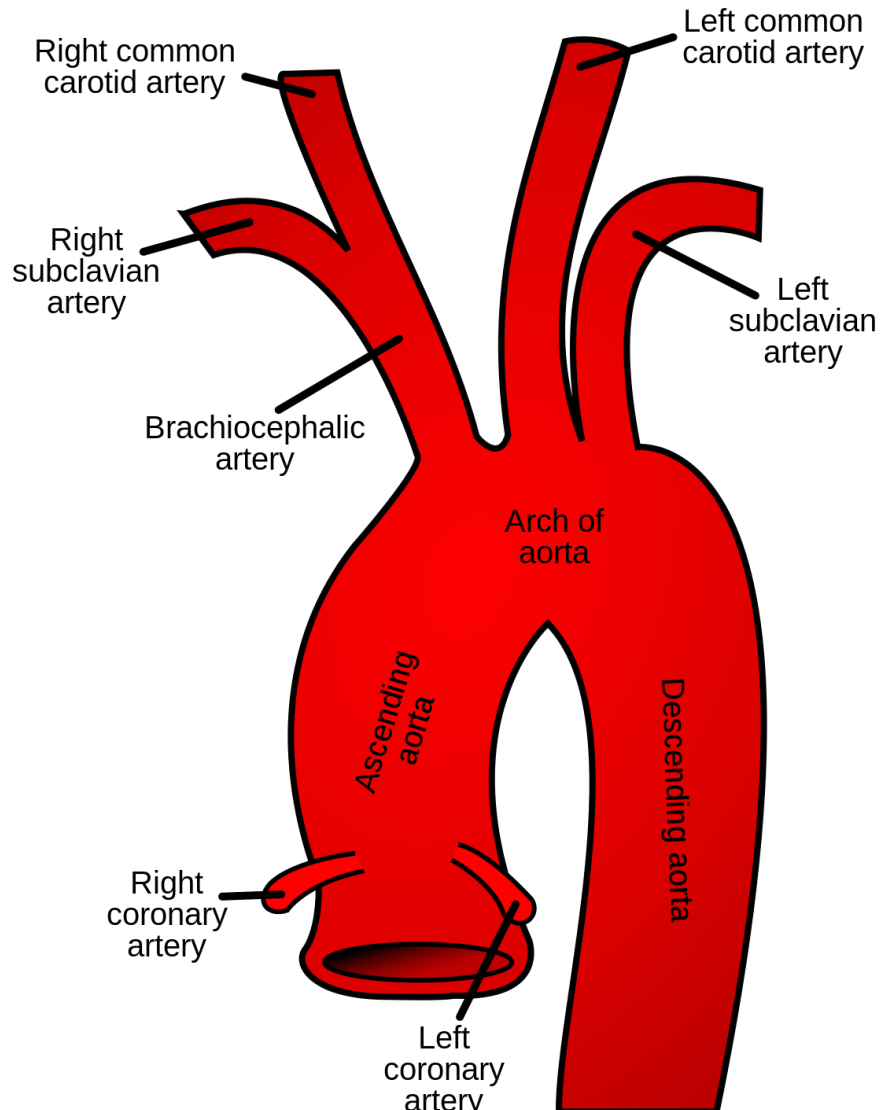
- ✓ The lymphatics of the heart drain back along the coronary arteries
- ✓ Emerge from the fibrous pericardium along with the aorta and pulmonary trunk,
- ✓ Empty into the tracheobronchial & brachiocephalic lymph nodes.



MAJOR ARTERIES AND VEINS IN THE BODY

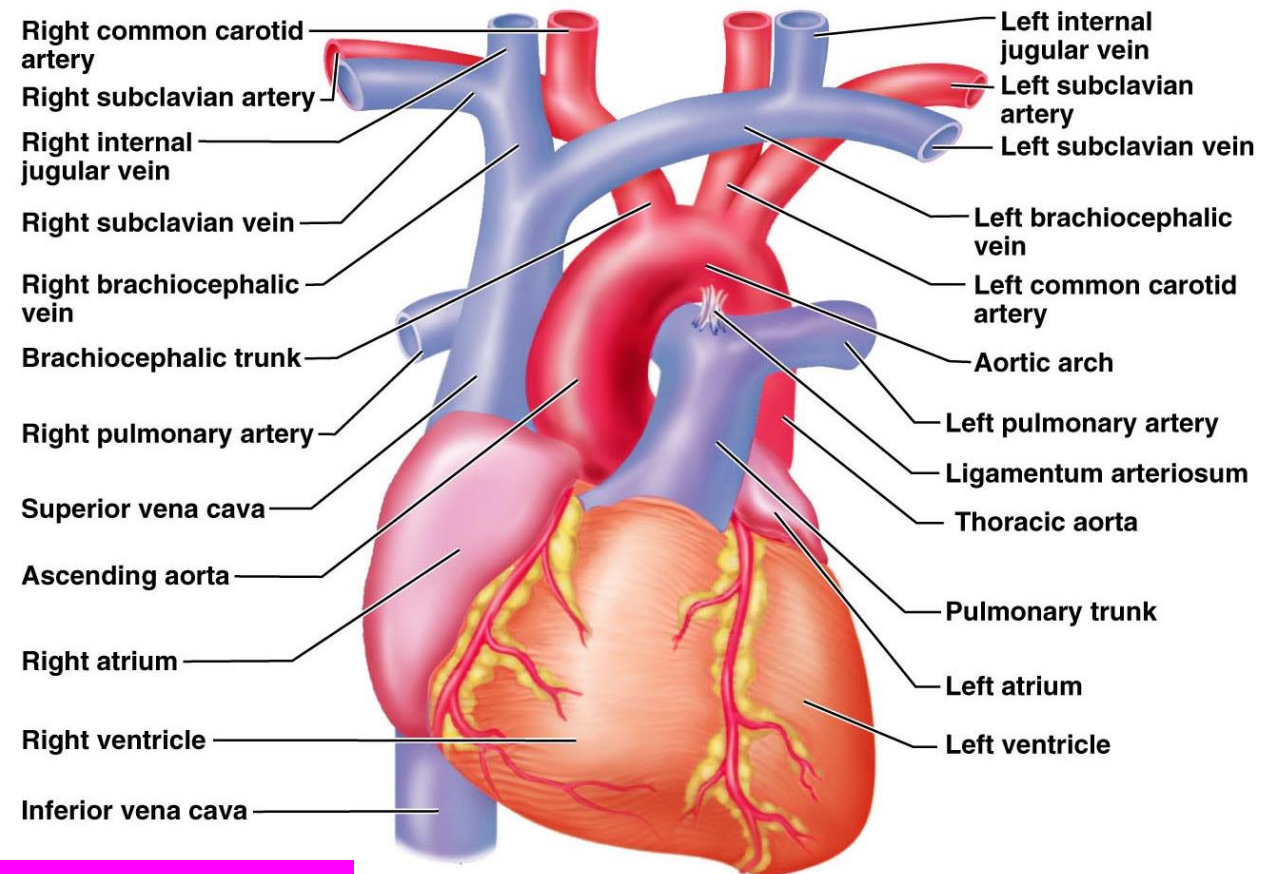
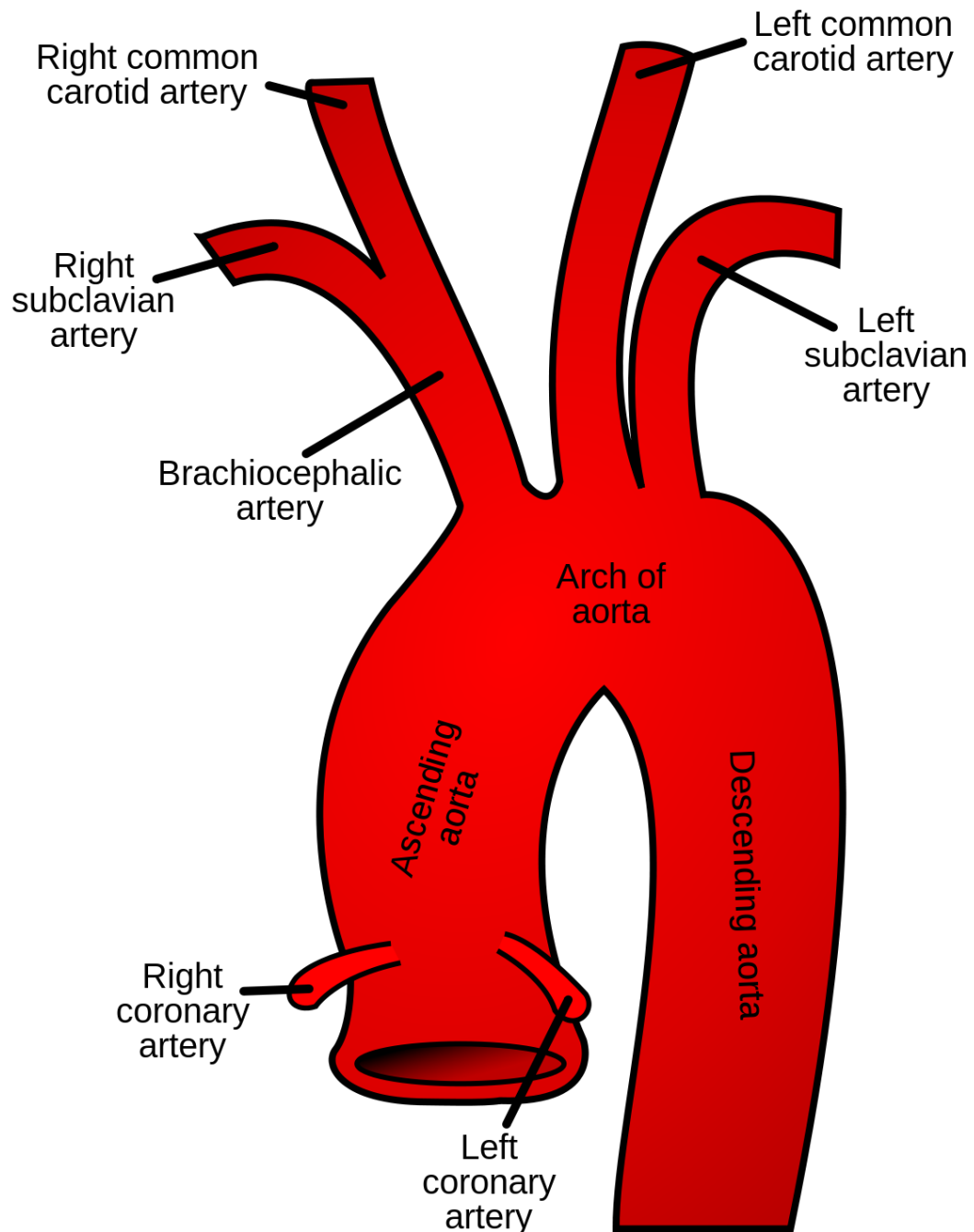
Aorta

✓ largest artery in the body



Ascending aorta

- ✓ Give rise to
 - i. right coronary artery
 - ii. left coronary artery
- ✓ supply the heart

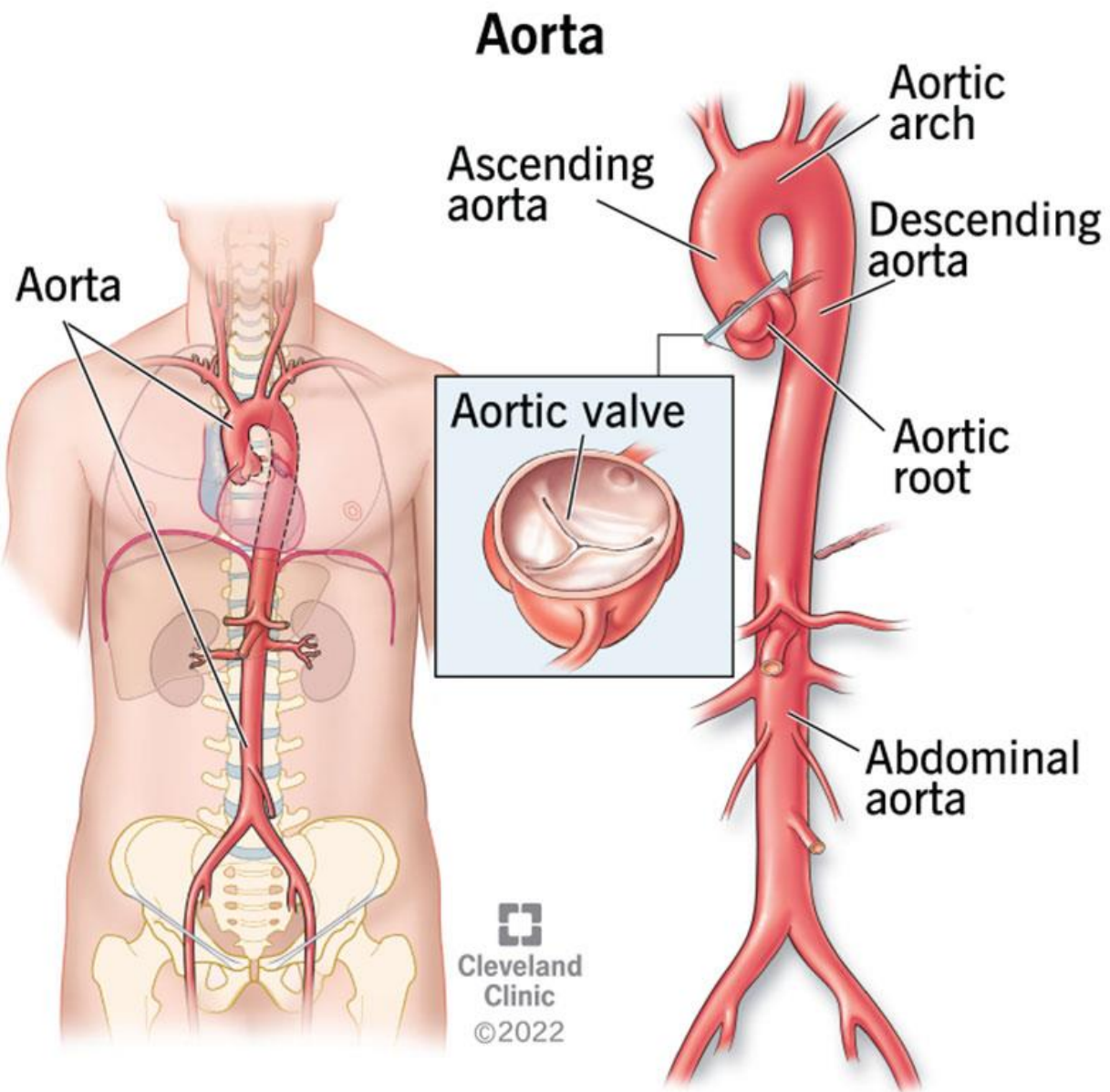


Arch of aorta

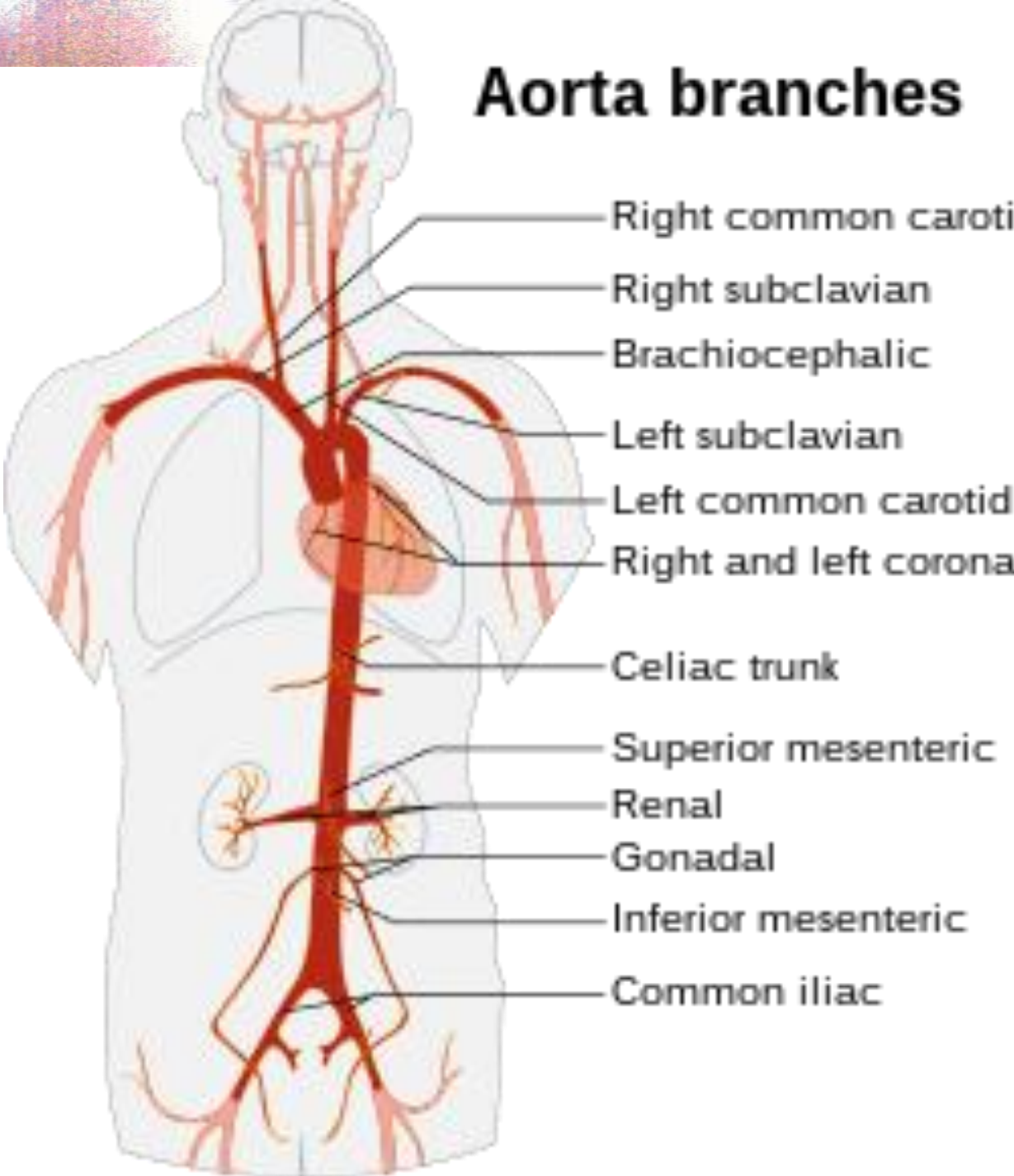
✓ 3 main branches

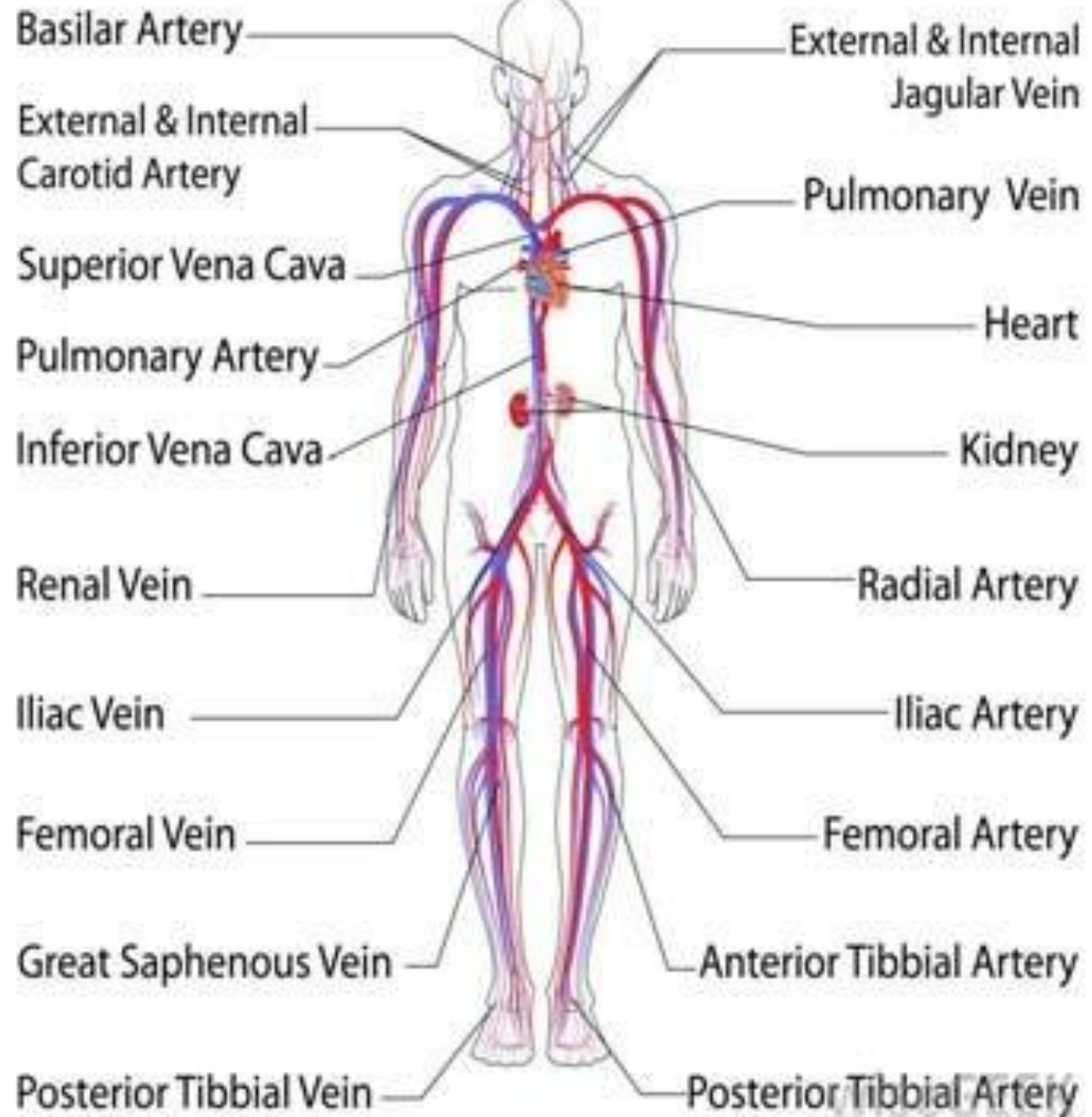
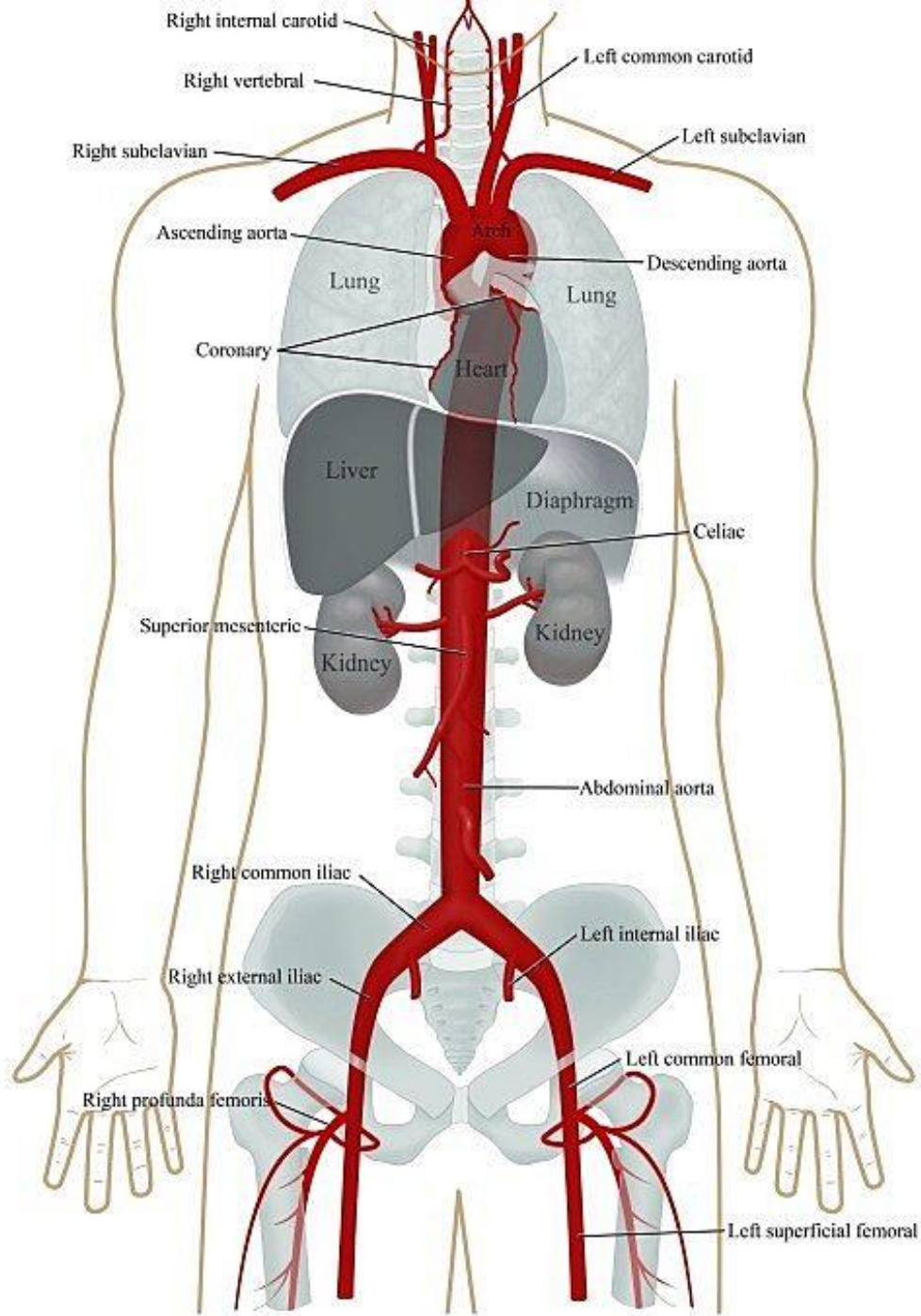
- i. **Brachiocephalic** trunk (innominate artery)
- ii. Left common carotid artery
- iii. Left subclavian artery

Descending aorta



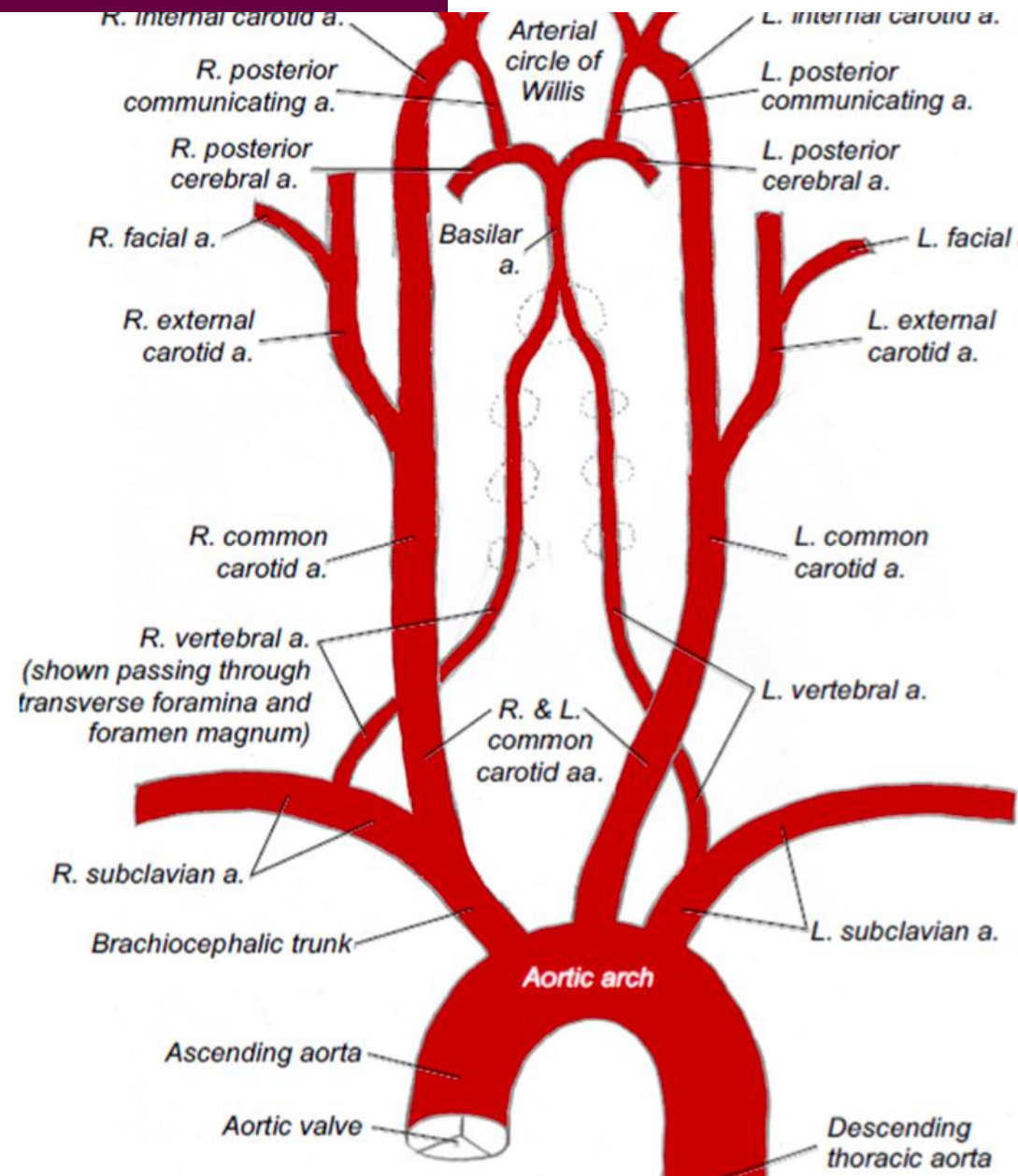
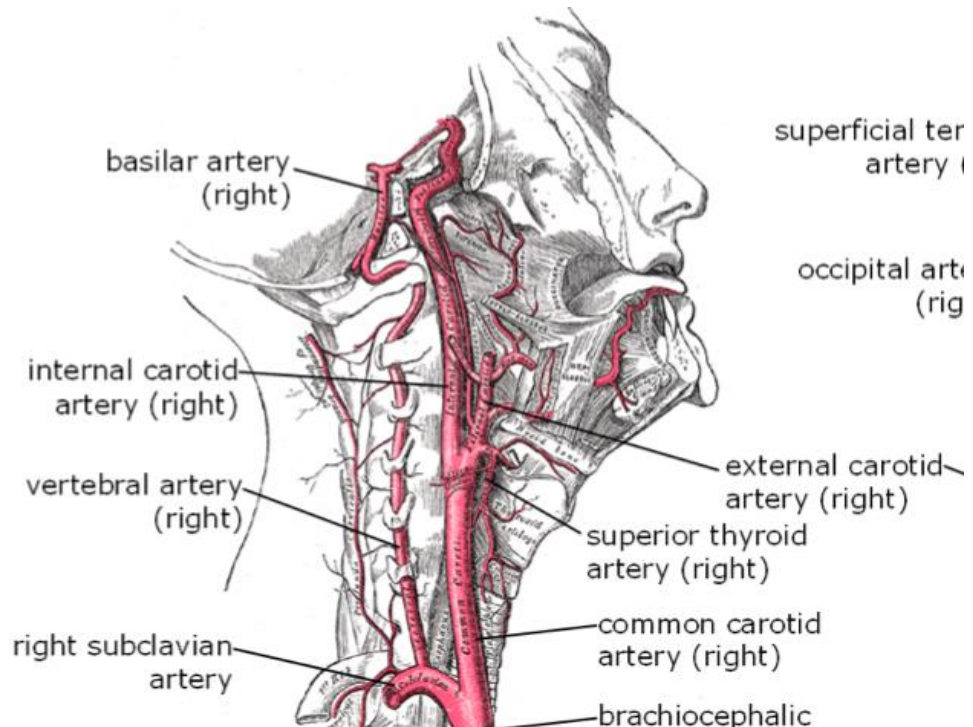
Aorta branches

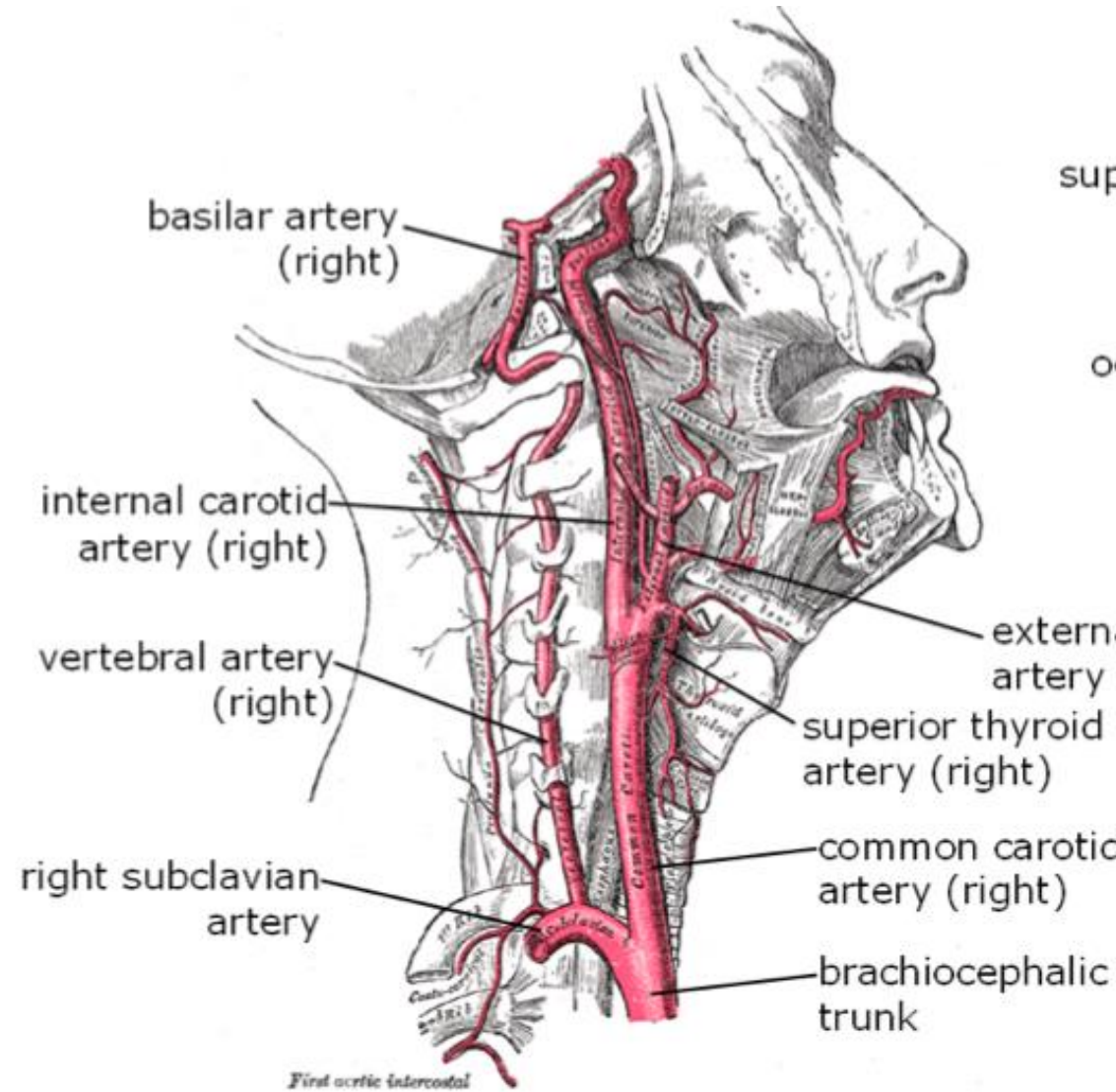
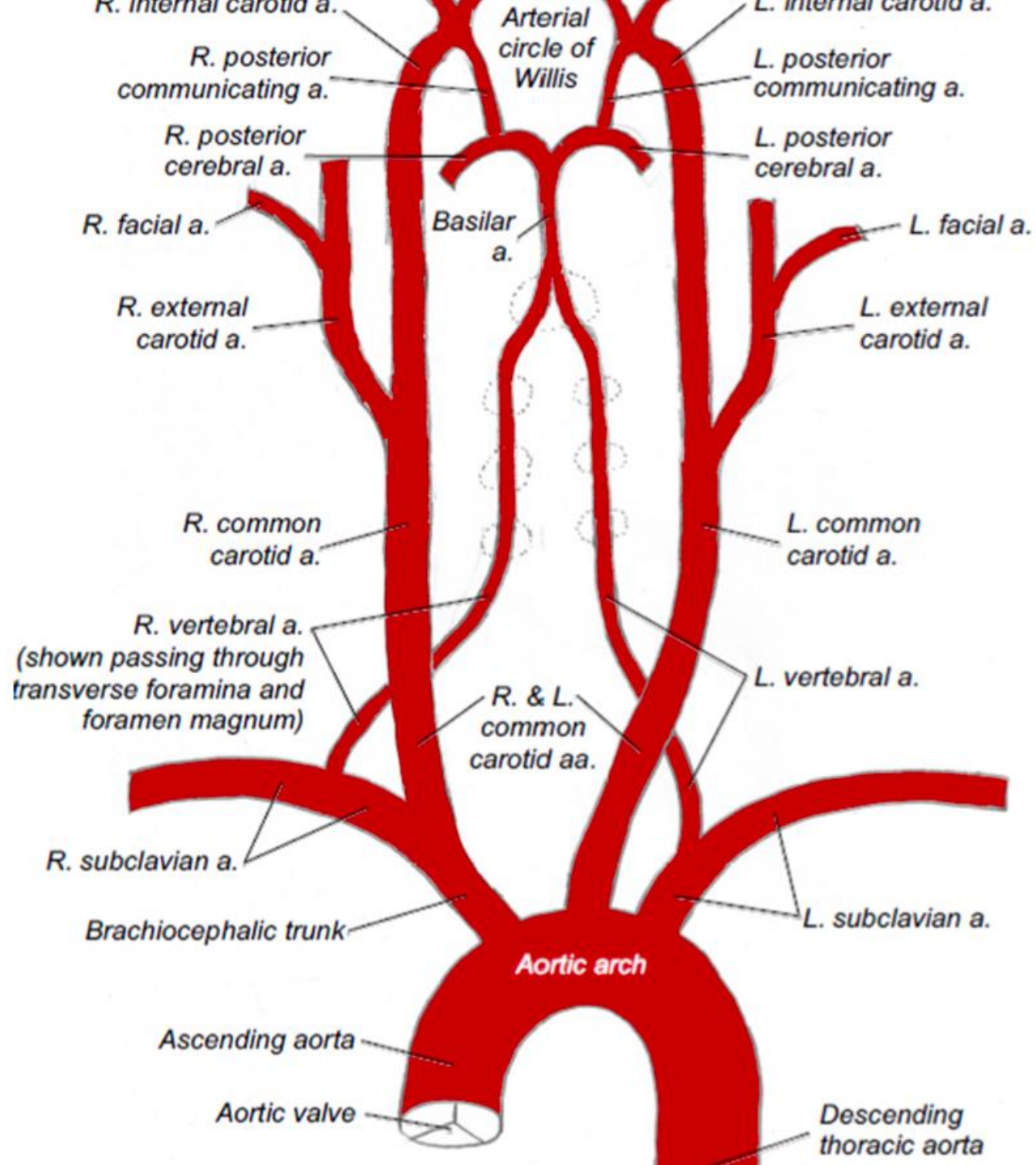




Common Carotid Artery

- ✓ Artery to head and neck
- ✓ **Right** from the brachiocephalic trunk;
- ✓ **left** from aortic arch; bifurcates at the upper border of the thyroid cartilage into:
 - Internal carotid artery
 - External carotid artery





Common Carotid Artery

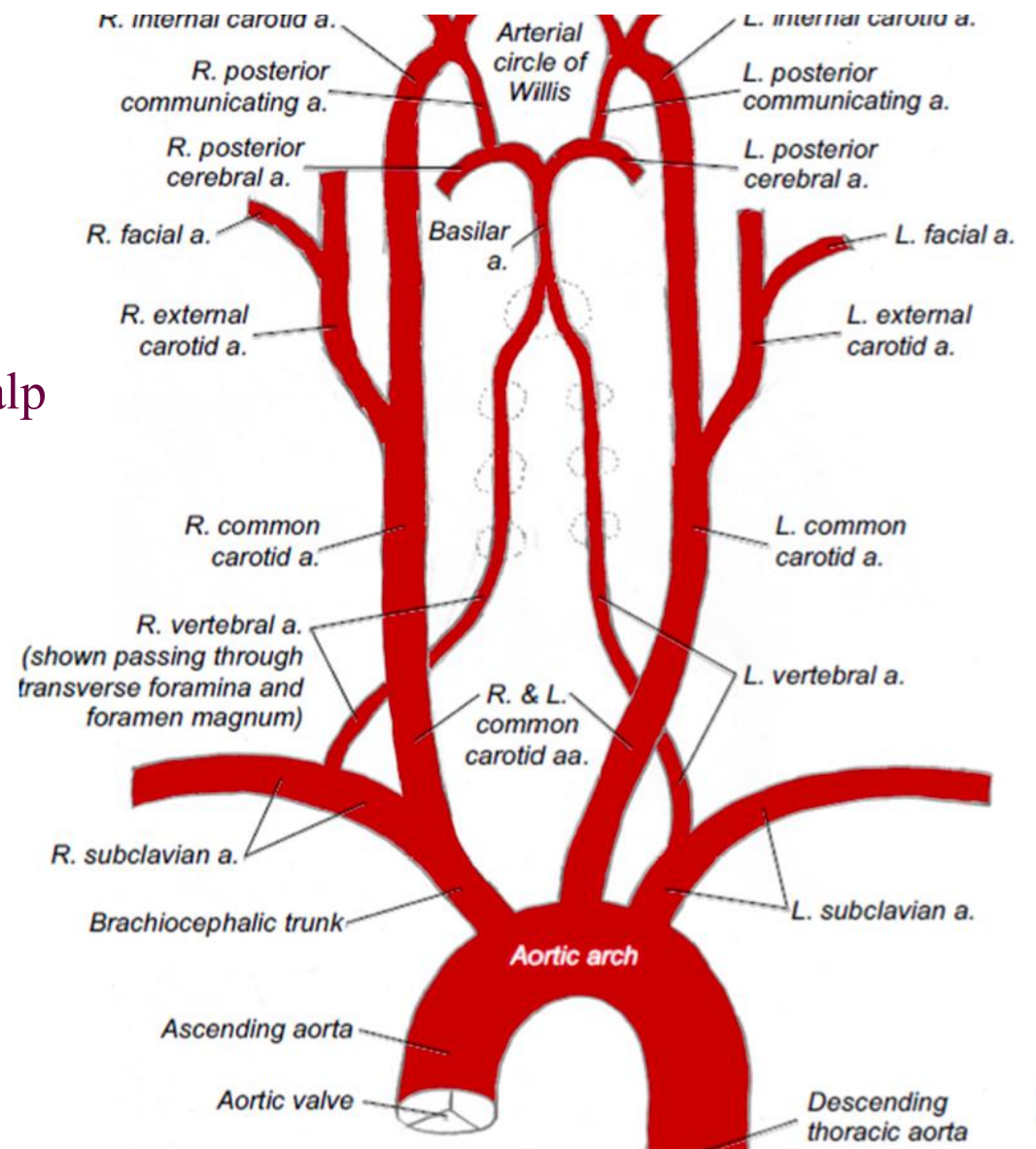
Internal carotid artery

- provides vascular supply to the brain (circle of Willis).
- Also, supply the orbit and the scalp.
- Important branches- **Anterior cerebral, middle cerebral, posterior communicating, and ophthalmic arteries.**

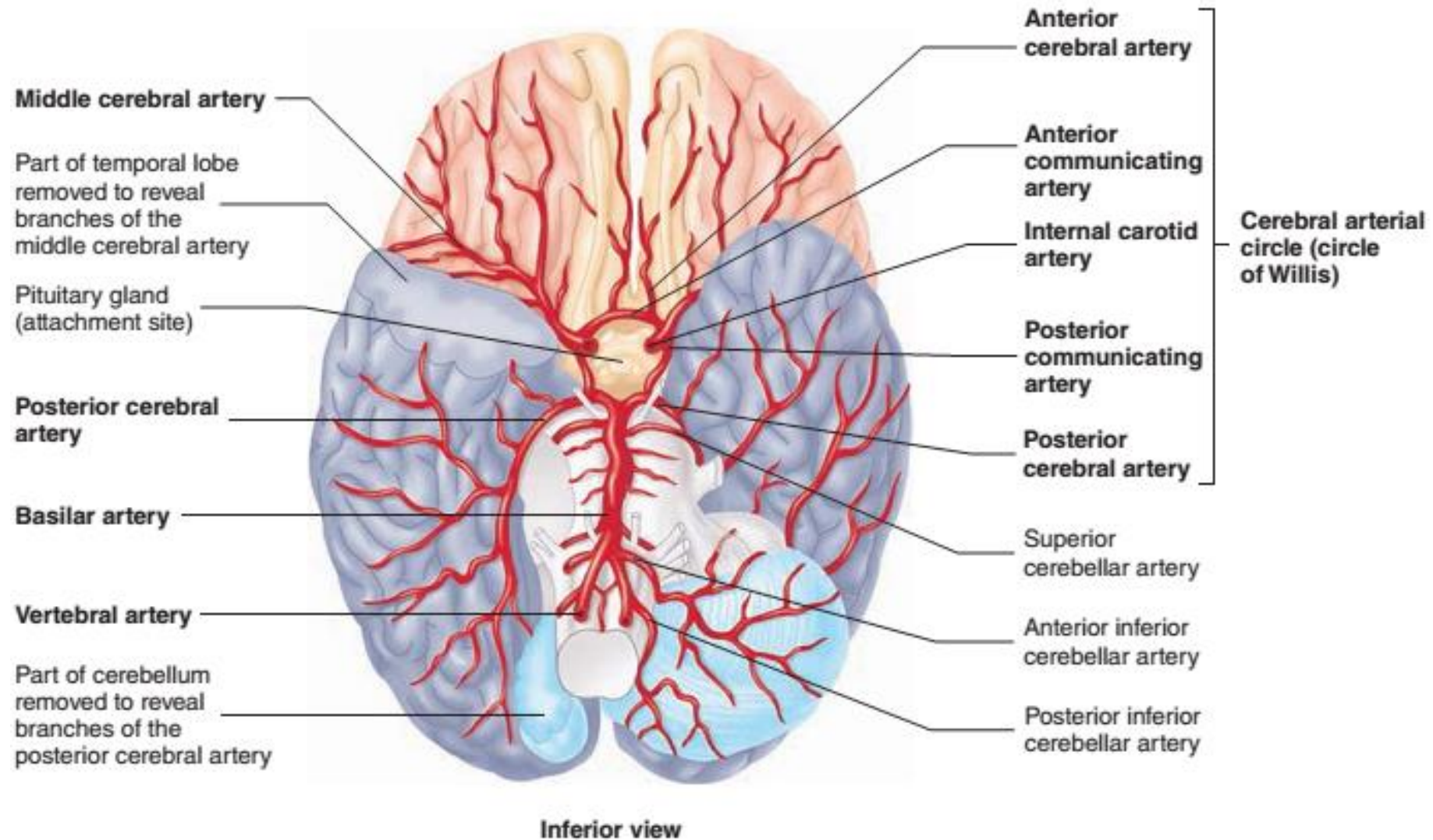
■ External carotid artery - supplies the neck, face and scalp

- Branches: superior thyroid, lingual, facial, ascending pharyngeal, occipital artery, posterior auricular, maxillary and superficial temporal arteries

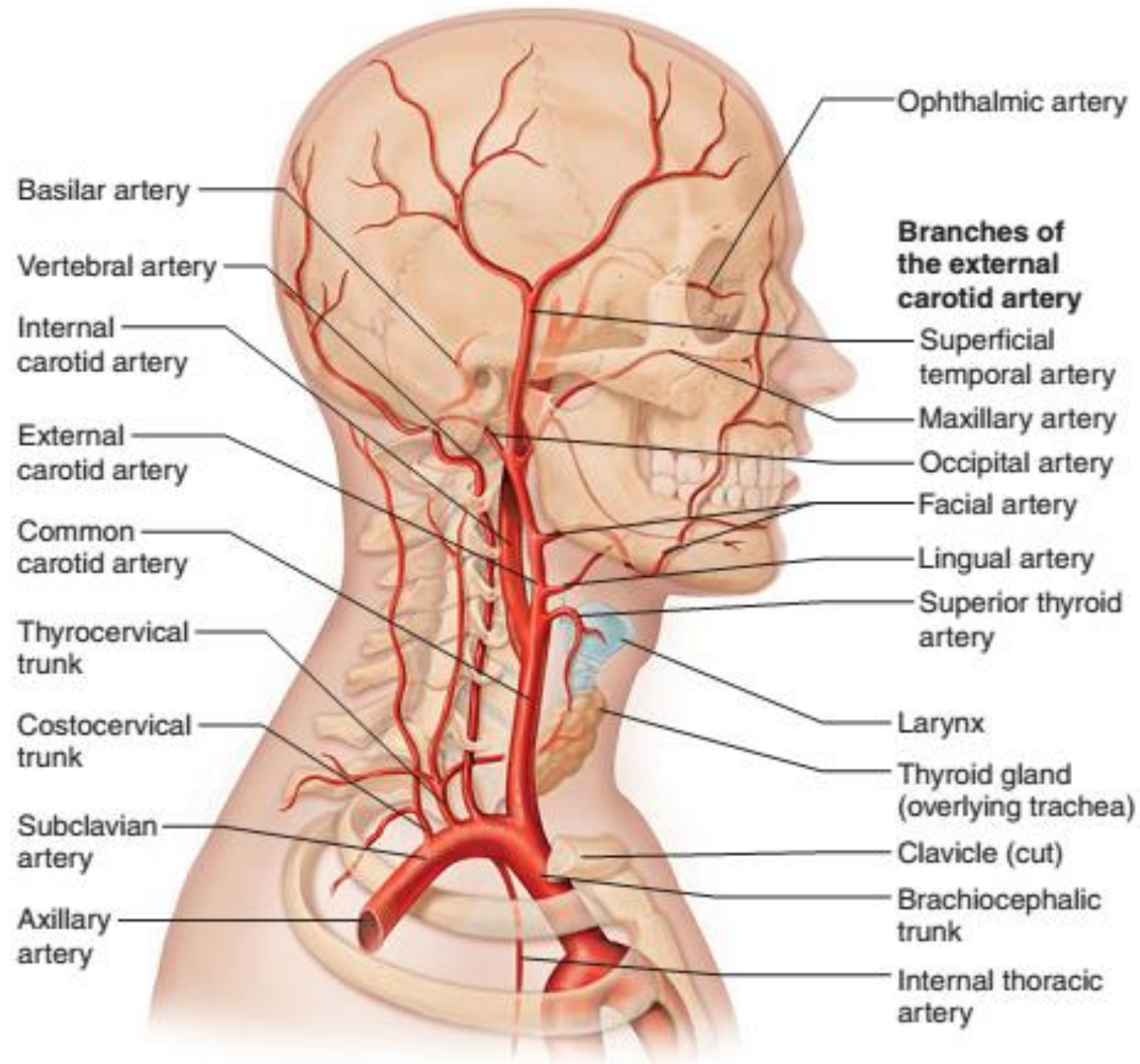
The middle meningeal artery is a branch of the maxillary artery



Blood Supply to the Brain

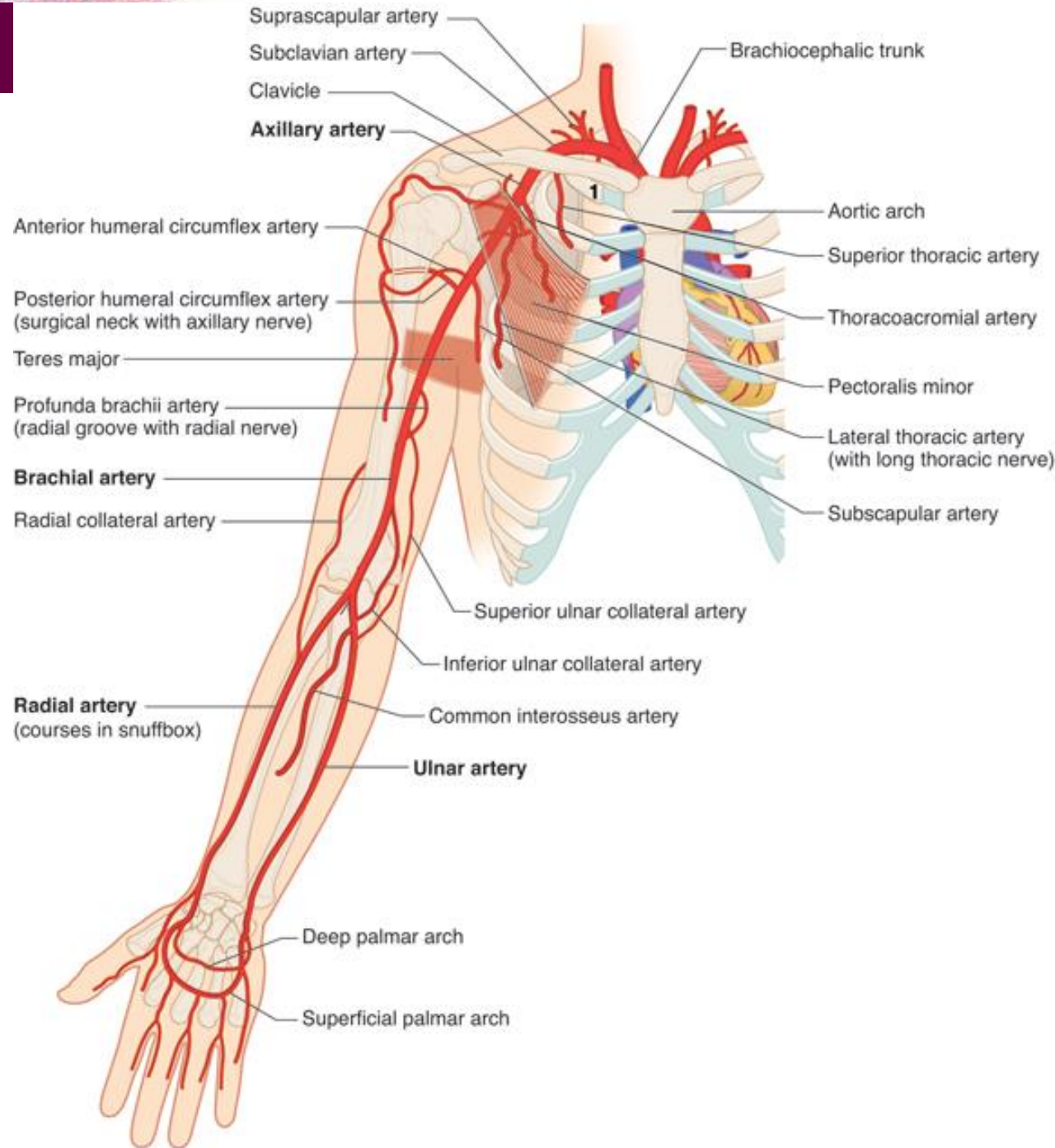


Arteries of the Head and Neck



Subclavian arteries

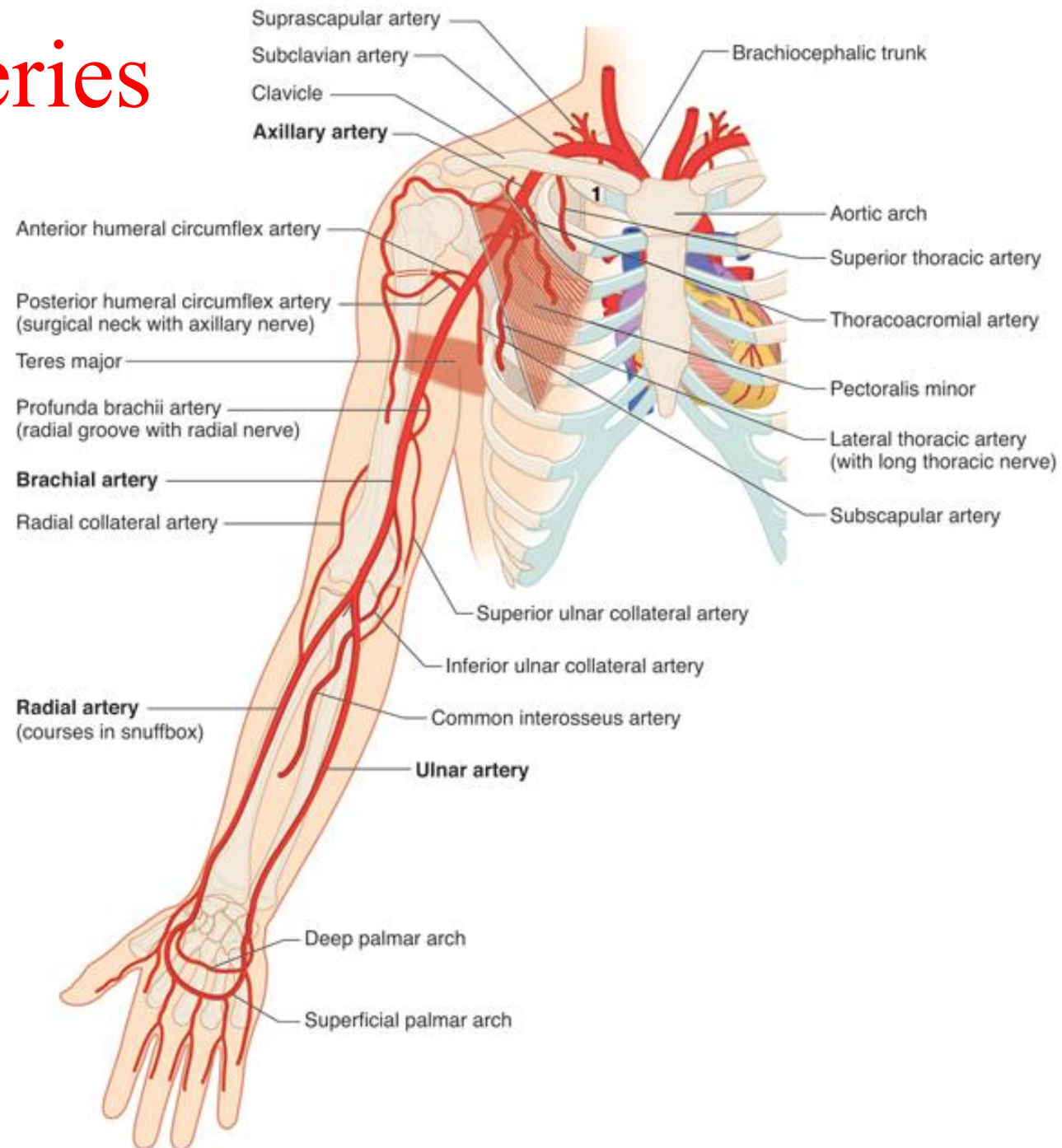
- Right from the brachiocephalic trunk; left from aortic arch
- Supply upper limbs, thoracic wall, shoulders, back, brain and spinal cord.



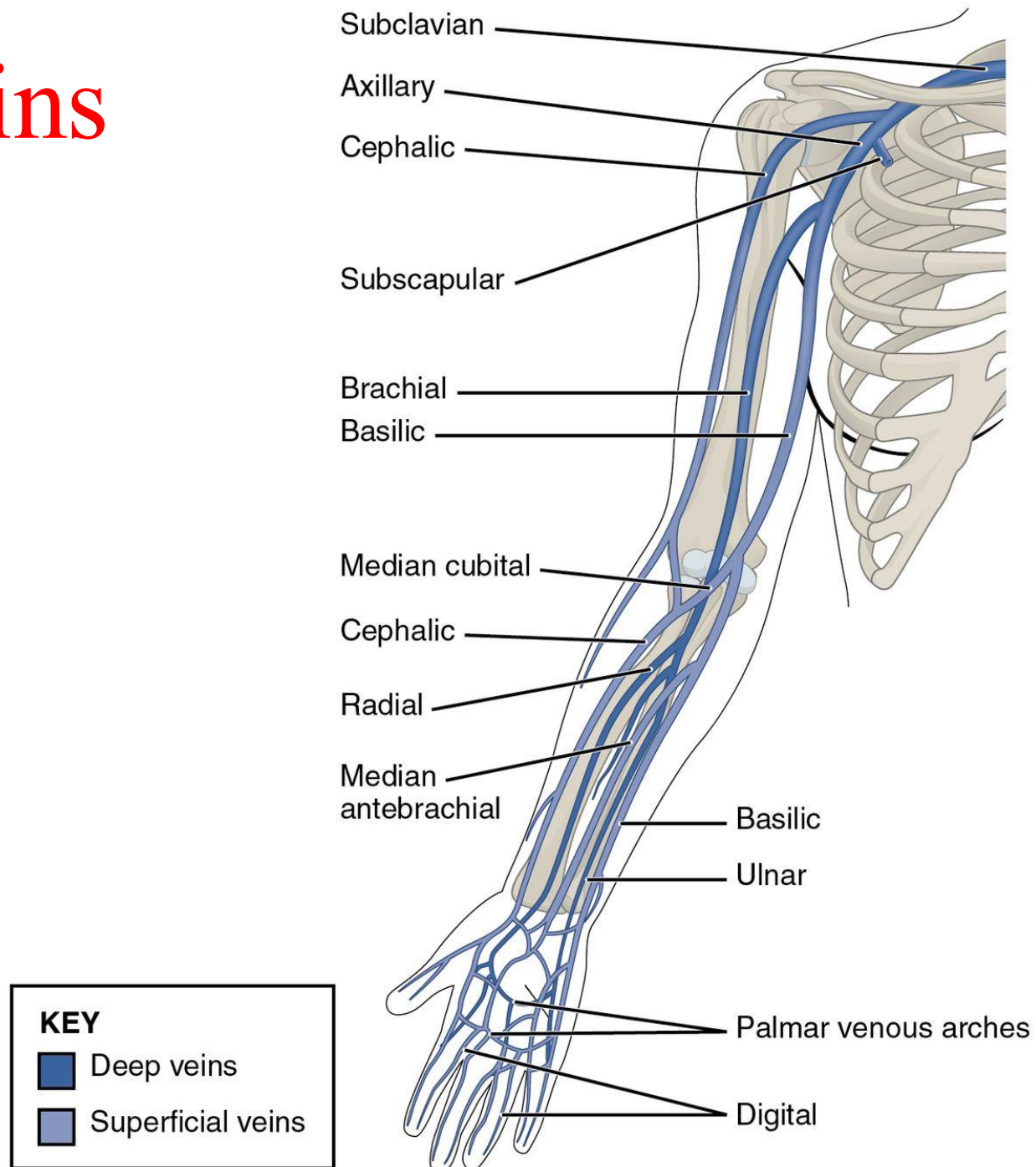
Subclavian arteries

- Main branches:
 - **Vertebral artery**
 - passes the transverse foramen of cervical vertebrae to enter cranial cavity via foramen magnum,
 - left and right join to form basilar artery - supply cerebellum and spinal cord
 - **Internal thoracic (mammary) artery**
 - supplies the pericardium, mediastinum, anterior thoracic wall, breasts, anterior abdominal wall.
- becomes the **axillary artery** at the outer border of the 1st rib
- Axillary artery becomes the **brachial artery** at the lower border of teres major

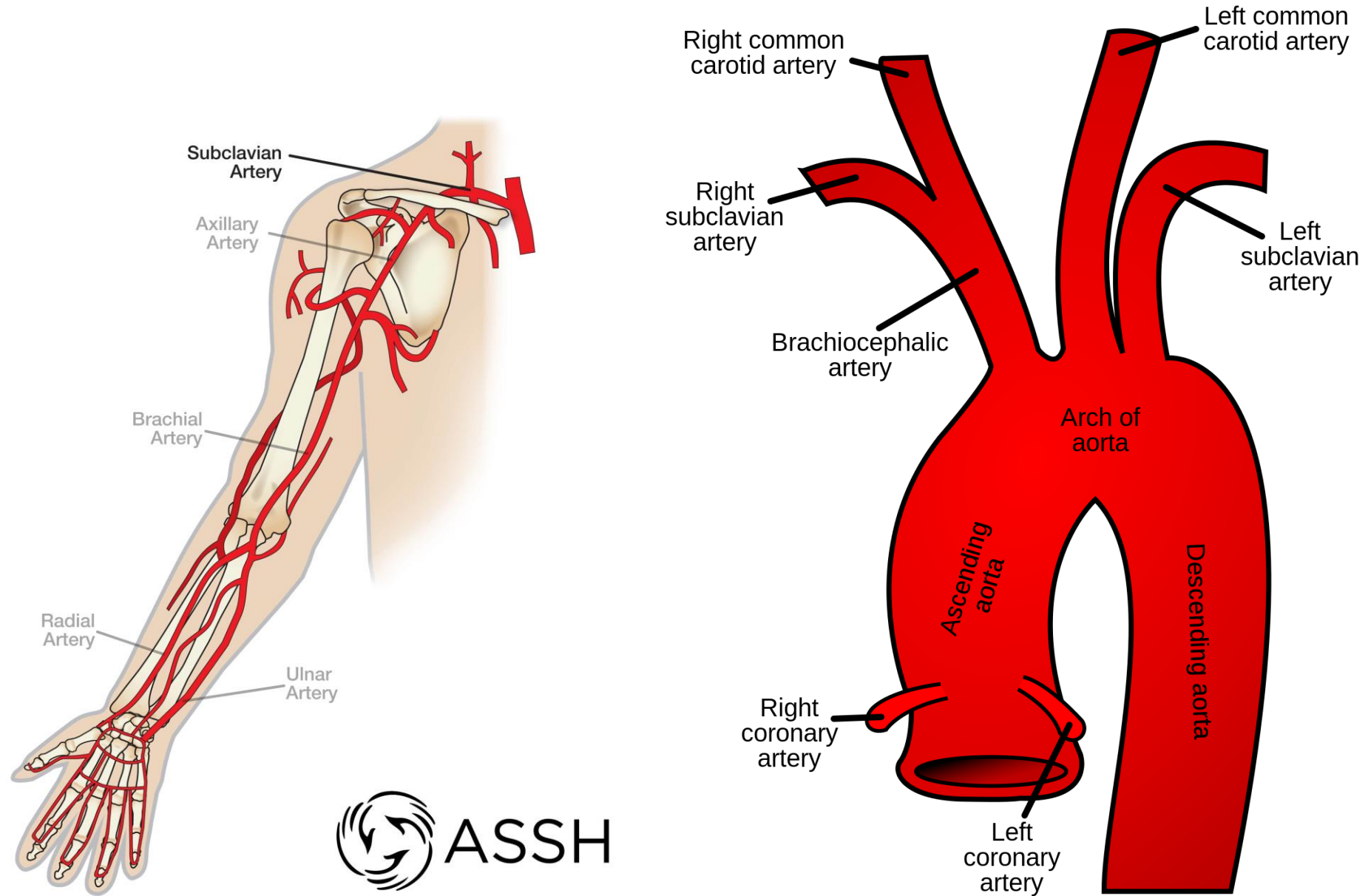
Arteries



Veins



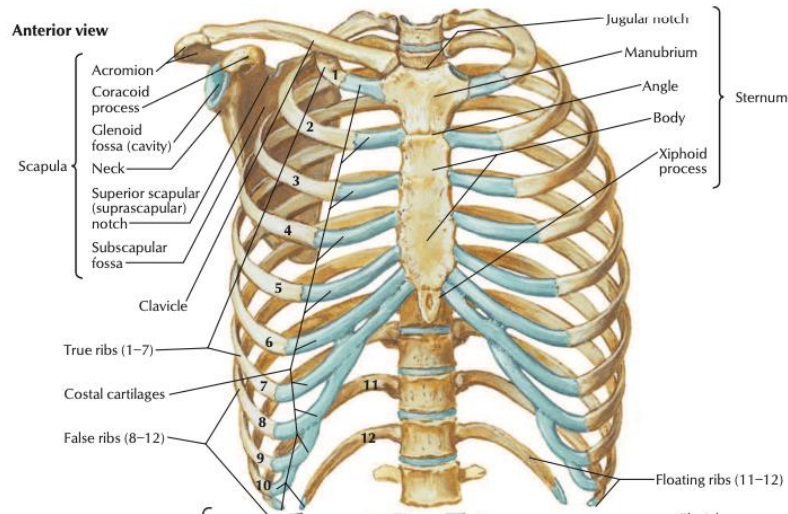
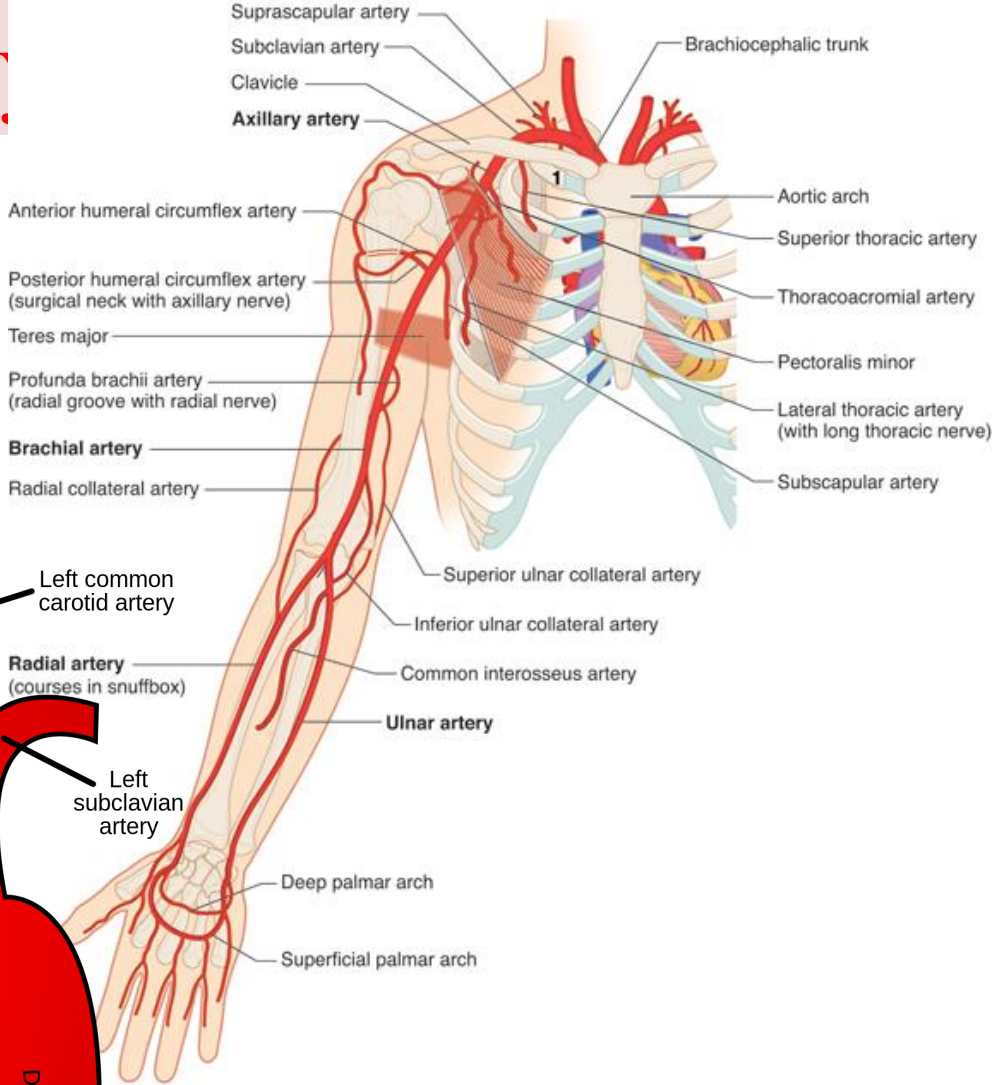
Arteries of the the upper limbs



The Subclavian Artery

❑ The right artery originates from the brachiocephalic artery.

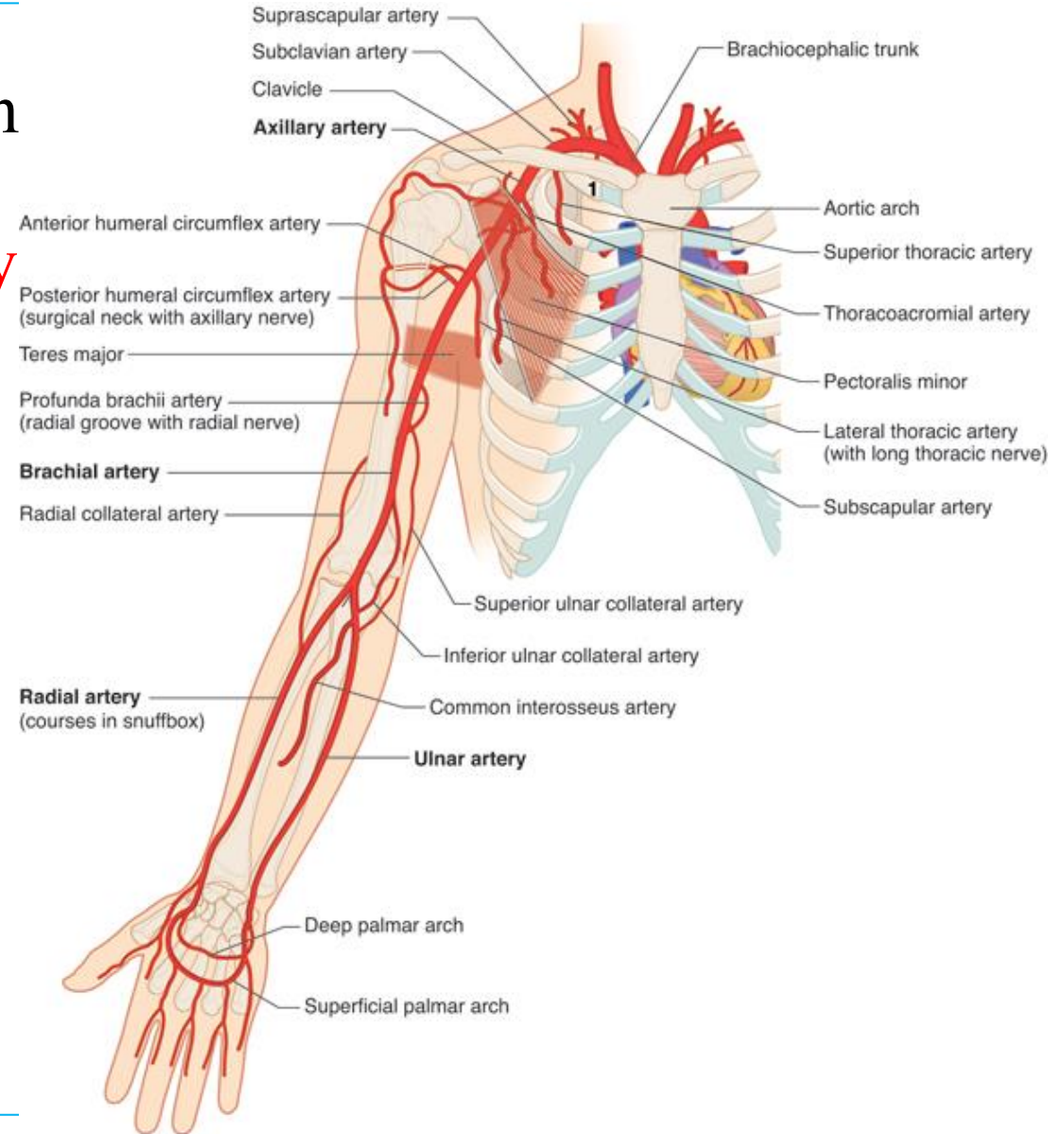
❑ The left artery originates from the arch of the aorta



Axillary artery at the lateral border of the 1st rib

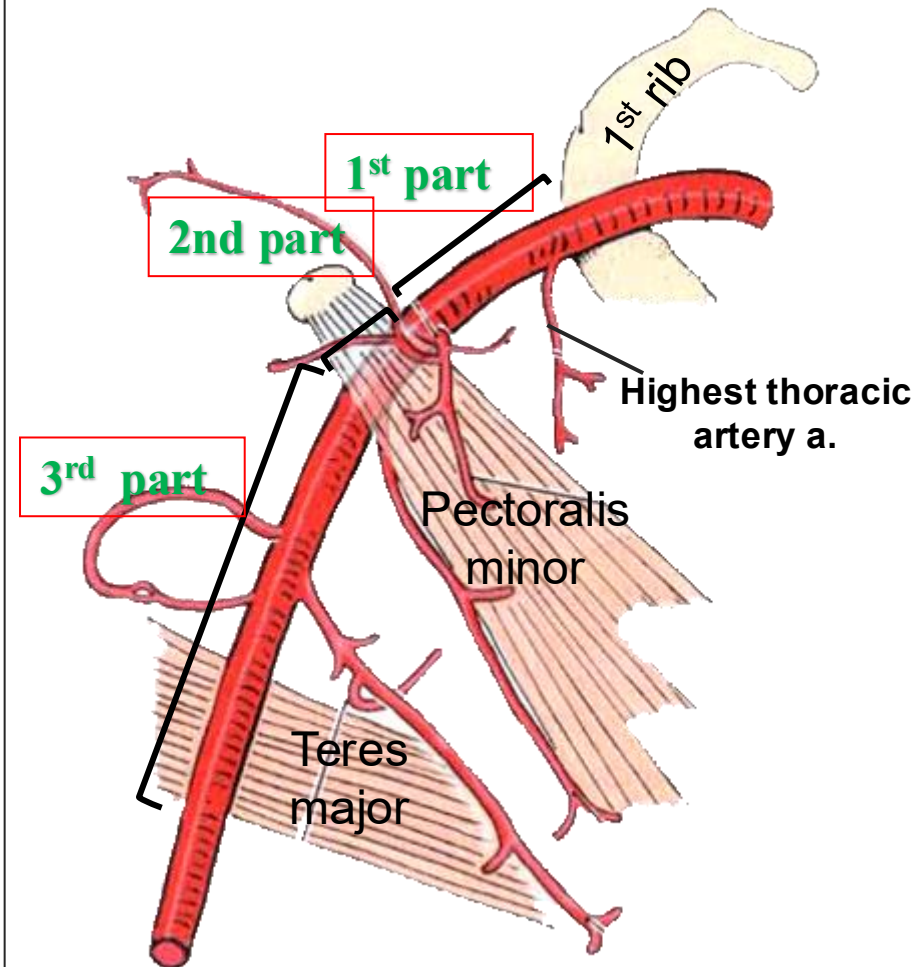
The Axillary Artery

- Begins at the lateral border of the 1st rib as continuation of the **subclavian artery**.
- Continues as **brachial artery** at lower border of teres major muscle.
- Is closely related to the cords of **brachial plexus** & their branches
- Is enclosed within the **axillary sheath**.
- Is crossed anteriorly by the **pectoralis minor** muscle & is divided into three parts; **1st, 2nd & 3rd**.



The 1st part of the axillary artery

- Extends from the **lateral border of 1st rib** to **upper border of the pectoralis minor muscle**.
- Related:
 - **Anteriorly:** to the pectoralis major muscle
 - **Laterally:** to the cords of the brachial plexus.
- ONE branch: **Highest thoracic artery**
- ✓ Supply blood to pectoralis major muscle



The 2nd part of the axillary artery

- Lies **behind the pectoralis minor muscle.**
- It is related medially, laterally, and posteriorly to the corresponding **cord** of the brachial plexus.

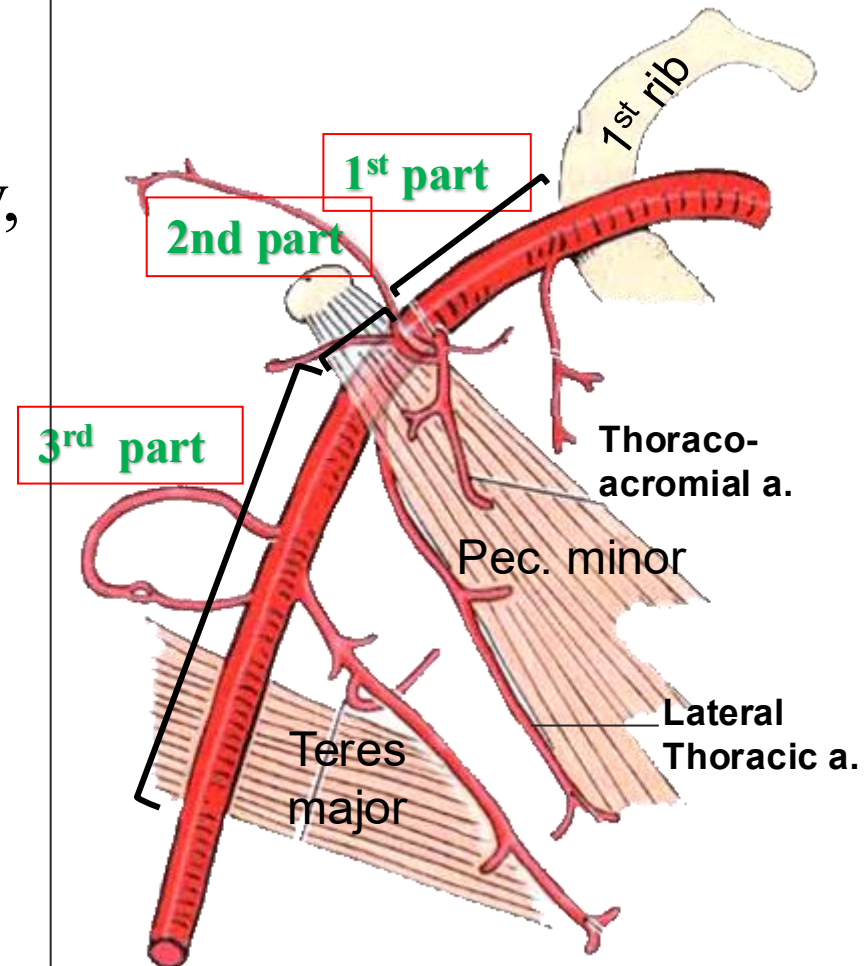
- TWO branches:

1. Thoracoacromial a.

- ✓ Clavicle, deltoid, acromial, pectoral

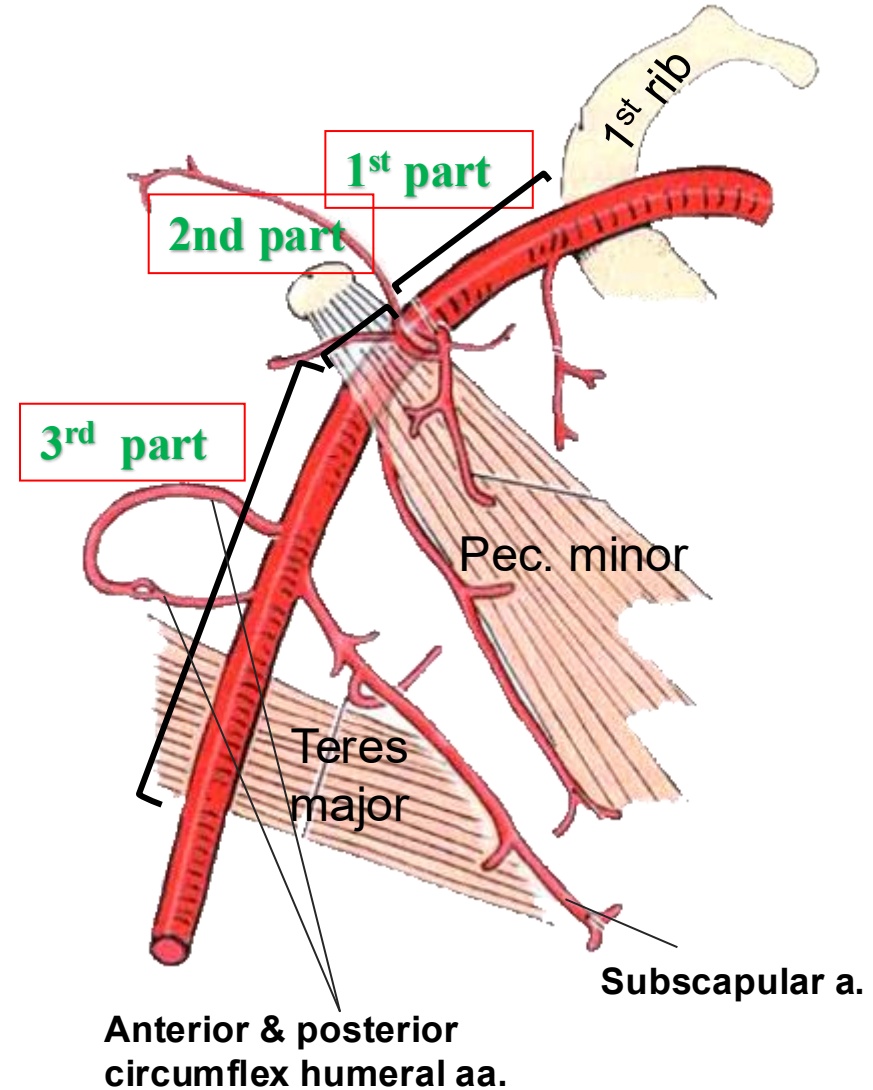
2. Lateral thoracic a.

- ✓ P. minor
- ✓ P. major

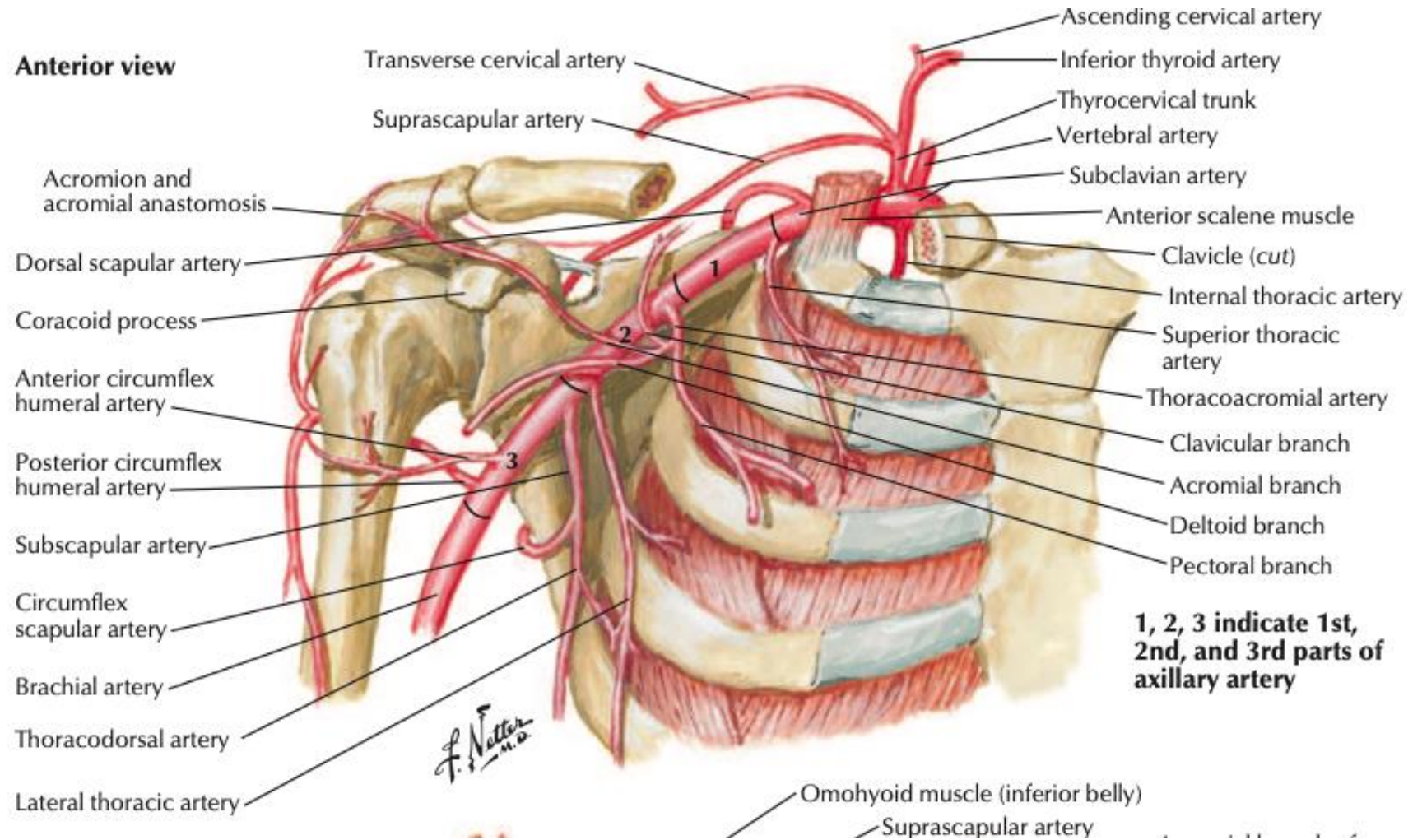


The 3rd part of the axillary artery

- ✓ Extends from the **lower border of pectoralis minor muscle** to the **lower border teres major muscle**.
- ✓ Related medially, laterally, and posteriorly, to the **branches** of the cords of the brachial plexus
- ✓ Gives THREE Branches:
 - Subscabular**
 - Anterior circumflex humeral**
 - Posterior circumflex humeral.**

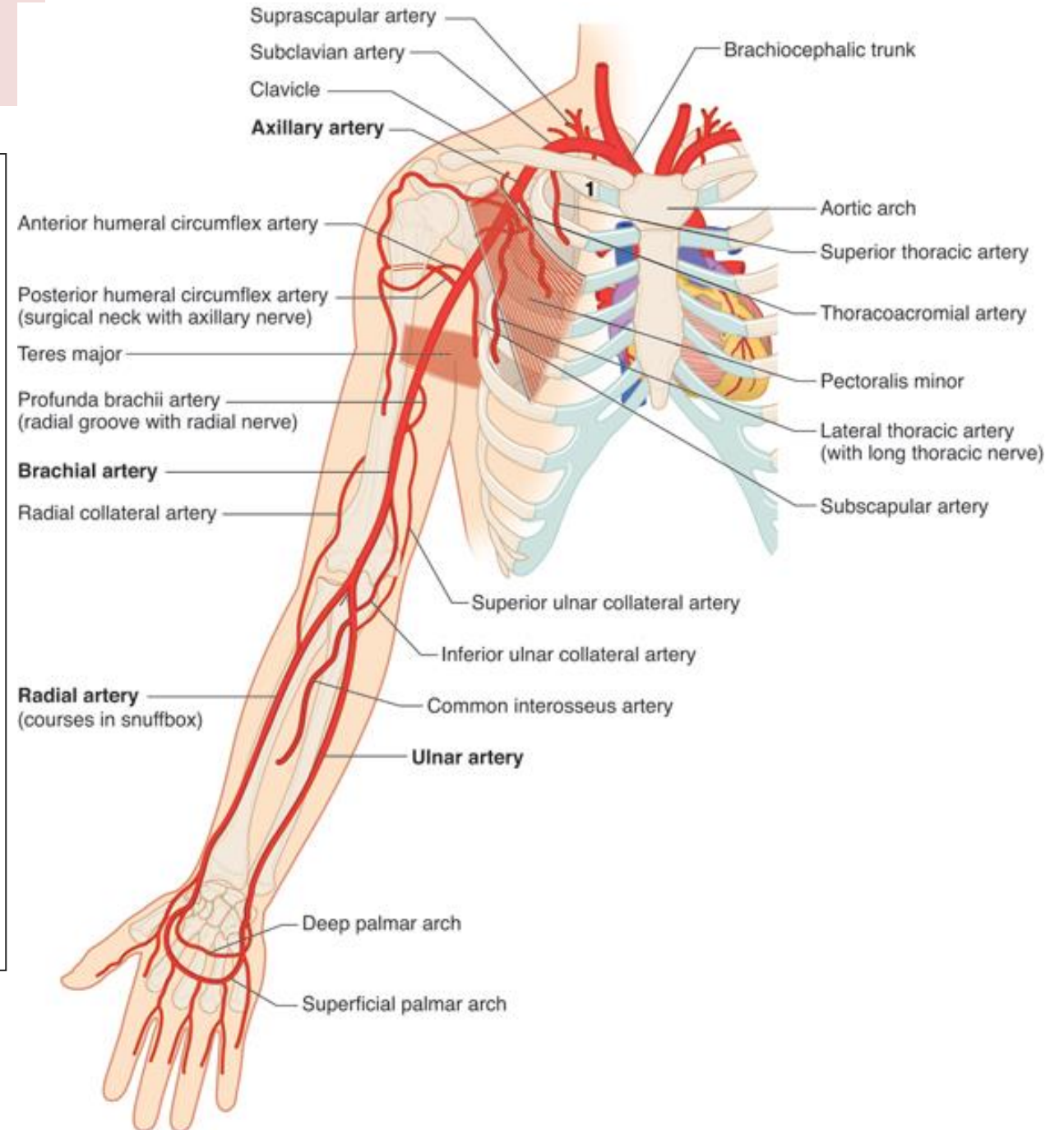


Anterior view



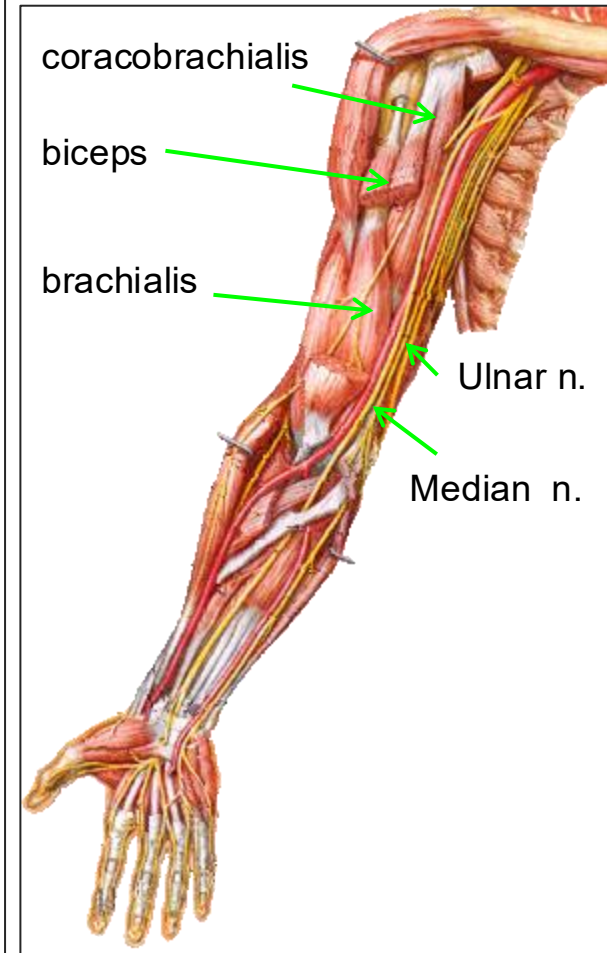
The Brachial Artery

- Is a continuation of the **axillary artery** at the lower border of teres major muscle.
- Provides main arterial supply for the arm.
- **Terminates** opposite neck of radius by dividing into **radial** & **ulnar** arteries.



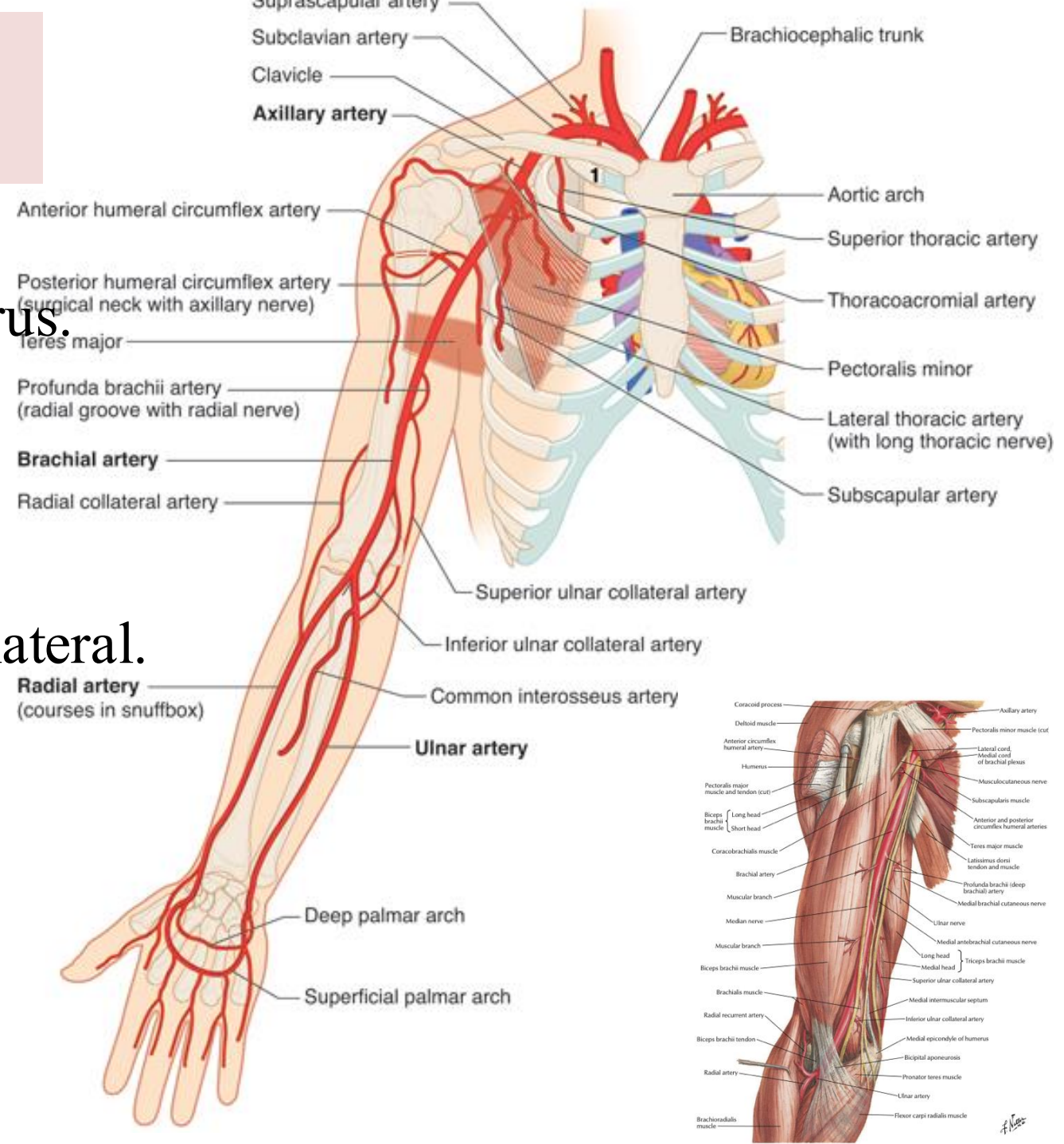
Brachial Artery: Relations

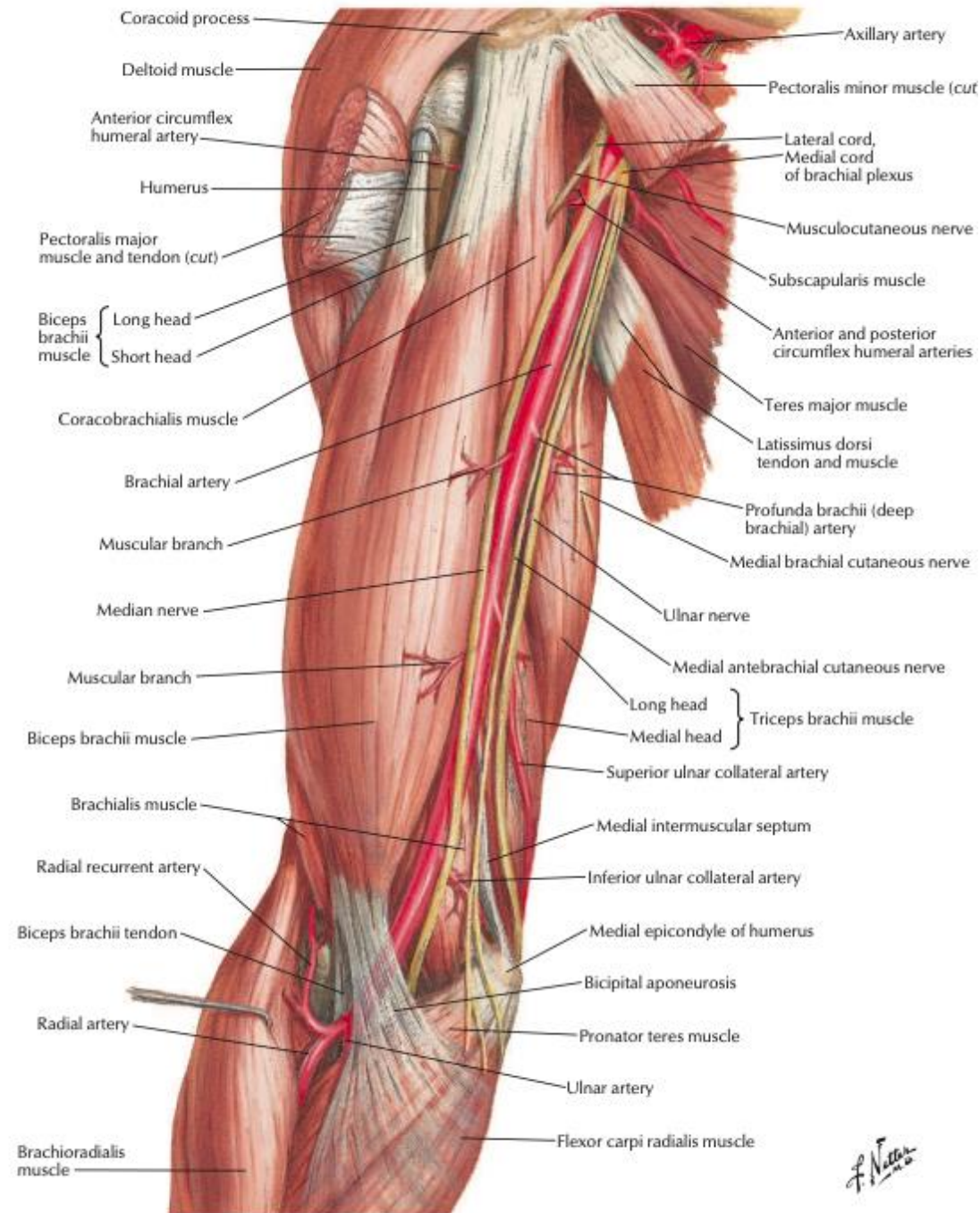
- **Anteriorly:** crossed from above downward by medial cutaneous nerve of the forearm, median nerve & bicipital aponeurosis.
- **Posteriorly:** triceps, coracobrachialis and brachialis.
- **Medially:** basilic vein, ulnar and median nerves.
- **Laterally:** coracobrachialis and biceps muscles.



Brachial Artery: Branches

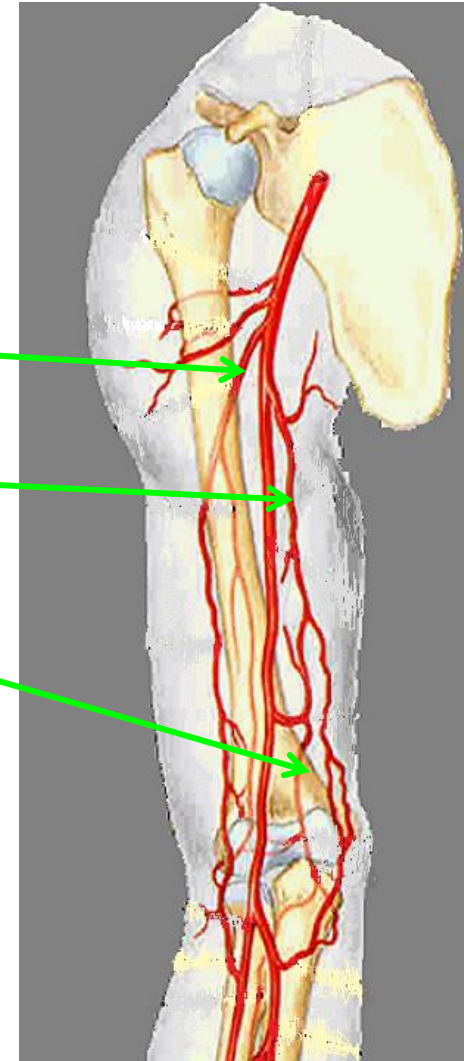
- Muscular.
- Nutrient to humerus.
- Profunda brachii
- Superior ulnar collateral.
- Inferior ulnar collateral.





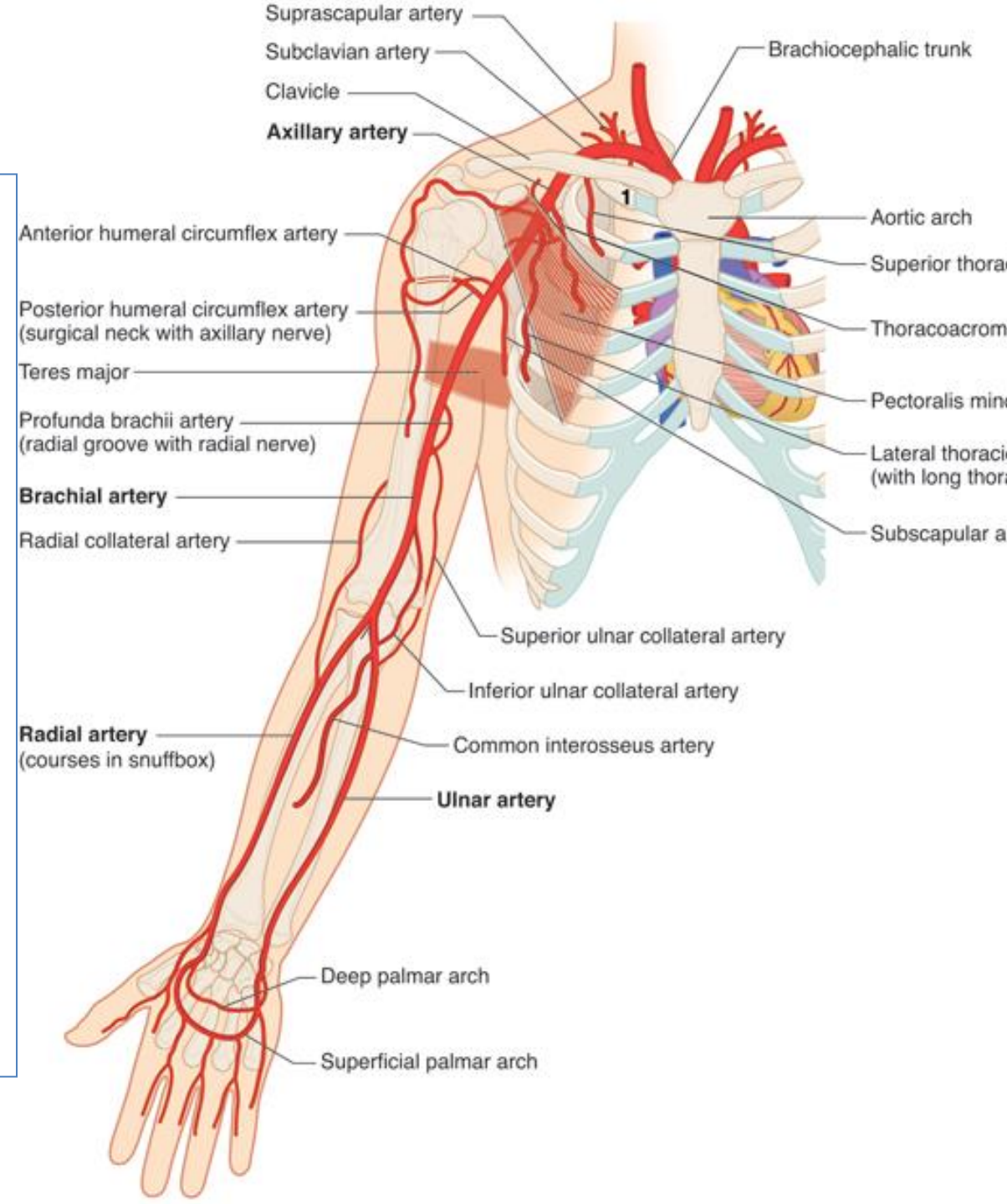
Brachial Artery: Branches

- Muscular.
- Nutrient to humerus.
- Profunda brachii
- Superior ulnar collateral.
- Inferior ulnar collateral.



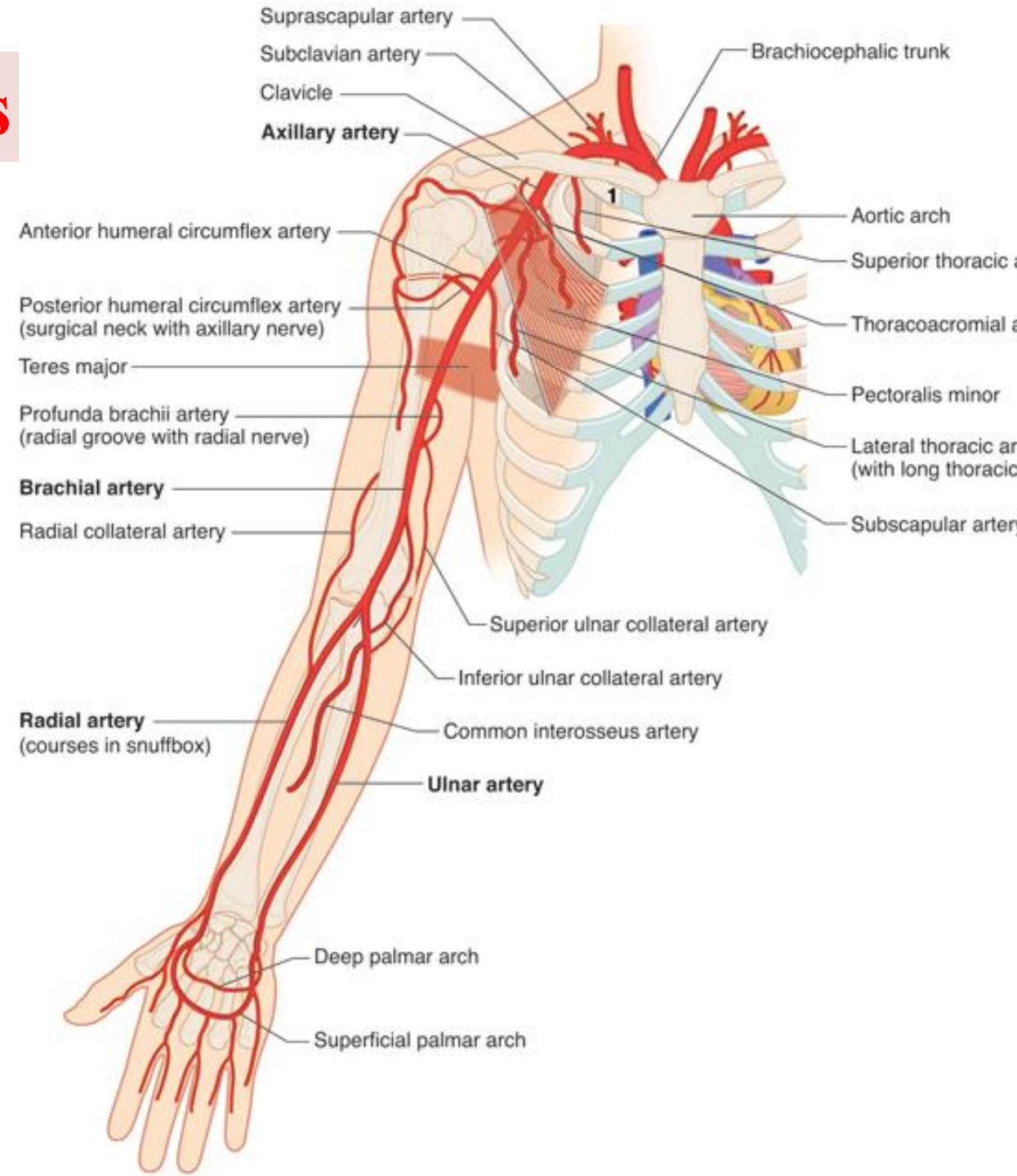
The Ulnar Artery

- The **larger** of the two terminal branches of the **brachial artery**.
- Begins in the cubital fossa at the level of **neck of radius**.
- Descends through the **anterior compartment** of the forearm.
- Enters the palm, **in front of the flexor retinaculum**, with the ulnar nerve.
- **Ends** by forming the **superficial palmar arch**, by anastomosing with superficial palmar branch of radial artery.



Ulnar Artery: Branches

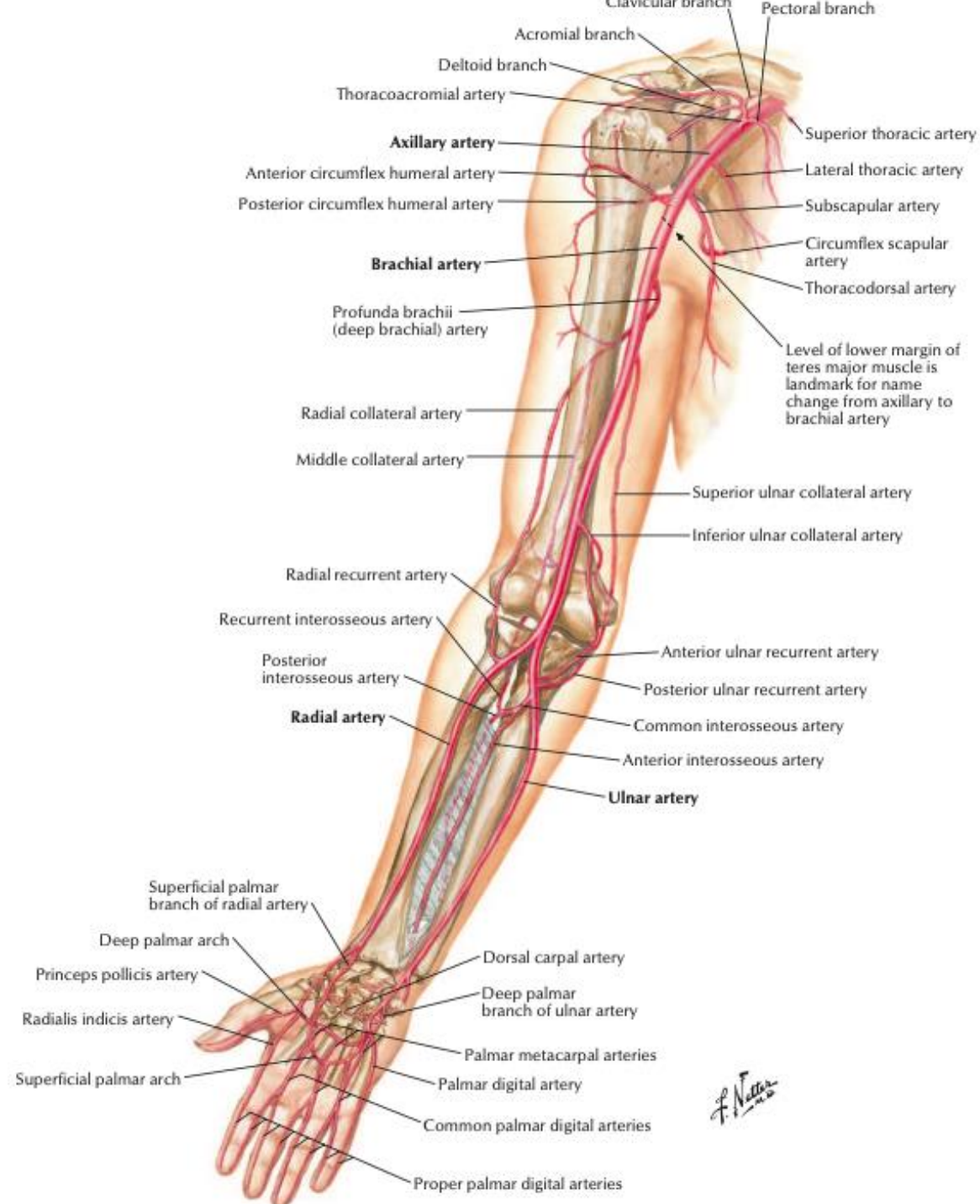
- Muscular .
- Recurrent branch for anastomosis around the elbow joint.
- Common interosseous artery, which gives **anterior and posterior interosseous arteries**.
- Branch to anastomoses around the wrist joint.



Ulnar Artery: Branches

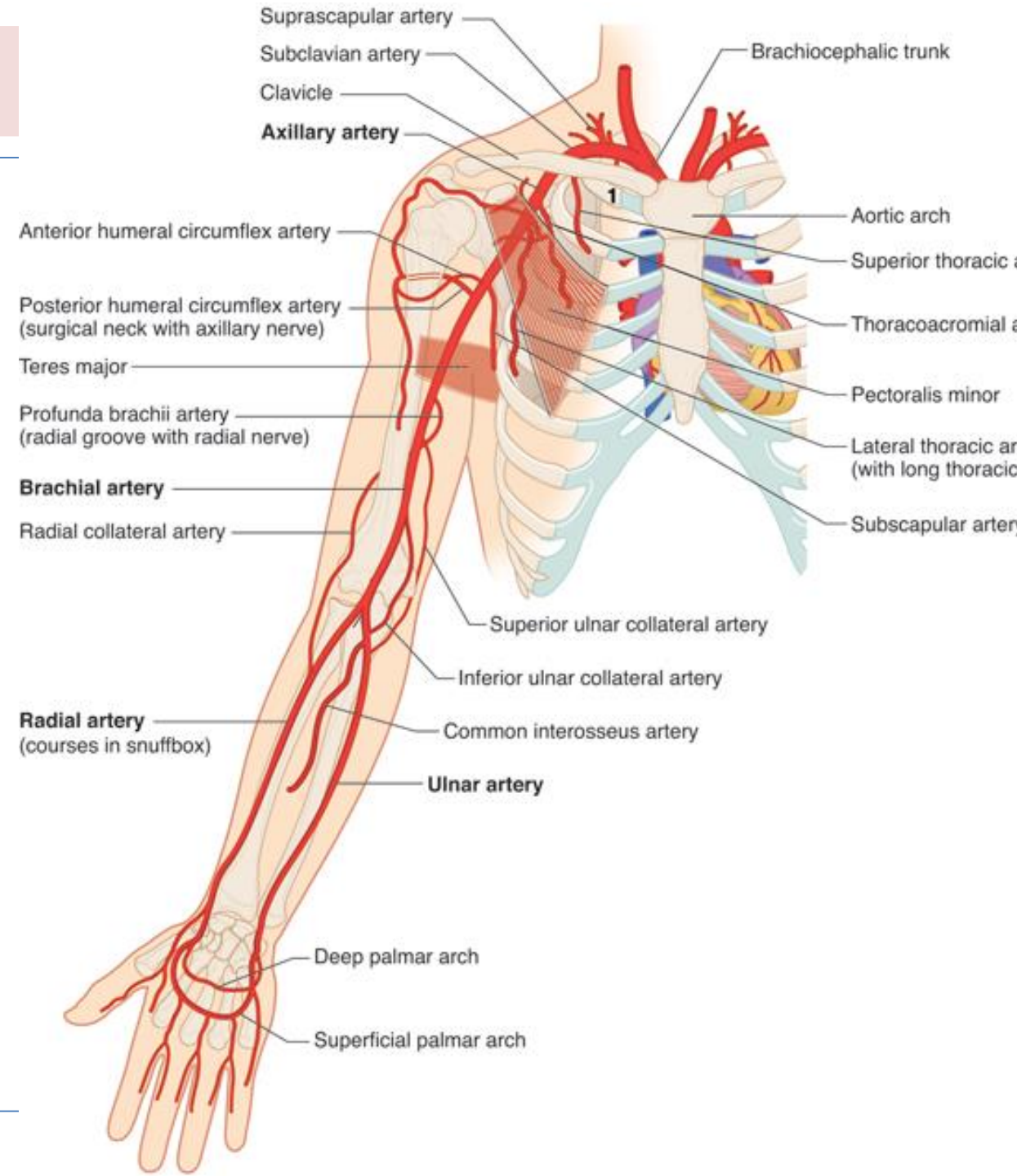


- Muscular .
- Recurrent branch for anastomosis around the elbow joint.
- Common interosseous artery, which gives anterior and posterior interosseous arteries.
- Branch to anastomoses around the wrist joint.



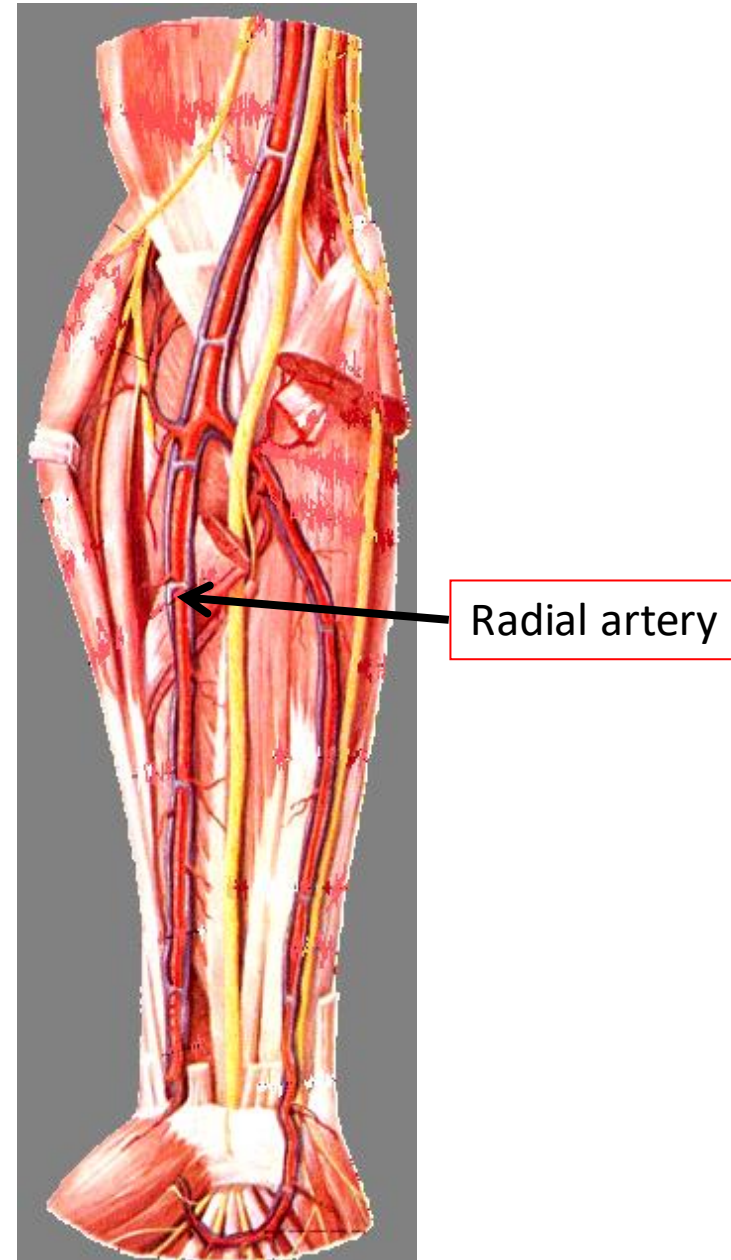
The Radial Artery

- The **smaller** of the two terminal branches of the **brachial artery**.
- Begins in the cubital fossa at the level of **neck of radius**.
- Descends downward and laterally.
- Leaves the forearm by **winding** around the lateral aspect of the wrist to reach the **dorsum** of the hand.



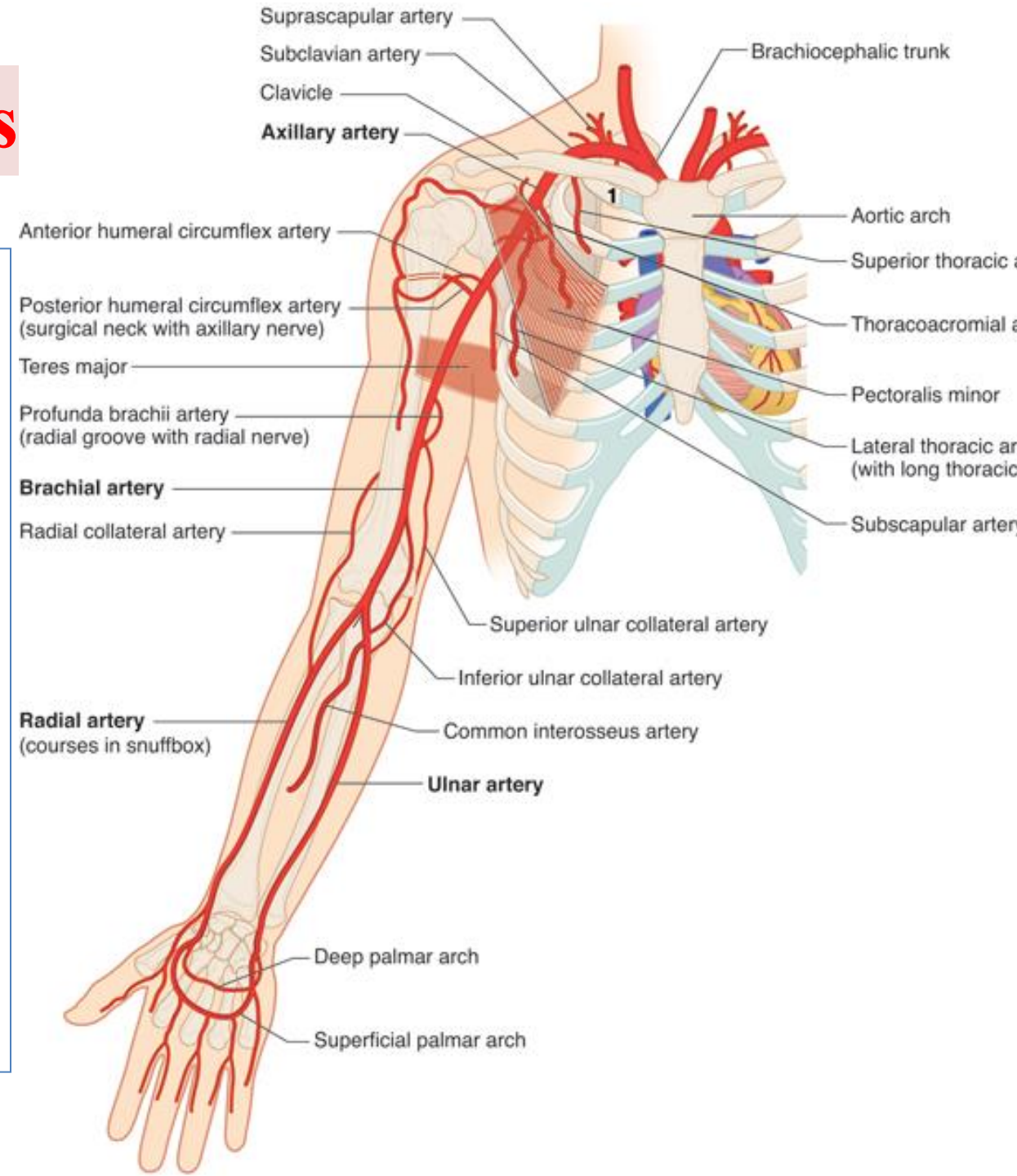
The Radial Artery

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- Begins in the cubital fossa at the level of **neck of radius**.
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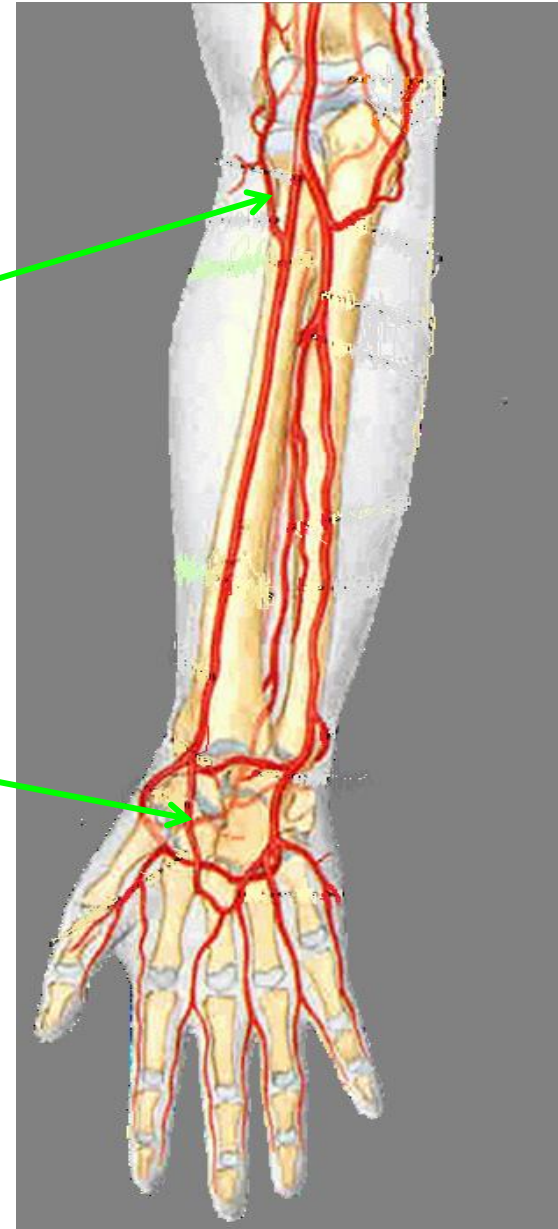
Radial Artery: Branches

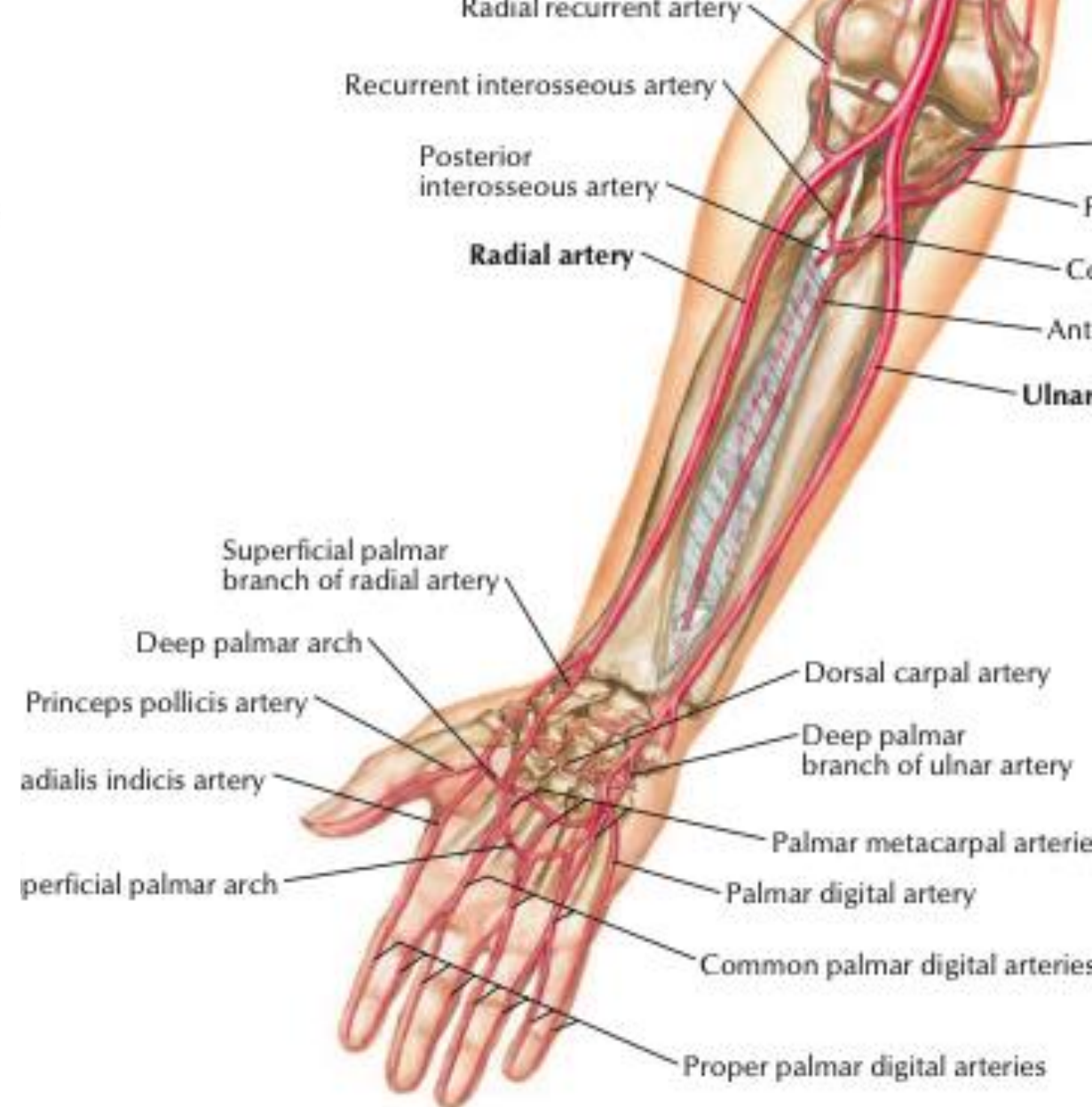
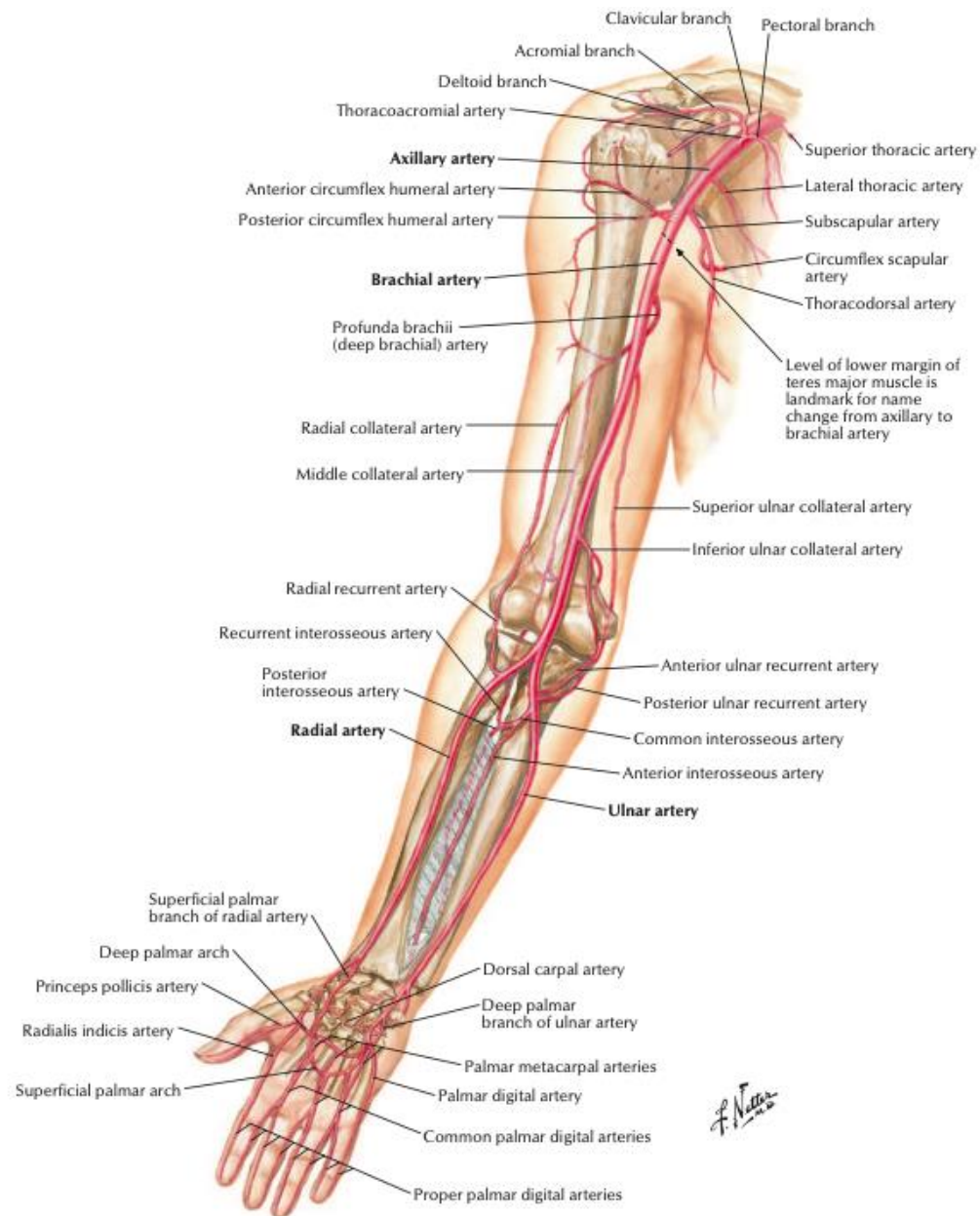
- Muscular
- Recurrent branch for anastomosis around the elbow joint.
- Superficial palmar branch, joins the ulnar artery to form the superficial palmar arch.



Radial Artery: Branches

- Muscular
- Recurrent branch for anastomosis around the elbow joint.
- Superficial palmar branch, joins the ulnar artery to form the superficial palmar arch.



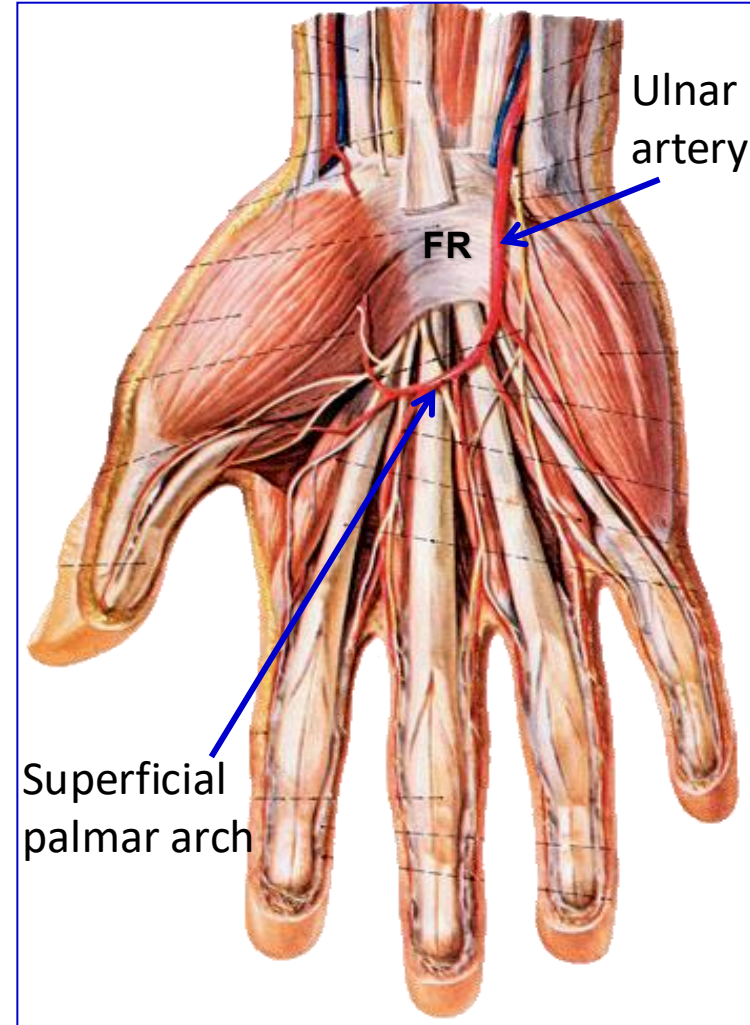




Arteries of the Palm

➤ Ulnar artery:

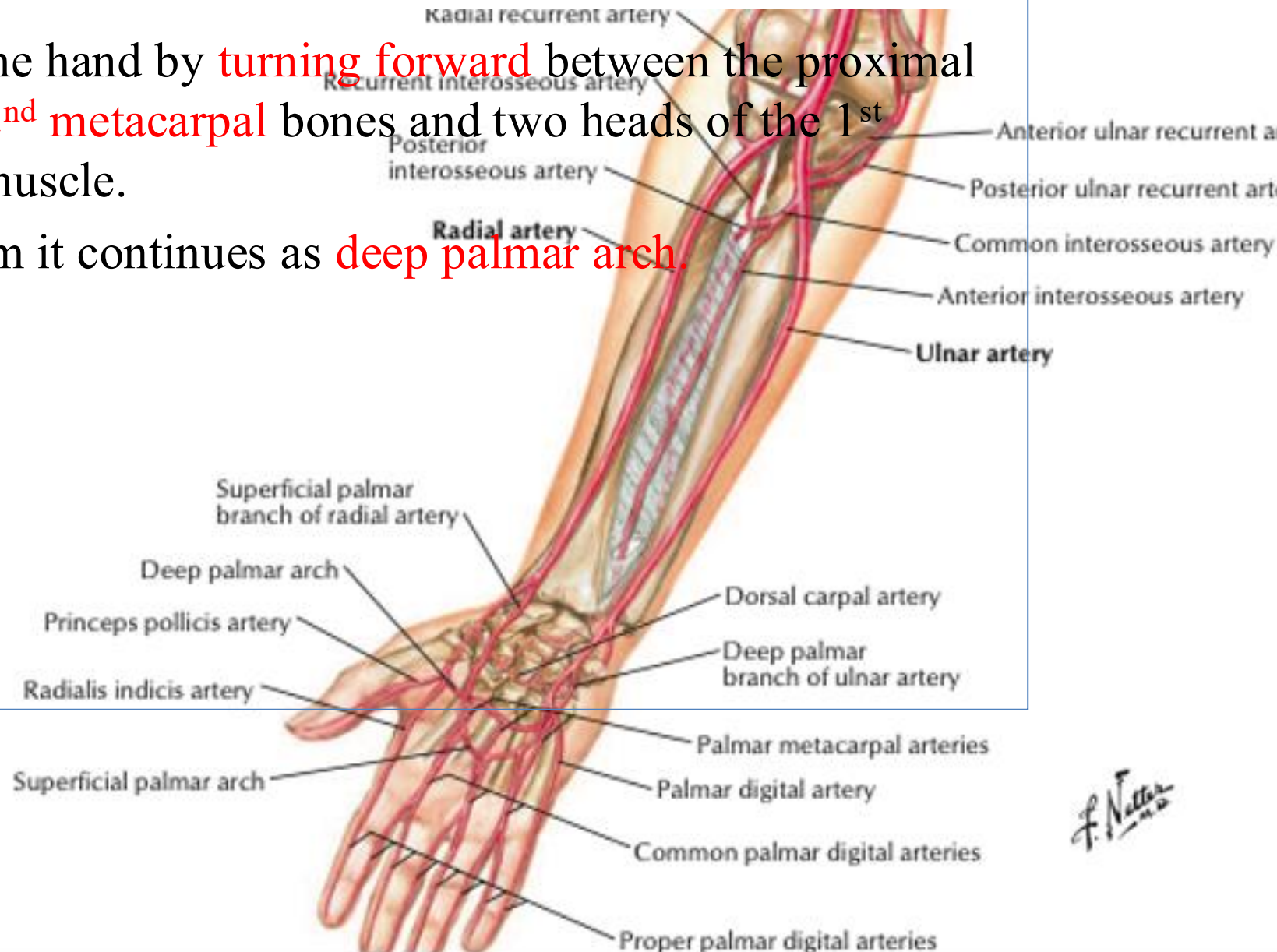
- Enters the hand:
 - anterior to the flexor retinaculum,
 - on the lateral side of the ulnar nerve and pisiform bone.
- Gives a deep branch.
- Continue as the superficial palmar arch.



Arteries of the Palm

➤ Radial artery;

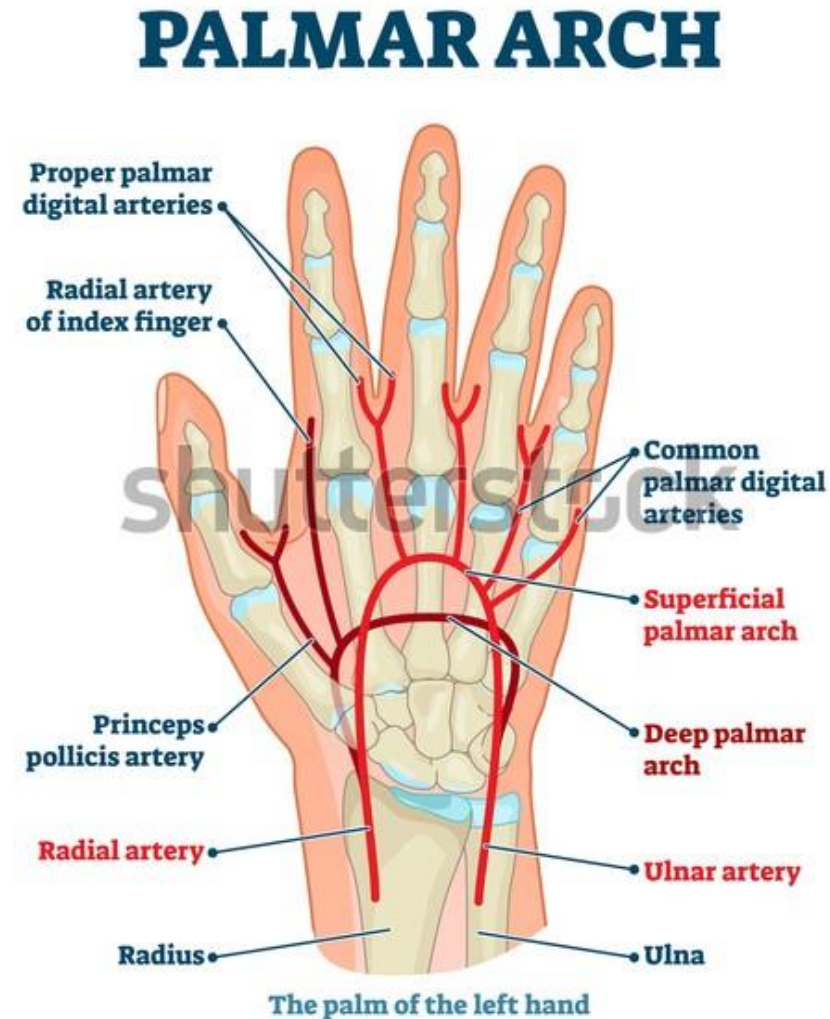
- Leaves dorsum of the hand by **turning forward** between the proximal ends of the **1st and 2nd metacarpal** bones and two heads of the 1st dorsal interosseous muscle.
- On entering the palm it continues as **deep palmar arch**.
- It gives; arteria
 - i. **radialis indicis a.**
 - ii. **princeps policis a**



F. Netter M.D.

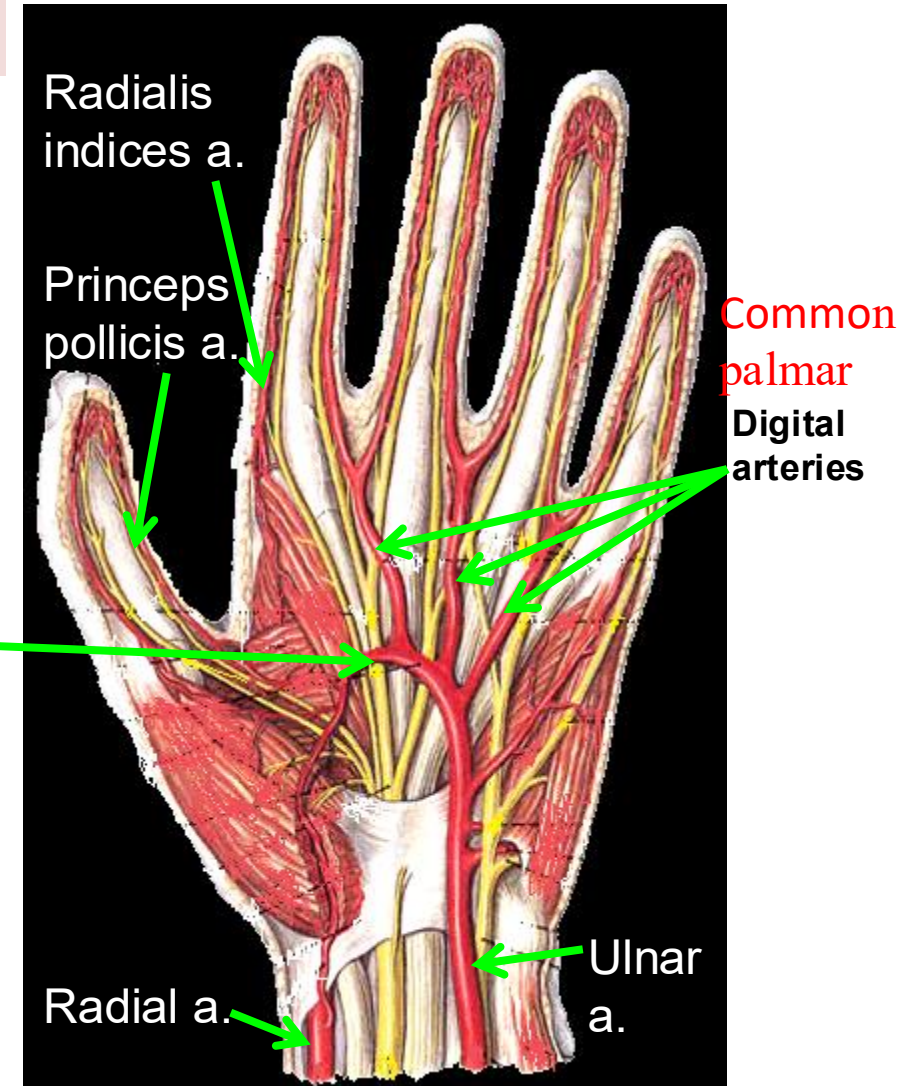
The Superficial Palmar Arch

- Is the direct continuation of the **ulnar artery**, as it curves laterally behind the palmar aponeurosis.
- Is completed by **branch from the radial artery**.
- Lies approximately at the level of the **distal border of the extended thumb**.
- Gives: **digital arteries** from its convexity to supply the fingers.



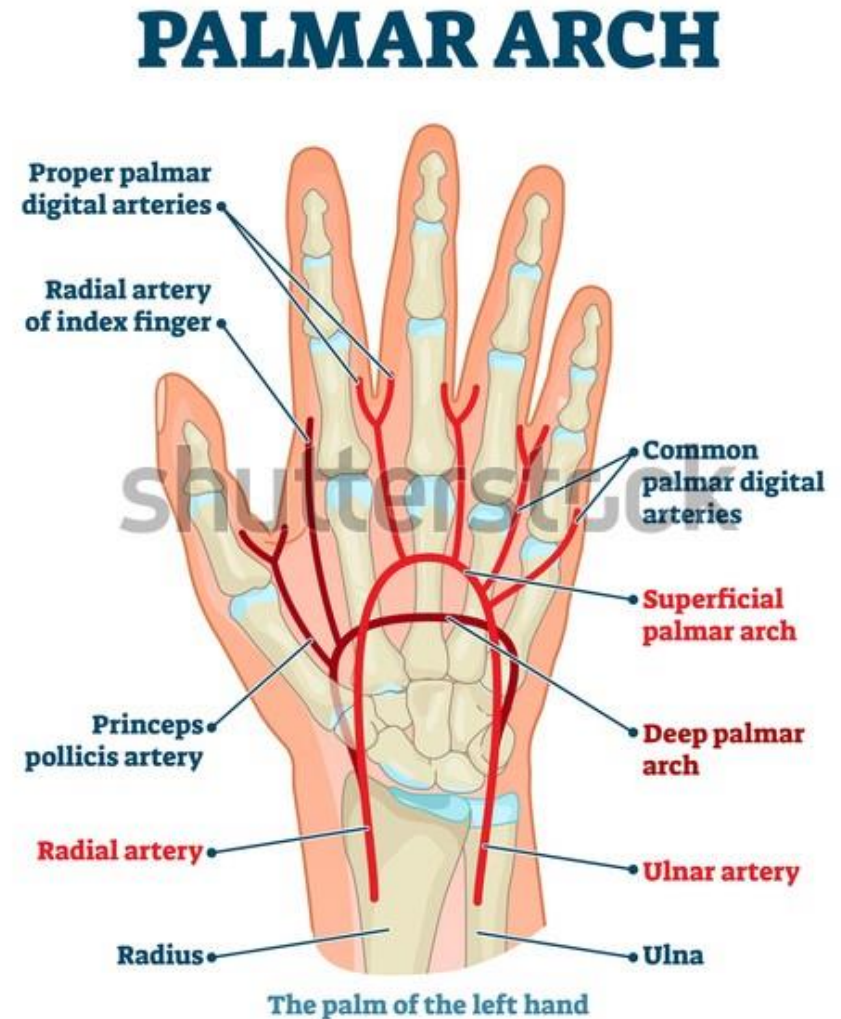
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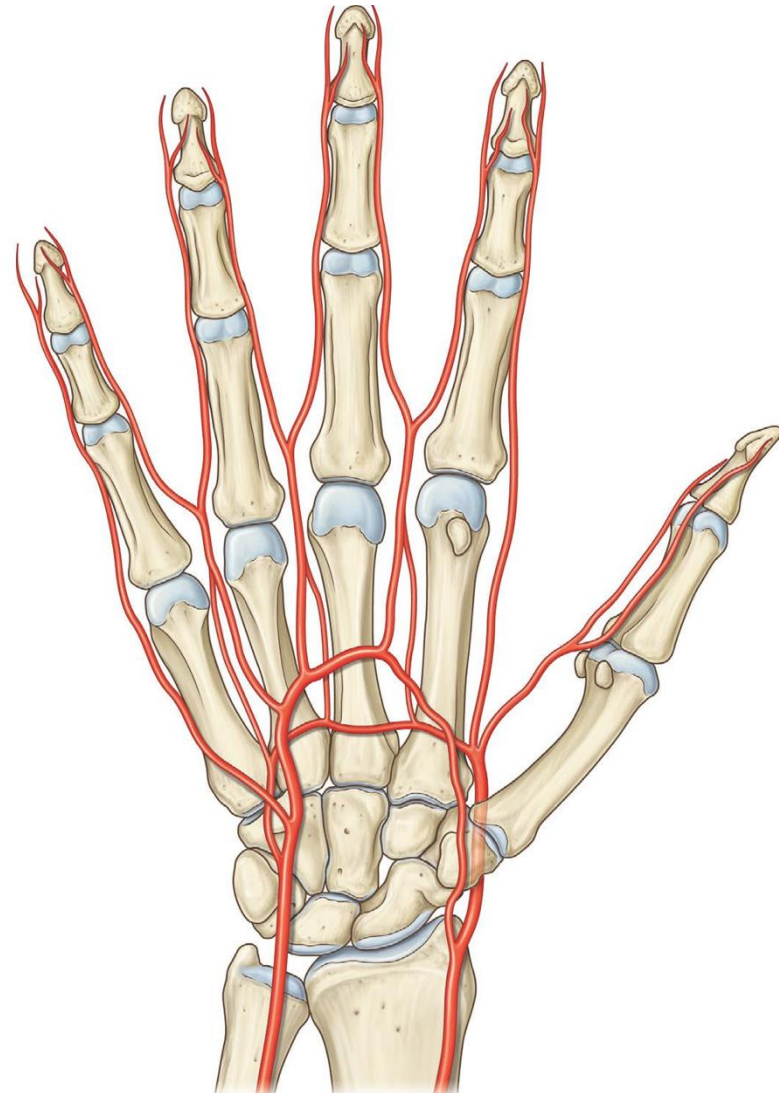
- Is a continuation of the **radial artery** as it curves medially beneath long flexor tendons, in front of the metacarpal bones and interosseous muscles.
- Is completed on the medial side by **deep branch of ulnar artery**.
- Lies at a level of the **proximal border of extended thumb**.
- It sends branches:
 - **superiorly** to share in anastomosis around the wrist joint &
 - **inferiorly** to join branches of the superficial palmar arch.

The Deep Palmar Arch



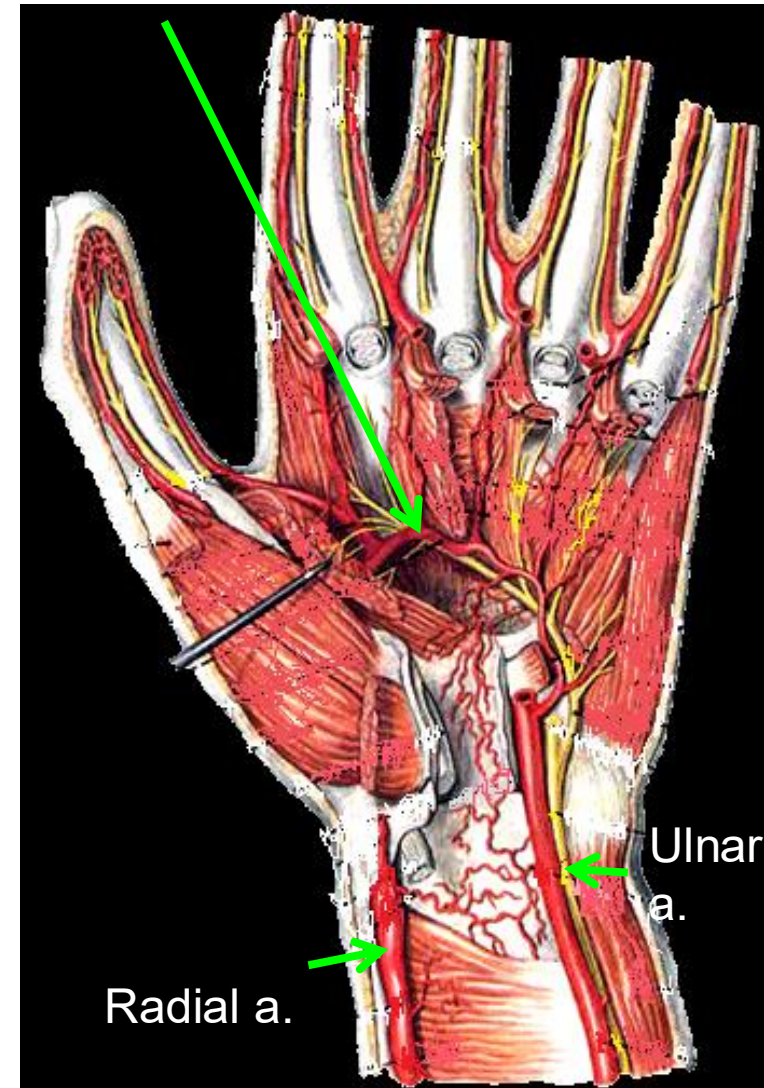
The Deep Palmar Arch

- Is a continuation of the **radial artery** as it curves medially beneath long flexor tendons, in front of the metacarpal bones and interosseous muscles.
- Is completed on the medial side by **deep branch of ulnar artery**.
- Lies at a level of the **proximal border of extended thumb**.
- It sends branches:
 - **superiorly** to share in anastomosis around the wrist joint &
 - **inferiorly** to join branches of the superficial palmar arch.

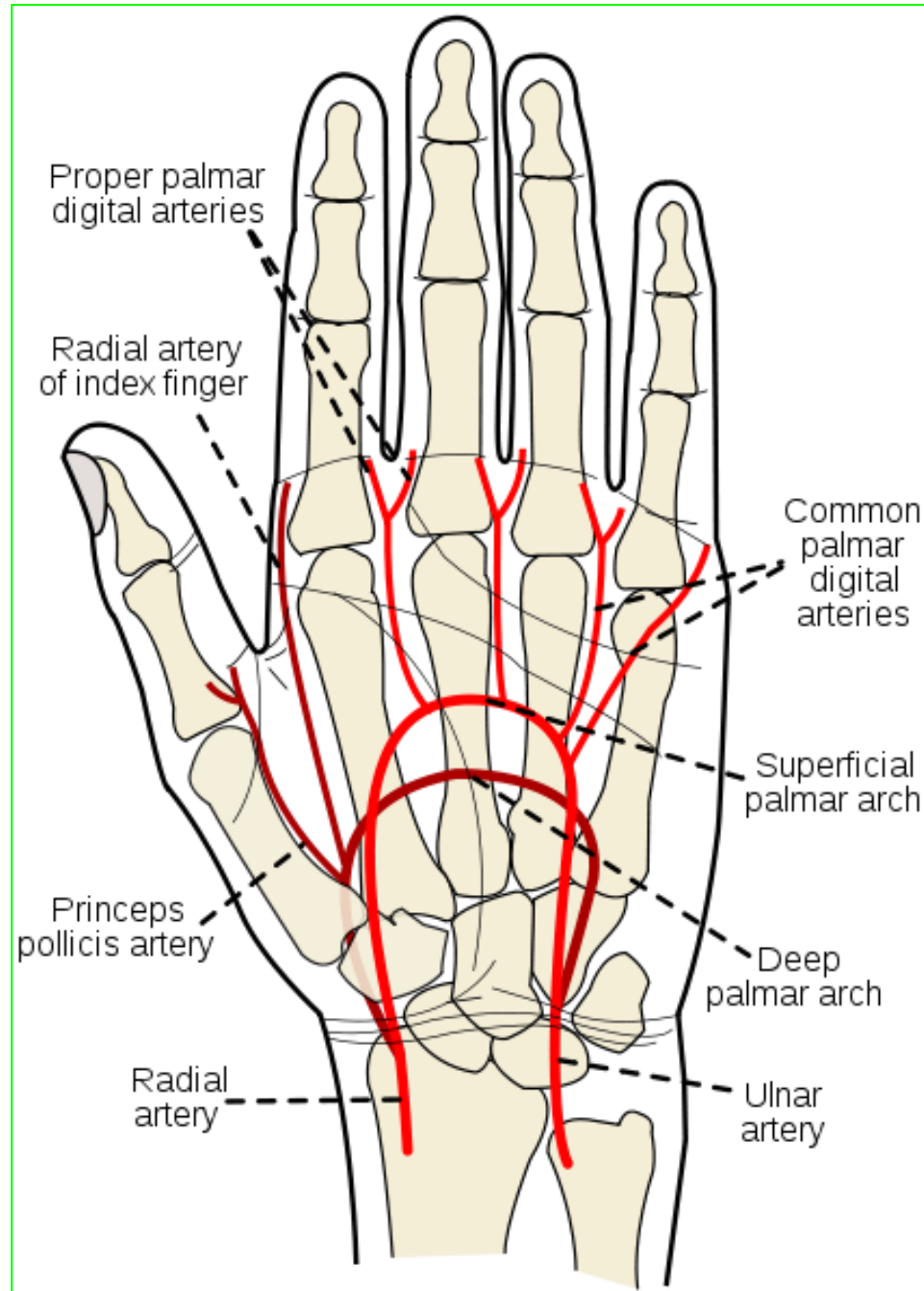


- Is a continuation of the **radial artery** as it curves medially beneath long flexor tendons , in front of the metacarpal bones and interosseous muscles.
- Is completed on the medial side by **deep branch of ulnar artery**.
- Lies at a level of the **proximal border of extended thumb**.
- It sends branches:
 - **superiorly** to share in anastomosis around the wrist joint &
 - **inferiorly** to join branches of the superficial palmar arch.

The Deep Palmar Arch



The superficial palmar arch is more distal than the deep palmar arch.



Abdominal Aorta

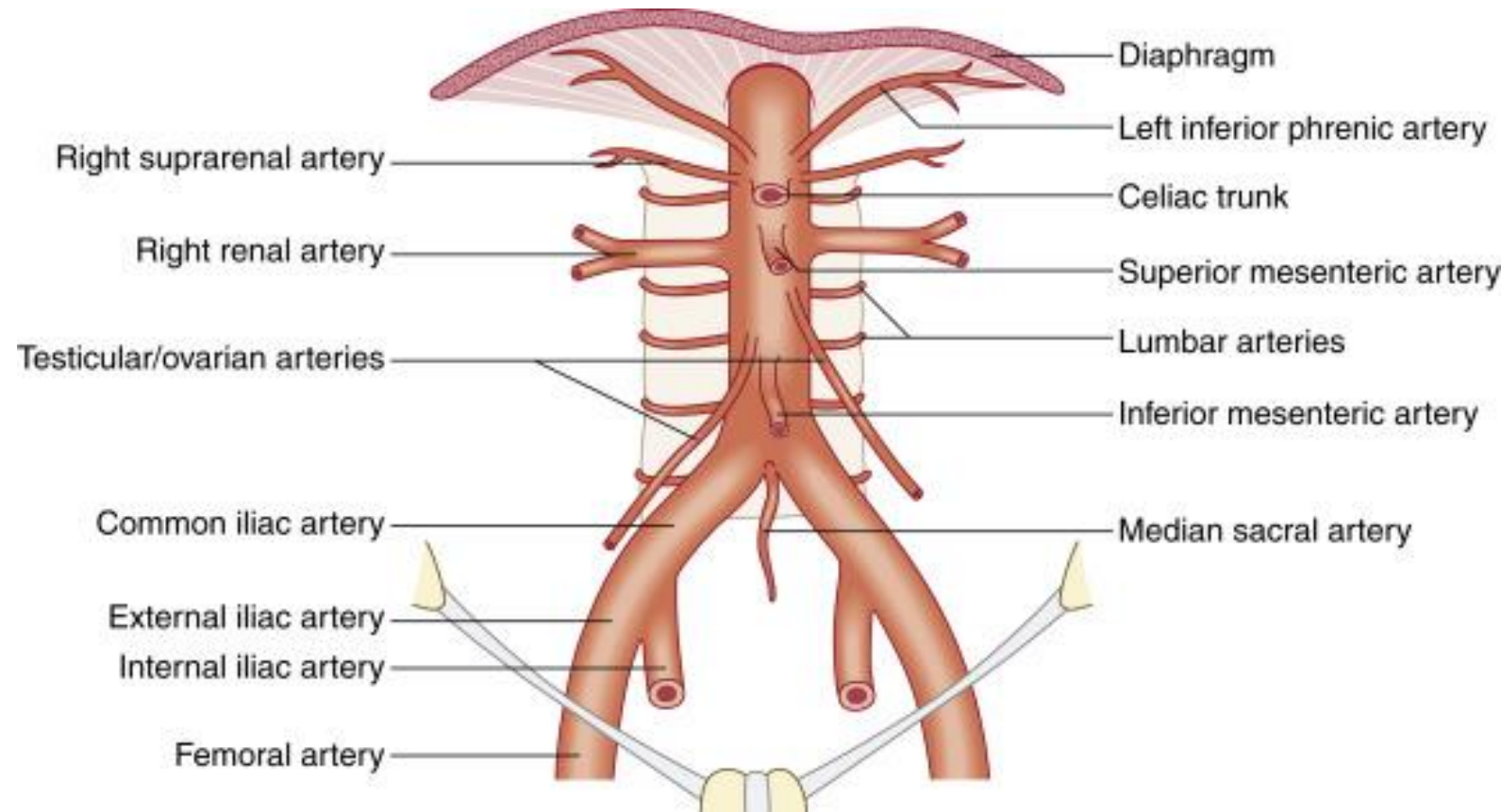
- ✓ Thoracic aorta passes behind the median arcuate ligament at T12 to become the **abdominal aorta**

Unpaired branches

- Celiac trunk (T12)
- Superior mesenteric artery (L1)
- Inferior mesenteric artery (L3)
- Median sacral artery

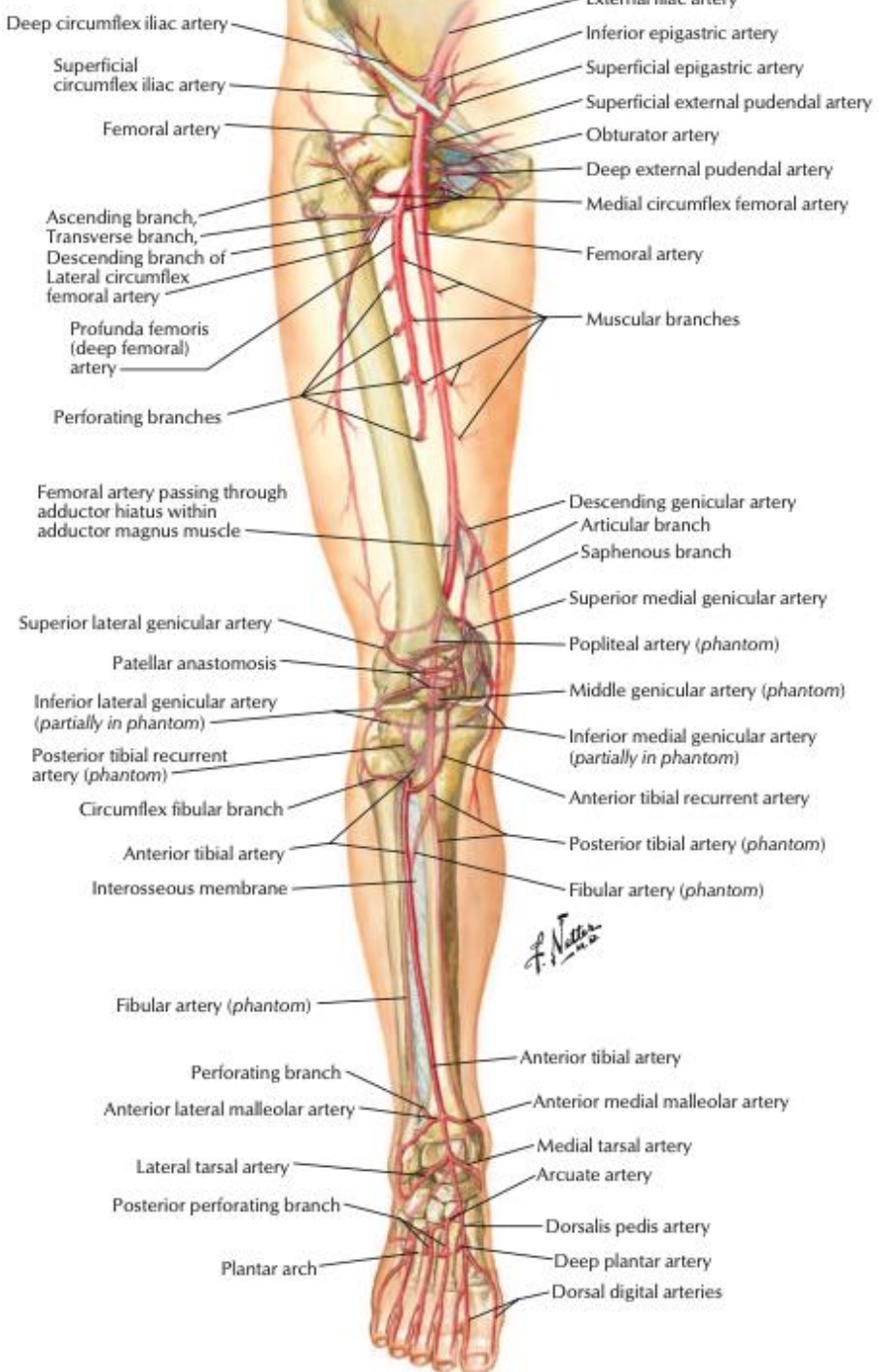
Paired branches

- Inferior phrenic aa.
- Superior suprarenal a.
- Middle suprarenal aa.
- Renal aa.
- Inferior suprarenal a.
- Gonadal (ovarian/testicular) aa.
- Lumbar (4 pairs) aa.



Arteries of the lower limb

- ✓ Femoral Artery
- ✓ Popliteal Artery
 - i. anterior tibial a.
 - ii. posterior tibial a.
- ✓ Dorsalis Pedis Artery
- ✓ Posterior Tibial Artery



Arteries of the lower limb

✓ Femoral Artery

✓ Popliteal Artery

i. anterior tibial a.

ii. posterior tibial a.

✓ Dorsalis Pedis Artery

✓ Posterior Tibial Artery

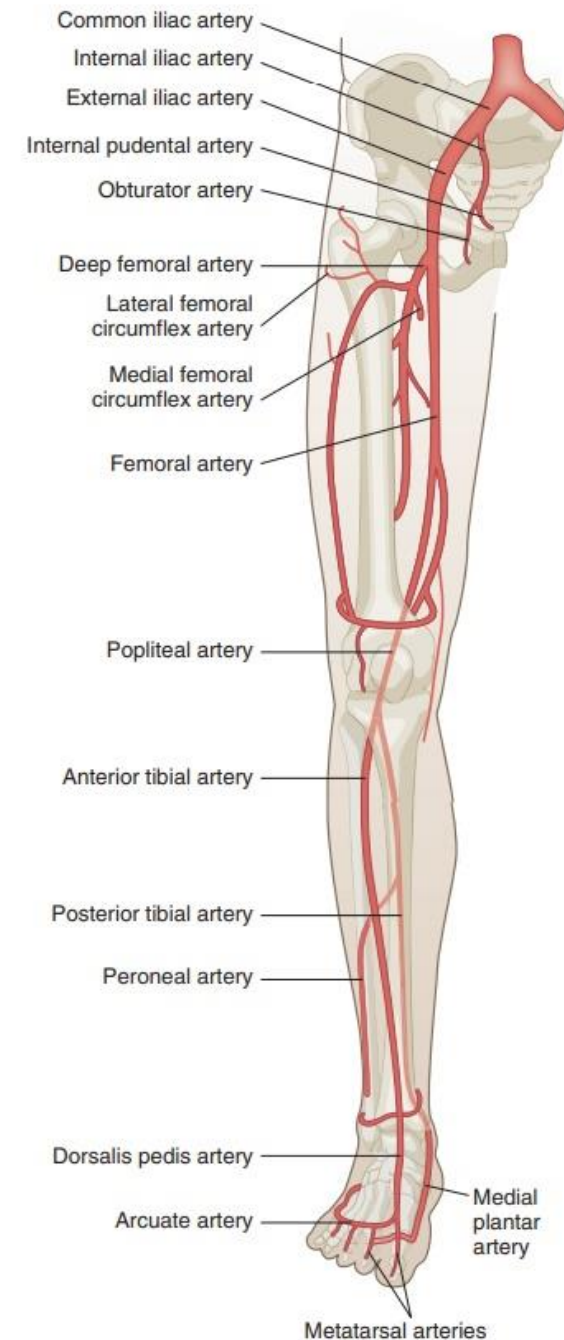
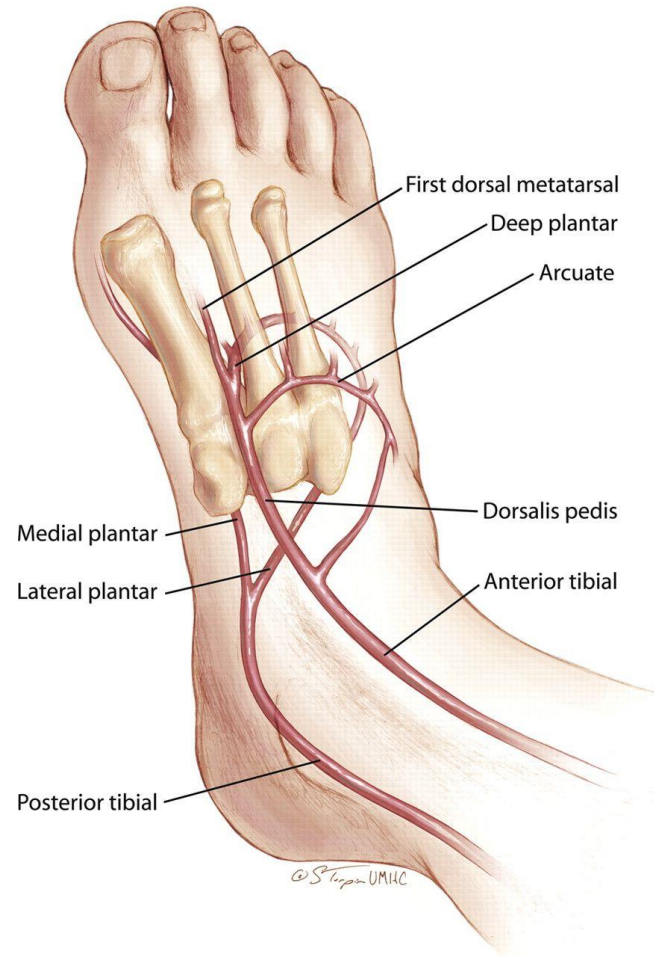
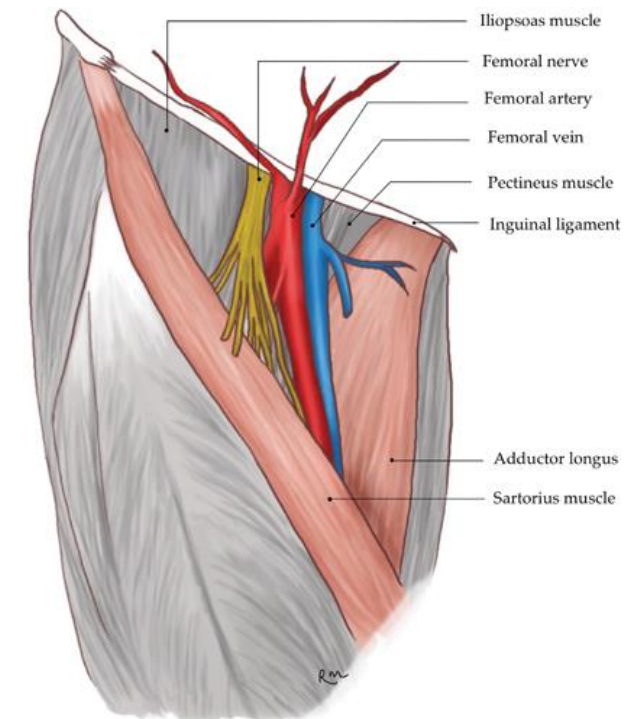
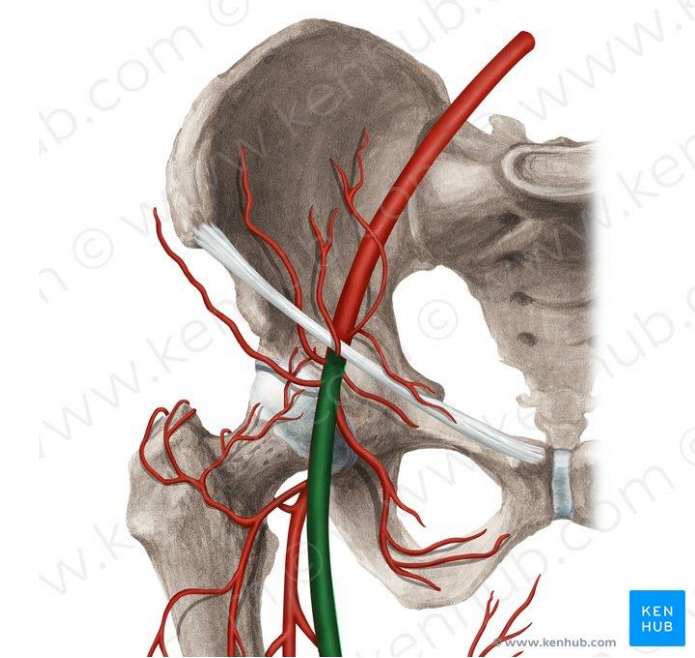


FIGURE 19-13 Pelvic and lower limb arteries.

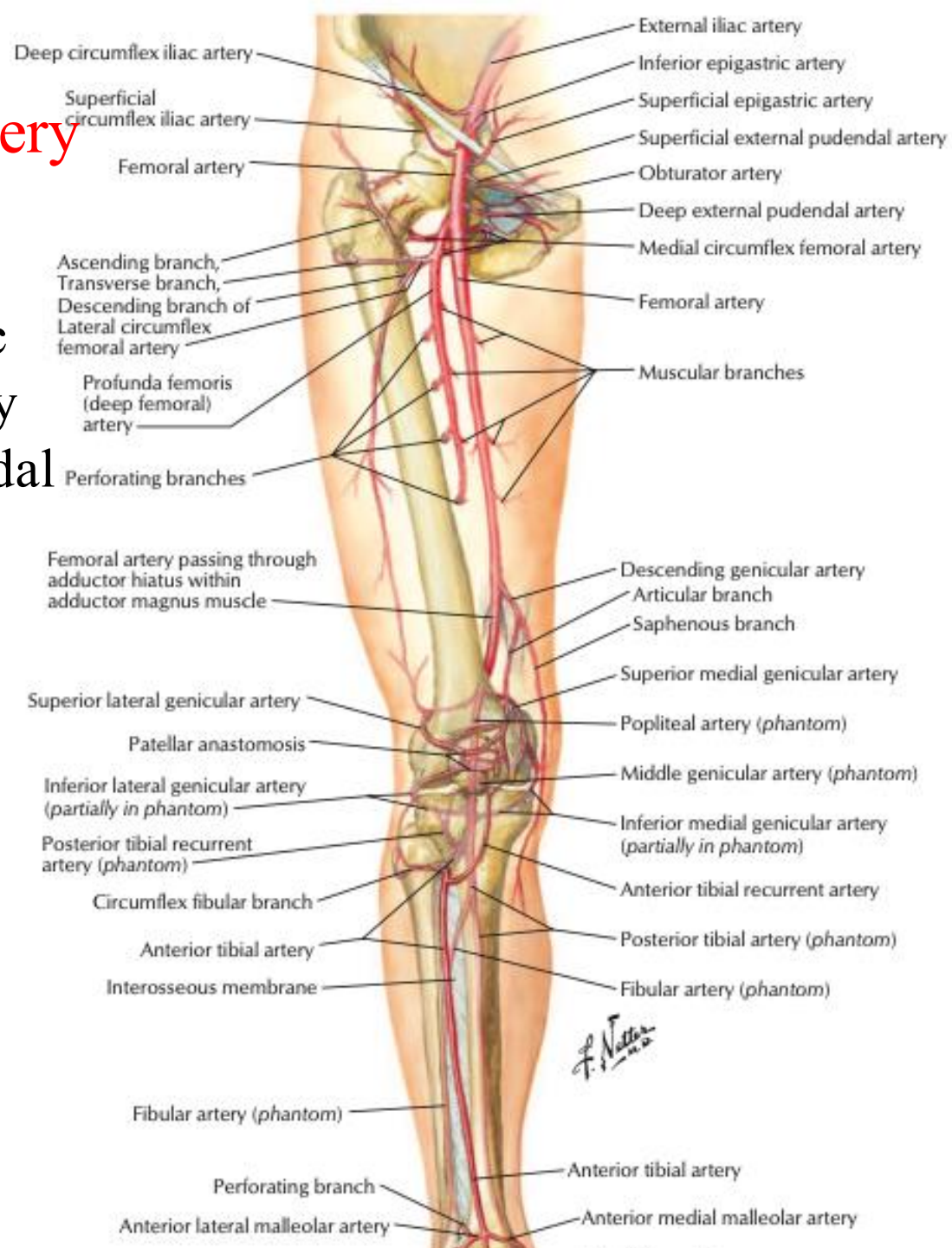
Femoral Artery

- ✓ Continuation of the external iliac artery.
- ✓ It begins behind the inguinal ligament, where it lies midway between the anterior superior iliac spine and the symphysis pubis.
- ✓ Descends through the femoral triangle & the adductor canal
- ✓ And it leaves the front of the thigh by passing thru the opening in the adductor magnus & then entering the popliteal space as the popliteal artery



Branches of Femoral Artery

- ✓ Superficial circumflex iliac
- ✓ Superficial epigastric artery
- ✓ Superficial external pudendal
- ✓ Deep external pudendal
- ✓ Profunda femoris artery
- ✓ Medial & lateral femoral circumflex
- ✓ Four perforating arteries.
- ✓ Descending genicular



Popliteal Artery

- ✓ Is a continuation of the femoral artery.
- ✓ It extends from the opening in the adductor magnus to the lower border of the popliteus muscle.
- ✓ It divides into:
 - i. anterior tibial a.
 - ii. posterior tibial a.

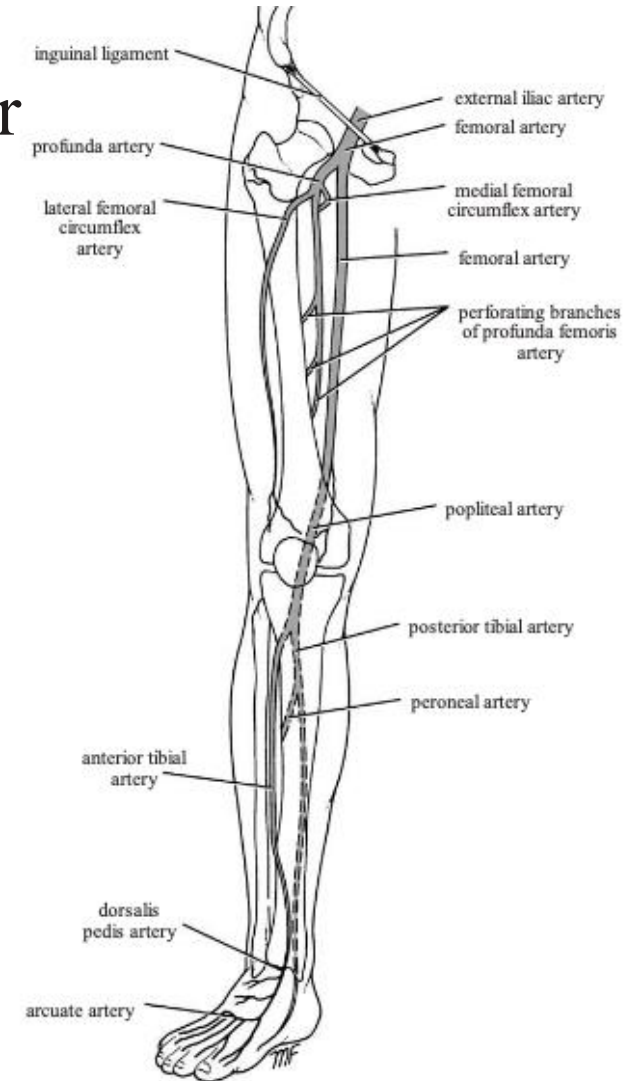
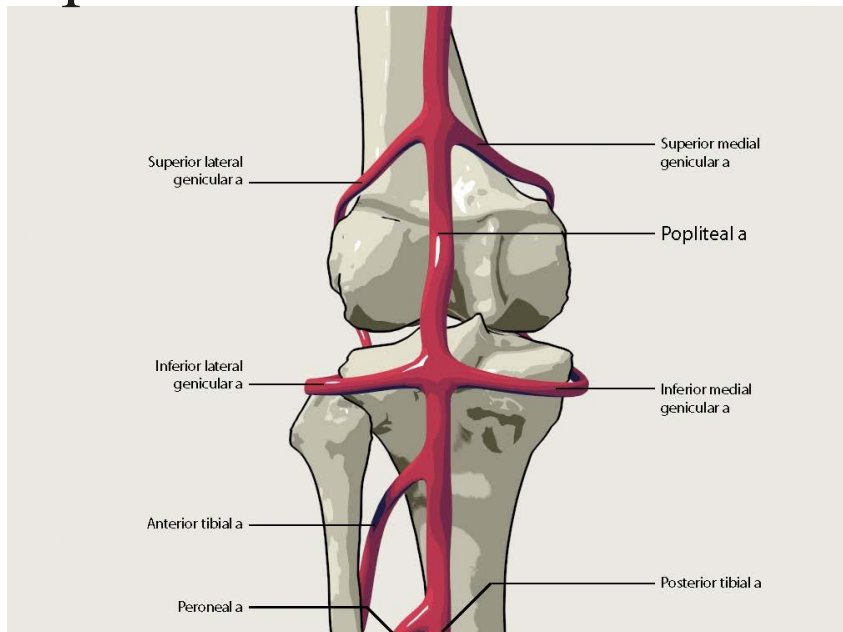
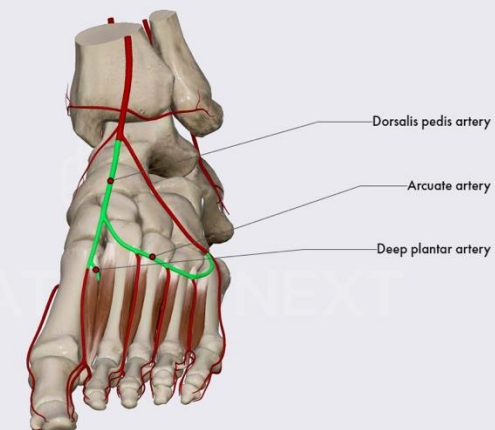
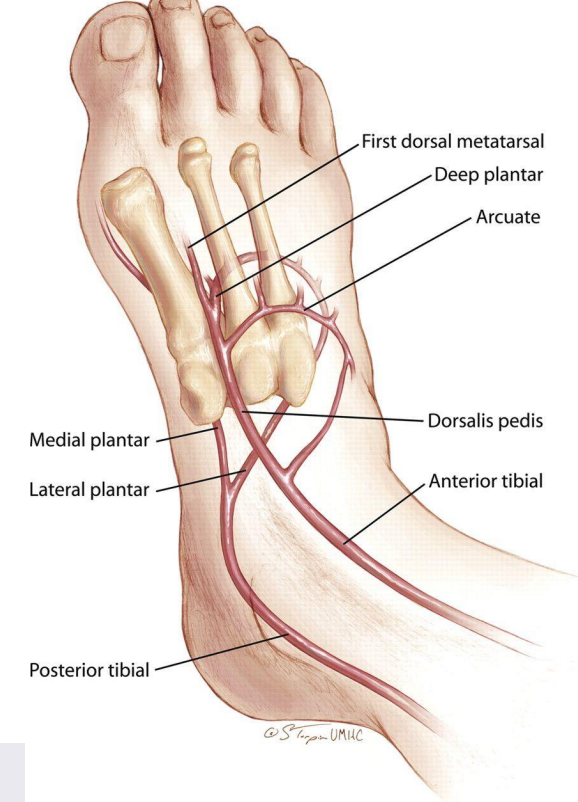


Figure 5-18 Major arteries of the lower limb.

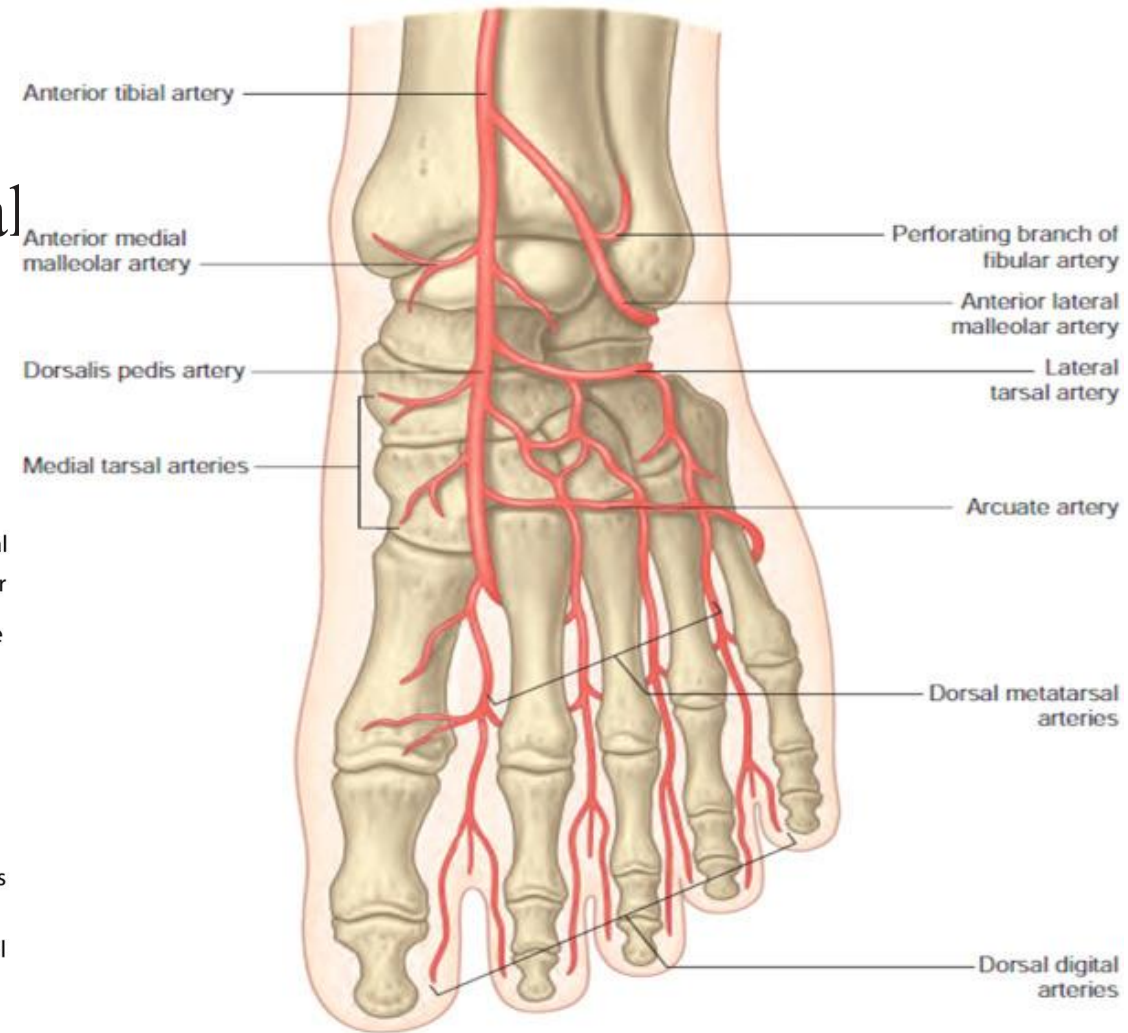
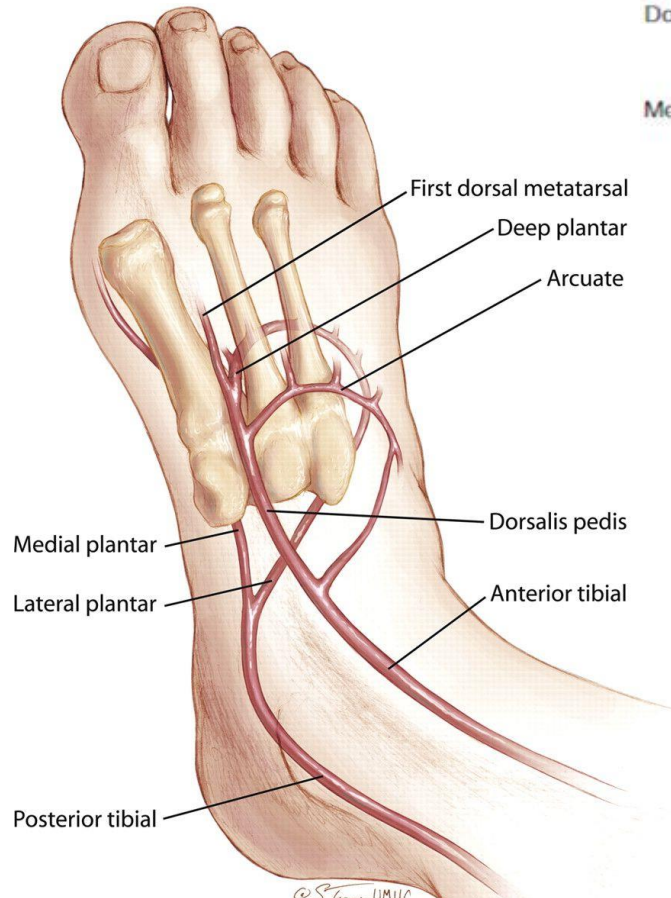
Dorsalis Pedis Artery

- ✓ Begins in front of the ankle joint midway between the malleoli.
- ✓ Is a continuation of the anterior tibial artery.
- ✓ It ends by entering the sole through the proximal part of the space between the 1st & 2nd metatarsal bones.
- ✓ it joins the lateral plantar artery and completes the plantar arch.



Branches of the Dorsalis Pedis Artery

- ✓ Lateral tarsal artery
- ✓ Arcuate artery
- ✓ First dorsal metatarsal arteries



Posterior Tibial Artery

- ✓ Arises at the bifurcation of the popliteal artery in the popliteal fossa.
- ✓ It descends in the posterior compartment of the leg & is accompanied by the tibial nerve.
- ✓ It terminates behind the medial malleolus by dividing into:
 - i. medial plantar a..
 - ii. lateral plantar a.

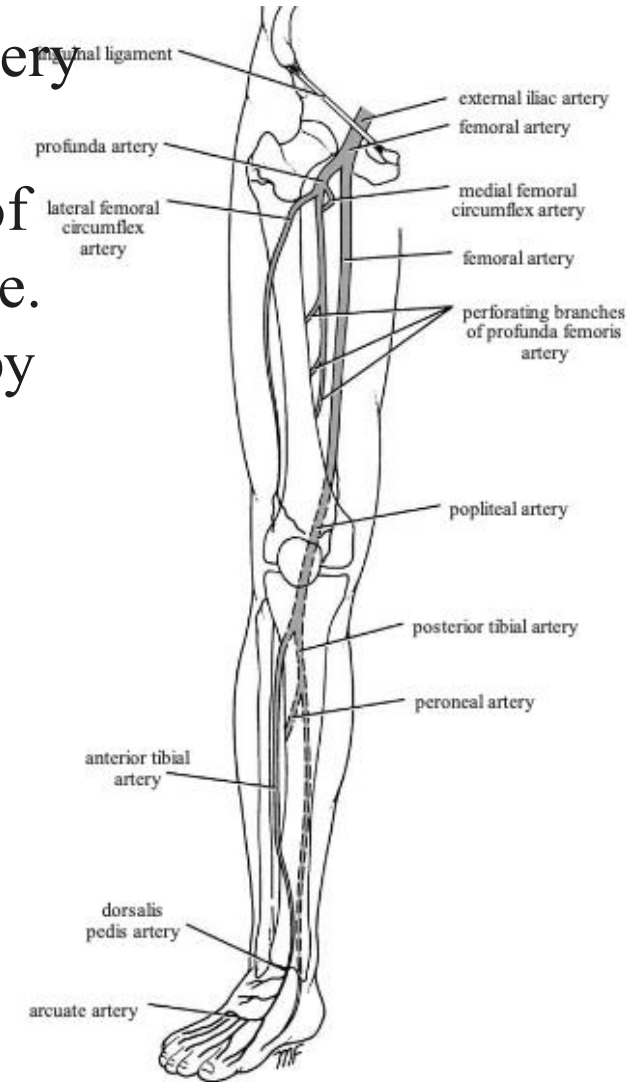
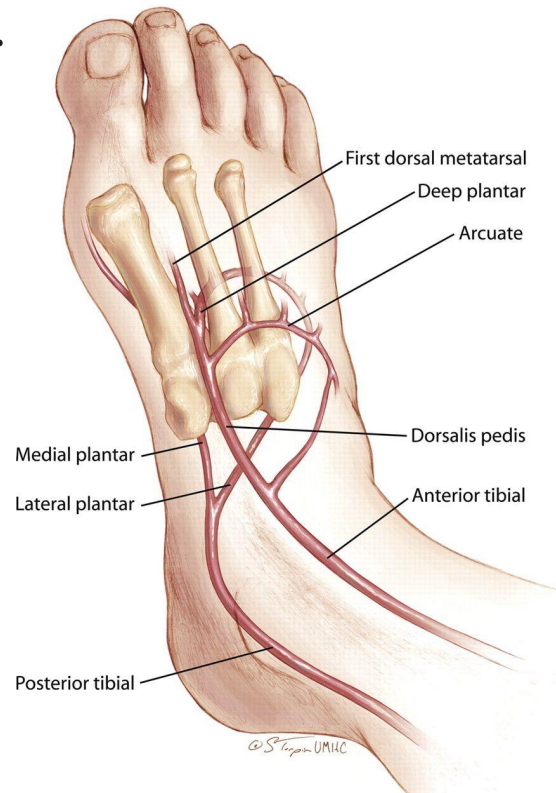
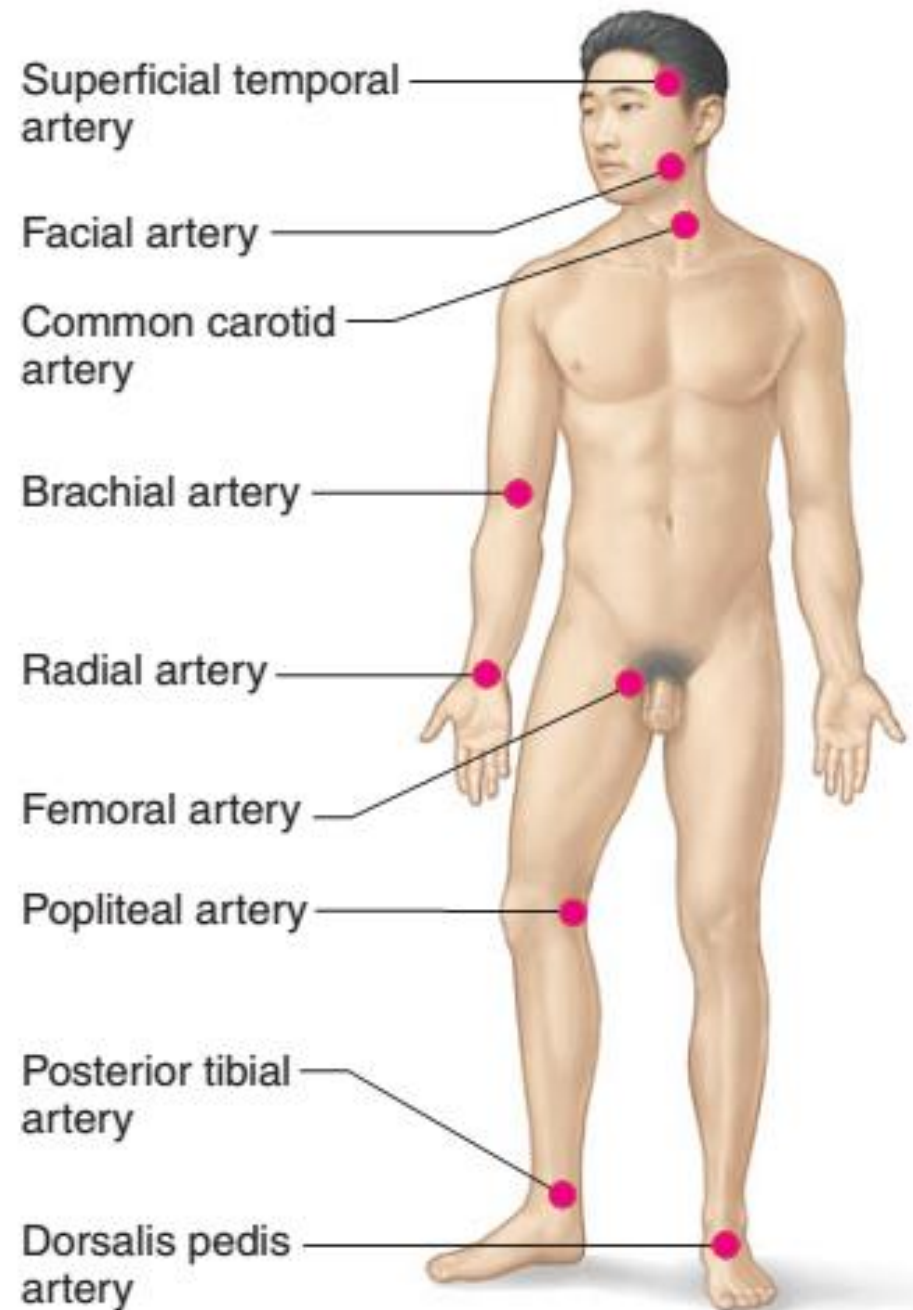


Figure 5-18 Major arteries of the lower limb.

At certain points along the pathway of the systemic arterial circulation, large and medium-sized arteries lie near the body's surface and can be used to take a **pulse** by compressing the artery against a hard underlying structure (usually a bone).

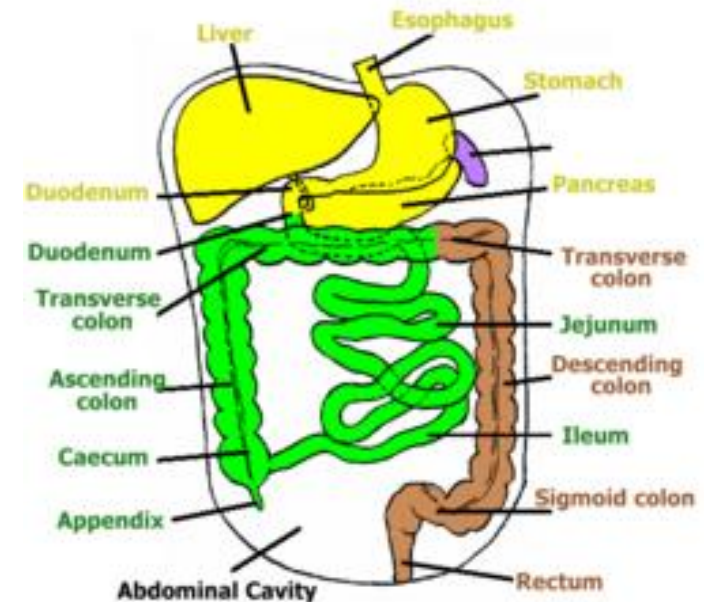
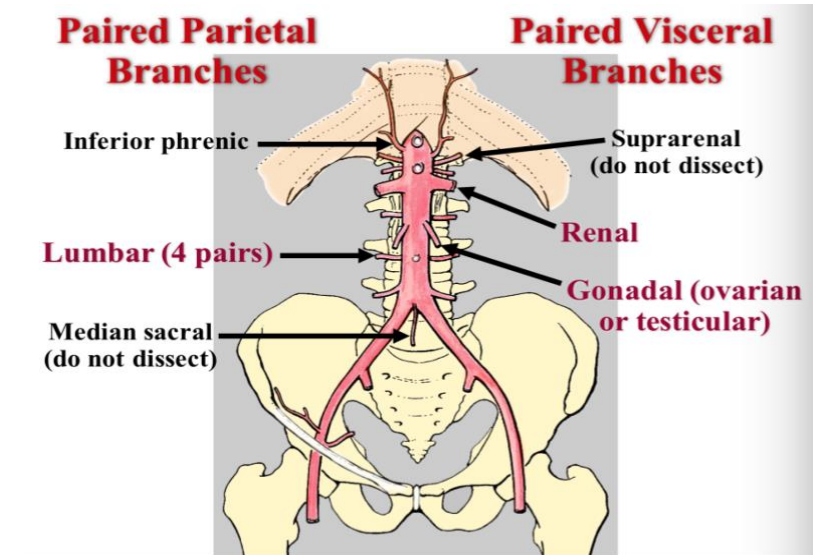


Abdominal Aorta

- Thoracic aorta passes behind the median arcuate ligament at T12 to become the **abdominal aorta**

Unpaired branches of abdominal aorta

- **Celiac trunk (T12)** - supplies the lower esophagus, stomach, duodenum (superior to the major duodenal papilla), liver, pancreas, gallbladder.
- **Superior mesenteric artery (L1)** - supplies the duodenum (inferior to the major duodenal papilla) up to the lateral $\frac{2}{3}$ of transverse colon (i.e. ileum, jejunum, caecum, appendix, ascending colon & $\frac{2}{3}$ of TC)
- **Inferior mesenteric artery (L3)** - supplies the distal $\frac{1}{3}$ of the transverse colon up to the upper part of the anal canal (pectinate line).
- **Median sacral artery** – supplies lower lumbar vertebrae, sacrum and coccyx

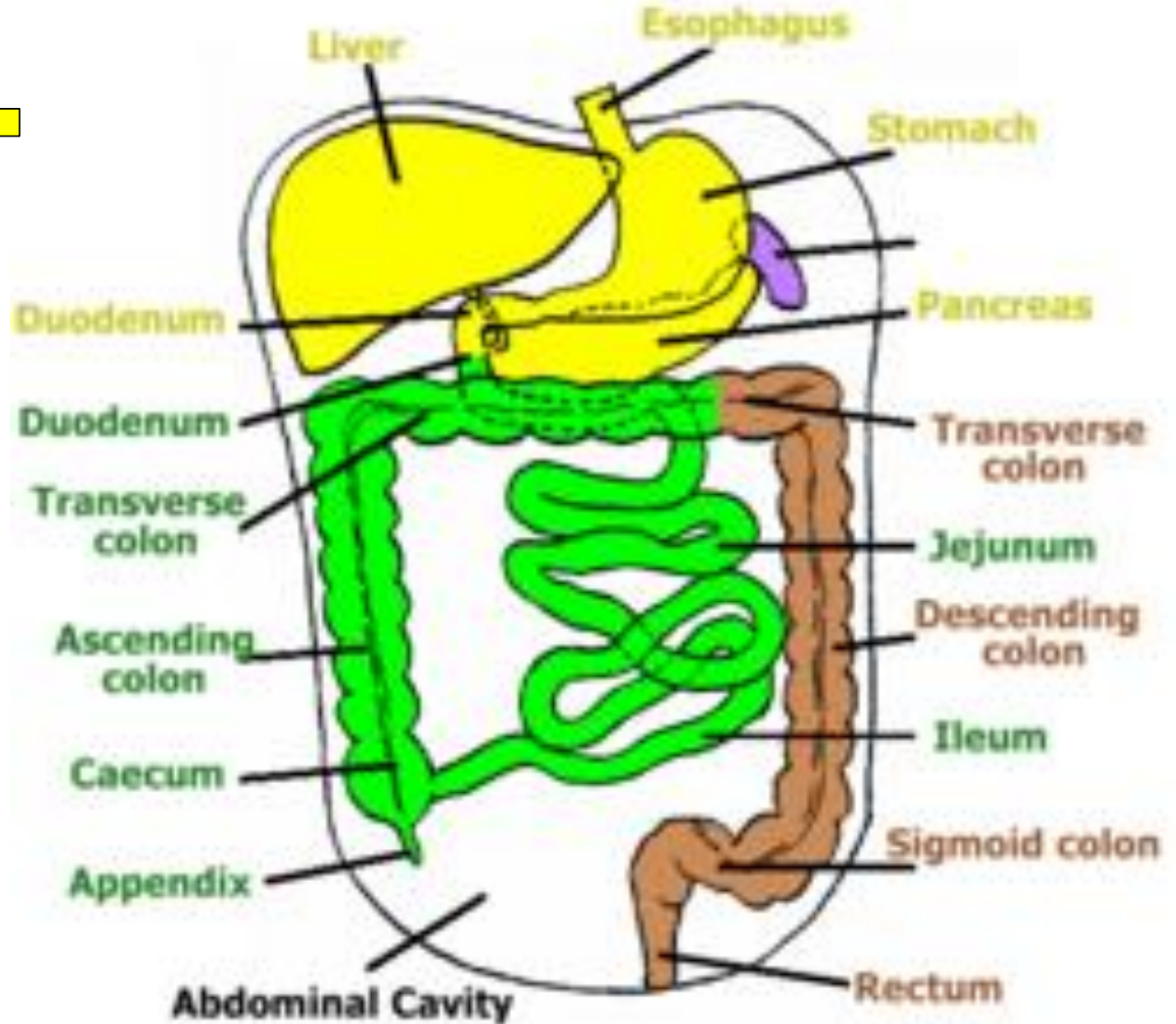


Abdominal Aorta

Area supplied by
celiac trunk

Area supplied by
SMA

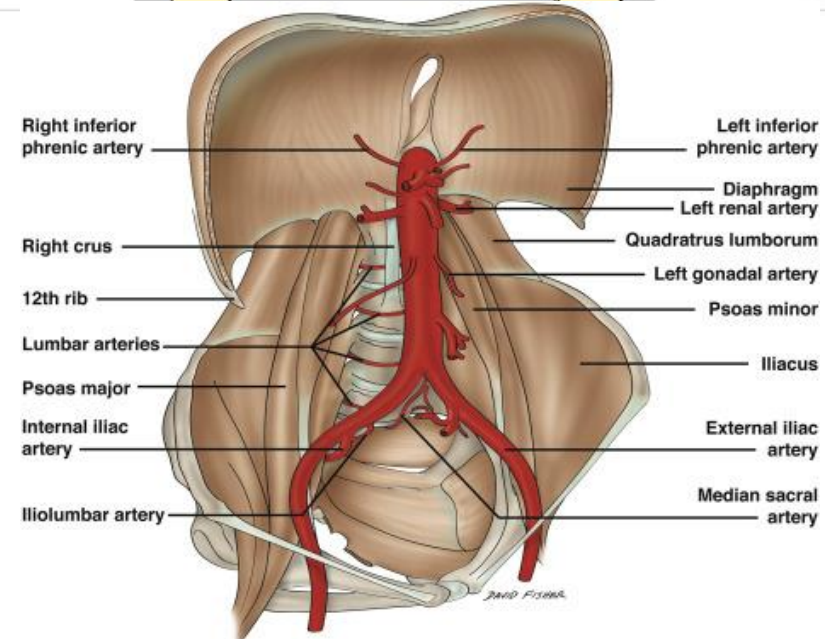
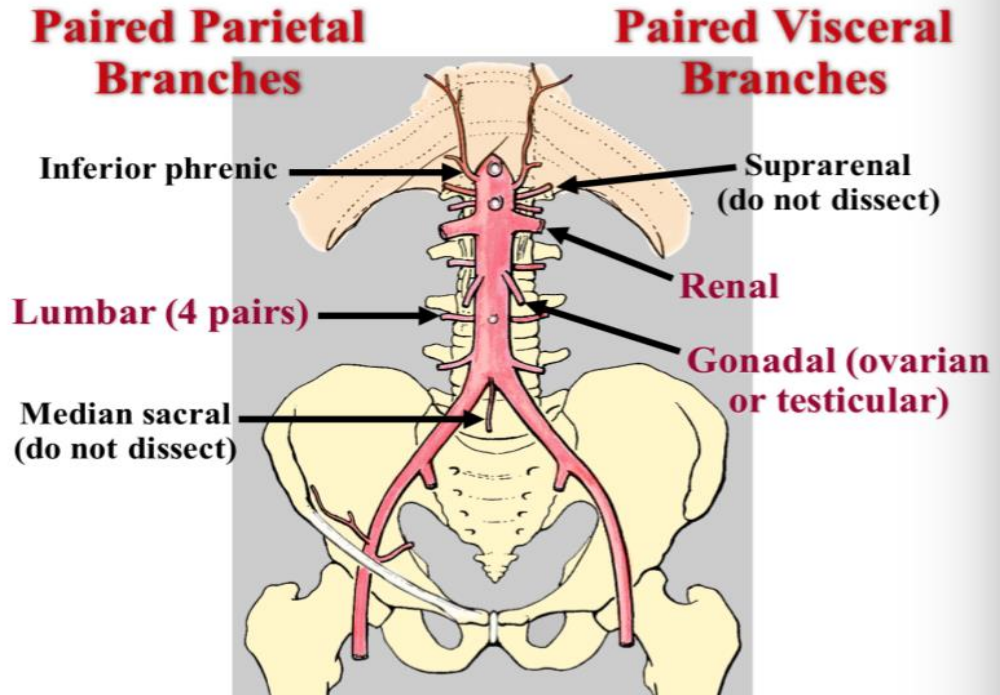
Area supplied by
IMA



Arteries of the Abdominal Aorta

Paired branches of the abdominal aorta

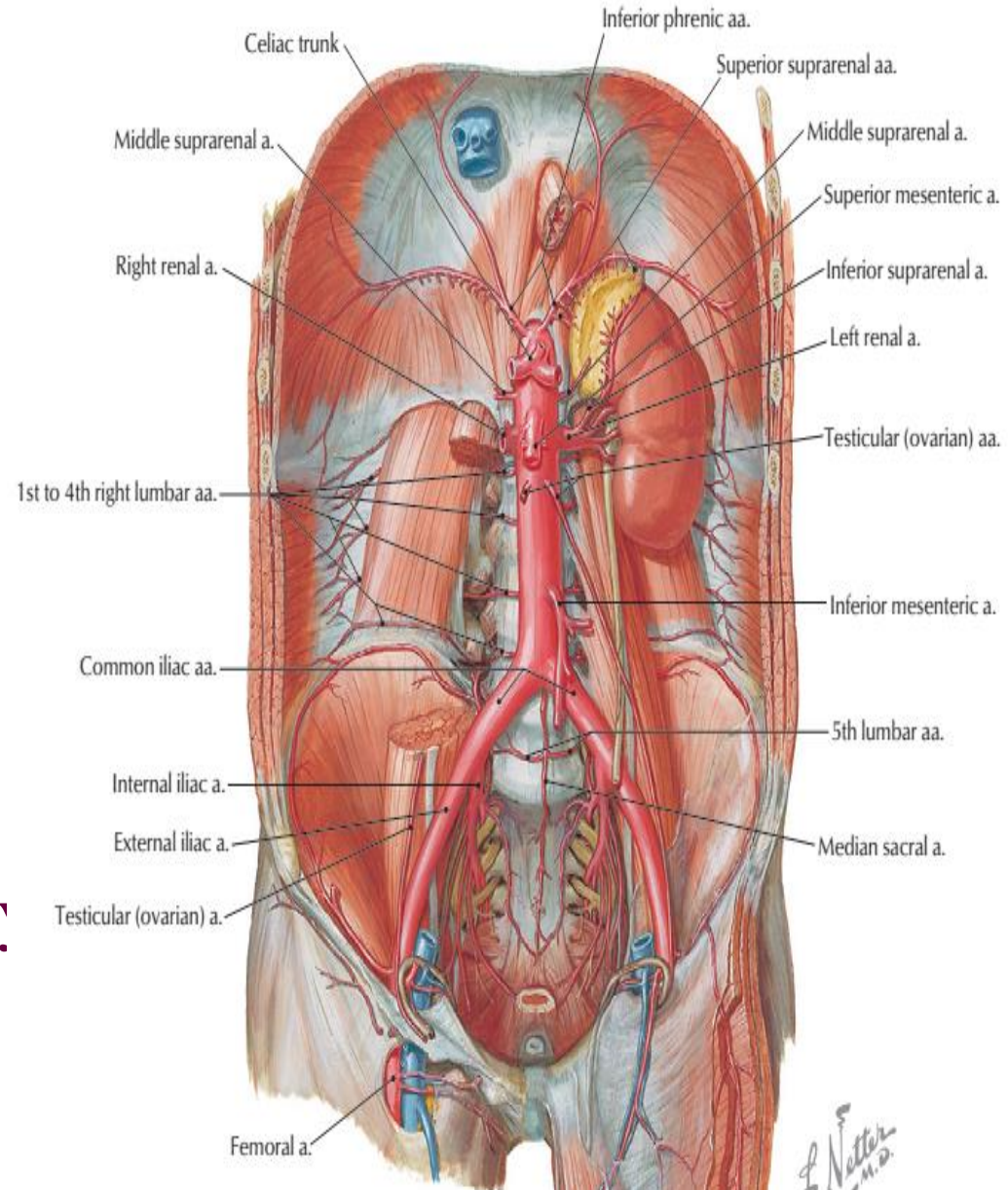
- **Inferior phrenic a.**
 - Superior suprarenal a.
- **Middle suprarenal a.**
- **Renal a.**
 - Inferior suprarenal a.
- **Gonadal**
(ovarian/testicular)
- **Lumbar (4 pairs)** - supply the posterior abdominal wall



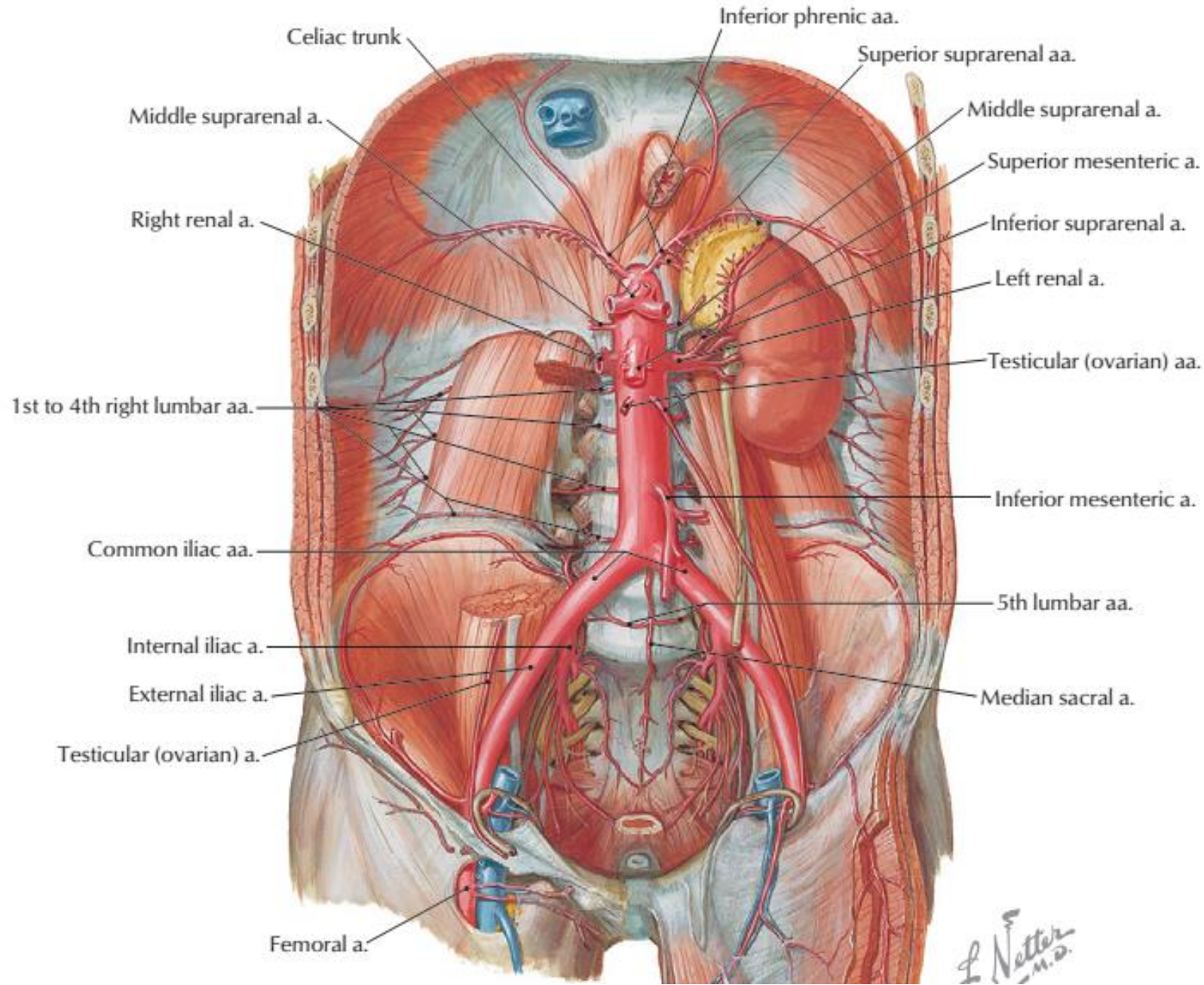
Arteries of the Abdominal Aorta

Paired branches of the abdominal aorta

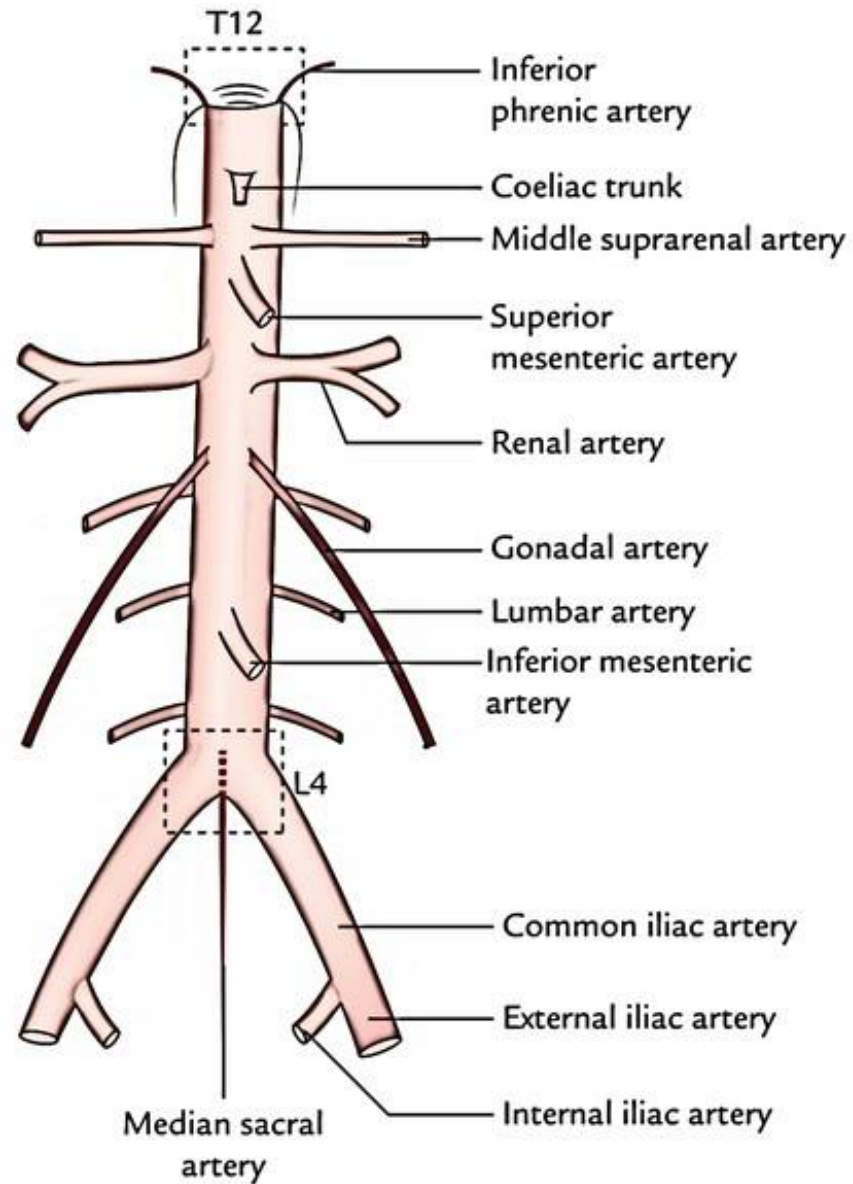
- **Inferior phrenic a.**
 - Superior suprarenal a.
- **Middle suprarenal a.**
- **Renal a.**
 - Inferior suprarenal a.
- **Gonadal**
(ovarian/testicular)
- **Lumbar (4 pairs)** - supply the posterior abdominal wall



Arteries of the Abdominal Aorta



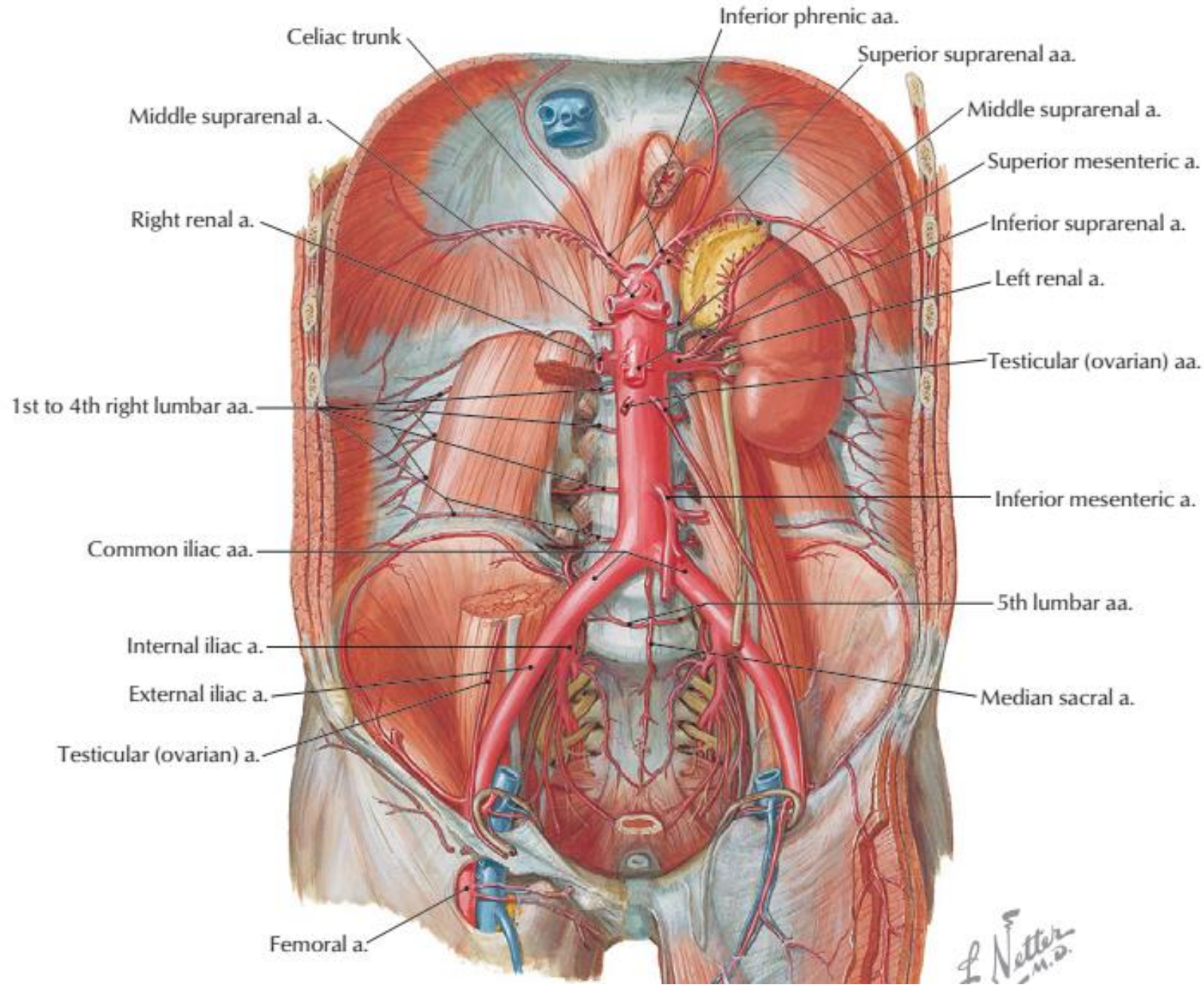
Arteries of the Abdominal Aorta



Arteries of the Abdominal Aorta

- Abdominal aorta ends at L4 where it bifurcates in front of the **sacroiliac joint** into right and left **common iliac arteries**. The **common iliac artery** bifurcates into:
 - **external iliac artery**
 - Gives rise to **inferior epigastric artery** and **deep circumflex iliac artery**.
 - Becomes the **femoral a.** after passing behind the inguinal ligament
 - **internal iliac artery**

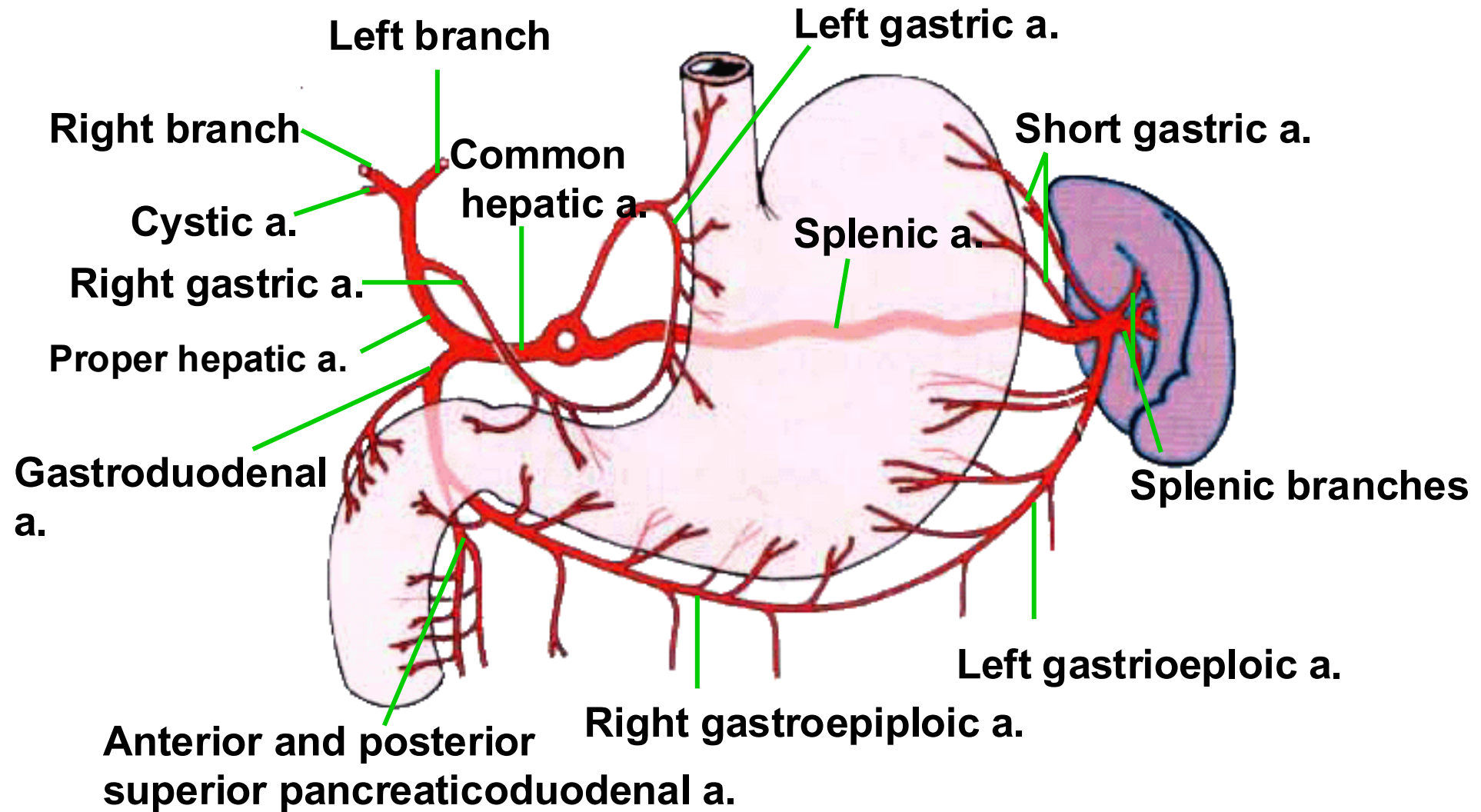
Arteries of the Abdominal Aorta

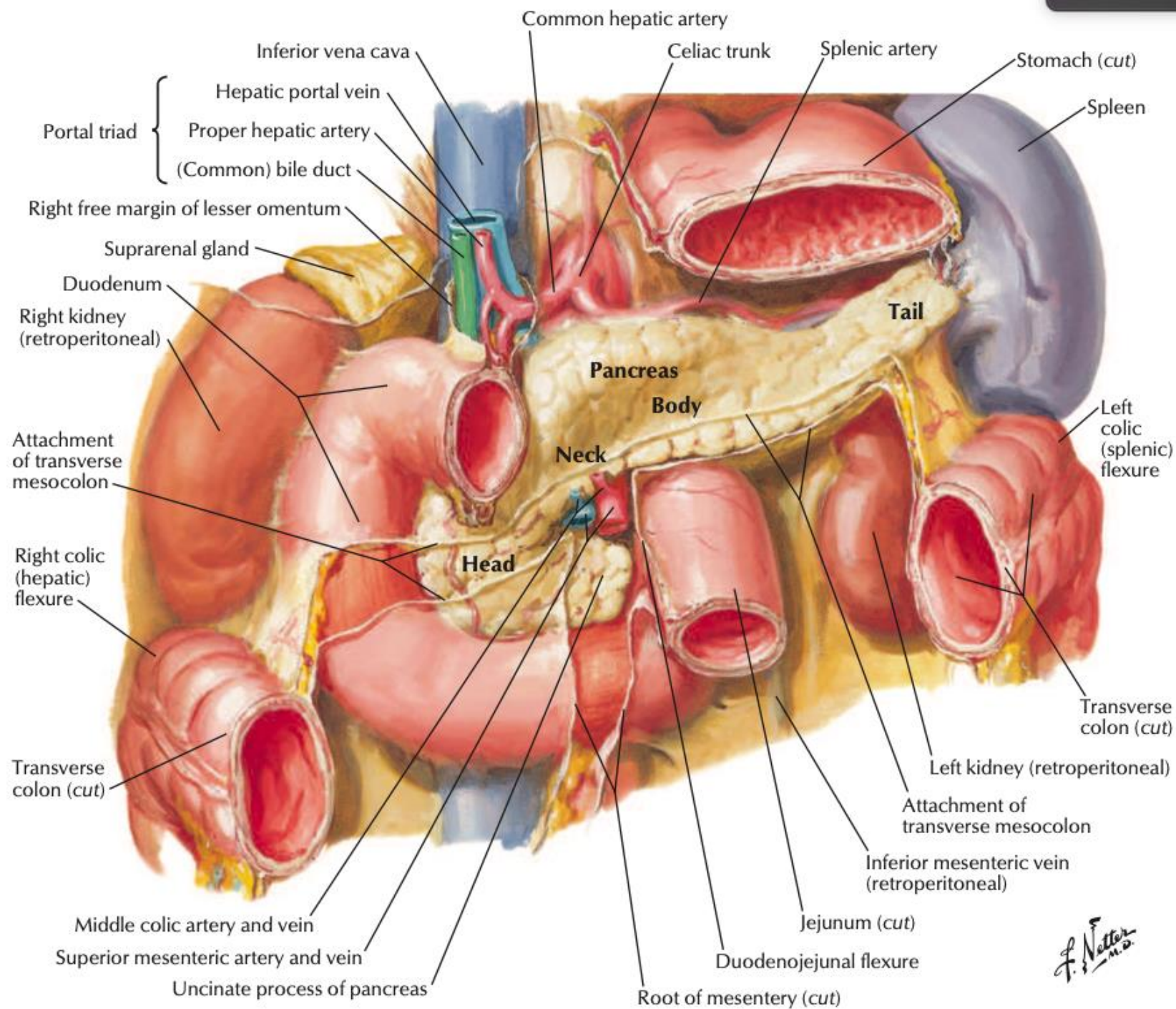


Celiac Trunk (foregut derivatives and spleen)

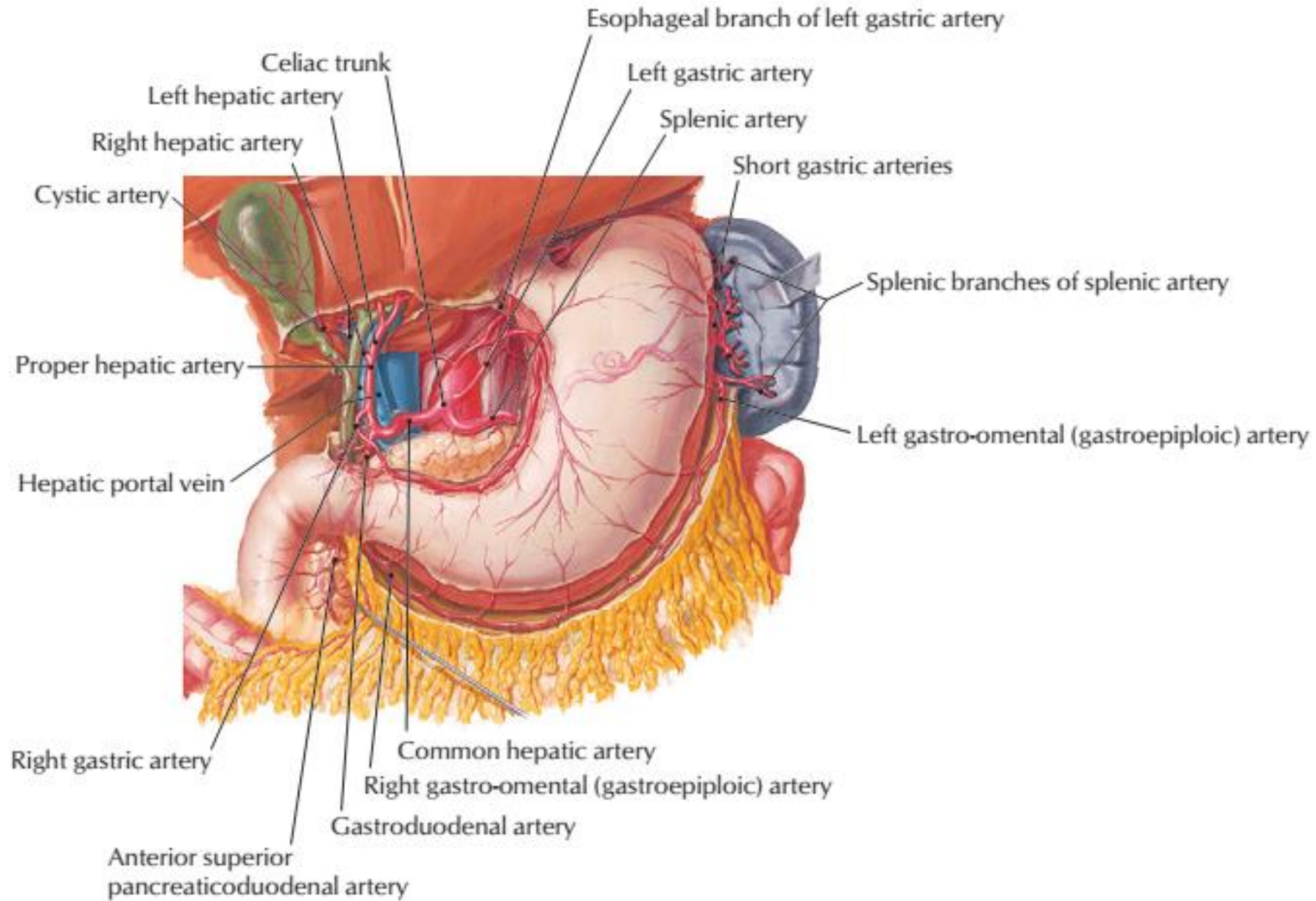
- **Splenic a.** - supplies the spleen, stomach, and pancreas (neck, body, and tail)
 - Short gastric arteries (6#)
 - Left gastroepiploic/gastro-omental a
 - Pancreatic branches – dorsal pancreatic a, great pancreatic a, posterior gastric a (occasionally)
- **Left gastric a.** — supplies distal esophagus and lesser curvature of stomach
- **Common hepatic a.** - supplies the liver, gallbladder, stomach, duodenum, and pancreas (head and neck)
 - Hepatic artery proper
 - Right and left hepatic aa. The R often gives the cystic a.
 - Right gastric a
 - Gastroduodenal a
 - Right gastroepiploic/gastro-omental
 - Superior pancreaticoduodenal a (smaller anterior and larger posterior branches)
- **Supraduodenal aa.** — may arise from the common hepatic artery or its branches

Celiac trunk





F. Netter
M.D.



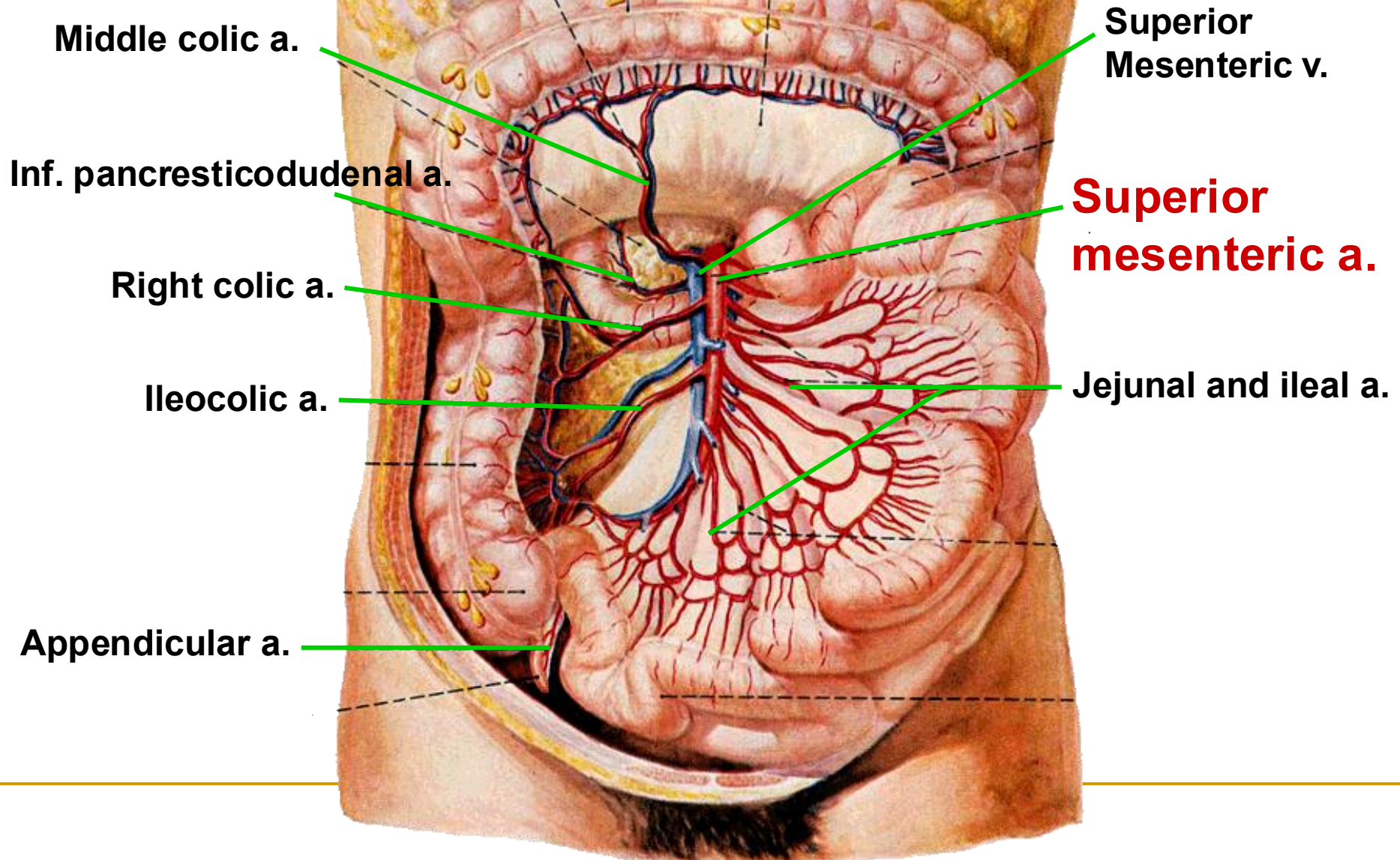
Pringle's manoeuvre?

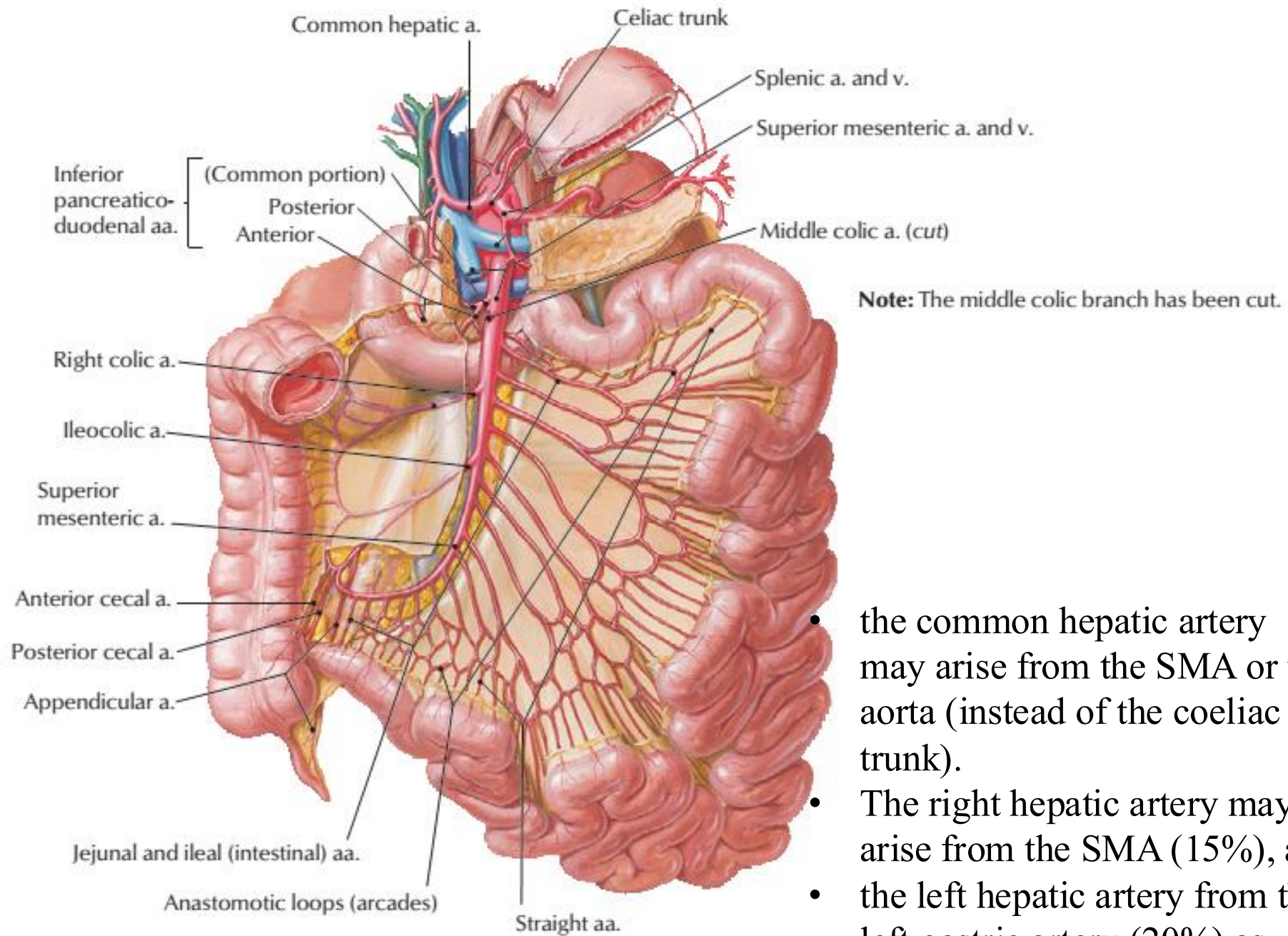


Abdominal Aorta

- **Superior mesenteric artery (L1), artery to midgut**
 - Inferior pancreaticoduodenal a. (anterior and posterior branches)
 - supplies duodenum (distal to entry of bile duct), pancreas, and spleen
 - Jejunal and ileal branches
 - Middle colic a.
 - Right colic a.
 - Ileocolic a.- supplies the ileum, cecum, appendix, and proximal ascending colon
 - superior division - anastomose with the right colic artery
 - inferior division
 - anterior and posterior caecal arteries. Larger posterior branch gives appendicular a. (end artery)
- **Inferior mesenteric artery (L3), artery to hindgut**
 - Left colic artery
 - Sigmoid aa.
 - Superior rectal a (terminal branch) - supplies the distal sigmoid colon and proximal rectum

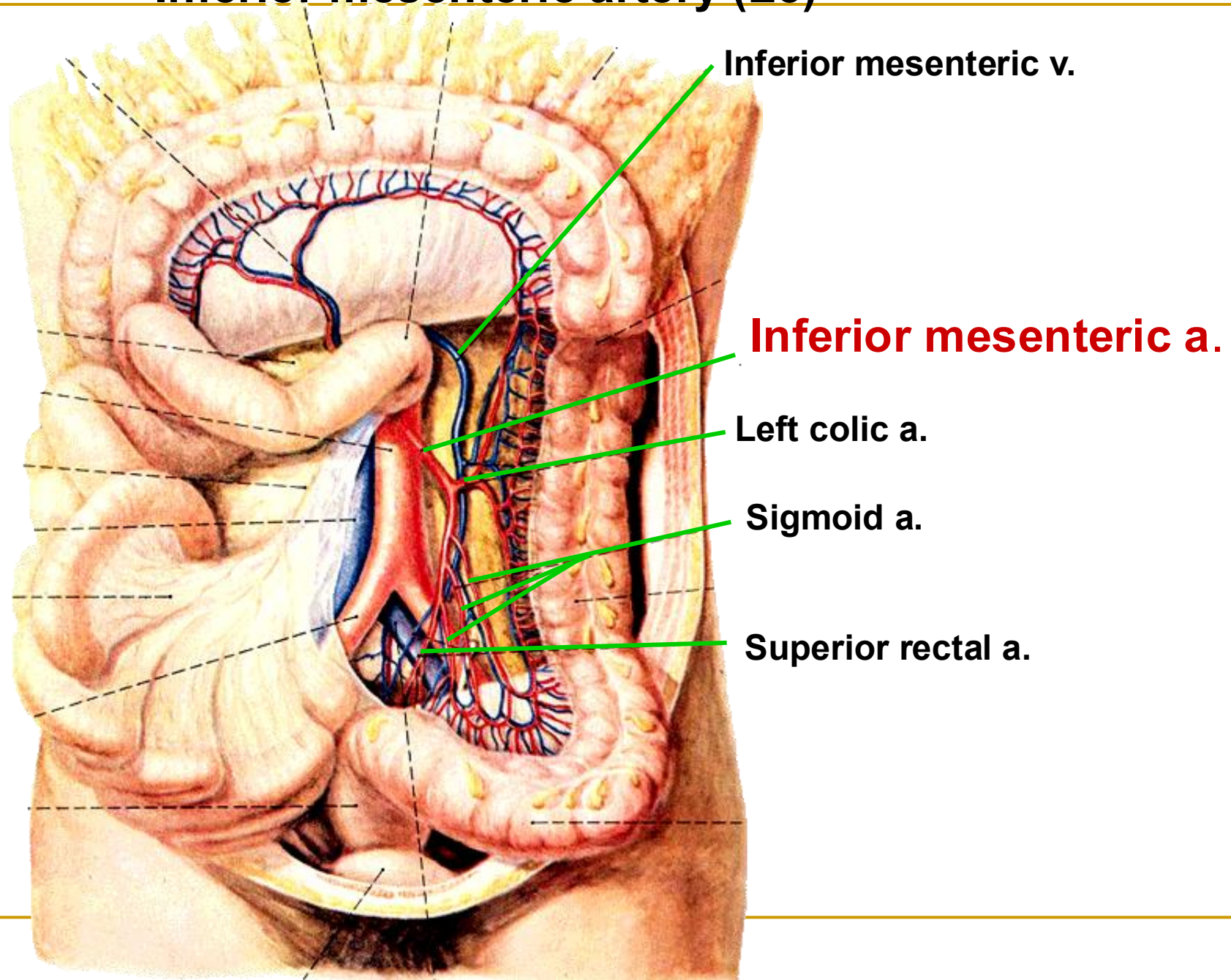
Superior mesenteric artery (L1), artery to midgut





- the common hepatic artery may arise from the SMA or the aorta (instead of the coeliac trunk).
- The right hepatic artery may arise from the SMA (15%), and the left hepatic artery from the left gastric artery (20%) as aberrant or accessory arteries

Inferior mesenteric artery (L3)



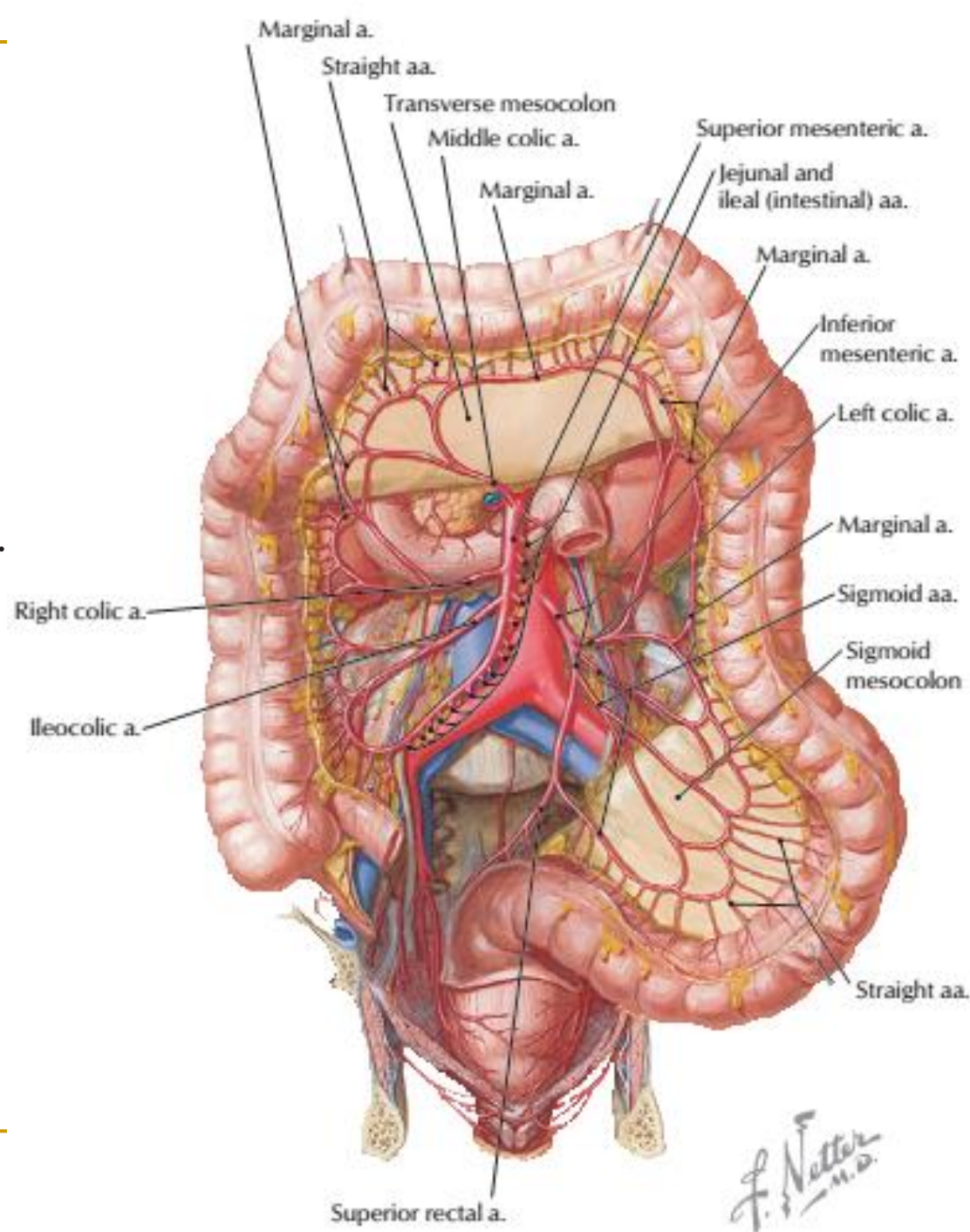
Marginal artery (of Drummond) anastomosis of branches of the ileocolic, right colic, middle colic, left colic, and sigmoid arteries. It forms a collateral circulatory circuit for the large intestine. .

Griffith's point?

Sudeck's point?

Arc of Rolan?

Watershed area of the colon?



Distribution of the Superior and Inferior Mesenteric Arteries

